Notes on Basic Category Theory

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Preface

I take ‘Basic Category Theory’ to include, roughly speaking, the topics of the first five of the twelve chapters in Mac Lane’s classic *Categories for the Working Mathematician*. So after initial definitions and examples of categories (and of ways of constructing new categories from old) we need to talk – in some order or other – about functors between categories and natural transformations between functors, about limits, about adjoints, about representables and the Yoneda Lemma, and about the relations between these things. These are also the topics of (for example) all but the last chapter of Awodey’s admired though uneven *Category Theory* and of the whole of Tom Leinster’s quite terrific – and appropriately titled – *Basic Category Theory*.

But then, if there are some excellent texts out there, not to mention various sets of notes on category theory available online (see here), why produce *these* notes?

I didn’t intend to! My goal all along was to understand better how category theory, set theory, and logic fit together (and to understand what light the first throws on the second and third). But I decided that I initially needed to get a lot more securely on top of basic category theory if I was to pursue these more philosophical issues further. And so these ‘Notes on Basic Category Theory’ began life as detailed jottings for myself, to help fix ideas. They were prompted in particular by following a course of 24 lectures given for Part III of the Maths Tripos by Rory Lucyshyn-Wright in the Michaelmas Term, 2014. These notes very quickly departed from the details of those lectures in both style and content, especially as I read around on the topics being covered. And I’m taking advantage of the lack of time-constraints to be much more discursive. However, for better or worse, some of the overall structure of the course is still there, in particular in the order in which topics are currently tackled.

So what’s distinctive, for good or ill, about these Notes is that they are written by someone who doesn’t pretend to be a fully-formed expert in category theory. That’s why I go slowly over ideas that initially gave me pause, and have generally tried to be as clear as possible in an introductory way which I hope might help others in the same boat when starting to get to grips with categories. Despite the length, however, the level of sophistication we reach here is rather less than that eventually required of Part III students, and the coverage falls somewhat short even of the books by Awodey and Leinster, let alone the typical Part III syllabus (to see what we are missing, you could take a look at the excellent notes, Goedecke 2013).

These Notes are now in a complete version – complete at least in the sense that (at least for the moment) I do not intend to press on to add further chapters on significant new topics like monads or abelian categories. Rather the plan is to leave these notes in more or less their present form, for all their shortcomings: I hope in due course to write a differently organized, more discursive, bigger and better version. Still, I would very much like to hear about errors of one kind or another. And I may issue occasional 'maintenance upgrades' of this version when I hear about mistakes or spot passages which really won't do.

1 Introduction

1.1 Why categories?

Here is a fundamental insight: we can think of a family of mathematical structures equipped with some structure-preserving maps/morphisms/functions between them as itself forming a further, non-trivial, mathematical structure.

For example: a particular individual group is a structure (some objects equipped with an operation on them obeying some familiar axioms). But we can also think of the family of groups taken together with maps between them – the homomorphisms which preserve group structure – as themselves forming a further structure-of-structures.

We can then investigate such structures-of-structures, and indeed go on to think about structure-preserving maps – as they say, functors – between *them*. Going up another level, we can talk in turn about maps between such functors.

These layers of increasing abstraction are the topic of category theory. So if modern mathematics already abstracts (moving, for example, from concrete geometries to abstract metric and topological spaces), category theory abstracts again, and then again. What, if anything, do we gain by going up another level of abstraction or two? To quote Tom Leinster,

> Category theory takes a bird's eye view of mathematics. From high in the sky, details become invisible, but we can spot patterns that were impossible to detect from ground level. (Leinster, 2014, p. 1)

So that suggests one justification for ascending to the dizzying heights: we are told that we will find some large-scale but still illuminating patterns in mathematics which aren't otherwise discernible. Which is perhaps more than enough reason for mathematicians, or at least those mathematicians with a certain generalizing cast of mind, to be interested in category theory.

What about others? For a start, category theory provides new ways of framing physical theories in a structural form; so some interested in the foundations of physics have found themselves working in a categorial framework. And it isn't surprising that category theory, a sort of generalised theory of maps or functions, should turn out to have close links to the lambda calculus, another sort of generalized theory of functions, already of concern to logic and computer science. There turn out to be other connections too which make at least some aspects of category theory of more general interest to logicians and theoretic computer scientists.

Again, philosophers – at least those interested in questions about the sense in which modern pure mathematics is, in some basic way, a study of abstract structures – will find category theory illuminating. Indeed, more generally, the theory should appeal to a certain philosophical temperament. Thus many philosophers, pressed for a short characterization of their discipline, like to quote a famous remark by the profound American philosopher Wilfrid Sellars,

> The aim of philosophy, abstractly formulated, is to understand how things in the broadest possible sense of the term hang together in the broadest possible sense of the term. (Sellars, 1963, p. 1)

This conception sees philosophy, not as the self-indulgent pursuit of linguistic games or of obscurely edifying discourse, but as continuous with other serious enquiries, seeking to develop an overview of how their various subject-matters are interrelated. And category theory helps us to understand part of the story, as it explores the logical geography of aspects of abstract mathematics.

There are solid reasons enough, then, for those pursuing various disciplines to take an interest in category theory. However, various advocates have made much grander claims for the novelty and fundamental importance of categorial insights. For example, some proponents have suggested that category theory should replace set theory in the role of, so to speak, the official foundations of mathematics. I certainly make no such claims here at the outset. Indeed, we will only be in a position to assess the ambitions of the enthusiasts after we have made a decent start on understanding category theory for its own sake. These Notes aim to make a start at helping you to work towards such an understanding.

1.2 What do you need to bring to the party?

Legend has it that over the doorway to Plato's Academy was written 'Let no one ignorant of mathematics enter here'. And perhaps the doorway to category theory should be similarly inscribed. After all, you will not be well placed to appreciate how category theory gives us a story about how parts of mathematics hang together if you don't already know some mathematics.

But in fact – comfortingly! – in these Notes on the basics, we need not presuppose very much. And if a some of the examples pass you by because you lack the necessary background, then don't panic. I usually try to give multiple examples of important concepts and constructs; so simply skip those examples that don't work for you. True, if you are e.g. a philosopher whose prior exposure to mathematics comprises only some logic and introductory set-theory, then you might want to do small amounts of homework when you encounter less familiar notions like monoid, poset, topological space, etc. But I can emphasize the word 'small'! A bit of judicious browsing in Wikipedia may well provide enough for your purposes.

1.3 Theorems as exercises

In the current version of these Notes, there are no exercises – or at least, there are none explicitly labeled as such.

However, almost all the proofs of theorems in basic category theory are easy, and often they are *very* easy – just a matter of applying definitions and then following the only obvious route forward to the conclusion. Even when a result like the Yoneda Lemma takes a bit more work, I try to break things down using intermediate lemmas so that each stage is straightforward to prove. So as a reader you can – and perhaps should! – think of almost every statement of a theorem as in fact presenting you with an exercise which you should pause to attempt to help fix ideas; and then the ensuing proof which I spell out is the answer (or at least, *an* answer) to the exercise.

2 Categories defined

We said we are going to be interested in the idea of a family of structures with structure-preserving maps between them. Such a family is a paradigm of a category. So without further ado, let's dive in to give our first definition of categories and present some initial examples. We also introduce the important – and labour-saving – notion of duality.

2.1 The very idea of a category

A category $\mathscr{C}$ consists in certain data (in the sense of 'givens', rather than of 'information'!), governed by a couple of axioms.

Definition 1. The data for a category $\mathscr{C}$ comes in two sorts:

(1) *Objects* (which we will typically notate by 'A', 'B', 'C', ...).

(2) *Arrows* (which we typically notate by 'f', 'g', 'h', ...).

Further,

(3) For each arrow f, there are unique associated objects $src(f)$ and $tar(f)$, respectively the *source* and *target* of f. We write '$f\colon A \to B$' or '$A \xrightarrow{f} B$' to notate that f is an arrow with $src(f) = A$ and $tar(f) = B$.

(4) For any two arrows $f\colon A \to B$, $g\colon B \to C$, where $src(g) = tar(f)$, there exists an arrow $g \circ f\colon A \to C$, 'g following f', which we call the *composite* of f with g.

(5) Given any object A, there is an arrow $1_A\colon A \to A$ called the *identity arrow* on A.

The axioms then require

Identity. The identity arrows do behave like identities! So for any $f\colon A \to B$ we have $f \circ 1_A = f = 1_B \circ f$.

Associativity. For any $f\colon A \to B$, $g\colon B \to C$, $h\colon C \to D$, we have $h \circ (g \circ f) = (h \circ g) \circ f$.

Five quick remarks on our terminology and notation:

(a) The objects of a category (the first type of data) are very often not 'naked' objects, so to speak, but rather – as promised – are typically entities that come equipped with operations and relations etc., i.e. are structures.

(b) It should also be noted that the objects in categories needn't be objects in the usual logician's strict sense, i.e. in the sense which contrasts objects with entities like relations or functions. In some categories, for example, the objects in the sense of the first type of data are in fact relations or functions.

(c) Borrowing familiar functional notation '$f\colon A \to B$' for arrows in categories is entirely natural given that arrows in many categories indeed *are* functions. But as we'll soon see, not all arrows are functions. Which means that not all arrows are morphisms either, in the usual sense of that term. Which is why I rather prefer the colourless 'arrow' to the equally common term 'morphism' for the second sort of data in a category. (Not that that will stop me often talking of morphisms when context makes that natural!)

(d) In keeping with the functional paradigm, the source and target of an arrow are very often called, respectively, the 'domain' and 'codomain' of the arrow. But again that usage has the potential to mislead when arrows aren't (the right kind of) functions, which is again why I prefer our terminology.

(e) Note the order in which we write the components of a composite arrow (some from computer science do things the other way about). Our notational convention is again suggested by the functional paradigm. In a category where $f\colon A \to B$, $g\colon B \to C$ are both functions, then $g \circ f(x) = g(f(x))$.

    Occasionally, to reduce clutter, we may allow ourselves to write simply 'gf' rather than $g \circ f$.

Now, our axioms suffice to ensure our first mini-result:

Theorem 1. *Identity arrows on a given object are unique; and the identity arrows on distinct objects are distinct.*

Proof. Suppose A has identity arrows 1_A and $1'_A$. Then $1_A = 1_A \circ 1'_A = 1'_A$.

For the second part, we simply note that $A \neq B$ entails $src(1_A) \neq src(1_B)$ which entails $1_A \neq 1_B$. □

(As this illustrates, we go along I will cheerfully call the most trivial of lemmas, run-of-the-mill propositions, interesting corollaries, and the weightiest of results all 'theorems' without distinction.)

So every object in a category is associated with one and only one identity arrow. And we can pick out identity arrows by the way they interact with other arrows. Hence we could in principle define categories initially just in terms of arrows. For an account of how to do this, see Adámek et al. (2009, pp. 41–43). But I find this technical trickery rather unhelpful. (The central idea of category theory is perhaps best understood as the idea that we should probe objects by considering the morphisms between them; but that doesn't mean writing the objects out of the story altogether!)

2.2 Examples

Naively, we might think of a structure as comprising *some objects* (plural) equipped with functions and/or relations on them. But a structure nowadays is much more commonly thought of as being *a set* (singular) equipped with functions and/or relations on it. But is there a difference? If so, does it matter?

As we said, we are having to shelve such questions for now. So, going along with the usual modern line, we are going to expect some paradigm examples of categories to have as objects *sets*-equipped-with-some-functions/relations, and then the arrows between such objects will then be suitable functions between the carrier sets which in a good sense 'preserve structure' (or perhaps we should say 'respect structure', for 'preservation' sounds like a matter of producing a full copy, which is more than we usually require).

We start with an extremal case, where the sets in fact come with no additional structure:

(1) **Set** is a category with

> Objects: all the sets (just naked sets, equipped with no extra structure).
>
> Arrows: given sets X, Y, any (total) set-function $f\colon X \to Y$ is an arrow.

There's an identity function on any set; set-functions $f\colon A \to B$, $g\colon B \to C$ (where the domain of g is the codomain of f) compose; and the axioms for being a category are evidently satisfied.

Four comments on this initial case. (i) Annoyingly, there is already a substantive issue here about just what category we have in mind here in talking about **Set**. What kind of sets are these? Pure sets or do we allow sets with urelements (members which aren't themselves sets)? Are they constrained by e.g. the axioms of ZFC or is the universe of sets better described by a set theory like NF? But again perhaps we need not pause over that now, or we'd never get started. For whatever your favoured conception of the universe of sets, its objects and functions should constitute a category. So for the moment, you can just interpret talk of sets in your preferred way. For some more introductory remarks, see here. And do eventually take a look at Leinster 2014, Ch. 3, 'Interlude on sets'. But let's here press on!

(ii) Note that arrows in **Set**, like any arrows, must come with determinate targets/codomains. But a common way of treating functions set-theoretically is simply to identify a set-function f with its graph, the set of pairs $\langle x, y\rangle$ such that $f(x) = y$. That kind of definition is lop-sided in that it fixes a domain, which will be the set of first elements in the pairs, but it doesn't determine the function's codomain (two functions $f\colon A \to B$ and $f'\colon A \to C$ where $B \subsetneq C$ could have the same graph). We then have two options in talking about arrows in **Set**. Either keep the usual lopsided definition of a set-function f as its graph, but identify an arrow $f\colon A \to B$ in **Set** with a triple $\langle A, f, B\rangle$. Or revise the way we understand the notion of a function in set theory, using such a triple there, and then identify arrows in the category with set-functions newly understood. There's plainly nothing to choose.

(iii) Perhaps we should remind ourselves why there *is* an identity arrow for $\emptyset$ in **Set**. Vacuously, for any set Y, there is exactly one set-function $f\colon \emptyset \to Y$ – i.e. one appropriate set of ordered pairs – namely the empty set, and hence in particular there is a function $1_\emptyset\colon \emptyset \to \emptyset$.

Note that in **Set**, the empty set is in fact the only set such that there is exactly one arrow *from* it to any other set. This gives us a nice first example of how we can characterize a significant object in a category not by its internal constitution, so to speak, but by what arrows it has to and from other objects. We will return to this.

(iv) The function id_A defined on a set A by $id_A(x) = x$ will evidently serve in the category **Set** as the (unique) identity arrow 1_A.

We can't say that, however, in pure category-speak. Still, we can do something that comes to the same. Glancing ahead to ideas which we'll again return to, note first that we can define singletons in **Set** by relying on the observation that there is exactly one arrow from any object *to* a singleton. So fix on a singleton, call it simply 1. Then consider the possible arrows (i.e. set-functions) $\vec{x}\colon 1 \to A$. There is exactly one such arrow for every $x \in A$. So we can think of talk of arrows $\vec{x}\colon 1 \to A$ as our category-speak surrogate for talking about elements x of A. Then instead of saying $id_A(x) = x$ for all members x of A, we can say that for any $\vec{x}\colon 1 \to A$, we have $1_A \circ \vec{x} = \vec{x}$. (More on this sort of thing in due course: but it gives us another glimpse ahead of how we might trade in talk of sets-and-their-elements for categorial talk of sets-and-arrows-between-them.)

(2) To continue with our examples of categories, there is also a category **FinSet** of hereditarily finite sets (i.e. sets with at most finite numbers of members, these members in turn having at most finite numbers of members, which in turn …) and the functions between *them*.

(3) **Pfn** is the category of sets and *partial* functions. Here, the objects are naked sets again, but an arrow $f\colon A \to B$ is a function not necessarily everywhere defined on A (one way to think of such an arrow is as a total function $f\colon A' \to B$ where $A' \subseteq A$). Given arrows-qua-partial-functions $f\colon A \to B$, $g\colon B \to C$, their composition $g \circ f\colon A \to C$ is defined in the obvious way, though you need to check that this indeed makes composition associative.

(4) $\mathbf{Set}_\star$ is a category (of 'pointed sets') with

objects: all the sets [apart from the empty set!], each set A equipped with a zero-place function that picks out a distinguished object $\star_A$ in the set,

arrows: for A, B among the sets, any (total) function $f\colon A \to B$ which maps the distinguished object $\star_A$ to the distinguished object $\star_B$ is an arrow.

As we'll see later, **Pfn** and $\mathbf{Set}_\star$ are in a good sense equivalent categories (can you already see why we should expect that)?

Picking out a distinguished object, as in $\mathbf{Set}_\star$, is about the least structure we can put on a set. We now consider cases with rather more structure:

(5) Recall the usual definition of a monoid $(M, \cdot, 1_M)$. This is a set M equipped with a two-place function mapping elements to elements and with a designated element (or if you like, it is a pointed set equipped with a function). And it is required that (i) the function is associative, i.e. for all elements $a, b, c \in M$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$, and (ii) the designated element 1_M acts as a unit, i.e. is such that for any $a \in M$, $1_M \cdot a = a = a \cdot 1_M$. Then **Mon** is a category with

objects: all the monoids,

arrows: for $(M, \cdot, 1_M), (N, \times, 1_N)$ among the monoids, any monoid homomorphism $f\colon M \to N$ is an arrow. Here, $f\colon M \to N$ is a monoid homomorphism when it preserves monoidal structure, i.e. for any $x, y \in M$, $f(x \cdot y) = f(x) \times f(y)$, and f preserves identity elements, i.e. $f(1_M) = 1_N$.

In this case, the identity arrows are the identity functions on the carrier sets for the monoids, and composition of arrows is composition of homomorphisms.

(6) **Mon** is just the first of a family of similar algebraic cases, where the objects are sets equipped with some functions and the arrows are functions preserving that structure. We also have:

a) **Grp**, the category of groups (the family of structures we mentioned at the very outset – except we now officially think of a group as comprising a *set* of objects equipped with a suitable operation), with
objects of **Grp**: all the groups,
arrows: group homomorphisms (preserving group structure)

b) **Ab**, the category of abelian groups, with
objects: all abelian groups,
arrows: group homomorphisms.

c) **Rng** the category of rings, with
objects: all rings,
arrows: ring homomorphisms.

And so it goes!

(7) The category **Rel** again has naked sets as objects, but this time an arrow $A \to B$ in **Rel** is (not a function but) any relation R between A and B, which we can take to mean any subset of $A \times B$.

The identity arrow on A is then the diagonal relation $\{\langle x, x\rangle \mid x \in A\}$. And composition of arrows $R\colon A \to B$ and $S\colon B \to C$ is then defined by putting $S \circ R\colon A \to C$ to be the set of pairs $\langle a, c\rangle \in A \times C$ such that $\exists b(\langle a, b\rangle \in R \wedge \langle b, c\rangle \in S)$. It is easily checked that this is associative.

So here we have our first example where the arrows in a category are *not* functions.

(8) The monoids as objects together with all the monoid homomorphisms form a (very large!) category **Mon**. But now we should observe the important fact that any single monoid taken just by itself can be thought of as corresponding to a (perhaps very small) category.

Thus take the monoid $(M, \cdot, 1_M)$. And now define $\mathscr{M}$ to be the corresponding category with data

i. the sole object of $\mathscr{M}$: just some object – choose whatever you like, and dub it '$\star$'.

ii. arrows of $\mathscr{M}$: the elements of the monoid M.

iii. the identity arrow $1_\star$ of $\mathscr{M}$ is the identity element 1_M of the monoid M.

iv. the composition $m \circ n\colon \star \to \star$ of two arrows $m\colon \star \to \star$ and $n\colon \star \to \star$ (those arrows being just the elements $m, n \in M$), is $m \cdot n$.

It is trivial that the axioms for being a category are satisfied. So we can think of a monoid as a one-object category. (Conversely, then, we can think of categories as, in a sense, generalized monoids.)

Note in this case, unless the elements of the original monoid M are themselves functions, the arrows of the associated category $\mathscr{M}$ are again not themselves functions or morphisms in any natural sense.

(9) **Ord** is a category, with

objects: the pre-ordered sets. Recall, a set S is pre-ordered iff equipped with an order $\preccurlyeq$ where for all $x, y \in S$, $x \preccurlyeq x$, and $x \preccurlyeq y \wedge y \preccurlyeq z \to x \preccurlyeq z$. We represent the resulting pre-ordered set $(S, \preccurlyeq)$.

arrows: monotone maps – i.e. maps $f\colon S \to T$, from the carrier set of $(S, \preccurlyeq)$ to the carrier set of $(T, \sqsubseteq)$, such that if $x \preccurlyeq y$ then $f(x) \sqsubseteq f(y)$.

(10) Note too that any single pre-ordered set can also itself be regarded as a category (this time, a category with at most one arrow between objects). For corresponding to the pre-ordered set $(S, \preccurlyeq)$, we will have a category $\mathscr{S}$, where

i. The objects of $\mathscr{S}$ are just the members of S.

ii. An arrow from $\mathscr{S}$ from source C to target D is just an ordered pair of objects $\langle C, D\rangle$ such that $C \preccurlyeq D$.

iii. $1_C = \langle C, C\rangle$.

iv. Composition is defined by setting $\langle D, E\rangle \circ \langle C, D\rangle = \langle C, E\rangle$.

It is easily checked that this satisfies the identity and associatively axioms. Conversely, any category $\mathscr{S}$ whose objects form a set S and where there is at most one arrow between objects can be regarded as a pre-ordered set $(S, \preccurlyeq)$ where for $C, D \in S$, $C \preccurlyeq D$ just in case there is an arrow from C to D of $\mathscr{S}$.

We can thus call a category with at most one arrow between objects a *pre-order category.*

An aside here on a point of notation. I'm falling into the practice of using old-school angle brackets, as in '$\langle C, D\rangle$', when we are indeed definitely talking about an ordered-pair treated as a single object (however exactly we want to implement that idea). By contrast, we can in many contexts take ordinary parentheses as simply helpful punctuation, not as a constructor for a new object. For example, when talking informally of the pre-ordered set $(S, \preccurlyeq)$ as we just did, we are talking about the set S and about the ordering $\preccurlyeq$ defined over S, and not – or at least, not yet – about some further pair-object. Likewise, in talking of the monoid $(M, \cdot, 1_M)$ we are talking about a set, an operation on it and a selected element and not – or at least, not yet – about some further object. The rule for the moment, then, will be that angle brackets are always used to denote ordered-pairs-as-objects. While the interpretation of expressions with parentheses may be up for grabs: case by case, read as noncommittally as possible. But we'll have to see how this notational policy works out. To continue:

(11) A closely related case to Ex. (9): **Pos** is a category with

objects: the posets – S is a poset iff equipped with a partial order $\preccurlyeq$, i.e. with a pre-order which satisfies the additional anti-symmetry constraint, i.e. for $x, y \in S$, $x \preccurlyeq y \wedge y \preccurlyeq x \to x = y$.

arrows: monotone maps.

And exactly as each pre-ordered set can be regarded as category, so each individual poset can be regarded as a category, a *poset category.*

(12) **Top** is a category with

objects: all the topological spaces,

arrows: the continuous maps.

(13) **Met** is a category with

objects: metric spaces, any set of points S equipped with a real metric d, arrows: the non-expansive maps, where – in an obvious shorthand notation – $f\colon (S, d) \to (T, e)$ is non-expansive iff $d(x, y) \geq e(f(x), f(y))$.

(14) $\mathbf{Vect_k}$ is a category with

objects: vector spaces over the field k (each such space is a set of vectors, equipped with vector addition and multiplication by scalars in the field k),

arrows: linear maps between the spaces.

For those who know about such beasts, we could also mention categories of sheaves, of schemes, of simplicial sets, and so on, in each case with the relevant objects equipped with predictable structure-preserving maps as arrows. But we won't pause over those: instead let's finish with a few much simpler cases:

(15) For any collection of objects M, there is a *discrete category* on those objects. This is the category whose objects are just the members of M, and which has as few arrows as possible, i.e. just the identity arrow for each object in M.

(16) The smallest discrete category is $\mathbf{1}$ which has exactly one object and one arrow (the identity arrow on that object). Let's picture it in all its glory!

$$\begin{tikzcd} \bullet \arrow[loop above] \end{tikzcd}$$

But should we talk about *the* category $\mathbf{1}$? Won't a different choice of object make for different one-object categories?

Yes and no! We can have, in our mathematical universe, different cases of single objects equipped with an identity arrow – *but they will be indiscernible from within category theory.* So as far as category theory is concerned, they are all 'the same'. (Compare a probably familiar sort of case from elsewhere in mathematics. There will be many concrete groups which have the right structure to be e.g. a Klein four-group. But they are group-theoretically indiscernible. So we talk of *the* Klein four-group.)

(17) And having mentioned the one-object category $\mathbf{1}$ here's another very small category, this time with two objects, the necessary identity arrows, and one further arrow between them. We can picture it like this:

$$\begin{tikzcd} \bullet \arrow[loop left] \arrow[r] & \star \arrow[loop right] \end{tikzcd}$$

Call this category $\mathbf{2}$. (We can think of the von Neumann ordinal 2, i.e. the set $\{\emptyset, \{\emptyset\}\}$, as giving rise to this category when it is considered as a poset with the subset relation as the ordering relation. Other von Neumann ordinals, finite and infinite, similarly give rise to other poset categories.)

And that will do for the moment as an introductory list. There is no shortage of categories, then!

Indeed we might well begin to wonder whether it is rather *too* easy to be a category. If such very different sorts of structures as monoids and posets can be extremal cases of categories, how much mileage can there be in theorizing about categories in general? Well, that's the story that emerges over the coming chapters.

2.3 Diagrams

We can graphically represent categories – and in particular, represent facts about the equality of arrows – in a very natural way, using so-called commutative diagrams. We've just seen a couple of trivial mini-examples. We'll be using diagrams a great deal: so we'd better say something about them straight away.

Talk of diagrams is in fact used by category theorists in three different (but very closely related) ways. For the moment, we'll mention two. First, we have the initial, most natural, sense of diagrams-as-pictures:

Definition 2. A *(representational) diagram* is a 'graph' with nodes representing objects from a given category $\mathscr{C}$, and drawn arrows between nodes representing arrows of $\mathscr{C}$. Nodes and drawn arrows are usually labelled.

Two nodes in a diagram can be joined by zero, one or more drawn arrows. A drawn arrow labelled 'f' from the node labeled 'A' to the node labeled 'B' of course represents the arrow $f\colon A \to B$ of $\mathscr{C}$. There can also be arrows looping from a node to itself, representing the identity arrow on an object or other 'endomorphisms' (i.e. other arrows whose source and target is the same).

Definition 3. A *diagram in a category $\mathscr{C}$* is what is represented by a representational diagram – i.e. some $\mathscr{C}$-objects and $\mathscr{C}$-arrows between some of them.

I'm being a little pernickety in distinguishing the two ideas here, the diagram-as-picture, and the diagram-as-what-is-pictured. But having made the distinction, we needn't fuss except when it matters, and will let context determine a sensible reading of claims about diagrams.

An important point is that diagrams (in either sense) needn't be full. That is to say, a diagram-as-a-picture need only show some of the objects and arrows in a category; and a diagram-as-what-is-pictured need only be a fragment of the category in question.

Now, within a drawn diagram, we may be able to follow a directed path through more than two nodes, walking along the connecting drawn arrows as we go. So a path in a category diagram from node A to node E (for example) might look like this

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D \xrightarrow{j} E$$

And then we'll call the composite arrow $j \circ h \circ g \circ f$ the *composite along the path.* (We know that, because of the associativity of composition, we needn't fuss about bracketing here. And henceforth we freely insert or omit brackets, doing whatever promotes local clarity.) Then:

Definition 4. A category diagram *commutes* if (i) for any two directed paths along edges in the diagram from a node X to the distinct node Y, the composite arrow along the first path is equal to the composite arrow along the second path, and (ii) for any closed path with more than one node that loops around from a node X to itself, the composite arrow along that path is equal to the (perhaps undiagrammed) identity arrow on the object at X.

The main clause here is (i): it is useful to add (ii) to cover some additional special cases.

Hence, for example, the associativity law can be represented by saying that the following diagram commutes:

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[dr, "g \circ f"'] & B \arrow[d, "g"] \arrow[dr, "h \circ g"] & \\
& C \arrow[r, "h"] & D
\end{tikzcd}$$

Each triangle commutes, and hence so does the whole diagram. So, in particular, the two outer paths from A to D give equal composites. (Note, just drawing a diagram with different routes from e.g. A to D doesn't mean that we have a commutative diagram – the equality of the composites along the paths in each case has to be proved!)

2.4 Duality

We have already noted a good initial range of examples of categories. And now here's a cheap way of getting new categories from old: reverse all the arrows. More carefully:

Definition 5. Given a category $\mathscr{C}$, then $\mathscr{C}^{op}$ is the category with the data

(1) The objects of $\mathscr{C}^{op}$ are just the objects of $\mathscr{C}$ again.

(2) If f is an arrow of $\mathscr{C}$ with source A and target B, then f is an arrow of $\mathscr{C}^{op}$ but now with source B and target A.

(3) Identity arrows remain the same, i.e. $1_A^{op} = 1_A$.

(4) Composition-in-$\mathscr{C}^{op}$ is defined in terms of composition-in-$\mathscr{C}$ by putting $f \circ^{op} g = g \circ f$.

It is trivial to check that this definition is in good order and that $\mathscr{C}^{op}$ is indeed a category. And it is trivial to check that $(\mathscr{C}^{op})^{op}$ is $\mathscr{C}$. So *every* category is the opposite of some category.

Do be careful here. Take for example $\mathbf{Set}^{op}$. An arrow $f: A \to B$ in $\mathbf{Set}^{op}$ is the same thing as an arrow $f: B \to A$ in $\mathbf{Set}$, which is of course a set-function from B to A. But this means that f in $\mathbf{Set}^{op}$ typically *won't* be a function from *its* source to its target – it's an arrow in that direction but usually only a function in the other! (So this is one of those cases where talking of 'domains' and 'codomains' instead of 'sources' and 'targets' could initially encourage confusion.)

Take $\mathscr{L}$ to be the elementary pure language of categories. This will be a two-sorted first-order language with identity, with one sort of variable for objects, $A, B, C \ldots$, and another sort for arrows $f, g, h, \ldots$. It has built-in function-expressions '*src*' and '*tar* (denoting two maps from arrows to objects), a built-in relation, in effect '... is the identity arrow for ...', and a two place function-expression '$\ldots \circ -$' we expresses the function which takes two composable arrows to another arrow.

Definition 6. Suppose φ is a wff of $\mathscr{L}$. Then its *dual* φ^{op} is the wff you get by swapping (i) '*src*' and '*tar*' and swapping (ii) '$f \circ g$' for '$g \circ f$', etc.

Now, the claim that $\mathscr{C}^{op}$ is a category just reflects the fact that the duals of the axioms for a category are also axioms. And *that* observation gives us the following *duality principle*:

Theorem 2. *Suppose φ is an $\mathscr{L}$-sentence with no free variables – so is a general claim about objects/functions in an arbitrary category. Then if the axioms of category theory entail φ, they also entail the dual claim φ^{op}.*

Syntactic proof. If there's a first-order proof of φ from the axioms of category theory, then by taking the duals of every wff in the proof we'll get a proof of φ^{op} from the duals of the axioms of category theory. But those duals of axioms are themselves axioms, so we have a proof of φ^{op} from the axioms of category theory. □

Semantic proof. If φ always holds, i.e. holds in every category $\mathscr{C}$, then φ^{op} will hold in every $\mathscr{C}^{op}$ – but the $\mathscr{C}^{op}$s just comprise every category again, so φ^{op} also holds in every category. □

The duality principle captured by this theorem is a very simple but also very labour-saving result; we'll see this time and time again, starting in the next chapter.

3 Kinds of arrows

This chapter characterizes a number of kinds of arrows in categories in terms of how they interact with other arrows. This will give us some elementary but characteristic examples of arrow-theoretic (re)definitions of familiar notions.

3.1 Monomorphisms, epimorphisms

(a) Take a set-function $f\colon A \to B$ living as an arrow in **Set**: how could we say that it is injective, i.e. one-one, using just category-speak about arrows?

We noted that we can think of elements x of f's domain A as arrows $\vec{x} = \vec{x}\colon 1 \to A$ (where 1 is some singleton). So then injectiveness comes to this: $f \circ \vec{x} = f \circ \vec{y}$ implies $\vec{x} = \vec{y}$, for any element-arrows $\vec{x}, \vec{y}$. Hence if a function is more generally 'left-cancellable' in **Set** – meaning that, for any g, h, $f \circ g = f \circ h$ implies $g = h$ – then it certainly has to be an injection.

And conversely: if f is injective as a set-function, then for all x, $f(g(x)) = f(h(x))$ implies $g(x) = h(x)$ – which is to say that if $f \circ g = f \circ h$ then $g = h$, i.e. f is left-cancellable.

So that motivates introducing a notion with the following definition (you have to learn to live with the new-fangled terminology):

Definition 7. An arrow $f\colon C \to D$ in the category $\mathscr{C}$ is a *monomorphism* (is *monic*) if and only if it is left-cancellable, i.e. for every pair of maps $g\colon B \to C$ and $h\colon B \to C$, if $f \circ g = f \circ h$ then $g = h$.

We have just proved

Theorem 3. *The monomorphisms in* **Set** *are exactly the injective functions.*

The same applies in many, but not all, other categories where arrows are functions. For example, we have:

Theorem 4. *The monomorphisms in* **Grp** *are exactly the injective group homomorphisms.*

Proof. We show as before that the injective group homomorphisms are monomorphisms in **Grp**.

For the other direction, suppose $f\colon C \to D$ is a group homorphism between $(C, \cdot, e_C)$ and $(D, \star, e_D)$ and *not* an injection.

We must have $f(c) = f(c')$ for some $c, c' \in C$ where $c \neq c'$. Note then that

$$f(c^{-1} \cdot c') = f(c^{-1}) \star f(c') = f(c^{-1}) \star f(c) = f(c^{-1} \cdot c) = f(e_C) = e_D.$$

So $c^{-1} \cdot c'$ is an element in K, the kernel of f (the set of elements that f sends to the unit e_D of $(D, \star)$). Since $c \neq c'$, we have $c^{-1} \cdot c' \neq e_C$, and hence K has more than one element.

Now define $g\colon K \to C$ to be the inclusion map, while $h\colon K \to C$ sends everything to e_C. Since K has more than one element, $g \neq h$. But obviously, $f \circ g = f \circ h$ (both send everything in K to e_D). So f isn't left-cancellable.

Hence, contraposing, if f is monic in **Grp** it is injective. □

(b) Next, here is a companion definition:

Definition 8. An arrow $f\colon C \to D$ in the category $\mathscr{C}$ is an *epimorphism* (is *epic*) if and only if it is right-cancellable, i.e. for every pair of maps $g\colon D \to E$ and $h\colon D \to E$, if $g \circ f = h \circ f$ then $g = h$.

Predictably, *the notion of an epimorphism is dual to that of a monomorphism* in the following strict sense: if f is right-cancellable and so epic in $\mathscr{C}$ if and only if it is left-cancellable and hence monic in $\mathscr{C}^{op}$. And, again predictably, just as monomorphisms in the category **Set** are injective functions, we have:

Theorem 5. *The epimorphisms in* **Set** *are exactly the surjective functions.*

Proof. Suppose $f\colon C \to D$ is surjective. And consider two functions $g, h\colon D \to E$ where $g \neq h$. Then for some $d \in D$, $g(d) \neq h(d)$. But by surjectivity, $d = f(c)$ for some $c \in C$. So $g(f(c)) \neq h(f(c))$, whence $g \circ f \neq h \circ f$. So the surjectivity of f in **Set** implies that if $g \circ f = h \circ f$, then $g = h$, i.e. f is epic.

Conversely, suppose $f\colon C \to D$ is not surjective, so $f[C] \neq D$. Consider two functions $g\colon D \to E$ and $h\colon D \to E$ which agree on $f[C] \subset D$ but disagree on the rest of D. Then $g \neq h$, even though by hypothesis $g \circ f$ and $h \circ f$ will agree everywhere on C, so f is not epic. Contraposing, if f is epic in **Set**, it is surjective. □

A similar result holds in many other categories, but in §3.3, Ex. (2), we'll encounter a case where we have an epic function which is *not* surjective.

As the very gentlest of exercises, let's add for the record a mini-theorem:

Theorem 6. *(1) Identity arrows are always monic. Dually, they are always epic too.*

(2) If f, g are monic, so is $f \circ g$. If f, g are epic, so is $f \circ g$.

(3) If $f \circ g$ is monic, so is g. If $f \circ g$ is epic, so is f.

Proof. (1) is trivial. For (2) suppose $(f\circ g)\circ j = (f\circ g)\circ k$, then $f\circ(g\circ j) = f\circ(g\circ k)$. Whence, assuming f is monic, $g\circ j = g\circ k$. Whence, assuming g is monic, $j = k$. So $(f\circ g)$ is monic. The proof for epics is the dual.

For (3) assume $f\circ g$ is monic. Temporarily suppose $g\circ j = g\circ k$. Then $f\circ(g\circ j) = f\circ(g\circ k)$, and hence $(f\circ g)\circ j = (f\circ g)\circ k$, so $j = k$. Therefore if $g\circ j = g\circ k$ then $j = k$; i.e. g is monic. Dually again for epics. ☐

(c) We should, I guess, note in passing a common convention that you may very well encounter elsewhere, though we won't use it here:

Notation. $f\colon C \rightarrowtail D$ represents a monomorphism f with source C and target D. While $f\colon C \twoheadrightarrow D$ represents an epimorphism f with source C and target D.

As a useful mnemonic (well, it works for me!) just think (ML) a *m*onomorphism is *l*eft cancellable and its representing arrow has an extra fletch on the *l*eft, while (PR) an e*p*imorphism is *r*ight cancellable and its representing arrow has an extra fletch on the *r*ight.

3.2 Inverses

(a) We define some more types of arrow:

Definition 9. Given an arrow $f\colon C \to D$ in the category $\mathscr{C}$,

(1) $g\colon D \to C$ is a *right inverse* of f (in $\mathscr{C}$, of course) iff $f\circ g = 1_D$.
(2) $g\colon D \to C$ is a *left inverse* of f iff $g\circ f = 1_C$.
(3) $g\colon D \to C$ is an *inverse* of f iff it is both a right inverse and a left inverse of f.

Three remarks. First, on the use of 'left' and 'right': if we represent the situation in (1) like this

$$\begin{tikzcd} D \arrow[r, "g"] \arrow[rr, bend right, "1_D"'] & C \arrow[r, "f"] & D \end{tikzcd}$$

then f's right inverse g appears on the left! It is just a matter of convention that we describe handedness by reference to the representation '$f\circ g = 1_D$' rather than by reference to our diagram. (Similarly, of course, in defining left-cancellability, etc.)

Second, note that $g\circ f = 1_C$ in $\mathscr{C}$ iff $f\circ^{op} g = 1_C$ in $\mathscr{C}^{op}$. So a left inverse in $\mathscr{C}$ is a right inverse in $\mathscr{C}^{op}$. And vice versa. The ideas of a right inverse and left inverse are therefore, exactly as you would expect, dual to each other; and the idea of an inverse is dual to itself.

Third, if f has a right inverse g, then it is itself a left inverse (of g, of course!). Dually, if f has a left inverse, then it is a right inverse.

It is obvious that an arrow f need not have a left inverse: just consider, for example, those arrows in **Set** which are many-one functions. An arrow f can also have many left inverses: for a toy example in **Set** again, consider $f\colon \{0,1\} \to \{0,1,2\}$ where $f(n) = n$. Then the map $g\colon \{0,1,2\} \to \{0,1\}$ is a left inverse so long as $g(0) = 0$ and $g(1) = 1$, which leaves us two choices for $g(2)$, and hence we have two left inverses. By the duality principle, an arrow can also have zero or many right inverses. However,

Theorem 7. *If an arrow has both a right inverse and a left inverse, then these are the same and are the arrow's unique inverse.*

Proof. Suppose $f\colon C \to D$ has right inverse $r\colon D \to C$ and left inverse $s\colon D \to C$. Then

$$r = 1_C \circ r = (s \circ f) \circ r = s \circ (f \circ r) = s \circ 1_D = s.$$

Or, in a commuting diagram,

$$\begin{tikzcd}
D \arrow[r, "r"] \arrow[rrr, bend left=40, "r"] \arrow[rr, bend right, "1_D"'] \arrow[rrr, bend right=40, "s"'] & C \arrow[r, "f"] \arrow[rr, bend left, "1_C"] & D \arrow[r, "s"] & C
\end{tikzcd}$$

Hence r, i.e. s, is an inverse.

Suppose now that f has inverses r and s. By definition r will be a right inverse and s a left inverse, so as before $r = s$. So in fact inverses are unique. □

(b) Now, how does talk of an arrow as a right inverse/left inverse hook up to talk of an arrow as monic/epic?

Theorem 8. *(1) In general, not every monomorphism is a right inverse; and dually, not every epimorphism is a left inverse.*

(2) But every right inverse is monic, and every left inverse is epic.

Proof. (1) can be shown by a toy example. Take the category **2** which we met back in §2.2, Ex. (17) – i.e. take as a category the two-element pre-ordered set which has just one non-identity arrow. That non-identity arrow is trivially monic and epic, but it lacks both a left and a right inverse.

For (2), suppose f is a right inverse for e, which means that $e \circ f = 1$ (suppressing unnecessary labellings of domains and codomains). Now suppose $f \circ g = f \circ h$. Then $e \circ f \circ g = e \circ f \circ h$, and hence $1 \circ g = 1 \circ h$, i.e. $g = h$, so indeed f is monic. Similarly for the dual. □

So monics need not in general be right inverses nor epics left inverses. But how do things pan out in the particular case of the category **Set**? Here's the answer:

Theorem 9. *In* **Set***, every monomorphism is a right inverse apart from arrows of the form* $\emptyset \to D$*. Also in* **Set***, the proposition that every epimorphism is a left inverse is (a version of) the Axiom of Choice.*

Proof. Suppose $f\colon C \to D$ in **Set** is monic. It is therefore one-to-one between C and $f[C]$, so consider a function $g\colon D \to C$ that reverses f on $f[C]$ and somehow or other maps $D - f[C]$ into C. Such a g is always possible to find in **Set** unless C is the empty set. So $g \circ f = 1_C$, and hence f is a right inverse.

Now suppose $f\colon C \to D$ in **Set** is epic, and hence a surjection. Assuming the Axiom of Choice, there will be a function $g\colon D \to C$ which maps each $d \in D$ to some chosen one of the elements c such that $f(c) = d$. And assuming g exists, $f \circ g = 1_D$, so f is a left inverse.

Conversely, suppose we have a partition of C into disjoint subsets indexed by (exactly) the elements of D. Let $f\colon C \to D$ be the function which sends an object in C to the index of the partition it belongs to. f is surjective, hence epic. Suppose f is a left inverse, so for some $g\colon D \to C$, $f \circ g = 1_D$. Then g is evidently a choice function, picking out one member of each partition. So the claim that all epics have a left inverse gives us (one version of) the Axiom of Choice. □

(c) There is an oversupply of other jargon hereabouts, also in quite common use. We should note alternatives for the record.

Assume we have a pair of arrows in opposite directions, $f\colon C \to D$, and $g\colon D \to C$.

Definition 10. If $g \circ f = 1_C$, then f is also called a *section* of g, and g is a retraction of f. (In this usage, f *is* a section iff it *has* a retraction, etc.)

Definition 11. If f has a left inverse, then f is a *split monomorphism*; if g has a right inverse, then g is a *split epimorphism.*

In the latter usage, we can say that the claim that every epimorphism splits in **Set** is the categorial version of the Axiom of Choice. Note that Theorem 8 tells us that right inverses are monic, so a split monomorphism is indeed properly called a monomorphism. Dually, a split epimorphism is an epimorphism.

3.3 Isomorphisms

(a) In familiar cases, we say that two structures 'look just the same' – are isomorphic – if there is a structure-preserving bijection between them. So we might expect the isomorphisms in a category to be the bijective arrows, at least when the arrows are functions. But as we have seen, in the category **Set**, the arrows which are both monic and epic are indeed the bijective functions. So shall we, more generally, define isomorphisms to be arrows which are monic and epic?

No. Because isomorphisms need to have inverses (we want being isomorphic to be an equivalence relation). But being monic and epic doesn't always imply having an inverse. We can use again the toy case of **2**, or here's a generalized version of the same idea:

(1) Take the category $\mathscr{S}$ corresponding to the pre-ordered set $(S, \preccurlyeq)$. Then there is at most one arrow between any given objects of $\mathscr{S}$. But if $f \circ g = f \circ h$, then g and h must share the same object as domain and same object as codomain, hence $g = h$, so f is monic. Similarly it is epic. But no arrows other than identities have inverses.

The arrows in that example aren't functions, however. So next, here's a revealing case where the arrows *are* functions but where being monic and epic still doesn't imply having an inverse:

(2) Consider the category **Mon** of monoids. Among its objects are $N = (\mathbb{N}, +, 0)$ and $Z = (\mathbb{Z}, +, 0)$ – i.e. the monoid of natural numbers equipped with addition and the monoid of positive and negative integers equipped with addition).

The natural inclusion map $i\colon N \to Z$ obviously does not have an inverse in **Mon**. It is however both monic and epic.

First, suppose $\overline{M} = (M, \cdot, 1_M)$ is some monoid and we have two arrows $g, h\colon \overline{M} \to N$, where $g \neq h$. So there is some element $m \in M$ such that the natural numbers $g(m)$ and $h(m)$ are different, which means that the corresponding integers $i(g(m))$ and $i(h(m))$ are different, so $i \circ g \neq i \circ h$. Contraposing, this means i is monic in the category.

Second, again take a monoid $\overline{M}$ and this time consider any two monoid homomorphisms $g, h\colon Z \to \overline{M}$ such that $g \circ i = h \circ i$. Then g and h must agree on all integers from zero up. But then note

$$\begin{aligned} g(-1) = g(-1) \cdot 1_M = g(-1) \cdot h(0) &= g(-1) \cdot h(1 + -1) \\ &= g(-1) \cdot h(1) \cdot h(-1) = g(-1) \cdot g(1) \cdot h(-1) \\ &= g(-1 + 1) \cdot h(-1) = g(0) \cdot h(-1) = 1_M \cdot h(-1) = h(-1). \end{aligned}$$

But if $g(-1) = h(-1)$, then

$$g(-2) = g(-1 + -1) = g(-1) \cdot g(-1) = h(-1) \cdot h(-1) = h(-1 + -1) = h(-2),$$

and so it goes. So we have $g(j) = h(j)$ for all $j \in \mathbb{Z}$, positive and negative, and hence $g = h$ and i is right-cancellable, i.e. epic.

So: we can't define an isomorphism as an epic monic if we want isomorphisms to have the essential feature of invertibility. Which all goes to motivate the following key definition:

Definition 12. An *isomorphism* (in category $\mathscr{C}$) is an arrow which has an inverse.

So isomorphisms are monic and epic but not always vice versa. However, we do have

Theorem 10. *If f is both monic and split epic (or both epic and split monic), then f is an isomorphism.*

Proof. If f is a split epimorphism, it has a right inverse, i.e. there is a g such that $f \circ g = 1$. Then $(f \circ g) \circ f = f$, whence $f \circ (g \circ f) = f \circ 1$. Hence, given that f is also mono, $g \circ f = 1$, and g is both a left and right inverse for f, i.e. f has an inverse. Dually for the other half of the theorem. □

Notation. It is standard to decorate arrows which are isomorphisms thus: $\xrightarrow{\sim}$.

(b) From what we have already seen, we know or can readily check that

Theorem 11. *(1) Identity arrows are isomorphisms.*

(2) An isomorphism $f\colon C \xrightarrow{\sim} D$ has a unique inverse which we can call $f^{-1}\colon D \xrightarrow{\sim} C$, such that $f^{-1} \circ f = 1_C$, $f \circ f^{-1} = 1_D$, $(f^{-1})^{-1} = f$, and f^{-1} is also an isomorphism.

(3) If f and g are isomorphisms, then $g \circ f$ is an isomorphism if it exists, whose inverse will be $f^{-1} \circ g^{-1}$.

Definition 13. If there is an isomorphism $f\colon C \xrightarrow{\sim} D$ in $\mathscr{C}$ then C, D are said to be *isomorphic* (in $\mathscr{C}$), and we write $C \cong D$.

Then, from the ingredients of Theorem 11, we immediately get the desired result that

Theorem 12. *Isomorphism between objects in a category is an equivalence relation.*

As you would expect, in the category **Set** isomorphisms are bijections, in the category **Grp** isomorphisms are group isomorphisms, in the category of vector spaces isomorphisms are invertible linear maps. And so it goes.

(c) Let's remark that, just as we can consider a particular monoid as a category, in the same way we can consider a particular group as a category. Thus take a group $(G, \cdot, e)$. And now define $\mathscr{G}$ to be the corresponding category whose sole object is whatever you like, dub it '$\star$', and who arrows $g\colon \star \to \star$ are the elements g of G, with $e\colon \star \to \star$ the identity arrow, and with composition of arrows in $\mathscr{G}$ defined as group-multiplication of elements in G. Since every element in the group has an inverse, it follows immediately that every arrow in the corresponding category has an inverse, i.e. is an isomorphism.

So in sum, a group-as-a-category has one object and every arrow is an isomorphism. (And for the record, generalizing, a category with perhaps more than one object but whose arrows are all isomorphisms is called a *groupoid*.)

4 Functors

Our definition of an isomorphism characterizes a type of arrow not – as it were – 'internally', by how it operates input-by-input (assuming it is indeed a function), but 'externally', by its interaction with another arrow, its inverse. This is typical of a category-theoretic (re)definition of a familiar notion: we look for external, relational, characterizations of arrows and/or structured objects.

Here's Awodey, offering some similarly arm-waving quick

> ... remarks about category-theoretical definitions. By this I mean characterizations of properties of objects and arrows in a category in terms of other objects and arrows only, that is, in the language of category theory. Such definitions may be said to be abstract, structural, operational, relational, or external (as opposed to internal). The idea is that objects and arrows are determined by the role they play in the category via their relations to other objects and arrows, that is, by their position in a structure and not by what they 'are' or 'are made of' in some absolute sense. (Awodey, 2006, p. 25)

So a very natural way forward from where we've got to would be to proceed to give further 'external' category-theoretic definitions of familiar notions.

For example, take the usual 'internal' way of dealing with Cartesian products, sets of ordered pairs, by treating them as collections of (unordered!) Kuratowski-pairs of the kind $\{\{x\}, \{x, y\}\}$. A familiar complaint is that this construction is arbitrary; there are endless alternatives. And the equally familiar riposte is: "Don't worry. The construction *works*!" Yes: but what does 'working' come to here?

It means that there is a pairing function which takes us from sets X and Y to their Cartesian product, and a couple of matched unpairing functions which take a product and extract the original sets, and these pairing/upairing functions behave nicely (in ways we can and will spell out). But then, so long as we have suitable product-constructing and deconstructing functions, the 'internal' constitution of the product (e.g. as a collection of Kuratowski-pairs) doesn't really matter at all. It's the pattern of interrelations between various functions that matters. Which is just the sort of thing that category theory highlights.

For an approach to category-theory which starts by exploring categorial accounts of products and similar constructions, see e.g. Goldblatt (2006, Ch. 3) or McLarty (1992, Part I). But in these notes, we will turn to consider products and the like rather later – indeed, not until Chapter 14. First, we go even more abstract and start considering not just the arrows/maps that can obtain between objects *inside* a category, but also maps *between* categories. There are pluses and minuses to taking this route via the stratosphere: but I'm following what seems to be something of a Cambridge tradition in the Part III course (that has its roots, perhaps, in the history of category theory: compare also the canonical text Mac Lane 1997). So let's see how things work out.

4.1 Functors defined

The usual term for a map between categories is a *functor*. A category $\mathscr{C}$ has two kinds of data, its objects and its arrows. So a functor F from category $\mathscr{C}$ to category $\mathscr{D}$ will need to have two components, one that operates on objects, one that operates on arrows. Hence, to start our definition, we will say

Definition 14. Given categories $\mathscr{C}$ and $\mathscr{D}$, a *functor* $F\colon \mathscr{C} \to \mathscr{D}$ comprises the following data:

(1) A mapping F_{ob} whose value at the $\mathscr{C}$-object A is some $\mathscr{D}$-object we can represent simply as $F(A)$ or indeed often just as FA.
(2) A mapping F_{arw} whose value at the $\mathscr{C}$-arrow $f\colon A \to B$ is a $\mathscr{D}$-arrow from $F(A)$ to $F(B)$ which we can represent as $F(f)\colon F(A) \to F(B)$, or indeed often just as $Ff\colon FA \to FB$.

But there's more. If a functor is to preserve the most basic categorial structure, its component mappings must obey two obvious conditions. First they must map identity arrows to identity arrows. Second they should respect composition. That is to say, since the commutative diagram $\begin{tikzcd} & B \arrow[dr, "g"] & \\ A \arrow[ur, "f"] \arrow[rr, "g\circ f"'] & & C \end{tikzcd}$ gets sent by F to $\begin{tikzcd} & FB \arrow[dr, "F(g)"] & \\ FA \arrow[ur, "F(f)"] \arrow[rr, "F(g\circ f)"'] & & FC \end{tikzcd}$ the second diagram should also commute. Hence,

Definition 14 continued. The data in F must satisfy the following conditions:

Preserving identities: for any $\mathscr{C}$-object A, $F(1_A) = 1_{FA}$;
Respecting composition: for any $\mathscr{C}$-arrows f, g which compose, $F(g \circ f) = Fg \circ Ff$.

These conditions on F are often called, simply, *functoriality*.

4.2 Some elementary examples of functors

Our first example illustrates a broad class of cases:

(F1) There is a functor $F : \mathbf{Mon} \to \mathbf{Set}$ with the following data:

i. F_{ob} sends the monoid $(M, \cdot, 1_M)$ to its carrier set M.

ii. F_{arw} sends an arrow $f\colon (M, \cdot, 1_M) \to (N, \times, 1_N)$ (which is in fact a monoid homomorphism acting on elements on M) to the same function $f\colon M \to N$.

So defined, F trivially obeys the axioms for being a functor. All it does is 'forget' about the structure carried by a monoid. It's a *forgetful functor*, for short.

There are other equally forgetful functors. For example, there is the functor $F\colon \mathbf{Grp} \to \mathbf{Set}$ that sends groups to their underlying carrier sets, and sends group homorphisms to their underlying functions, but forgets about the group structure. Sometimes a forgetful functor such as this is called an *underlying* functor (and hence the common practicce, which we shall occasionally adopt, of using the letter 'U' to denote such a functor).

To continue:

(F2) There are also slightly less forgetful functors, such as the functor from **Rng** to **Grp** that sends a ring to the additive group it contains, forgetting the rest of the ring structure.

(F3) Take monoids $(M, \cdot, 1_M)$ and $(N, \times, 1_N)$ and considered the corresponding categories $\mathscr{M}$ and $\mathscr{N}$ in the sense of §2.2.

So $\mathscr{M}$ has a single object $\star_{\mathscr{M}}$, and its arrows are elements of M, where the composition of the arrows m_1 and m_2 is just $m_1 \cdot m_2$, and the identity arrow is the identity element of the monoid, 1_M.

Likewise $\mathscr{N}$ has a single object $\star_{\mathscr{N}}$, and arrows are elements of N, where the composition of the arrows n_1 and n_2 is just $n_1 \times n_2$, and the identity arrow is the identity element of the monoid, 1_N.

So now we see that a functor $F : \mathscr{M} \to \mathscr{N}$ will need to do the following:

i. F must send $\star_{\mathscr{M}}$ to $\star_{\mathscr{N}}$.

ii. F must send the identity arrow 1_M to the identity arrow 1_N.

iii. F must send $m_1 \circ m_2$ (i.e. $m \cdot n$) to $Fm_1 \circ Fm_2$ (i.e. $Fm \times Fn$).

Apart from the trivial first condition, that just requires F to be a monoid homomorphism. So any homomorphism between two monoids induces a corresponding functor between the corresponding monoids-as-categories.

(F4) Take the posets $(S, \preccurlyeq)$ and $(T, \sqsubseteq)$ considered as categories $\mathscr{S}$ and $\mathscr{T}$. It is easy to check that a monotone function $f\colon S \to T$ induces a functor $F\colon \mathscr{S} \to \mathscr{T}$ which sends an $\mathscr{S}$-object s to the $\mathscr{T}$-object $f(s)$, and sends an $\mathscr{S}$-arrow $\langle s, s' \rangle$ to the $\mathscr{T}$-arrow $\langle f(s), f(s') \rangle$.

(F5) Next, consider $\mathscr{G}$, the group $(G, \cdot, e)$ considered as a category (see the end of §3.3), and suppose $F\colon \mathscr{G} \to \mathbf{Set}$ is a functor. Then F must send $\mathscr{G}$'s unique object $\star$ to some set X. And F must send a $\mathscr{G}$-arrow $m\colon \star \to \star$ (that's just a member of the group G) to a function $F(m)\colon X \to X$. Functoriality requires that $F(e) = 1_X$ and $F(m \cdot m') = F(m) \circ F(m')$. But those are just the conditions for F to constitute a group action of G on X. Conversely, a group action of G on X amounts to a functor from $\mathscr{G}$ to $\mathbf{Set}$.

(F6) There is a list functor $List\colon \mathbf{Set} \to \mathbf{Set}$, where $List(X)$ is the set of all finite lists or sequences of elements of X, including the empty one. And given a function $f\colon X \to Y$, $List(f)\colon List(X) \to List(Y)$ is the function which sends the list $x_0^\frown x_1^\frown x_2^\frown \ldots {}^\frown x_n$ to $fx_0^\frown fx_1^\frown fx_2^\frown \ldots {}^\frown fx_n$ (where ${}^\frown$ is concatenation).

Now, we have seen cases of functors that simply forget (some of the) structure put on structured sets. We can also have a functor on **Set** which forgets that arrows are functions:

(F7) There is a functor $F : \mathbf{Set} \to \mathbf{Rel}$ which sends a set, considered as an object of **Set**, to the same set, now thought of as an object of **Rel**, and sends and arrow $f\colon A \to B$ in **Set** to its graph, the set of pairs $\langle a, f(a) \rangle$ in $A \times B$.

There will be other functors which – so to speak – collapse a category in on itself by forgetting distinctions between objects or arrows. Here's an example.

(F8) Consider a category $\mathscr{C}$, and consider the corresponding slimmed-down category $\mathscr{S}$ which has the same objects but, whenever there are some $\mathscr{C}$-arrows from A to B, $\mathscr{S}$ has just one of them. (We'll assume we have a strong enough choice principle to be able to do this while keeping $\mathscr{S}$ a category.)

Evidently there is a functor $F\colon \mathscr{C} \to \mathscr{S}$ which sends an object to itself, and sends a $\mathscr{C}$-arrow from A to B to the unique $\mathscr{S}$-arrow from A to B.

$\mathscr{S}$ is a pre-order category in the sense of §2.2, Ex. (8), and we could call F a 'pre-orderfication' functor on $\mathscr{C}$.

For a more extreme case, for any non-empty category $\mathscr{C}$ there is what we might call the 'total collapse' functor $F_c\colon \mathscr{C} \to \mathbf{1}$ which sends every object of $\mathscr{C}$ to the sole object of $\mathbf{1}$, and sends every arrow in $\mathscr{C}$ to the sole arrow of $\mathbf{1}$.

We have just mentioned the unique functor from (non-empty) $\mathscr{C}$ to $\mathbf{1}$. Now go in the other direction:

(F9) For each object C in $\mathscr{C}$ there is a corresponding functor – overloading notation, we can usefully call it $C\colon \mathbf{1} \to \mathscr{C}$ – which sends the sole object of $\mathbf{1}$ to C, and sends the sole arrow of $\mathbf{1}$ to 1_C.

And finally in this introductory list we should also mention an even more trivial case:

(F10) For any category $\mathscr{C}$ there is the trivial functor $1_{\mathscr{C}}\colon \mathscr{C} \to \mathscr{C}$ which sends each object in $\mathscr{C}$ to itself and sends each arrow in $\mathscr{C}$ to itself!

4.3 A functor from Set to Mon

We are next going to construct a functor going in the reverse direction to the forgetful functor in (F1), i.e. a functor $F : \mathbf{Set} \to \mathbf{Mon}$.

Let's consider how to send a set to a monoid *making as few assumptions as we can* about the monoid that a given set M gets mapped to.

Starting, then, from a set M, if we are going to get a monoid from it we need to equip it with a two-place associative function $*$. But we are assuming as little as we can about $*$, so we don't even yet know that $*$ keeps us inside the original set M! So M will need to get associated with a set M^* that contains not only the original members of M, e.g. $x, y, z, \ldots$, but all the likes of $x * x,\ x * y,\ y * z,\ x*y*x,\ x*y*x*z,\ x*x*y*y*z\ldots$, etc., etc. But even that's not enough, for (in our assumption-free state) we don't know that any of these elements of M^* act as an identity for the $*$-function. So to get a monoid, we will have to throw in some identity element 1.

Now, here's a neat way to model $(M^*, *, 1)$. Represent a monoid element (such as $x * x * y * y * z$) as a *finite list of members of* M, so M gets represented by $List(M)$ and treat the $*$-function as just concatenation $^\frown$. The identity element will then be modelled by the null list.

Which all goes to motivate the following construction:

(F11) There is a functor $F : \mathbf{Set} \to \mathbf{Mon}$ with the following data:

i. F_{ob} sends the set M to the monoid $(List(M), ^\frown, 1)$, where $^\frown$ is concatenation and the unit is the empty list. This monoid is said to be the *free monoid* on M.

ii. F_{arw} sends the arrow $f\colon M \to N$ to $List(f)$, now treated as an arrow from $(List(M), ^\frown, 1)$ to $(List(N), ^\frown, 1)$.

As long as we remember about the null sequence, it is trivial to check that Ff is a monoid homomorphism, and that F is then a functor.

Similarly, there are other functors that send sets to *freely generated* structures on the set. For example there is a functor from **Set** to **Ab** which sends a set X to the freely generated abelian group on X, i.e. the direct sum of X-many copies of $\mathbb{Z}$ (where we can think of $\mathbb{Z}$ as the free abelian group on one generator). [For more, see http://en.wikipedia.org/wiki/Free_abelian_group.]

4.4 Subcategories and inclusion functors

To introduce our next type of functor, we need an obvious definition:

Definition 15. Given a category $\mathscr{C}$, if $\mathscr{S}$ consists of the data

(1) Objects: some or all of the $\mathscr{C}$-objects,
(2) Arrows: some or all of the $\mathscr{C}$-arrows,

subject to the conditions

(3) for each $\mathscr{S}$-object C, the $\mathscr{C}$-arrow 1_C is also an $\mathscr{S}$-arrow,
(4) for any $\mathscr{S}$-arrows $f\colon C \to D$, $g\colon D \to E$, the arrow $g \circ f\colon C \to E$ is also an $\mathscr{S}$-arrow,

then $\mathscr{S}$ is a *subcategory* of $\mathscr{C}$.

Plainly, the conditions in the definition – containing identity arrows and being closed under composition – are there to ensure that $\mathscr{S}$ is indeed still a category.

Some examples:

(1) $\mathscr{S}$ is a subcategory of $\mathscr{C}$ if $\mathscr{S}$ is a pre-orderfication of $\mathscr{C}$.
(2) The discrete category on the objects of $\mathscr{C}$ is a subcategory of $\mathscr{C}$ for any category.
(3) **Set** is a subcategory of **Pfn**.
(4) **FinSet** is a subcategory of **Set**,
(5) **Ab** is a subcategory of **Grp**.

So we can shed objects and/or arrows in moving from a category to a subcategory. The first three examples are cases where we shed some or all of the non-identity arrows while keeping all the objects. But the last two examples are ones where drop some objects while keeping all the arrows between those objects retained in the subcategory, and there is a standard label for such cases:

Definition 16. If $\mathscr{S}$ is a subcategory of $\mathscr{C}$ where, for all $\mathscr{S}$-objects A and B, the $\mathscr{S}$-arrows from A to B are all the $\mathscr{C}$-arrows from A to B, then $\mathscr{S}$ is said to be a *full subcategory* of $\mathscr{C}$

And now note that

(F8) If $\mathscr{S}$ is a subcategory of $\mathscr{C}$, then there is an (obviously defined) inclusion functor $F\colon \mathscr{S} \to \mathscr{C}$. (We reserve the notation $F\colon \mathscr{S} \hookrightarrow \mathscr{C}$ for the case when $\mathscr{S}$ is a *full* subcategory of $\mathscr{C}$.)

4.5 What do functors preserve, reflect, create?

Definition 17. Suppose $F : \mathscr{C} \to \mathscr{D}$ and P is some property of arrows. Then

(1) F *preserves* P iff, for any $\mathscr{C}$-arrow f, if f has property P, so does $F(f)$.
(2) F *reflects* P iff, for any $\mathscr{C}$-arrow f, if $F(f)$ has property P, so does f.
(3) F *creates* P iff any $\mathscr{D}$-arrow which is P is $F(f)$ for some unique $\mathscr{C}$-arrow f, and f is P.

We also say that F preserves Xs if F preserves the property of being an X, etc. There are analogous notions of preserving (reflecting, creating) properties of objects and of other categorial entities.

In this section, we concentrate on preserving and reflecting. First,

Theorem 13. *Functors preserve left inverses, right inverses and isomorphisms.*

Proof. Easy! Recall, $f : C \to D$ is a right inverse in the category $\mathscr{C}$ iff for some arrow g, $g \circ f = 1_C$. Let F be a functor mapping the category $\mathscr{C}$ to $\mathscr{D}$. Then $F(g \circ f) = F(1_C)$. By the functorial nature of F, that implies $F(g) \circ F(f) = 1_{FC}$. So $F(f)$ is a right inverse in the category $\mathscr{D}$.

Duality gives the result for left inverses. And putting the two results together gives us the result for isomorphisms. □

Now we said, arm-wavingly, that functors respect basic categorial structure. But that doesn't mean that they preserve *all* structural properties defined in categorial terms:

Theorem 14. *Functors do not necessarily preserve monomorphisms and epimorphisms.*

Proof. Example (i). Consider the inclusion map $i_M : (\mathbb{N}, +, 0) \to (\mathbb{Z}, +, 0)$ in **Mon**. We saw in §3.3, Ex. (2) that this is epic.

But plainly the inclusion map $i_S : \mathbb{N} \to \mathbb{Z}$ in **Set** is not epic (as it isn't surjective).

Hence the forgetful functor $F : \mathbf{Mon} \to \mathbf{Set}$ therefore maps an epic map (i_M) to a non-epic one (i_S). □

Theorem 15. *Functors do not necessarily reflect left inverses, right inverses or isomorphisms.*

Proof. It is enough to produce a functor that doesn't reflect isomorphisms (since an isomorphism is both a left- and a right inverse).

Let the poset A be $\{2, 4, 6\}$ ordered by divisibility, and the poset B be $\{2, 4, 6\}$ naturally ordered. The map $f : A \to B$ which sends a number to itself is order-preserving but evidently not an order-isomorphism. But the forgetful functor $F : \mathbf{Mon} \to \mathbf{Set}$ sends f to the set function $F(f)$, the identity map on $\{2, 4, 6\}$, which is trivially an isomorphism in **Set**. □

4.6 An example from algebraic topology

We'll now briefly consider another example of a functor, from algebraic topology, which also allows us to apply Theorem 13. This example can readily be skipped – though in fact, to follow it will be enough to have read a few early pages from e.g. Allen Hatcher's *Algebraic Topology*. Even more briefly, to get a glimmer of what's going on, you just need the idea of the fundamental group of a topological space (at a point).

Thus, given a space and a chosen base point in it, consider all directed paths that start at this base point, wander around and eventually loop back to their starting point. Such directed loops can be "added" together in an obvious way: you traverse the "sum" of two loops by going round the first loop, then round the second. Every loop has an "inverse" (you go round the same path in the opposite direction). Two loops are considered 'homotopically' equivalent if one can be continuously deformed into the other. Consider, then, the set of all such equivalence classes of loops – so-called homotopy equivalence classes – and define "addition" for these classes in the obvious derived way. This set, when equipped with addition, evidently forms a group: it is the *fundamental group* for that particular space, with the given basepoint. (Though for many spaces, the group is independent of the basepoint.)

Suppose, therefore, that $\mathbf{Top}_*$ is the category of pointed topological spaces: an object in the category is a topological space X equipped with a distinguished base point x_0, and the arrows in the category are continuous maps that preserve basepoints. Then:

(F9) There is a functor $\pi_1 \colon \mathbf{Top}_* \to \mathbf{Grp}$, to use its standard label, with the following data

i. π_1 sends a pointed topological space (X, x_0) – i.e. X with base point x_0 – to the fundamental group $\pi_1(X, x_0)$ of X at x_0.

ii. π_1 sends a basepoint-preserving continuous map $f \colon (X, x_0) \to (Y, y_0)$ to a corresponding group homomorphism $f_* \colon \pi_1(X, x_0) \to \pi_1(Y, y_0)$. [For arm-waving motivation: f maps a continuous loop based at x_0 to a continuous loop based at y_0; and since f is continuous it maps a continuous deformation of a loop in (X, x_0) to a continuous deformation of a loop (Y, y_0) – and that induces a corresponding association f_* between the homotopy equivalence classes of (X, x_0) and (Y, y_0), and this will respect the group structure.]

Suppose we do the work of checking that π_1 is indeed functorial. Then here's an application. We'll prove Brouwer's Fixed Point Theorem (and at last, you might say, we get something worth calling a *theorem*!):

Theorem 16. *Any continuous map of the closed unit disc to itself has a fixed point.*

Proof. Suppose that there is a continuous map f on the two-dimensional disc D (considered as a topological space) such that we always have $f(x) \neq x$.

Let the boundary of the disc be the circle S (again considered as a topological space). Then we can define a map that sends the point x in D to the point in S at which the ray from $f(x)$ through x intersects the boundary of the disc.

This map sends a point on boundary to itself. Pick a boundary point to be the base point of the pointed space D_* and also of the pointed space S_*, then our map induces a map $r\colon D_* \to S_*$. Moreover, this map is evidently continuous (intuitively: nudge a point x and since f is continuous that just nudges $f(x)$, and hence the ray from $f(x)$ through x is only nudged, and the point of intersection with the boundary is only nudged). And r is a left inverse of the inclusion map $i\colon S_* \hookrightarrow D_*$ in $\mathbf{Top}_*$, since $r \circ i = 1$.

Functors preserve left inverses by Theorem 13, so $\pi_1(r)$ will be a left inverse of $\pi_1(i)$ in the category **Grp**, which means that $\pi_1(i)\colon \pi_1(S_*) \to \pi_1(D_*)$ is right-inverse in **Grp**, hence by Theorem 8 is monic, and hence by Theorem 4 is an injection.

But that's impossible. $\pi_1(S_*)$, the fundamental group of S_*, is [equivalent to] the group $\mathbb{Z}$ of integers under addition (think of looping round a circle, one way or another, n times – each positive or negative integer corresponds to a different path); while $\pi_1(D_*)$, the fundamental group of D_*, is just a one element group (for every loop in D_* can be smoothly shrunk to a point). And there is no injection between the integers and a one-element set! $\square$

What, if anything, do we gain from putting the proof in category theoretic terms? It might be said: the proof crucially depends on facts of algebraic topology – continuous maps preserve homotopic equivalences in a way that makes π_1 a functor (not entirely trivial), and the fundamental groups of S^* and D^* are respectively $\mathbb{Z}$ and the trivial group. And we could run the whole proof without actually mentioning categories at all. But what we've done is, so to speak, very clearly demarcate those bits of the proof that *do* specifically depend on facts of algebraic topology and those bits which depend on more general proof-ideas about kinds of maps (inverses, monics, injections) which are thoroughly *portable* to other contexts. And *that* surely counts as a gain in understanding.

5 More about functors and categories

5.1 Functors compose

Theorem 17. *Suppose we have functors $F\colon \mathscr{C} \to \mathscr{D}$, $G\colon \mathscr{D} \to \mathscr{E}$, then there is a functor $G \circ F\colon \mathscr{C} \to \mathscr{E}$ with the following data:*

(1) A mapping $(G \circ F)_{ob}$ which sends a $\mathscr{C}$-object A to the $\mathscr{E}$-object GFA – i.e., if you prefer that with brackets, to $G(F(A))$.

(2) A mapping $(G \circ F)_{arw}$ which sends a $\mathscr{C}$-arrow $f\colon A \to B$ to the $\mathscr{E}$-arrow $GFf\colon\ GFA \to GFB$ – i.e. to $G(F(f))$.

An elementary check confirms that the axioms for being a functor are satisfied. It is also elementary to check that we have:

Theorem 18. *Composition of functors is associative.*

5.2 Covariant vs contravariant functors

(a) How do functors interact with the operation of taking the opposite category? Well, first we note:

Theorem 19. *A functor $F\colon \mathscr{C} \to \mathscr{D}$ induces a functor $F^{op}\colon \mathscr{C}^{op} \to \mathscr{D}^{op}$.*

Proof. Recall, the objects of $\mathscr{C}^{op}$ are exactly the same as the objects of $\mathscr{C}$. We can therefore define the object-mapping component of F^{op} as acting on $\mathscr{C}^{op}$-objects exactly as the object-mapping component of F acts on $\mathscr{C}$-objects. And then, allowing for the fact that taking opposites reverses arrows, we can define the arrow-mapping component of F^{op} as acting on the $\mathscr{C}^{op}$-arrow $f\colon C \to D$ exactly as the arrow-mapping component of F acts on the $\mathscr{C}$-arrow $f\colon D \to C$.

F^{op} will evidently obey the axioms for being a functor because F does. $\square$

It is trivial, by the way, that $(F^{op})^{op} = F$.

But now for a new departure: we introduce a new kind of functor:

Definition 18. $F\colon \mathscr{C} \to \mathscr{D}$ is a *contravariant* functor from $\mathscr{C}$ to $\mathscr{D}$ if $F\colon \mathscr{C}^{op} \to \mathscr{D}$ is a functor in the original sense. So it comprises the following data:

(1) A mapping F_{ob} whose value at the $\mathscr{C}$-object A is some $\mathscr{D}$-object $F(A)$.
(2) A mapping F_{arw} whose value at the $\mathscr{C}$-arrow $f\colon B \to A$ is a $\mathscr{D}$-arrow $F(f)\colon FA \to FB$. (NB the directions of the arrows!)

And this data satisfies the two axioms:

Preserving identities: for any $\mathscr{C}$-object A, $F(1_A) = 1_{F(A)}$;
Respecting composition: for any $\mathscr{C}$-arrows f, g which compose, $F(g \circ f) = Ff \circ Fg$.

It would of course be equivalent to define a contravariant functor from $\mathscr{C}$ to $\mathscr{D}$ to be a functor (in the original sense) from $\mathscr{C}$ to $\mathscr{D}^{op}$. Note: a functor in our original sense, when the contrast needs to be stressed, is also called a *covariant* functor.

(b) Let's have an example of a naturally arising contravariant functor. Take $\mathbf{V}$, the category whose objects are the finite dimension vector spaces over the reals, and whose arrows are linear maps between spaces.

Now recall, the dual space of such a vector space V is V^*, the set of all linear functions $f\colon V \to \mathbb{R}$ (where this set is equipped with vectorial structure in the obvious way). V^* has the same dimension as V (so, a fortiori, is also finite dimensional and belongs to $\mathbf{V}$). We'll construct a dualizing functor-style map D from the category $\mathbf{V}$ to itself where D_{ob}, acting on objects, sends a vector-space to its dual.

So now, how is our functor D going to act on arrows in the category $\mathbf{V}$? Take spaces V, W and consider any linear map $g\colon W \to V$. Then, over on the dual spaces, there will be a naturally corresponding map $(- \circ g) : V^* \to W^*$ which maps $f\colon V \to \mathbb{R}$ to $f \circ g\colon W \to \mathbb{R}$: note the direction that g has to go in for a composition with f to work. This defines the action of a component D_{arw} for the dualizing map D: it sends a linear map g to the map $(- \circ g)$.

And these components D_{ob} and D_{arw} evidently do give us a contravariant functor.

5.3 Product categories, arrow categories, and associated functors

(a) 'Categories beget categories.' We've already seen one way they do this: for every category there is an opposite category. In the rest of this chapter we meet some other ways of getting new categories from old. However, this all gets rather abstract rather quickly, and I rather hesitate to cover the material at this stage. Do feel very free to skim and skip for moment, only returning when you actually need these ideas in later applications.

Definition 19. Given categories $\mathscr{C}$ and $\mathscr{D}$, $\mathscr{C} \times \mathscr{D}$ is the derived *product category* with the following data:

(1) Its objects are all the ordered pairs $\langle C, D\rangle$, for C a $\mathscr{C}$-object and D a $\mathscr{D}$-object .
(2) For any two such pairs $\langle C, D\rangle, \langle C', D'\rangle]$, the associated arrows from $\langle C, D\rangle$ to $\langle C', D'\rangle$ are the pairs $\langle f, f'\rangle$, where $f\colon C \to D$ and $f'\colon C' \to D'$.
(3) For each $\langle C, D\rangle$, there is an identity arrow $1_{\langle C,D\rangle}$ which is the pair $\langle 1_C, 1_D\rangle$.
(4) Composition of arrows is defined component-wise: $\langle f, f'\rangle \circ \langle g, g'\rangle = \langle f \circ g, f' \circ g'\rangle$.

It is immediate that this does indeed define a category.

Associated with a product category $\mathscr{C} \times \mathscr{D}$ there will be a pair of 'projection' functors $P^1\colon \mathscr{C} \times \mathscr{D} \to \mathscr{C}$, $P^2\colon \mathscr{C} \times \mathscr{D} \to \mathscr{D}$ defined in the obvious way. Thus

(1) P^1_{ob} maps $\langle C, D\rangle$ to C.
(2) P^1_{arw} maps a $\mathscr{C} \times \mathscr{D}$-arrow $\langle f, g\rangle$ to the $\mathscr{C}$-arrow f.

Similarly, of course, for the projection functor P^2.

Why might this be of interest? We have defined functors as mapping a single category to another category. Yet, as we'll see later, we might sometimes be interested in functors that are naturally construed as being functors 'in two variables', e.g. as mapping an object from $\mathscr{C}$ and an object from $\mathscr{D}$ to an object from $\mathscr{E}$. Well, we can treat such a functor as in fact being some $F\colon \mathscr{C} \times \mathscr{D} \to \mathscr{E}$ – and that allows us to keep our original official definition of functors as operating on one category at a time. A useful device (we'll return to this theme when we consider products more generally).

(b) Let's also quickly mention another example of a derived category (if only to illustrate that the objects of category need not be really object-like!):

Definition 20. Given a category $\mathscr{C}$, the derived *arrow category* $\mathscr{C}^{\to}$ has the following data:

(1) $\mathscr{C}^{\to}$'s objects, its first sort of data, are simply the *arrows* of $\mathscr{C}$,
(2) Given $\mathscr{C}^{\to}$-objects f_1, f_2 (i.e. $\mathscr{C}$-arrows $f_1\colon X_1 \to Y_1$, $f_2\colon X_2 \to Y_2$), an arrow $a\colon f_1 \to f_2$ is a pair $\langle j, k\rangle$ of $\mathscr{C}$-arrows such that the following diagram commutes:

$$\begin{tikzcd} X_1 \arrow[r, "f_1"] \arrow[d, "j"] & Y_1 \arrow[d, "k"] \\ X_2 \arrow[r, "f_2"] & Y_2 \end{tikzcd}$$

Again it is easily checked that, with composition defined in the only sensible way, this does define a category.

(Two questions to think about. First, consider the operation of sending a category $\mathscr{C}$ to the corresponding arrow category $\mathscr{C}^{\to}$. Is this functorial? Second,

$\mathscr{C}^{\rightarrow}$ by definition has as its objects $\mathscr{C}$'s arrows: there could similarly be a category $(\mathscr{C}^{\rightarrow})^{\rightarrow}$ which has as its objects $\mathscr{C}^{\rightarrow}$'s arrows. What should this new category's arrows be?)

5.4 Slice categories and associated functors

Suppose next that $\mathscr{C}$ is a category, and I a particular $\mathscr{C}$-object. We will define a new category from $\mathscr{C}$, this time the so-called 'slice' category $\mathscr{C}/I$ where – as in an arrow category – the new category's objects are in fact (some of) the original category's *arrows*. (To keep things clear but brief, let's use 'arrow$_{\mathscr{C}}$' to refer to the old arrows and reserve plain 'arrow' for the new arrows to be found in $\mathscr{C}/I$.)

We can now give the following definition:

Definition 21. Let $\mathscr{C}$ be a category, and I be a $\mathscr{C}$-object. Then the new category $\mathscr{C}/I$, the *slice category over* I, has the following data:

(1) The $\mathscr{C}/I$-objects are all the arrows$_{\mathscr{C}}$ $f\colon A \to I$, for any $\mathscr{C}$-object A.
(2) For $\mathscr{C}/I$-objects (yes, objects!) $f\colon A \to I$, $g\colon B \to I$, the associated arrows from f to g are just the arrows$_{\mathscr{C}}$ $j\colon A \to B$ such that $g \circ j = f$ in $\mathscr{C}$.
(3) For every $\mathscr{C}/I$-object $f\colon A \to I$, there is an identity arrow, namely the arrow$_{\mathscr{C}}$ $1_A\colon A \to A$.
(4) For any $\mathscr{C}/I$-objects $f\colon A \to I$, $g\colon B \to I$, $h\colon C \to I$, the composition of $j\colon f \to g$ and $k\colon g \to h$ is defined as follows: $k \circ j\colon f \to h$ is the arrow$_{\mathscr{C}}$ $k \circ j\colon A \to C$.

Of course, we need to check that these data do indeed together satisfy the axioms for constituting a category. So let's do that.

$$\begin{tikzcd} A \arrow[d, "j"'] \arrow[dr, "f"] & \\ B \arrow[r, "g"'] & I \end{tikzcd}$$

First, a diagram might help. Take the arrows$_{\mathscr{C}}$ $f\colon A \to I$, $f\colon B \to I$: they will be objects f, g of $\mathscr{C}/I$. And the arrows of $\mathscr{C}/I$ from f to g will be the arrows$_{\mathscr{C}}$ like $j\colon A \to B$ which make our first diagram commute. Note: the domain and co-domain of j as an arrow$_{\mathscr{C}}$ are respectively A, B. But the domain and co-domain of j as an arrow in the slice category $\mathscr{C}/I$ are respectively f and g.

$$\begin{tikzcd} A \arrow[d, "j"'] \arrow[dr, "f"] & \\ B \arrow[r, "g"] \arrow[d, "k"'] & I \\ C \arrow[ur, "h"'] & \end{tikzcd}$$

We now need to confirm that our definition of $k \circ j$ for $\mathscr{C}/I$ works. We are given in $\mathscr{C}$ that $j\colon A \to B$ is such that $g \circ j = f$, and also that $h \circ k = g$. Putting things together we get the second commutative diagram.

Or in equations, we have $(h \circ k) \circ j = f$ in $\mathscr{C}$, and therefore $h \circ (k \circ j) = f$. So $(k \circ j)$ does indeed count as an arrow in $\mathscr{C}/I$ from f to h, as we require.

The remaining checks to confirm $\mathscr{C}/I$ satisfies the axioms for being a category are then trivial.

Unsurprisingly, there's a dual notion, the idea of a *co-slice category*, $I/\mathscr{C}$ whose objects are the arrows$_{\mathscr{C}}$ $f\colon I \to A$, for any $\mathscr{C}$-object A and the rest of the definition is as you would expect.

Here are a couple of quick examples of slice and co-slice categories, one of each kind:

(1) First, we mentioned before the idea of a pointed topological space, i.e. a space X equipped with a selected base-point x (see §4.6). Now, we can think of a one-element set as trivial topological space, call it '1'. And we also mentioned before the idea that we can think of any element $x \in X$ as an arrow $\vec{x}\colon 1 \to X$ (a map from a singleton to X). So now think about the co-slice category $1/\mathbf{Top}$. Its objects are all the morphisms $\vec{x}\colon 1 \to X$ (for any space X, one morphism for every point in X): we can think of each such morphism as equipping some space X with a basepoint x. And the arrows in $1/\mathbf{Top}$ from some $\vec{x}\colon 1 \to X$ to some $\vec{y}\colon 1 \to Y$s are all the maps $f\colon X \to Y$ from the original category such that $f \circ x = y$, and we can think of such maps as the continuous maps which preserve basepoints. So we can think of $1/\mathbf{Top}$ as being (or at least, in some strong sense, as being 'the same as') the category $\mathbf{Top}_*$ of pointed topological spaces.

(2) Second, take an n-membered index set $I_n = \{c_1, c_2, c_3, \ldots, c_n\}$. Think of the members of the set as 'colours'. Then a morphism $S \to I_n$ is an n-colouring of the set S. So we can think of $\mathbf{FinSet}/I_n$ as the category of n-coloured finite sets, which is the sort of thing that combinatorialists might be interested in.

(More generally, we can think of a slice category $\mathbf{Set}/I$ as a category of 'indexed' or 'typed' sets, with I providing the indices/types.)

And now here a couple of examples of *functors* involving slice categories (and of course there will be corresponding constructions for co-slice categories):

(1) There is another forgetful functor, $F\colon \mathscr{C}/I \to \mathscr{C}$, which sends a $\mathscr{C}/I$-object $f\colon A \to I$ back to A, and sends an arrow j in $\mathscr{C}/I$ back to the original arrow j in $\mathscr{C}$.

For example, the forgetful functor $F\colon \mathbf{FinSet}/I_n \to \mathbf{FinSet}$ forgets about the 'colourings' provided by functions from a set S to the index set I_n!

(2) Next, let's show how we can use an arrow $k\colon I \to J$ (for $I, J \in \mathscr{C}$) to generate a corresponding functor $K\colon \mathscr{C}/I \to \mathscr{C}/J$.

The functor needs to act on *objects* in $\mathscr{C}/I$ and send them to objects in $\mathscr{C}/J$. That is to say, K_{ob} needs to send an arrow$_{\mathscr{C}}$ $f\colon X \to I$ to an arrow$_{\mathscr{C}}$ with codomain J. The obvious thing to do is to put $K_{ob}(f) = k \circ f$.

$$\begin{tikzcd}
A \arrow[dd, "j"'] \arrow[dr, "f"] \arrow[drr, "k\circ f"] & & \\
 & I \arrow[r, "k"] & J \\
B \arrow[ur, "g"] \arrow[urr, "k\circ g"'] & &
\end{tikzcd}$$

And how will K act on *arrows* of $\mathscr{C}/I$? Another diagram might help! The $\mathscr{C}/I$-arrows from $f\colon A \to I$ to $g\colon B \to I$, by definition, include any j which makes the left-hand inner triangle commute. But then such a j will also make the outer triangle commute, i.e. j is an arrow from $k \circ f\colon A \to J$ to $k \circ g\colon B \to J$ (which is therefore an arrow from $K(f)$ to $K(g)$). So we can simply put $K(j)$ (for $j\colon f \to g$ in $\mathscr{C}/I$) to be j (i.e. $j\colon K(f) \to K(g)$ in $\mathscr{C}/J$).

It is then readily checked that K is indeed a functor from $\mathscr{C}/I$ to $\mathscr{C}/J$.

5.5 Comma categories

The last section in this chapter again looks ahead. The notion of a so-called comma-category introduced in this section won't really be put to use until §12.4. This may be as good a place as any to first introduce the ideas, as it explains yet another way of getting new categories from old; but (as we said before) by all means just skim through for the moment.

Definition 22. Given functors $S\colon \mathscr{A} \to \mathscr{C}$ and $T\colon \mathscr{B} \to \mathscr{C}$, then the '*comma category*' $(S \downarrow T)$ is the category with the following data:

(1) The objects of $(S \downarrow T)$ are triples $\langle A, f, B\rangle$ where A is an $\mathscr{A}$-object, B is a $\mathscr{B}$-object, and $f\colon SA \to TB$ is an arrow in $\mathscr{C}$.

(2) An arrow of $(S \downarrow T)$ from $\langle A, f, B\rangle$ to $\langle A', f', B'\rangle$ is a pair $\langle a, b\rangle$, where $a\colon A \to A'$ is an $\mathscr{A}$-arrow, $b\colon B \to B'$ is an $\mathscr{B}$-arrow, and the following diagram commutes:

$$\begin{tikzcd} SA \arrow[r, "f"] \arrow[d, "Sa"] & TB \arrow[d, "Tb"] \\ SA' \arrow[r, "f'"] & TB' \end{tikzcd}$$

(3) The identity arrow on the object $\langle A, f, B\rangle$ is the pair $\langle 1_A, 1_B\rangle$.

(4) Composition in $(S \downarrow T)$ is induced by the composition laws of $\mathscr{A}$ and $\mathscr{B}$, thus: $\langle a', b'\rangle \circ \langle a, b\rangle = \langle a' \circ_{\mathscr{A}} a, b' \circ_{\mathscr{B}} b\rangle$.

It is easily checked that, so defined, $(S \downarrow T)$ is indeed a category. The standard label 'comma category' arises from an unhappy earlier notation '(S, T)'. And our use of 'S' and 'T' for the two functors here is supposed to be helpfully remind us which functor outputs the source and which outputs the target for the arrow f in the object $\langle A, f, B\rangle$.

The notion of a comma category generalizes a number of simpler constructions. And indeed, we have already met two comma categories in thin disguise.

(1) Take the minimal case where $\mathscr{A} = \mathscr{B} = \mathscr{C}$, and where both S and T are the identity functor on that category, $1_\mathscr{C}$.

Then the objects in this category $(1_\mathscr{C} \downarrow 1_\mathscr{C})$ are triples $\langle X, X \xrightarrow{f} Y, Y \rangle$ for X, Y both $\mathscr{C}$-objects. And an arrow from $\langle X, X \xrightarrow{f} Y, Y \rangle$ to $\langle X', X' \xrightarrow{f'} Y', Y' \rangle$ is a pair of $\mathscr{C}$-arrows $a\colon X \to X'$, $b\colon Y \to Y'$ such that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "a"] & Y \arrow[d, "b"] \\ X' \arrow[r, "f'"] & Y' \end{tikzcd}$$

So the only difference between $(1_\mathscr{C} \downarrow 1_\mathscr{C})$ and the arrow category $\mathscr{C}^\to$ is that we have now 'decorated' the objects of $\mathscr{C}^\to$, i.e. $\mathscr{C}$-arrows $f\colon X \to Y$, with explicit assignments of their sources and targets as $\mathscr{C}$-arrows, to give triples $\langle X, X \xrightarrow{f} Y, Y \rangle$. Hence $(1_\mathscr{C} \downarrow 1_\mathscr{C})$ and $\mathscr{C}^\to$, although not strictly identical, come to the just same. (Later, we'll have an official notion of isomorphic categories, and these categories are trivially isomorphic.)

(2) Take secondly the special case where $\mathscr{A} = \mathscr{C}$ with S the identity functor $1_\mathscr{C}$, and where $\mathscr{B} = \mathbf{1}$ (the category with a single object $\star$ and the single arrow $1_\star$). And take the functor $I\colon \mathbf{1} \to \mathscr{C}$ which sends $\star$ to some individual $\mathscr{C}$-object which we'll also call I – see §4.2, Ex. (F9).

Applying the definition, the objects of the category $(1_\mathscr{C} \downarrow I)$ are therefore triples $\langle A, A \xrightarrow{f} I, \star \rangle$, and an arrow between $\langle A, A \xrightarrow{f} I, \star \rangle$ and $\langle B, B \xrightarrow{g} I, \star \rangle$ will be a pair $\langle j, 1_\star \rangle$, with $j\colon A \to B$ an arrow such the diagram on the left commutes:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "j"] & I \arrow[d, "1_I"] \\ B \arrow[r, "g"] & I \end{tikzcd} \qquad \begin{tikzcd} A \arrow[dr, "f"] \arrow[dd, "j"] & \\ & I \\ B \arrow[ur, "g"'] & \end{tikzcd}$$

But the diagram on the left is equivalent to that on the right – which should look familiar! We've ended up with something tantamount to the slice category $\mathscr{C}/I$, the only differences being that (i) instead of the slice category's objects $f\colon A \to I$ (for $A \in \mathscr{C}$) we now have 'decorated' objects $\langle A, A \xrightarrow{f} I, \star \rangle$ which correspond one-to-one with them, and (ii) instead of the slice category's arrows $j\colon A \to B$ we have decorated arrows $\langle j, 1_\star \rangle$ which correspond one-to-one with *them*.

Again later, in §7.3, we'll be able to do better than saying we have 'something tantamount' to a slice category here: we will be able to say that the categories $(1_\mathscr{C} \downarrow I)$ and $\mathscr{C}/I$ are isomorphic categories.

While we looking at examples of comma categories, let's for the record add a third case which is pretty similar to the last one. It doesn't introduce any new ideas, but it will turn out to be useful – so we choose notation with an eye to a later application.

(3) Suppose we have a functor $G\colon \mathscr{B} \to \mathscr{A}$ and an object $A \in \mathscr{A}$. There is a functor $A\colon \mathbf{1} \to \mathscr{A}$ (which sends the sole object $\star$ in the one-object category $\mathbf{1}$ to the object A in $\mathscr{A}$). Then what is the comma category $(A \downarrow G)$? Flat-footedly applying the definitions, we get:

a) The objects of $(A \downarrow G)$ are triples $\langle \star, f, B\rangle$ where B is a $\mathscr{B}$-object, and $f : A \to GB$ is an arrow in $\mathscr{A}$.

b) An arrow of $(A \downarrow G)$ from $\langle \star, f, B\rangle$ to $\langle \star, f', B'\rangle$ is a pair of arrows, $\langle 1_\star, j\rangle$ with $j\colon B \to B'$ such the following square commutes:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "1_A"] & GB \arrow[d, "Gj"] \\ A \arrow[r, "f'"] & GB' \end{tikzcd}$$

But since the $\star$-component in all the objects of $(A \downarrow G)$ is doing no real work, our comma category is tantamount to the stripped-down category such that

(a′) the objects are, more simply, pairs $\langle B, f\rangle$ where B is a $\mathscr{B}$-object and $f : A \to GB$ is an arrow in $\mathscr{A}$,

(b′) an arrow from $\langle B, f\rangle$ to $\langle B', f'\rangle$ is, more simply, a $\mathscr{B}$-arrow $j\colon B \to B'$ making

$$\begin{tikzcd} & GB \arrow[d, "Gj"] \\ A \arrow[ur, "f"] \arrow[r, "f'"'] & GB' \end{tikzcd}$$

commute,

and with the obvious definitions for the identity arrows and for composition of arrows. And it is this stripped-down version which is in fact usually referred to by the label '$(A \downarrow G)$' (and we can read 'A' in the label here as just referring to an object, not to the corresponding functor).

6 Natural transformations

> Category theory is an embodiment of Klein's dictum that it is the maps that count in mathematics. If the dictum is true, then it is the functors between categories that are important, not the categories. And such is the case. Indeed, the notion of category is best excused as that which is necessary in order to have the notion of functor. But the progression does not stop here. There are maps between functors, and they are called natural transformations.
>
> (Freyd 1965, quoted in Marquis 2008.)

Natural transformations, as special morphisms-between-functors, were there from the very start. The founding document of category theory is the paper by Samuel Eilenberg and Saunders Mac Lane 'General theory of natural equivalences' (Eilenberg and Mac Lane, 1945). But the key idea there had already been introduced, three years previously, in a paper on 'Natural isomorphisms in group theory', before the categorial framework was invented precisely to provide a general setting for the account (Eilenberg and MacLane, 1942). So natural transformations are going to be central to our story.

6.1 Motivation

It might be helpful scene-setting, then, to pause to reflect in some detail about the kind of pre-category-theoretic problem that the account of natural transformations aims to solve. Let's take in stages an example used by Eilenberg and Mac Lane:

(a) Consider a finite dimensional vector space over the reals, V, and the corresponding dual space V^* of linear functions $f\colon V \to \mathbb{R}$. It is elementary to show that V is isomorphic to V^* (there's a bijective linear map between the spaces).

Proof sketch: Take a basis $B = \{v_1, v_2, \ldots, v_n\}$ for V. Define the functions $v_i^*\colon V \to \mathbb{R}$ by putting $v_i^*(v_j) = 1$ if $i = j$ and $v_i^*(v_j) = 0$ otherwise. Then $B^* = \{v_1^*, v_2^*, \ldots, v_n^*\}$ is a basis for V^*, and the linear function $\varphi^B\colon V \to V^*$ generated by putting $\varphi^B(v_i) = v_i^*$ is an isomorphism.

Note, however, that the isomorphism we have arrived at here depends on the initial choice of basis B.[1] But no choice of basis is more 'natural' than any other; none of the isomorphisms from V to V^* of the kind we've just defined is to be especially preferred.

To get a contrasting case, now consider the double dual of V, i.e. V^{**} the space of functionals $g\colon V^* \to \mathbb{R}$. We can again get an isomorphism by selecting a basis B for V, defining a derived basis B^* for V^* as we just did, and then using *that* basis in turn to define a basis B^{**} for V^{**} by repeating the same construction. And then we can define an isomorphism from V to V^{**} by mapping the elements of B to the corresponding elements of B^{**}.

However, *we don't have to go through the palaver of choosing bases.* Suppose we define ψ_V as acting on an element $v \in V$ to give as output the functional $g\colon V^* \to \mathbb{R}$ which sends a function $f\colon V \to \mathbb{R}$ to the value $f(v)$. So, in short, $\psi_V(v)(f) = f(v)$. It is readily checked that this is an isomorphism (we rely on the fact that V is finite-dimensional). And obviously we get this isomorphism independently of any arbitrary choice of basis.

Interim summary: it is very natural(!) to say that the isomorphisms of the kind we just described between V and V^* are not intrinsic, are not 'natural' to the spaces involved. By contrast there *is* a 'natural' isomorphism between V and V^{**}, generated by a general procedure that applies to any vector space.

Now, there are many other cases where we might want to contrast intuitively 'natural' maps from more arbitrarily cooked-up maps between structured objects. The story goes that such talk was already bandied about quite a bit e.g. by topologists in the 1930s, but without any account of what it meant. Eilenberg and Mac Lane were aiming to provide the missing story.

(b) To continue with our example, the idea is that the isomorphism $\psi_V : V \xrightarrow{\sim} V^{**}$ which we constructed is natural because the only information about V it relies on is that V has the structure of a finite dimensional vector space. That implies that the construction will work the same way on other such vector spaces, so we get a corresponding isomorphism $\psi_W : W \xrightarrow{\sim} W^{**}$.

Now, we will expect such naturally constructed isomorphisms to be respected by structure-preserving between the spaces V and W (and the induced maps from V^{**} to W^{**}). Putting that more carefully, we want the following diagram to commute, whatever vector spaces we take and for any structure-preserving $f\colon V \to W$ (i.e. for any linear map f):

$$\begin{tikzcd}
V \arrow[r, "f"] \arrow[d, "\psi_V"] & W \arrow[d, "\psi_W"] \\
V^{**} \arrow[r, "DD(f)"] & W^{**}
\end{tikzcd}$$

[1] Look at it like this: our map φ^B in fact defines an inner product for the space V, where for $e, e' \in V$, $\langle e, e' \rangle = \varphi^B(e)(e')$. Now recall that inner products need not be preserved by a change of basis.

where $DD(f)$ is the double-dual correlate of f.

Recall, back in §5.2, we saw that the correlate Df of $f\colon V \to W$ is the functional $(-\circ f)\colon W^* \to V^*$; and then moving to the double dual, the correlate DDf will be the functional we can notate $(—\circ(-\circ f))\colon V^{**} \to W^{**}$. And then our diagram does indeed commute, both paths sending an element $v \in V$ to the functional that maps a function $k\colon V \to \mathbb{R}$ to the value $k(f(x))$. Think about it!

(c) So far, so good. Now let's consider why there can't be a similarly 'natural' isomorphism from V to V^*. (The isomorphisms based on an arbitrary choice of basis aren't natural: but we want to show that there is no other 'natural' isomorphism either.)

Suppose then that there were a construction which gave us an isomorphism $\varphi_V : V \xrightarrow{\sim} V^*$ which again does not depend on information about V other than that it has the structure of a finite dimensional vector space. So again we will want the construction to work the same way on other such vector spaces, and to be preserved by structure-preserving maps between the spaces. This time, therefore, we want the following diagram to commute for any structure-preserving f between vector spaces (note we have to reverse an arrow for things to make sense, given our definition of the contravariant functor D):

$$\begin{tikzcd} V \arrow[r, "f"] \arrow[d, "\varphi_V"] & W \arrow[d, "\varphi_W"] \\ V^* & W^* \arrow[l, "D(f)"] \end{tikzcd}$$

So $D(f) \circ \varphi_W \circ f = \varphi_V$.

But by hypothesis, the φs are isomorphisms, so in particular φ_V has an inverse. So we have $(\varphi_V^{-1} \circ D(f) \circ \varphi_W) \circ f = 1_V$. Therefore f has a left inverse. But it is obvious that in general, a linear map $f\colon V \to W$ need not have a left inverse. Hence there can't in general be isomorphisms $\varphi_V, \varphi_W\colon V \to V^*$ making that diagram commute.

(d) 'Ah! Perhaps what the last argument shows is that we were asking too much. Perhaps intuitive naturality requires only that the diagram commutes for any mapping which *fully* preserves structure, i.e. it requires that we get a commuting diagram only for arrows $f\colon V \xrightarrow{\sim} W$ which are isomorphisms.' Nice try! But this revised proposal doesn't rescue the situation. For any isomorphism f, we will still have $\varphi_V(v_1) = D(f) \circ ((\varphi_W \circ f)(v)) = ((\varphi_W) \circ f)(v_1) \circ f$, for $v_1 \in V$. The left and right sides here are functions from V to $\mathbb{R}$. Evaluate them both at some $v_2 \in V$ and we get

$$\varphi_V(v_1)(v_2) = ((\varphi_W) \circ f)(v_1) \circ f(v_2).$$

But while the left-hand side is constant, the right-hand side can vary as the isomorphism f varies. Take the two dimensional case, and suppose v_1 and v_2

form a basis for V. Then if f is an isomorphism, so is f', the mapping generated by setting $f'(v_1) = f(v_1), f'(v_2) = 2f(v_2)$. But then substituting f' for f, the l.h.s. of the equation stays fixed and the value of the r.h.s. doubles. Impossible! So even the proposed revised criterion of naturalness can't make an isomorphism between V and V^* natural.

(e) Here's another interim summary. We started off by saying that, intuitively, there's a 'natural', instrinsic, isomorphism between a (finite dimensional) vector space and its double dual, one that depends only on their structures are vector spaces. And we've now suggested that this intuitive idea can be formally captured by saying that a certain diagram always commutes, for any choice of vector spaces and structure-preserving maps between them. We've also seen that we can't get analogous always-commuting diagrams for the case of isomorphisms between a vector space and its dual – which chimes with the intuition that (at least the obvious examples) are *not* 'natural' isomorphisms. Promising!

So let's now try to generalize. First, the claim that the diagram

$$\begin{tikzcd}
V \arrow[r, "f"] \arrow[d, "\psi_V"] & W \arrow[d, "\psi_W"] \\
V^{**} \arrow[r, "DD(f)"] & W^{**}
\end{tikzcd}$$

always commutes can be put a slightly different way, using category-speak. For we have in effect been talking about the category we called $\mathbf{V}$ (of finite spaces and the structure-preserving maps between them), and about a functor we can call $DD\colon \mathbf{V} \to \mathbf{V}$ which takes a vector space to its double dual, and maps each arrow between vector spaces to its double-dual correlate as explained. So, the claim that the previous diagram commutes is just the claim that, for every arrow $f\colon V \to W$ in $\mathbf{V}$, there are isomorphisms ψ_V and ψ_W in $\mathbf{V}$ such the following commutes:

$$\begin{tikzcd}
V \arrow[r, "f"] \arrow[d, "\psi_V"] & W \arrow[d, "\psi_W"] \\
DD(V) \arrow[r, "DD(f)"] & DD(W)
\end{tikzcd}$$

NB: the isomorphisms depend only on the domain and codomain of f, and are independent of what f actually does.

Now, consider the trivial identity functor $1\colon \mathbf{V} \to \mathbf{V}$ that maps each vector space to itself and each $\mathbf{V}$-arrow to itself. Then we can re-express that last claim as follows. For every arrow $f\colon V \to W$ in $\mathbf{V}$, there are isomorphisms ψ_V and ψ_W in $\mathbf{V}$ such *this* diagram commutes:

$$\begin{array}{ccc} 1(V) & \xrightarrow{1(f)} & 1(W) \\ \Big\downarrow{\scriptstyle \psi_V} & & \Big\downarrow{\scriptstyle \psi_W} \\ DD(V) & \xrightarrow{DD(f)} & DD(W) \end{array}$$

And what's nice about this way of putting things is that it transmutes the claim about the availability of a 'natural' isomorphism between a space and its double dual into a claim about the naturalness of a certain map between *functors* on the category of spaces – in this case, between the functors 1 and DD. And this seems a natural(!) way to go: for talking about functors is our way of category-theoretic way of talking at once both about relations between objects and about associated relations between arrows.

The two functors in our example so far relate a category to itself. But there is now an obvious (and again surely 'natural'!) generalization:

Definition 23. Let $\mathscr{C}$ and $\mathscr{D}$ be categories, let $\mathscr{C} \underset{G}{\overset{F}{\rightrightarrows}} \mathscr{D}$ be functors, and let ψ be a whole family of $\mathscr{D}$-isomorphisms $\psi_C\colon F(C) \xrightarrow{\sim} G(C)$ indexed by the objects $\mathscr{C}$-objects. Then ψ is said to be a *natural isomorphism* between F and G if for every $f\colon A \to B$ in $\mathscr{C}$, the following diagram commutes in $\mathscr{D}$:

$$\begin{array}{ccc} F(A) & \xrightarrow{F(f)} & F(B) \\ \Big\downarrow{\scriptstyle \psi_A} & & \Big\downarrow{\scriptstyle \psi_B} \\ G(A) & \xrightarrow{G(f)} & G(B) \end{array}$$

So our intuitive claim that there is a natural isomorphism between a vector space and its double dual becomes reflected in the claim that there is a natural isomorphism in our official sense between the identity and double-dual functors from the category $\mathbf{V}$ to itself.

6.2 Natural transformations defined

We've now introduced one kind of mapping between functors. But just as isomorphisms are just one kind of morphism between objects within categories, so natural isomorphisms are just one kind of natural morphism between functors. We are next going to introduce the more general idea of a natural transformation (to use the standard term).

We can motivate the definition we arrive at as the obvious generalization from the isomorphism case. But here's a slightly different way of thinking about things (following e.g. some remarks of Goldblatt 2006, pp. 198-199). Take a pair of functors $F\colon \mathscr{C} \to \mathscr{D}$ and $G\colon \mathscr{C} \to \mathscr{D}$. Each projects the objects and arrows of $\mathscr{C}$ into $\mathscr{D}$ giving, so to speak, two 'pictures' or 'diagrams' of $\mathscr{C}$ within $\mathscr{D}$. Now, the thought is that we can think of a smooth or natural transformation from F to G as a way of smoothly 'superimposing' in $\mathscr{D}$ the picture of $\mathscr{C}$ which results from F onto the picture which results from G. This 'superimposing' will have to be done, then, using morphisms available in $\mathscr{D}$. So diagrammatically, the situation should in part look as below. On the left we have a couple of objects and an arrow between them in $\mathscr{C}$ (a fragment of what is going on in the category); and the two functors F and G yield two pictures of this fragmentary situation in $\mathscr{D}$, one at the top and one at the bottom, with the squiggly arrows then representing local 'superimposing' maps between the objects:

$$\begin{tikzcd} & & F(A) \arrow[r, "F(f)"] \arrow[d, squiggly, "\alpha_A"] & F(B) \arrow[d, squiggly, "\alpha_B"] \\ A \arrow[r, "f"] & B \arrow[ur, dashed] \arrow[dr, dashed] & & \\ & & G(A) \arrow[r, "G(f)"] & G(B) \end{tikzcd}$$

And now the thought is: if the 'pictures' are to indeed to properly 'superimpose', we need that diagram to commute.

So, all this motivates the following key definition (making one just crucial change from the previous definition, in moving to the more general case where the transformation doesn't have to be an isomorphism). We revert to our earlier bracket-free notation:

Definition 24. Let $\mathscr{C}$ and $\mathscr{D}$ be categories, let $\mathscr{C} \underset{G}{\overset{F}{\rightrightarrows}} \mathscr{D}$ be (covariant) functors, and let α be a family of $\mathscr{D}$-arrows $\alpha_C\colon FC \to GC$ indexed by the $\mathscr{C}$-objects C. Then α is said to be a *natural transformation* between F and G, written $\alpha\colon F \Rightarrow G$, if for every $f\colon A \to B$ in $\mathscr{C}$, the following *naturality square* commutes in $\mathscr{D}$:

$$\begin{tikzcd} FA \arrow[r, "Ff"] \arrow[d, "\alpha_A"] & FB \arrow[d, "\alpha_B"] \\ GA \arrow[r, "Gf"] & GB \end{tikzcd}$$

Arrows α_A, α_B etc. are said to be *components* of the natural transformation α, and we call α_A the component of α at A. (A natural isomorphism is thus a natural transformation each of whose components is an isomorphism.)

Alternative notation to indicate a natural transformation is $\alpha\colon F \dot{\to} G$ (with a dotted arrow) or simply $\alpha\colon F \to G$: but, whatever the arrow used, a Greek letter label is the conventionally reliable give-away, signalling a natural transformation. A further bit of notation is very useful. When we have functors $F\colon \mathscr{C} \to \mathscr{D}$, $G\colon \mathscr{C} \to \mathscr{D}$, together with a natural transformation $\alpha\colon F \Rightarrow G$, we can neatly represent the whole situation thus:

$$\begin{tikzcd} \mathscr{C} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathscr{D} \end{tikzcd}$$

We will also use the such a diagram just to denote the natural transformation: context will tell.

Note: Defn. 24 defines a natural transformation between covariant functors. Since a contravariant functor $F\colon \mathscr{C} \to \mathscr{D}$ is a covariant functor $F\colon \mathscr{C}^{op} \to \mathscr{D}$, we get for free a definition of a natural transformation between two contravariant functors. It comes to this:

Definition 25. Let $\mathscr{C} \overset{F}{\underset{G}{\rightrightarrows}} \mathscr{D}$ be *contravariant* functors, and let α be a family of $\mathscr{D}$-arrows $\alpha_C\colon FC \to GC$. Then α is a natural transformation between F and G if for every $f\colon B \to A$ in $\mathscr{C}$ (note the reversal!), the following diagram commutes in $\mathscr{D}$:

$$\begin{tikzcd} FA \arrow[r, "Ff"] \arrow[d, "\alpha_A"] & FB \arrow[d, "\alpha_B"] \\ GA \arrow[r, "Gf"] & GB \end{tikzcd}$$

Can we also talk about some kind of naturality of a transformation between a covariant functor F and a contravariant functor G from $\mathscr{C}$ to $\mathscr{D}$? The diagram in §6.1 (c) gives us a pointer of where we might try to go with this idea. But it turns out not to be of use for our purposes, so we won't pause over it.

6.3 Examples of natural transformations

We have already mentioned the natural transformation between a vector space and its double dual. Now for a few more examples:

(1) Given a group $G = (G, *, e)$ [to abuse notation in a familiar way] we can define its opposite $G^{op} = (G, *^{op}, e)$, where $a *^{op} b = b * a$.

We can now define a functor $Op\colon \mathbf{Grp} \to \mathbf{Grp}$ which sends an object in the category, i.e. a group G, to its opposite G^{op}, and sends an arrow f in the category, i.e. a group homomorphism $f\colon G \to H$, to $f^{op}\colon G^{op} \to H^{op}$ where $f^{op}(a) = f(a)$. f^{op} so defined is indeed a group homomorphism, since

$$f^{op}(a *^{op} b) = f(b * a) = f(b) * f(a) = f^{op}(a) *^{op} f^{op}(b)$$

Claim: there is a natural transformation $\alpha\colon 1 \Rightarrow Op$ (where 1 is the trivial identity functor in $\mathbf{Grp}$).

We need to find a family of arrows α in **Grp** such that the following diagram always commutes. [Careful! 'G', 'H' here are now groups, not functors, . . .]

$$\begin{tikzcd} G \arrow[r, "f"] \arrow[d, "\alpha_G"] & H \arrow[d, "\alpha_H"] \\ G^{op} \arrow[r, "f^{op}"] & H^{op} \end{tikzcd}$$

Now, since taking the opposite *between* groups involves reversing the order of multiplication and taking inverses *inside* a group in effect does the same, a good bet is to choose $\alpha_G(a) = a^{-1}$ for $a \in G$, the same taking-inverses operation for each G. It is easy to check that with this choice of components, α is a natural transformation.

Indeed, in this case, we have another natural isomorphism – reflecting the intuition that there a group and its opposite really 'come to the same'.

(2) To get an interesting example of a natural-transformation-which-isn't-an-isomorphism, recall the idea of the abelianization of a group G. Officially, this is the quotient of a group by its commutator subgroup $[G, G]$ (but you can think of it as sending a group to the 'biggest' Abelian group A for which there is a surjective homomorphism from G onto A). There is a functor Ab which sends a group to its abelianization, and sends an arrow $f\colon G \to H$ to the arrow $Ab(f)\colon Ab(G) \to Ab(H)$ in the obvious way.

So we have functors, $\mathbf{Grp} \underset{Ab}{\overset{1}{\rightrightarrows}} \mathbf{Grp}$, and this diagram always commutes,

$$\begin{tikzcd} G \arrow[r, "f"] \arrow[d, "\alpha_G"] & H \arrow[d, "\alpha_H"] \\ Ab(G) \arrow[r, "Ab(f)"] & Ab(H) \end{tikzcd}$$

where $\alpha_G = G/[G, G]$. So we have a natural transformation, but not usually a natural isomorphism, between the functors 1 and Ab.

(3) Recall from §4.2 the functor $List\colon \mathbf{Set} \to \mathbf{Set}$ which sends a set X to the set of finite lists of members of X. There is a natural transformation $\rho\colon List \to List$ whose component ρ_X is the map $List(X) \to List(X)$ whose action is to reverse the order of the elements of a list of X-elements.

(4) Recall from §4.2 again the functor $F\colon \mathbf{Set} \to \mathbf{Mon}$ which sends a set X to the free monoid on X. We also mentioned there the functor $G\colon \mathbf{Set} \to \mathbf{Ab}$ that sends a set X to the freely generated abelian group with generators in X; and by 'forgetting' the group structure in order to leave us just a monoid, we can get a derived functor $G'\colon \mathbf{Set} \to \mathbf{Mon}$. [*Aside* If we are pressing the metaphor of a functor from $\mathscr{C}$ to $\mathscr{D}$ giving us a 'picture' of $\mathscr{C}$

in $\mathscr{D}$, we'd have to say that the functors F and G' give us much decorated pictures of sets!]

There is then a natural transformation from F to G':

$$\begin{tikzcd}
\text{Free monoid on } X \arrow[r, "Ff"] \arrow[d, "\alpha_A"] & \text{Free monoid on } Y \arrow[d, "\alpha_B"] \\
\text{Abelian group on } X \arrow[r, "G'f"] & \text{Abelian group on } Y
\end{tikzcd}$$

where α_X sends a string $x_1x_2x_3\ldots x_n$ to the group element $x_1+x_2+x_3+\ldots+x_n$.

(5) Here are the bare bones of an example for topologists. We met in §4.6 a functor π_1 which sends pointed topological spaces $\mathbf{Top}_*$ to $\mathbf{Grp}$. There's another functor H_1 from **Top** to **Ab** which sends a space to its first homology group. Making those functors a bit forgetful (about base points and about the fact that homology groups are abelian), we can get a derived pair of functors from **Top** to **Grp**. It is a significant fact of topology that there is a natural transformation between them.

6.4 Functor categories

Suppose $F\colon \mathscr{C} \to \mathscr{D}$. Then quite trivially, the following diagram commutes for every $f\colon A \to B$ in $\mathscr{C}$:

$$\begin{tikzcd}
F(A) \arrow[r, "F(f)"] \arrow[d, "1_{FA}"] & F(B) \arrow[d, "1_{FB}"] \\
F(A) \arrow[r, "F(f)"] & F(B)
\end{tikzcd}$$

So we have a natural transformation

$$\begin{tikzcd}
\mathscr{C} \arrow[r, bend left, "F"{name=U}] \arrow[r, bend right, "F"'{name=D}] & \mathscr{D} \arrow[from=U, to=D, Rightarrow, "1_F"]
\end{tikzcd}$$

where the components $(1_F)_A$ of the transformation are the identity morphisms $1_{(FA)}$.

Next, suppose we have functors F, G and H between the categories $\mathscr{C}$ and $\mathscr{D}$, with a natural transformation between F and G, and another between G and H. In a diagram, we have:

$$\begin{tikzcd} \mathscr{C} \arrow[rr, bend left, "F"] \arrow[rr, "G" description] \arrow[rr, bend right, "H"'] \arrow[rr, phantom, bend left=20, "\Downarrow\alpha"] \arrow[rr, phantom, bend right=20, "\Downarrow\beta"] & & \mathscr{D} \end{tikzcd}$$

Then we can compose the natural transformations to give us a natural transformation between F and H: i.e. we get

$$\begin{tikzcd} \mathscr{C} \arrow[rr, bend left, "F"] \arrow[rr, bend right, "H"'] \arrow[rr, phantom, "\Downarrow\beta\circ\alpha"] & & \mathscr{D} \end{tikzcd}$$

We simply compose component-wise, i.e. put $(\beta \circ \alpha)_A = \beta_A \circ \alpha_A$. Why does this work? Consider the following diagram which pastes together two naturality squares for some $f\colon A \to B$:

$$\begin{tikzcd} FA \arrow[r, "Ff"] \arrow[d, "\alpha_A"] \arrow[dd, bend right=60, "(\beta\circ\alpha)_A"'] & FB \arrow[d, "\alpha_B"'] \arrow[dd, bend left=60, "(\beta\circ\alpha)_B"] \\ GA \arrow[r, "Gf"] \arrow[d, "\beta_A"] & GB \arrow[d, "\beta_B"'] \\ HA \arrow[r, "Hf"] & HB \end{tikzcd}$$

Evidently, we have

$$(\beta \circ \alpha)_B \circ Ff = \beta_B \circ \alpha_B \circ Ff = \beta_B \circ Gf \circ \alpha_A = Hf \circ \beta_A \circ \alpha_A = Hf \circ (\beta \circ \alpha)_A$$

and so the outer bent rectangle commutes giving another naturality square, so that $(\beta \circ \alpha)$ is indeed a natural transformation from F to H.

Right: we now have identity transformations and composition of transformations (with composition evidently defined in such a way that it is associative). So, lo and behold, the following definition must be in good order!

Definition 26. The *functor category* from $\mathscr{C}$ to $\mathscr{D}$, denoted $[\mathscr{C}, \mathscr{D}]$ (alternative: $\mathscr{D}^{\mathscr{C}}$) is the category whose objects are the functors $F\colon \mathscr{C} \to \mathscr{D}$, with the natural transformations between them as arrows.

Let's have a couple of easy examples, which illustrate how functor categories can be (intuitively equivalent to) interesting structures.

(1) Recall the category $\mathbf{2}$. Omitting identity arrows, we can diagram this as $\bullet \longrightarrow \star$. Now ask: what is the functor category $[\mathbf{2}, \mathscr{C}]$?

An object in this category is a functor $F\colon \mathbf{2} \to \mathscr{C}$, where (i) F_{ob} will send $\bullet$ to some $\mathscr{C}$-object X and send $\star$ to an object Y, and (ii) F will map the arrow from $\bullet$ to $\star$ to an arrow, $f\colon X \to Y$.

And what about the arrows in our new category? A natural transformations between two such functors $\mathbf{2} \overset{F_1}{\underset{F_2}{\rightrightarrows}} \mathscr{C}$ can have as components any two $\mathscr{C}$-arrows, j, k, which makes this a commutative square:

$$\begin{tikzcd} X_1 \arrow[r, "f_1"] \arrow[d, "j"] & Y_1 \arrow[d, "k"] \\ X_2 \arrow[r, "f_2"] & Y_2 \end{tikzcd}$$

Hence in sum, objects in $[\mathbf{2}, \mathscr{C}]$ correspond exactly to the $\mathscr{C}$-arrows $f: X \to Y$, and the arrows of the new category between two such objects correspond exactly to pairs of $\mathscr{C}$-arrows which make the relevant diagram commute. So $[\mathbf{2}, \mathscr{C}]$ corresponds exactly to the arrow category $\mathscr{C}^{\to}$ (see §5.3).

(2) Consider the twosome-category $\mathscr{T}$ which we can diagram as $\bullet \rightrightarrows \star$ (i.e. the category with just two objects, and two parallel arrows in addition to the identity arrows).

So now let's think about the functor category $[\mathscr{T}, \mathbf{Set}]$. An object in this category is a functor $F: \mathscr{T} \to \mathbf{Set}$. So F_{ob} will send $\bullet$ to some set E_F, and send $\star$ to a set V_F. And F will map the two arrows from $\bullet$ to $\star$ to two functions, $f_F^1, f_F^2: E_F \to V_F$.

However, a pair of sets E, V together with a pair of arrows $f^1, f^2: E \to V$ can be diagrammed as a directed graph, if we think of E as the set of edges, and the two functions mapping as an edge to its beginning and end vertices. So a functor $F: \mathscr{T} \to \mathbf{Set}$ 'pictures' $\mathscr{T}$ by a directed graph.

Now, what about natural transformations α between functors $\mathscr{T} \overset{F}{\underset{G}{\rightrightarrows}} \mathbf{Set}$? By definition they make the following diagram commute for any f^i:

$$\begin{tikzcd} E_F \arrow[r, "f_F^i"] \arrow[d, "\alpha_E"] & V_F \arrow[d, "\alpha_V"] \\ E_G \arrow[r, "f_G^i"] & V_G \end{tikzcd}$$

So a natural transformation is a pair of functions which preserves the assignments of vertices to edges. It is thus a homomorphism for directed graphs.

Thus, in short, the functor category $[\mathscr{T}, \mathbf{Set}]$ is or (in some good sense – we'll need to return to this) is 'the same as' the category of directed graphs.

6.5 Horizontal composition of natural transformations

Composing two natural transformations $\begin{tikzcd} \mathscr{C} \arrow[r, bend left, "F"] \arrow[r, "G" description] \arrow[r, bend right, "H"'] & \mathscr{D} \end{tikzcd}$ (with $\Downarrow\alpha$ and $\Downarrow\beta$) to get $\begin{tikzcd} \mathscr{C} \arrow[r, bend left, "F"] \arrow[r, bend right, "H"'] \arrow[r, phantom, "\Downarrow\beta\circ\alpha"] & \mathscr{D} \end{tikzcd}$

is naturally enough called *vertical composition*. But we can also put things together *horizontally* in various ways.

First, there is so-called *whiskering*(!) where we combine a functor with a natural transformation between functors to get a new natural transformation. Thus, what happens when we 'add a whisker' on the left of a diagram for a natural transformation?

The situation $\begin{tikzcd} \mathscr{C} \arrow[r, "F"] & \mathscr{D} \arrow[r, bend left, "J"] \arrow[r, bend right, "K"'] \arrow[r, phantom, "\Downarrow\beta"] & \mathscr{E} \end{tikzcd}$ gives rise to $\begin{tikzcd} \mathscr{C} \arrow[r, bend left, "J\circ F"] \arrow[r, bend right, "K\circ F"'] \arrow[r, phantom, "\Downarrow\beta F"] & \mathscr{E} \end{tikzcd}$

where the component of βF at A is the component of β at FA – i.e. $(\beta F)_A = \beta_{FA}$ (which is why the suggestive notation 'β_F' is quite often preferred to 'βF'). Why does this hold? Consider the function $Ff\colon FA \to FB$ in $\mathscr{D}$ (where $f\colon A \to B$ is in $\mathscr{C}$). Now apply the functors J and K, and since β is a natural transformation we get the commutative 'naturality square'

$$\begin{tikzcd} J(FA) \arrow[r, "J(Ff)"] \arrow[d, "\beta_{FA}"] & J(FB) \arrow[d, "\beta_{FB}"] \\ K(FA) \arrow[r, "K(Ff)"] & K(FB) \end{tikzcd}$$

and we can read that as giving a natural transformation between $J \circ F$ and $K \circ F$.

Likewise, adding a whisker on the right,

the situation $\begin{tikzcd} \mathscr{C} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathscr{D} \arrow[r, "J"] & \mathscr{E} \end{tikzcd}$ gives rise to $\begin{tikzcd} \mathscr{C} \arrow[r, bend left, "J\circ F"] \arrow[r, bend right, "J\circ G"'] \arrow[r, phantom, "\Downarrow J\alpha"] & \mathscr{E} \end{tikzcd}$

where the component of $J\alpha$ at X is $J(\alpha_X)$.

For future use, by the way, we should note the following mini-result:

Theorem 20. *Whiskering a natural isomorphism yields a natural isomorphism.*

Proof. Retaining the same notation as above, but now taking α and β to be isomorphisms, we saw that 'post-whiskering' α by J to get $J\alpha$ yields a transformation whose components are $J\alpha_X$, and since functors preserve isomorphisms, these components are all isomorphisms, hence so is $J\alpha$. 'Pre-whiskering' β by F to get βF yields a transformation whose components are (some of the) components of β and therefore are isomorphisms, hence again so is βF. □

Second, we can *horizontally compose* two natural transformations:

$$\text{We take } \begin{tikzcd} \mathscr{C} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathscr{D} \arrow[r, bend left, "J"] \arrow[r, bend right, "K"'] \arrow[r, phantom, "\Downarrow\beta"] & \mathscr{E} \end{tikzcd} \text{ and get } \begin{tikzcd} \mathscr{C} \arrow[r, bend left, "J\circ F"] \arrow[r, bend right, "K\circ G"'] \arrow[r, phantom, "\Downarrow\beta*\alpha"] & \mathscr{E} \end{tikzcd}.$$

How do we define $\beta * \alpha$? Consider any arrow $f\colon A \to B$ in $\mathscr{C}$. Given J is a functor, then since
$$\begin{tikzcd} FA \arrow[r, "Ff"] \arrow[d, "\alpha_A"] & FB \arrow[d, "\alpha_B"] \\ GA \arrow[r, "Gf"] & GB \end{tikzcd}$$
commutes,
$$\begin{tikzcd} J(FA) \arrow[r, "J(Ff)"] \arrow[d, "J(\alpha_A)"] & J(FB) \arrow[d, "J(\alpha_B)"] \\ J(GA) \arrow[r, "J(Gf)"] & J(GB) \end{tikzcd}$$
also commutes. And since $Gf\colon GA \to GB$ is a map in $\mathscr{D}$, and β is a natural transformation between $\mathscr{D} \underset{G}{\overset{F}{\rightrightarrows}} \mathscr{E}$, we have
$$\begin{tikzcd} J(GA) \arrow[r, "J(Gf)"] \arrow[d, "\beta_{GA}"] & J(GB) \arrow[d, "\beta_{GB}"] \\ K(GA) \arrow[r, "K(Gf)"] & K(GB) \end{tikzcd}$$
commutes.

Gluing together those last two commutative diagrams in the obvious ways, gives a natural transformation from $J \circ F$ to $K \circ G$, if we set the component of $\beta * \alpha$ at X to be $\beta_{GX} \circ J\alpha_X$.

Three remarks:

(1) That definition for $\beta * \alpha$ looks surprisingly asymmetric. But note that instead of first applying J and then using the fact that β is natural, we could have similarly first used the fact that β is natural and then applied K, showing that we could alternatively have defined the natural transformation as having the components $K\alpha_X \circ \beta_{FX}$. So we get do at least get two accounts which mirror each other.

(2) We can think of whiskering as a special case of the horizontal composition of two natural transformations where one of them is the identity natural transformation. For example
$$\begin{tikzcd} \mathscr{C} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathscr{D} \arrow[r, bend left, "J"] \arrow[r, bend right, "J"'] \arrow[r, phantom, "\Downarrow 1_J"] & \mathscr{E} \end{tikzcd}$$
produces
$$\begin{tikzcd} \mathscr{C} \arrow[r, bend left, "J\circ F"] \arrow[r, bend right, "K\circ G"'] \arrow[r, phantom, "\Downarrow 1_J*\alpha"] & \mathscr{E} \end{tikzcd},$$
and the component of $1_J * \alpha$ at X is an identity com-

posed with $J\alpha_X$. So this is the same as taking the left-hand natural transformation and simply whiskering with J on the right.

(3) We could now go on to consider the case of horizontally composing a couple of pairs of vertical compositions – and show that it comes to the same if we construe the resulting diagram as the result of vertically composing a couple of horizontal compositions. But we won't now pause over this, but return to the point if and when we ever need the construction. (Or see Leinster 2014, p. 38.)

6.6 More on natural isomorphisms

(a) Suppose $[\mathscr{C}, \mathscr{D}]$ is a functor category, then there will be isomorphisms in the category (if only the identity arrows). What are these isomorphisms?

Theorem 21. *The isomorphisms in the functor category* $[\mathscr{C}, \mathscr{D}]$ *are the natural isomorphisms* $\psi\colon F \Rightarrow G$*, where* $\mathscr{C} \overset{F}{\underset{G}{\rightrightarrows}} \mathscr{D}$.

Proof. Suppose $\psi\colon F \Rightarrow G$ is a natural isomorphism between $\mathscr{C} \overset{F}{\underset{G}{\rightrightarrows}} \mathscr{D}$ in the sense of Defn. 23. Then, by definition, all the components ψ_A, ψ_B, etc., are isomorphisms, and hence have inverses ψ_A^{-1}, ψ_B^{-1}, etc., and these will therefore are available as components of a natural isomorphism $\psi^{-1}\colon G \Rightarrow F$. This is readily seen to be an inverse of ψ in the category $[\mathscr{C}, \mathscr{D}]$. And hence, because it has an invoice, ψ counts as an isomorphism in $[\mathscr{C}, \mathscr{D}]$.

Conversely, suppose the natural transformation $\psi\colon F \Rightarrow G$ has an inverse ψ^{-1} in the category $[\mathscr{C}, \mathscr{D}]$, i.e. $\psi^{-1} \circ \psi = 1_F$. But composition is defined componentwise, so this requires for each component $\psi_X^{-1} \circ \psi_X = 1_{FX}$, i.e. each component of ψ has an inverse, so is an isomorphism, and hence ψ is a natural isomorphism. □

(b) Next, suppose A is a $\mathscr{C}$-object, and we have functors $F, G\colon \mathscr{C} \to \mathscr{D}$. Then there will be objects FA and GA in category $\mathscr{D}$. And there may perhaps be some isomorphism (some $\mathscr{D}$-arrow with an inverse) between these objects, so that we have $FA \cong GA$. But we are going to most interested in a special case, when the relevant isomorphism belongs to a natural family. There is standard terminology for this case:

Definition 27. Given functors $F, G\colon \mathscr{C} \to \mathscr{D}$, we say that $FA \cong GA$ *naturally in A* (or *naturally in A in* $\mathscr{C}$) just if F and G are *naturally* isomorphic.

Note, to have $FA \cong GA$ naturally in A depends on a lot more than what happens at the object A. It requires a natural isomorphism between F and G, and that's a condition which depends on how the functors F and G behave right across the category they act on. To help fix ideas, let's note a useful little result about this notion of an isomorphism between objects holding naturally:

Theorem 22. *Given functors $F, G, H\colon \mathscr{C} \to \mathscr{D}$, an object $A \in \mathscr{C}$, and a functor $K\colon \mathscr{B} \to \mathscr{C}$, then*

(1) if $FA \cong GA$ naturally in A, then for all $A' \in \mathscr{C}$, $FA' \cong GA'$ naturally in A'.

(2) if $FA \cong GA$ and $GA \cong HA$, both naturally in A, then $FA \cong HA$ naturally in A.

(3) if $FA \cong GA$ naturally in A, then $FKB \cong GKB$ naturally in $B \in \mathscr{B}$.

Proof. (1) is immediate, for if $FA \cong GA$ naturally in A, F is naturally isomorphic to G, so there is a component of the natural isomorphism at A' making $FA' \cong GA'$.

For (2), just note that natural isomorphisms vertically compose.

For (3), just note that, if there is a natural isomorphism α between F and G, then (by 'whiskering') there is a natural isomorphism between FK and GK, whose component at B is α_{KB}. □

(c) It is important to see that there are cases of categories and functors where we have $F, G\colon \mathscr{C} \to \mathscr{D}$, and for *all* relevant A, $FA \cong GA$, but still *not* naturally in A. (A slogan: *pointwise isomorphism doesn't entail natural isomorphism.*) Let's have a couple of examples.

(1) (A toy example, to make the point of principle.) Suppose $\mathscr{C}$ is a category with exactly one object A, and two arrows, the identity arrow 1_A, and another arrow f, where $f \circ f = f$. And now consider two functors, the identity functor $1_{\mathscr{C}}\colon \mathscr{C} \to \mathscr{C}$, and the functor $F\colon \mathscr{C} \to \mathscr{C}$ which sends the only object to itself, and sends both arrows to the identity arrow. Then, quite trivially, we have $1_{\mathscr{C}}(A) = F(A)$ for the one and only object in $\mathscr{C}$. But there isn't a natural isomorphism between the functors, because by hypothesis $1_A \neq f$, and the square

$$\begin{tikzcd} F(A) \arrow[r, "F(f)"] \arrow[d, "id_A"] & F(A) \arrow[d, "id_A"] \\ 1_{\mathscr{C}}(A) \arrow[r, "1_{\mathscr{C}}(f)"] & 1_{\mathscr{C}}(A) \end{tikzcd}, \text{ which is none other than } \begin{tikzcd} A \arrow[r, "id_A"] \arrow[d, "id_A"] & A \arrow[d, "id_A"] \\ A \arrow[r, "f"] & A \end{tikzcd},$$

cannot commute.

(2) (A standard illustrative example which is perhaps worth thinking through as a reality check.) We'll work in the category $\mathscr{F}$ of finite sets and *bijections* between them.

There is a functor $Sym\colon \mathscr{F} \to \mathscr{F}$ which (i) sends a set A in $\mathscr{F}$ to the set of permutations on A (treating permutation functions as sets, this is a finite set), and (ii) sends a bijection $f\colon A \to B$ in $\mathscr{F}$ to the bijection that sends the permutation p on A to the permutation $f \circ p \circ f^{-1}$ on B. Note: if A has n members, there are $n!$ members of the set of permutations on A.

There is also a functor $Ord\colon \mathscr{F} \to \mathscr{F}$ which (i) sends a set A in $\mathscr{F}$ to the set of linear orderings on A (you can identify an order-relation with a set, so we can think of this too as a finite set), and (ii) sends a bijection $f\colon A \to B$ in $\mathscr{F}$ to the bijection $Ord(f)$ which sends a total order on A to the total order on B where $x <_A y$ iff $f(x) <_B f(y)$. Again, if A has n members, there are also $n!$ members of the set of linear orderings on A.

Now, for any object A of $\mathscr{F}$, $Sym(A) \cong Ord(A)$ (since they are equinumerous finite sets). But there cannot be a natural transformation α between the functors Sym and Ord. For suppose otherwise, and consider the functors acting on a map $f\colon A \to A$. Then the following square would commute:

$$\begin{tikzcd}
Sym(A) \arrow[r, "Sym(f)"] \arrow[d, "\alpha_A"] & Sym(A) \arrow[d, "\alpha_A"] \\
Ord(A) \arrow[r, "Ord(f)"] & Ord(A)
\end{tikzcd}$$

But consider what happens to the identity permutation i in $Sym(A)$. It gets sent by $Sym(f)$ to $f \circ i \circ f^{-1}$, i.e. to itself. So the square tells us that the ordering $\alpha_A(i)$ (whatever) it is, gets sent by $Ord(f)$ to itself. But in general that won't be so – suppose f swaps around elements.

7 Equivalence of categories

7.1 Full, faithful, and other kinds of functors

Definition 28. A functor $F\colon \mathscr{C} \to \mathscr{D}$ is *full* if given any $\mathscr{C}$-objects C, C', then for any arrow $g\colon FC \to FC'$ there is an arrow $f\colon C \to C'$ such that $g = Ff$. (In other words, the map from $\mathscr{C}(C, C')$ to $\mathscr{D}(FC, FC')$ induced by F is surjective.)

A functor is *faithful* if given any $\mathscr{C}$-objects C, C', and any pair of parallel arrows $f, g\colon C \to C'$, then if $F(f) = F(g)$, then $f = g$. (In other words, the map from $\mathscr{C}(C, C')$ to $\mathscr{D}(FC, FC')$ induced by F is injective.)

Think of it like this. F puts a picture of $\mathscr{C}$ inside $\mathscr{D}$. It's a faithful picture if it has *enough* arrows: different arrows get sent to different arrows. It's a full picture if it doesn't have *too many* arrows: every arrow between pictures of $\mathscr{C}$- objects is indeed a picture of a $\mathscr{C}$-arrow.

Some examples:

(1) The forgetful functor $F : \mathbf{Mon} \to \mathbf{Set}$ is faithful, as F sends a set-function which happens to be a monoid homorphism to itself, so different arrows in **Mon** get sent to different arrows in **Set**. But the functor is not full: there will be lots of arrows in **Set** that don't correspond to a monoid homomorphism.

(2) Consider next the free monoid functor $F : \mathbf{Set} \to \mathbf{Mon}$. Different set functions $f, g\colon X \to Y$ get sent to different functions $F(f), F(g)\colon \mathit{List}(X) \to \mathit{List}(Y)$ (just consider their action on one-element lists!). So F is faithful.

Now consider a singleton set 1. This gets sent by F to the monoid we could represent as $(\mathbb{N}, +, 0)$. The only set-function $f\colon 1 \to 1$, the identity, gets sent to the identity monoid homomorphism on $(\mathbb{N}, +, 0)$. But there are other monoid homomorphisms from $(\mathbb{N}, +, 0)$ to $(\mathbb{N}, +, 0)$, e.g. $n \to 2n$. So F is not full.

(3) The 'pre-orderfication' functor from §4.2 (F8), $F\colon \mathscr{C} \to \mathscr{S}$, is full but not faithful unless $\mathscr{C}$ is already a pre-order category.

(4) Suppose $\mathscr{M}$ and $\mathscr{N}$ are the categories that correspond to the monoids $(M, \cdot)$ and $(N, \times)$. And let f be a monoid homomorphism which is surjec-

tive but not injective. Then the functor F corresponding to f is full but not faithful.

(5) You might be tempted to say that e.g. the 'total collapse' functor $F\colon \mathbf{Set} \to \mathbf{1}$ is full but not faithful. But it isn't full. Take C, C' in **Set** to be respectively the singleton of the empty set and the empty set. Then there is a trivial identity map in $\mathbf{1}$, $1\colon FC \to FC'$; but there is no arrow in **Set** from C to C'.

(6) An inclusion functor $F\colon \mathscr{S} \to \mathscr{C}$ is faithful; if $\mathscr{S}$ is a full subcategory of $\mathscr{C}$, then the inclusion map is fully faithful.

We evidently have

Theorem 23. *The composition of full functors is full and the composition of faithful functors is faithful.*

More significantly, we have:

Theorem 24. *A faithful functor $F\colon \mathscr{C} \to \mathscr{D}$ reflects monomorphisms and epimorphisms. That is to say, if Ff is monic (epic) then f is monic (epic).*

Proof. Suppose $f \circ g = f \circ h$, then $F(f \circ g) = F(f \circ h)$, so $Ff \circ Fg = Ff \circ Fh$. Since Ff is monic, $Fg = Fh$. And since F is faithful (so injective), $g = h$. Hence f is monic.

Dually for epics. □

Definition 29. A functor $F\colon \mathscr{C} \to \mathscr{D}$ is *essentially surjective on objects* (e.s.o.) iff every $\mathscr{D}$-object D is isomorphic in $\mathscr{D}$ to some object FC, for some $\mathscr{C}$-object C.

Definition 30. A functor $F\colon \mathscr{C} \to \mathscr{D}$ is *conservative* iff it is 'isomorphism-reflecting', i.e. if f is a $\mathscr{C}$-arrow such that Ff is an isomorphism in $\mathscr{D}$, then f is an isomorphism in $\mathscr{C}$.

Continuing with our examples:

(7) The forgetful functor $F\colon \mathbf{Grp} \to \mathbf{Set}$ (which, recall, sends a group G to its underlying set, and sends a group homomorphism to its underlying set-function) is conservative – for a group homomorphism is an isomorphism if and only if its underlying function is.

(8) On the other hand, the forgetful functor $F'\colon \mathbf{Top} \to \mathbf{Set}$ – while also faithful but not full – is *not* conservative. Consider the continuous bijection from the half-open interval $[0, 1)$ to S^1. Treated as a topological map f, it is not a homeomorphism in **Top**: however, treated as a set-function $F'f$, it *is* an isomorphism in **Set**.

(9) The forgetful functor $F''\colon \mathbf{Ab} \to \mathbf{Grp}$ is faithful, full but not e.s.o.

(10) On the other hand, the abelianization functor $Ab\colon \mathbf{Grp} \to \mathbf{Ab}$ *is* e.s.o. [Exercise for those who know about groups.]

Theorem 25. *If a functor is fully faithful it is conservative.*

Proof. Suppose $F\colon \mathscr{C} \to \mathscr{D}$, and Ff is an isomorphism, with f an arrow in $\mathscr{C}$ with source A. Then Ff has an inverse, and since F is full, this inverse must be Fg for some arrow g in $\mathscr{C}$. Hence $Fg \circ Ff = 1_{FA}$, so $F(g \circ f) = 1_{FA}$. But F is faithful so can send nothing other that 1_A to 1_{FA}. Hence $g \circ f = 1_A$, and f has a left inverse. Dually, it has a right inverse. So f is an isomorphism. □

7.2 The categories $\mathbf{Pfn}$ and $\mathbf{Set}_\star$ are 'equivalent'

A logico-philosophical digression to motivate a claim about the 'equivalence' of the categories $\mathbf{Pfn}$ *and* $\mathbf{Set}_\star$

There is no getting away from the central importance of the notion of a partial function from $\mathbb{N}$ to $\mathbb{N}$ in the general theory of computation (thus, the computable function φ_e computed by the e-th Turing machine in a standard enumeration is typically partial).

How should we treat partial functions in logic? Suppose the partial computable function $\varphi\colon \mathbb{N} \to \mathbb{N}$ takes no value for n (the algorithm defining φ doesn't terminate gracefully for input n). Then the term '$\varphi(n)$' apparently lacks a denotation. *But in standard first-order logic, all terms are assumed to denote.* Two-valued logic requires every sentence to be determinately either true or false (truth-value gaps aren't allowed): but a sentence with a non-denoting term, on the standard semantics, will lack a truth-value. What to do?

Historically, there are three options on the market for dealing with empty terms in a regimented logical language, and hence for dealing with the partial functions which give rise to them:

(1) *Frege's proposal.* Treat apparently empty terms as not *really* empty, by providing a default 'rogue' object for them to denote. So (apparently) empty terms are still genuine terms, but with a deviant denotation.

How does this work for functional terms? Well, given what we naively think of as a partial function $\varphi\colon \mathbb{N} \to \mathbb{N}$, treat this as officially being a total function $f\colon \mathbb{N} \to \mathbb{N} \cup \{\star\}$, where $\star$ is any convenient non-number, and where $f(n) = \varphi(n)$ when φ takes a numerical value, and $f(n) = \star$ otherwise. Or, loving symmetry as we do, we could perhaps better talk about a function $f'\colon \mathbb{N} \cup \{\star\} \to \mathbb{N} \cup \{\star\}$ where $f'(n) = f(n)$ on numbers, and $f'(\star) = \star$. If you like, you can think of $\star$ as coding 'not numerically defined'.

So, on this approach there are no genuinely partial functions. All functions are total. And what we *really* meet in the theory of computation are total functions which are only partially numerical (not all their values are numbers).

(2) *Russell's proposal* Treat apparently empty terms as not *really* terms at all, by translating them away. Thus, to use his famous example, 'The present King of France is bald' (which has an apparently non-denoting term creating a truth-value gap) really says 'There is a least one present king of France, there is at most one present king of France, and whoever is a present king of France is bald' (which is plain false because the first conjunct is false). The apparently denoting expression '*the* present King of France' disappears when we analyse the true logical form of the statement (we are left are left with a predicate '(is a) present king of France' and quantifiers).

For a mathematical example, 'The rational square root of two is greater than one' (we might want to suppose such a thing in a reductio ad absurdum proof) really says 'At least one rational is a square root of 2, at most one rational is a square root of 2, and any rational which is a square root of 2 is greater than one' – which again is plain false.

The suggestion, then, is that we translate away empty terms, and that we can thereby avoid the truth-value gaps which are anathema to standard two-valued logic.

But you'll immediately spot a snag. It isn't a matter of pure *logic* that 'the present King of France' fails to denote (that's an empirical fact), nor indeed that 'the rational square root of two' fails to denote (that's a mathematical fact). And in other cases, we may not know whether a term 'the F' denotes or not. Yet surely the intrinsic logical form of a statement should not depend on extraneous non-logical facts which are perhaps beyond our ken. It seems, then, that if we plan to avoid *empty* denoting terms of the kind 'the F' by translating them away, we'll have to adopt a more general policy of translating away *all* such terms.

And this is exactly what Russell ended up doing. Roughly, given a sentence $G(t)$ where t is a complex term, the general proposal is that we first treat t, if it isn't already obviously in that form, as being short for a so-called definite description of the kind 'the F', and then we render 'G(the F)' as really having the logical form 'There is at least one F, there is at most one F, and whatever is F is G'. The original term t has disappeared from the analysis, via what's called Russell's Theory of Descriptions.

How does this pan out for functional terms? Well, Russell (i) trades in functions for relations, so (ii) a functional term becomes a definite description involving a relation, and then (iii) this gets translated away using the same general policy.

Taking this in stages, the idea (i) that a one-place function φ, for example, is just a special sort of binary relation R (one where if Rxy and Rxy' then $y = y'$) is now a commonplace. Which is not to say it is right, but let's not pause over that. The present point is that this treatment applies equally smoothly to total and to partial functions. Then the idea (ii) is that we replace an occurrence of e.g. '$\varphi(n)$', for a (perhaps partial) function φ, with an occurrence of 'the y such that Rny'. Then (iii) we apply Russell's Theory of Descriptions to sentences that embed the definite description 'the y such that Rny'; and we are left with something involving new quantifiers and the new relation R, but without the original possibly empty term. So we've again eliminated those pesky truth-value gaps. And this time we have kept partial functions alongside the total ones by transmuting them into certain relations.

(3) *Logical revisionism* Frege's proposal for avoiding truth-value gaps is simple but artificial. Russell's proposal for avoiding truth-value gaps is arguably more principled but gets horribly messy with all those introduced quantifiers. So a third option is to bite the bullet and live with truth-value gaps, as we surely already do in informal reasoning.

That means, when we come to adopt an official formalized logic, we'll want one which is free from the assumption that all terms denote, by adopting a *free logic* for short. Then, even in our official formal treatment, we can take partial functions and the empty terms they give rise to at face value.

However, this policy too has its own costs (logical revisionism has its complications) . . .

But we won't continue the story any further now. The debate about the best logical treatment of partial functions is the sort of thing that might grip some philosophically-minded logicians but really seems of little more general mathematical interest.

And that's exactly the point of this section! We'll set aside the Russellian option for the moment, but from a mathematical point of view there surely isn't anything much to choose between options (1) and (3).

On the small scale, we can think of a world of genuinely *partial* numerical functions $\varphi\colon \mathbb{N} \to \mathbb{N}$ (genuinely partial because not everywhere defined, and hence giving rise to empty terms), or we can equally think of a corresponding world of *total* functions $f \cup \{\star\}\colon \mathbb{N} \to \mathbb{N} \cup \{\star\}$, with $\star \notin \mathbb{N}$, and $f(\star) = \star$. Take your pick! More generally, on the large scale, we can think of other sets with partial functions between them, or of corresponding pointed sets (sets with a distinguished object as base point) and base-point preserving total functions between *them*. What's to choose, apart from familiarity? Mathematically, surely both approaches come to the same.

And so back, at last, to categories! There is a category **Pfn** whose objects are sets and whose arrows are (possibly) *partial* functions between them. And there is also the category $\mathbf{Set}_\star$ of pointed sets which we've already met, whose objects are sets with a distinguished base point, and whose arrows are (total) set-functions which preserve base points. So, putting the upshot of our reflections in this section in categorial terms, we get the following attractive

Desideratum *An account of what it is for two categories to be equivalent* *should surely count* $\mathbf{Set}_\star$ *and* $\mathbf{Pfn}$ *as being so, for mathematically they come to the same.*

And in the next section we will see that a natural first-shot account of what it is for categories to be equivalent fails to meet this desideratum.

7.3 Isomorphic categories

In this section we introduce the obvious definition of isomorphic categories.

Objects are isomorphic if they have an isomorphism between them, i.e. an arrow which has a two-sided inverse. Functors between categories are like arrows between objects. So it is natural to say:

Definition 31. Categories $\mathscr{C}$ and $\mathscr{D}$ are *isomorphic*, in symbols $\mathscr{C} \cong \mathscr{D}$, iff there is an isomorphism $F\colon \mathscr{C} \xrightarrow{\sim} \mathscr{D}$, where a functor F is an isomorphism iff it has an inverse, i.e. there is a functor $G\colon \mathscr{D} \to \mathscr{C}$ where $GF = 1_{\mathscr{C}}$ and $FG = 1_{\mathscr{D}}$.

Recall, $1_{\mathscr{C}}$ is the trivial functor that sends every bit of $\mathscr{C}$'s data to itself (see §4.2, (F10)).

Let's have a few examples:

(1) Take the toy two-object categories with different pairs of objects which we can diagram as

$$\begin{tikzcd} \bullet \arrow[loop left] \arrow[r] & \star \arrow[loop right] \end{tikzcd} \qquad\qquad \begin{tikzcd} a \arrow[loop left] \arrow[r] & b \arrow[loop right] \end{tikzcd}$$

Plainly they are isomorphic (and indeed there is unique isomorphic functor that sends the first to the second)! If we don't care about distinguishing copies of structures that are related by a unique isomorphism that we'll count these as the same in a strong sense. Which to that extent warrants our earlier talk about *the* category $\mathbf{2}$ (e.g. in §2.2, Ex (17)).

(2) There is a category **Bool** whose objects are algebras $(B, \neg, \wedge, \vee, 0, 1)$ constrained by the familiar Boolean axioms, and whose arrows are homomorphisms that preserve algebraic structure. There is also a category **BoolR** whose objects are Boolean rings, i.e. rings $(R, +, \times, 0, 1)$ where $x^2 = x$ for all $x \in R$, and whose arrows are ring homomorphisms.

There is a familiar way of marrying up Boolean algebras with corresponding rings, and vice versa. Thus if we start from $(B, \neg, \wedge, \vee, 0, 1)$, take the same carrier set and distinguished objects, put

(i) $x \times y =_{\mathrm{def}} x \wedge y$,
(ii) $x + y =_{\mathrm{def}} (x \vee y) \wedge \neg(x \wedge y)$ (exclusive 'or'),

then we get a Boolean ring. And if we apply the same process to two algebras B_1 and B_2, it is elementary to check that this will carry a homomorphism of algebras $f_a\colon B_1 \to B_2$ to a corresponding homomorphism of rings $f_r\colon R_1 \to R_2$. We can equally easily go from rings to algebras, by putting

(i) $x \wedge y =_{\text{def}} x \times y$,

(ii) $x \vee y =_{\text{def}} x + y + (x \times y)$

(iii) $\neg x =_{\text{def}} 1 + x$.

Note that going from a algebra to the associated ring and back again takes us back to where we started.

In summary, without going into more details, we can in this way define a functor $F\colon \mathbf{Bool} \to \mathbf{BoolR}$, and a functor $G\colon \mathbf{BoolR} \to \mathbf{Bool}$ which are inverses to each other. So, as we'd surely expect, **Bool** is isomorphic to **BoolR**.

(3) Revisit the example in §5.4 of the coslice category $1/\mathbf{Top}$. This category has as objects all the arrows $\vec{x}\colon 1 \to X$ for any $X \in \mathbf{Top}$. And the arrows from $\vec{x}\colon 1 \to X$ to $\vec{y}\colon 1 \to Y$ are just the arrows $j\colon X \to Y$ from **Top** such that $j \circ \vec{x} = \vec{y}$.

Now we said before that this is in some strong sense 'the same as' the category $\mathbf{Top}_*$ of pointed topological spaces. And indeed the categories are isomorphic. For take the function F_{ob} from objects in $1/\mathbf{Top}$ to objects $\mathbf{Top}_*$ which sends an object $\vec{x}\colon 1 \to X$ to the pointed topological space (X, x), i.e. X-equipped-with-the-basepoint-x, where x is the value of the function $\vec{x}$ for its sole argument. And take F_{arw} to send a $j\colon X \to Y$ such that $j \circ \vec{x} = \vec{y}$ to $j\colon (X, x) \to (Y, y)$, which preserves base points. Then it is easy to check that F is a functor $F\colon 1/\mathbf{Top} \to \mathbf{Top}_*$.

In the other direction, we can define a functor $G\colon \mathbf{Top}_* \to 1/\mathbf{Top}$ which sends (X, x) to the function $\vec{x}\colon 1 \to X$ which sends the sole object in 1 to the point x, and sends a basepoint-preserving continuous function from X to Y to itself.

And it is immediate that these two functors F and G are inverse to each other. Hence, as claimed, $\mathbf{Top}_* \cong 1/\mathbf{Top}$.

(4) For two more easy examples, we noted in §5.5 two closely linked pairs (starting from a base category $\mathscr{C}$ and a $\mathscr{C}$-object I:

a) the comma category $(1_{\mathscr{C}} \downarrow 1_{\mathscr{C}})$ and the arrow category $\mathscr{C}^{\to}$,

b) the comma category we called $(1_{\mathscr{C}} \downarrow I)$ and the slice category $\mathscr{C}/I$.

It is easily seen that in each case the mentioned categories are isomorphic to each other.

So far, so good then. We've given examples of pairs of categories which, intuitively, 'come to just the same' and are indeed isomorphic by our definition. However, we also have the following result:

Theorem 26. $\mathbf{Set}_\star$ *is not isomorphic to* **Pfn**.

We can remark that there *is* an obvious functor $F\colon \mathbf{Set}_\star \to \mathbf{Pfn}$. F sends a pointed set (X, x) to the set $X \setminus \{x\}$, and sends a base-point preserving total function $f\colon (X, x) \to (Y, y)$ to the partial function $\varphi\colon X \setminus \{x\} \longrightarrow Y \setminus \{y\}$, where $\varphi(x) = f(x)$ if $f(x) \in Y \setminus \{y\}$, and is undefined otherwise. But, nice though this is, F isn't an isomorphism (it could send distinct (X, x) and (X', x') to the same target object).

Again, there is a whole family of functors from $\mathbf{Pfn}$ to $\mathbf{Set}_\star$ which take any set X and add an element not yet in X to give as an expanded set with the new object as a basepoint. Here's a way of doing this in a uniform way without making arbitrary choices for each X. Define $G\colon \mathbf{Pfn} \to \mathbf{Set}_\star$ as sending a X to the pointed set $X_* =_{\mathrm{def}} (X \cup \{X\}, X)$, remembering that in standard set theories $X \notin X$! And then let G send a partial function $\varphi\colon X \to Y$ to the total basepoint-preserving function $f\colon X_* \to Y_*$, where $f(x) = \varphi(x)$ if $\varphi(x)$ is defined and $f(x) = \{Y\}$ otherwise. G is a natural choice, but isn't an isomorphism (it isn't surjective on objects).

Still, those observations don't yet rule out there being *some* pair of functors between $\mathbf{Set}_\star$ and $\mathbf{Pfn}$ which are mutually inverse. But there can't be any such pair.

Proof. A functor which is an isomorphism from $\mathbf{Pfn}$ to $\mathbf{Set}_\star$ must send objects in $\mathbf{Pfn}$ one-to-one to objects in $\mathbf{Set}_\star$, and must send isomorphisms to isomorphisms, so should preserve the cardinality of isomorphism classes. But the isomorphism class of the empty set in $\mathbf{Pfn}$ has just one member, while there is no one-membered isomorphism class in $\mathbf{Set}_\star$. So there can't be an isomorphism between the categories. $\square$

7.4 Equivalent categories

(a) The last two sections have together shown that there are categories $\mathbf{Pfn}$ and $\mathbf{Set}_\star$ which to all intents and purposes are mathematically equivalent but which aren't isomorphic (according to the obvious definition of isomorphism for categories).

We did, however, note an obvious choice of functors $F\colon \mathbf{Set}_\star \to \mathbf{Pfn}$ and $G\colon \mathbf{Pfn} \to \mathbf{Set}_\star$. And while GF isn't the identity on $\mathbf{Set}_\star$ it *does* map $\mathbf{Set}_\star$ to itself in a rather natural way (without arbitrary choices).

Reflecting on this case a bit more suggests that the following weakening of the definition of isomorphism between categories:

Definition 32. Categories $\mathscr{C}$ and $\mathscr{D}$ are equivalent iff there are functors $F\colon \mathscr{C} \to \mathscr{D}$ and $G\colon \mathscr{D} \to \mathscr{C}$, together with a pair of natural isomorphisms $\alpha\colon 1_{\mathscr{C}} \Rightarrow GF$ and $\beta\colon FG \Rightarrow 1_{\mathscr{D}}$. If $\mathscr{C}$ is equivalent to $\mathscr{D}$, we write $\mathscr{C} \simeq \mathscr{D}$.

And yes, we can now give a direct proof that $\mathbf{Pfn}$ and $\mathbf{Set}_\star$ are indeed equivalent in this way. But in fact we won't do that. Rather we'll first prove a result which yields an alternative characterization of equivalence which is easier to apply.

Theorem 27. *Assuming a sufficiently strong choice principle, a functor $F\colon \mathscr{C} \to \mathscr{D}$ is part of an equivalence between $\mathscr{C}$ and $\mathscr{D}$ iff F is faithful, full and e.s.o.*

Proof. First suppose F is part of an equivalence between $\mathscr{C}$ and $\mathscr{D}$, so that there is a functor $G\colon \mathscr{D} \to \mathscr{C}$, where $GF \cong 1_{\mathscr{C}}$ and $FG \cong 1_{\mathscr{D}}$. Then:

(i) Given an arrows $f, g\colon A \to B$ in $\mathscr{C}$, then by hypothesis, the following square commutes for f (α is the required natural isomorphism between the identity functor and the composite GF),

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\alpha_A"] & B \arrow[d, "\alpha_B"] \\ GFA \arrow[r, "GFf"] & GFB \end{tikzcd}$$

and hence $\alpha_B^{-1} \circ GFf \circ \alpha_A = f$. And of course $\alpha_B^{-1} \circ GFg \circ \alpha_A = g$. It immediately follows that if $Ff = Fg$ then $f = g$, i.e. F is faithful. A companion argument, interchanging the roles of $\mathscr{C}$ and $\mathscr{D}$, shows that G too is faithful.

(ii) Suppose we are given an arrow $h\colon FA \to FB$, then put $f = \alpha_B^{-1} \circ Gh \circ \alpha_A$. But we know that $f = \alpha_B^{-1} \circ GFf \circ \alpha_A$. So it follows that $GFf = Gh$, and since G is faithful, $h = Ff$. So every such h in $\mathscr{D}$ is the image under F of some arrow in $\mathscr{C}$. So F is full.

(iii) Recall, $F\colon \mathscr{C} \to \mathscr{D}$ is e.s.o. iff for any $D \in \mathscr{D}$ we can find some isomorphic object FC, for $C \in \mathscr{C}$. But we know that our natural isomorphism between $1_{\mathscr{D}}$ and FG means that that there is an isomorphism from D to FGD, so putting $C = GD$ gives the desired result that F is e.s.o.

Now for the argument in the other direction. Suppose, then, that $F\colon \mathscr{C} \to \mathscr{D}$ is faithful, full and e.s.o. We need to construct (i) a corresponding functor $G\colon \mathscr{D} \to \mathscr{C}$, and then a pair of natural isomorphisms (ii) $\beta\colon FG \Rightarrow 1_{\mathscr{D}}$ and (iii) $\alpha\colon 1_{\mathscr{C}} \Rightarrow GF$:

(i) By hypothesis, F is e.s.o., so by definition every object $D \in \mathscr{D}$ is isomorphic in $\mathscr{D}$ to some object FC, for $C \in \mathscr{C}$. Hence – and here we are invoking an appropriate choice principle – for a given $D \in \mathscr{D}$, we can choose a pair (C, β_D), with $C \in \mathscr{C}$ and $\beta_D\colon FC \to D$ an isomorphism in $\mathscr{D}$. Now define G_{ob} as sending an object $D \in \mathscr{D}$ to the chosen $C \in \mathscr{C}$ (so $GD = C$, and $\beta_D\colon FGD \to D$).

To get a functor, we need the component G_{arw} to act suitably on an arrow $g\colon D \to E$. Now, note

$$FGD \xrightarrow{\beta_D} D \xrightarrow{g} E \xrightarrow{\beta_E^{-1}} FGE$$

and since F is full and faithful, there must be some unique $f\colon GD \to GE$ which F sends to the composite $\beta_E^{-1} \circ g \circ \beta_D$. Put $G_{arw}g = f$.

Claim: *G, with components G_{ob}, G_{arw}, is indeed a functor.* We need to show that G (a) preserves identities and (b) respects composition:

For (a), note that $G1_D = e$ where e is the unique arrow from GD to GD such that $Fe = \beta_D^{-1} \circ 1_D \circ \beta_D = 1_{FGD}$. So $e = 1_{GD}$.

For (b) we need to show that, given $\mathscr{D}$-arrows $g\colon D \to E$ and $h\colon E \to F$, $G(h \circ g) = Gh \circ Gg$. But note that

$$\begin{aligned} FG(h \circ g) = \beta_F^{-1} \circ h \circ g \circ \beta_D &= (\beta_F^{-1} \circ h \circ \beta_E) \circ (\beta_E^{-1} \circ g \circ \beta_D) \\ &= FG(h) \circ FG(g) = F(G(h) \circ G(g)) \end{aligned}$$

Hence, since $FG(h \circ g) = F(G(h) \circ G(g))$ and F is faithful, $G(h \circ g) = G(h) \circ G(g)$, so G is indeed a functor. Phew!

(ii) By construction, β is natural isomorphism from FG to $1_{\mathscr{D}}$.

(iii) Note next that we have an isomorphism $\beta_{FA}^{-1}\colon FA \to FGFA$. As F is full and faithful, $\beta_{FA}^{-1} = F(\alpha_A)$ for some unique $\alpha_A\colon A \to GFA$. Since F is fully faithful it is conservative, i.e. reflects isomorphisms (by Theorem 25), hence α_A is also an isomorphism. Also, the naturality diagram

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\alpha_A"] & B \arrow[d, "\alpha_B"] \\ GFA \arrow[r, "GFf"] & GFB \end{tikzcd}$$

always commutes for any arrow $f\colon A \to B$ in $\mathscr{C}$. Why? Because

$$\begin{aligned} &F(\alpha_B \circ f) = F\alpha_B \circ Ff = \beta_{FB}^{-1} \circ Ff = \\ &\qquad FGFf \circ \beta_{FA}^{-1} = FGFf \circ F\alpha_A = F(GFf \circ F\alpha_A) \end{aligned}$$

relying on the naturality of β^{-1}. But if $F(\alpha_B \circ f) = (GFf \circ F\alpha_A)$ then since F is faithful, $\alpha_B \circ f = GFf \circ F\alpha_A$. Hence the α_A are the components of our desired natural isomorphism $\alpha\colon 1_{\mathscr{C}} \Rightarrow GF$.

So we are done! □

Our theorem enables us now to quickly show

Theorem 28. $\mathbf{Pfn} \simeq \mathbf{Set}_\star$

Proof. Choose the particular functor $G\colon \mathbf{Pfn} \to \mathbf{Set}_\star$ which sends a set X to a set $X_* =_{\text{def}} X \cup \{X\}$ with basepoint X, and sends a partial function $f\colon X \to Y$ to the total function $f_*\colon X_* \to Y_*$, where for $f_*(x) = f(x)$ if $f(x)$ is defined and $f_*(x) = Y$ otherwise.

G is faithful, as it is easily checked that it sends distinct functions to distinct functions. And it is equally easy to check that G is full, i.e. given any basepoint preserving function between sets X_* and Y_*, there is a partial function f which G sends to it.

But G is essentially surjective on objects. For every pointed set in $\mathbf{Set}_\star$ – i.e. every set which can be thought of as the union of a set X with $\{*\}$ where $*$ is an additional basepoint element (not in X) – is isomorphic in $\mathbf{Set}_\star$ to the set $X \cup \{X\}$ with X as basepoint. Hence G is part of an equivalence between **Pfn** and $\mathbf{Set}_\star$. □

(b) Now for another example. Recall **FinSet** is the category of finite sets and functions between them. Let **FinOrdn** be its full subcategory containing the empty set and all sets of the form $\{0, 1, 2, \ldots n-1\}$ and all functions between them. It doesn't really matter for present purposes how you think of the natural numbers; but to fix ideas, think of them set-theoretically as von Neumann ordinals, so the objects of **FinOrdn** are then the finite ordinals – hence the label for the category. We then have:

Theorem 29. FinOrdn $\simeq$ FinSet

Proof. **FinOrdn** is a full subcategory of **FinSet**, so the inclusion functor F is fully faithful. F is also essentially surjective on objects: for take any object in **FinSet**, which is some n-membered set: that is in bijective correspondence (and hence isomorphic in **FinSet**) with the finite ordinal n. Hence F is part of an equivalence, and **FinOrdn** $\simeq$ **FinSet**. □

How should we regard this last result? We saw that defining equivalence of categories in terms of isomorphism would be too strong, as it rules out our treating **Pfn** and $\mathbf{Set}_\star$ as in effect equivalent. But now we've seen that defining equivalence of categories as in Defn. 7.4 makes the seemingly very sparse category **FinOrdn** equivalent to the seemingly much more abundant **FinSet**. Is that a strike against the definition of equivalence, showing it to be too weak?

It might help to think of a toy example. Consider the two categories which we can diagram respectively as

$$\bullet \circlearrowleft \qquad\qquad \circlearrowright \bullet \rightleftarrows \star \circlearrowleft$$

On the left, we have the category $\mathbf{1}$; on the right we have a two-object category $\mathbf{2!}$ with arrows in *both* directions between the objects (in addition, of course, to the required identity arrows). These two categories are also equivalent. For the obvious inclusion functor $\mathbf{1} \hookrightarrow \mathbf{2!}$ is full and faithful, and it is trivially essentially surjective on objects as each object in the two-object category is isomorphic to the other.

What this toy example highlights is that our equivalence criterion counts categories as amounting to the same when (putting it very roughly) one is just the same as the other padded out with new objects and just enough arrows to make the new objects isomorphic to some old objects.

But on reflection that's fine. Taking a little bit of the mathematical world and adding indiscernable copies of structures it already contains won't, for many (most? nearly all?) purposes, give us a real enrichment. Therefore a criterion of equivalence of categories-as-mathematical-universes that doesn't care about surplus isomorphic copies is what we typically need. Hence the results that $\mathbf{1} \simeq \mathbf{2!}$ and $\mathbf{Finord} \simeq \mathbf{FinSet}$ are arguably welcome features, not bugs, of our account of equivalence.

7.5 Skeleton categories

Still, even if we treat categories as equivalent when one just has additional objects isomorphic to those already there in the other, shouldn't the usual sort of concern for Bauhaus elegance and lack of redundancy lead us to privilege categories which are as skeletal as possible? Let's say:

Definition 33. The category $\mathscr{S}$ is a *skeleton* of the category $\mathscr{C}$ if $\mathscr{S}$ is a full subcategory of $\mathscr{C}$ which contains exactly one object from each class of isomorphic objects of $\mathscr{C}$. A category is *skeletal* if it is a skeleton of some category.

For a toy example, suppose $\mathscr{C}$ is a category arising from a pre-order (as in §2.2, Ex. (10)). Then any skeleton of $\mathscr{C}$ will be a poset category. (Check that!)

Theorem 30. *If $\mathscr{S}$ is a skeleton of the category $\mathscr{C}$ then $\mathscr{S} \simeq \mathscr{C}$.*

Proof. The inclusion functor $\mathscr{S} \hookrightarrow \mathscr{C}$ is fully faithful, and by the definition of $\mathscr{S}$ is essentially surjective on objects. So we can apply Theorem 27. $\square$

So how about this for a programme? Take the usual universe of categories. But now slim it down by taking skeletons. Then work with these. And we can now forget bloated non-skeletal categories (and forget too about the notion of equivalence and revert to using the simpler notion of isomorphism, because equivalent skeletal categories are in fact isomorphic). What's not to like?

The trouble is that hardly any categories that occur in the wild (so to speak) are skeletal. And slimming down to be done by appeal to an axiom of choice. Indeed the following statements are each equivalent to a version of the axiom of choice:

(1) Any category has a skeleton.

(2) A category is equivalent to any of its skeletons

(3) Any two skeletons of a given category are isomorphic.

The choice of a skeleton is usually quite artificial – there typically won't be a canonical choice. So any gain in simplicity from concentrating on skeletal categories would be bought at the cost of having to adopt 'unnatural', non-canonical, choices of skeletons. Given that category theory is supposed to be all about natural patterns already occurring in mathematics, this perhaps isn't going to be such a good trade-off after all.

8 Categories of categories: issues of size

8.1 Categories of categories

(a) We noted in §4.2 that there is an identity functor $1_{\mathscr{C}} : \mathscr{C} \to \mathscr{C}$ for every category. We also noted in §5.1 that composable functors $F : \mathscr{C} \to \mathscr{D}$ and $G : \mathscr{D} \to \mathscr{E}$ (with the target of the first the source of the second) always do have a composite $G \circ F$, and moreover composition of functors is associative.

So the following definition looks to be in good order:

Definition 34. Suppose $\mathscr{X}$ comprises two sorts of data:

(1) *Objects*: some categories, $\mathscr{C}, \mathscr{D}, \mathscr{E}, \ldots,$

(2) *Arrows*: some functors, $F, G, H, \ldots,$ between those categories,

where the arrows include the identity functor on each category, and include $G \circ F$ for each included composable pair F and G: then $\mathscr{X}$ is a *category of categories*.

Let's have some examples of categories of categories:

(1) Trivially, there is a category of categories whose sole object is the category $\mathscr{C}$ and whose sole arrow is identity functor $1_{\mathscr{C}}$.

(2) We noted that every monoid can be thought of as itself a category. So the familiar category **Mon** can be thought as a category of categories.

(3) There is a category whose objects are the finite categories, and whose arrows are all the functors between finite categories.

So there certainly are *some* categories of categories. But there are limitations. Suppose we next say:

Definition 35. A category is *normal* iff it is not one of its own objects.

The categories which we have met in previous chapters have all been normal, and hence all the functors we have met have been normal functors too, i.e. functors between normal categories. Now ask: can all the normal categories be gathered together as the objects of one really big category whose arrows are some or all the normal functors?

The answer is given by

Theorem 31. *There is no category of normal categories.*

Proof. Suppose there is a category $\mathscr{N}$ whose objects are all the normal categories. Now ask, is $\mathscr{N}$ normal? If it is, then it is one of the objects of $\mathscr{N}$, so $\mathscr{N}$ is non-normal. So $\mathscr{N}$ must be non-normal. But then it is not one of the objects of $\mathscr{N}$, so $\mathscr{N}$ is normal after all. Contradiction. □

That argument just re-runs, in our new environment, the very familiar argument from Russell's Paradox to the conclusion that there is no set of all the normal sets (where a set is normal if it is not a member of itself). It is worth remarking, however, that the argument is not especially to do with sets. For at its core is a simple, purely logical, observation. Take *any* two-place relation R defined over some objects; then there can be no object r which is related by R to all and only those objects which are not R-related to themselves. In other words, it is a simple logical theorem that $\neg\exists r\forall x(Rxr \leftrightarrow \neg Rxx)$. Russell's original argument applies this elementary logical theorem to the set-theoretic relation R_1, '... is a set which is a member of the set ...', to show that there is no set of all normal (i.e. non-self-membered) sets. Our argument above applies the same logical theorem to the category-theoretic relation R_2, '... is a category which is an object of the category ...', to show that there is no category of all normal categories. There is no lurking presumption here that categories are sets.

(b) Russell's original argument that there is no set of all *normal* sets is usually taken to entail that, a fortiori, there is no universal set, no set of *all* sets. The reasoning being that if there were a universal set then we should be able carve out of it (via a separation principle) a subset containing just those sets which are normal, which we know can't be done.

To keep ourselves honest, however, we should note that this further argument can be, and has been, resisted. There are cogent set theories on the market which work by restricting seperation and blocking the idea that, if there is a universal set of all sets, we should be able to carve out from it a set of all normal sets: see Forster (1995). But we can't explore deviant set theories here: henceforth in these notes we'll have to assume our set theory is a standard one, one more like Zermelo-Fraenkel set theory with or without ur-elements.

Similarly to the argument about sets, the Russellian argument that there is no category of all normal categories is naturally taken to entail that there is no universal category in the obvious sense:

Definition 36. A category $\mathscr{U}$ is *universal* if it is a category of categories such that every category is an object of $\mathscr{U}$.

Theorem 32? *There is no universal category.*

The argument goes: suppose a universal category $\mathscr{U}$ exists. Then we could take the full subcategory whose objects are just the normal categories, to get a category of all normal categories. But we have shown there can be no such category.

Can this argument be resisted? Well, how could we justify saying that even if there is a category of *all* categories, it needn't have a full subcategory of *normal* categories? Perhaps some themes in debates about set theories with a universal set could be carried over to this case. But again, it would take us far too far away from mainstream concerns in category theory to try to explore this here.

8.2 'Small' and 'locally small' categories

(a) When we talk about sets, then, we are assuming we are working in a reasonably standard set theory according to which there can be entities (e.g. the sets themselves) which are too many to form a set. Likewise, the normal categories are too many to form a category. Still, there can it seems be some pretty capacious categories of categories.

Let's say

Definition 37. A category $\mathscr{C}$ is *small* iff it has overall only a 'set's worth' of arrows – i.e. the arrows of $\mathscr{C}$ can be put into one-one correspondence with the members of some set. We use **Cat** to denote the category whose objects are small categories and whose arrows are the functors between them.

A category $\mathscr{C}$ is *locally small* iff for every pair of $\mathscr{C}$-objects C, D there is only a 'set's worth' of arrows from C to D, i.e. those arrows can be put into one-one correspondence with the members of some set. We use **CAT** to denote the category, assuming there is one, whose objects are locally small categories and whose arrows are the functors between them.

Some comments and examples:

(1) The terms 'small' and 'locally small' are standard.

(2) It would be more usual to say that in a small category the arrows themselves form a set. But if our favoured set theory is a theory like ZFC where sets only have other sets as members, that would presuppose that arrows are themselves sets, and we might not want to make that assumption just yet. So, for smallness, let's only require that the arrows aren't too many to be represented by a set. Similarly for local smallness.

(3) Since for every object in $\mathscr{C}$ there is at least one arrow, namely the identity arrow on $\mathscr{C}$, if the objects of $\mathscr{C}$ are too many to be mapped into a set, then $\mathscr{C}$ has too many arrows to be small. Contraposing, if $\mathscr{C}$ is small, not only its arrows but its objects can be put into one-one correspondence with the members of some set.

(4) Among our examples in §2.2, tiny finite categories like **1** and **2** are of course small. But so too are the categories corresponding to an infinite but set-sized monoid or to an infinite pre-ordered set. Categories such as **Set** or **Mon**, however, have too many objects and arrows to be small.

(5) A discrete category only has as many arrows as objects. Which implies that the discrete category on any set is small. However it also implies that the category **Cat** of small categories has at least as many objects as there are sets, and hence is itself determinately *not* small. Since **Cat** is determinately not small, no Russellian paradox arises for **Cat** as it did for the putative category of normal categories.

(6) But while not small, categories such as **Set** or **Mon** and indeed all our other examples so far are locally small (some authors even build local smallness into the very definition of a category). In **Set**, for example, the arrows between objects C to D are members of a certain subset of the powerset of $C \times D$: which makes **Set** locally small.

(7) Take a one-element category $\mathbf{1}$, which is certainly locally small. Then a functor from $\mathbf{1}$ to **Set** will just map the object of $\mathbf{1}$ to some particular set: and there will be as many distinct functors $F \colon \mathbf{1} \to \mathbf{Set}$ as there are sets. In other words, arrows from $\mathbf{1}$ to **Set** in **CAT** are too many to be mapped one-to-one to a set. Hence **CAT** is determinately *not* locally small. So again no Russellian paradox arises for **CAT**.

Since no Russellian paradox arises for either **Cat** or **CAT**, it seems that we can indeed countenance these categories, and will henceforth assume that they do exist.

(b) Note however that we have the following negative result:

Theorem 33. *Smallness is not preserved by categorial equivalence.*

In other words, we can have $\mathscr{C}$ a small category, $\mathscr{C} \simeq \mathscr{D}$, yet $\mathscr{D}$ not small. This is a simple corollary of our observation in §7.4 that if we take a category, inflate it by adding lots of objects and just enough arrows to ensure that these objects are isomorphic to the original objects, then the augmented category is equivalent to the one we started with. For an extreme example, start with the one-object category $\mathbf{1}$, i.e. $\bullet \circlearrowleft$ (that's small)! Now add as new objects every set, and as new arrows an identity arrow for each set, and also for every set X a pair of arrows $\bullet \rightleftarrows X$ which composed to give identities. Then we get a new pumped-up category $\mathbf{1}^+$ (which is certainly not small). But $\mathbf{1}^+ \simeq \mathbf{1}$.

Categorial notions that are not invariant under equivalence are sometimes said to be 'evil'. That makes smallness an evil notion! If you fuss about these things, you can highlight a neighbouring notion which evidently is virtuous:

Definition 38. A category is *essentially small* if it is equivalent to category with a set's worth of arrows.

But we aren't going to fuss here.

There is, by the way, a companion positive result

Theorem 34. *Local smallness* is *preserved by categorial equivalence.*

Proof. An equivalence $\mathscr{C} \underset{G}{\overset{F}{\rightleftarrows}} \mathscr{D}$ requires F and G to be full and faithful functors. So in particular, for any $\mathscr{D}$-objects D, D', there are the same number of arrows between them as between the $\mathscr{C}$-objects GD, GD'. So that ensures that if $\mathscr{C}$ has only a set's worth of arrows between any pair of objects, the same goes of $\mathscr{D}$. You may come across the terminology. $\square$

8.3 A very brief aside on 2-categories

Having mentioned **Cat**, it is worth remarking that here we have a category with objects (small categories, $\mathscr{C}, \mathscr{D}, \ldots$), arrows between them (functors $F\colon \mathscr{C} \to \mathscr{D}, \ldots$) and where each hom-set of these arrows $\mathbf{Cat}(\mathscr{C}, \mathscr{D})$, i.e. each set of functors from $\mathscr{C}$ to $\mathscr{D}$, *itself* forms a category, the functor category $[\mathscr{C}, \mathscr{D}]$.

So we have here two layers of categorial structure, and **Cat** is the paradigm of a 2-category, i.e. a category where the hom-sets are themselves categories.

8.4 Classes, virtual and real

We are going need to make use in the next section and onwards of a notion of a class of entities, where this notion is to be distinguished from that of a set. This section therefore does some preliminary explaining: the points will probably be familiar from an introductory set theory course.

Recall the usual story about the cumulative hierarchy of sets. We start at level zero with some objects – or indeed, in pure ZFC set theory, we start with nothing at all. Then at each succeeding level, we add to whatever entities we have so far accumulated all the possible sets which can be formed with those entities as members. Every set, the story goes, is formed at some level. And we keep on going: at no level in the set-building process completed. We can always go up a level to form new sets – however many sets we've formed so far, we can always collect them together into new sets at the next level. Which means that the whole universe of sets never becomes available at any level to itself be collected into a set. Which is why there is no set of all sets.

However, even if there is no set of all sets (given that we are indeed conceiving of sets as living in the cumulative hierarchy), can't we still think of the universe of sets as in some sense forming a super-collection, albeit one too big to be a set, call it – to use the standard term – a (proper) *class*? But what does that really mean? There is a spectrum of positions available here, taking classes with increasing seriousness:

Virtual classes We begin by noting that in many contexts class-talk involves no more than a convenient logical fiction; i.e. the apparent reference to classes can simply be parsed away. In such a usage, to say that a belongs to the class of Fs is to say no more than that a is one of the Fs, which it turn says no more than that a is an F; to say that a belongs to the intersection of the class of Fs with the class of Gs is to say no more than that a is both an F and a G; to say that the class of Fs is identical to the class of Gs is to say no more than that for all x, x is an F iff x is a G. And so it goes.

We can formalize such talk of *virtual classes*, to use the standard term introduced by Quine (1963, §2). Extend a standard first-order theory T with what look like set-abstracts of the form '$\{z \mid C(z)\}$' and with what looks like a membership sign '$\in$' (perhaps we ought to use different symbolism here – but mostly authors recycle familiar set notation). We then explain the role of these class-abtracts by stipulating that $x \in \{z \mid C(z)\}$ is to be understood as a whole, as just *equivalent* to $C(x)$, for all contexts $C(\cdot)$ of the original unaugmented language (avoiding clash of variables): the first formula is just long-winded for the second formula. We can then define the further abstract $\alpha \cap \beta$ where α and β are class abstracts by stipulating that $x \in \alpha \cap \beta$ iff $x \in \alpha \wedge x \in \beta$; $\emptyset$ is introduced as shorthand for the class abstract $\{z \mid z \neq z\}$; the wff $\alpha \subseteq \beta$ is stipulated to be equivalent to $\forall x(x \in \alpha \to x \in \beta)$; we define $\alpha = \beta$ by stipulating it is equivalent to $\alpha \subseteq \beta \wedge \beta \subseteq \alpha$. And so it goes. Crucially, the class-abstracts are not genuine terms which can be unrestrictedly substituted in arbitrary contexts: they can only appear in (shorthand for) expressions of the form $x \in \{z \mid C(z)\}$. And note that class 'membership' always goes that way around. No sense is ascribed to a formula $\{z \mid C(z)\} \in x$: indeed, that is deemed ill-formed.

We can now add 'substitutional' quantification over our virtual classes: so if $\mathbf{C}$ is a class variable, '$\forall \mathbf{C}(\ldots\ z \in \mathbf{C}\ \ldots)$' is just treated as a compendious way of asserting every instance of a schema of the corresponding form '$\ldots\ C(z) \ldots$'.

In this way, we get something of the convenience of talking about classes as if they are set-like entities – indeed we can arguably reconstruct all or nearly all that elementary talk of sets or classes which was fashionable in 'new math' – but without taking on any ontological commitment to new entities over and above those postulated by our original theory T. "We can explain the sham [classes] away as a mere manner of speaking, by contextual definition, when the ontological reckoning comes" (Quine, 1970, p. 69).

Now, we can in particular add such eliminable talk of classes to a given theory of sets like ZFC. Indeed, this is exactly what happens in some classic texts. Of course, in this context we already have genuine set abstracts $\{z \mid C(z)\}$ to play with: but of course not every context C gives us a legitimate set abstract (e.g. there is no Russellian set $\{z \mid z \notin z\}$). The proposal is, in headline terms, that we can make up for those missing set abstracts by now introducing proper class abstracts for talking in the singular about objects which are too many to be a set (these class abstracts are standardly made to look just the same as set abstracts – so care is needed here!).

Thus there is no set of sets or set of ordinals in ZFC: but we can introduce virtual classes of sets or ordinals as does Kunen (1980, p. 24), putting

$$\mathbf{V} = \{x \mid x = x\}$$

$$\mathbf{ON} = \{x \mid x \text{ is an ordinal}\}$$

where 'x is an ordinal' abbreviates a suitable wff of the language of ZFC. Then Kunen adds:

> Formally, proper classes do not exist, and expressions involving them must be thought of as abbreviations for expressions not involving them. Thus, $x \in \mathbf{ON}$ abbreviates the formula expressing that x is an ordinal, and $\mathbf{ON} = \mathbf{V}$ abbreviates the (false) sentence (abbreviated by)
>
> $$\forall x(x \text{ is an ordinal} \leftrightarrow x = x).$$
>
> There is, in fact, no formal distinction between a formula and a class; the distinction is only in the informal presentation.

And what does adding these virtual classes buy us? Convenience. Kunen writes

> The abbreviations obtained with the class become very useful when discussing general properties of classes. Asserting that a statement is true of all classes is equivalent to asserting a theorem schema.

His example is the principle of transfinite induction for classes of ordinals, in the form 'For any class $\mathbf{C}$, if $\mathbf{C} \subseteq \mathbf{ON}$ and $\mathbf{C} \neq \emptyset$, then $\mathbf{C}$ has a least member'. This helpfully wraps up into a single claim the effect of asserting every instance of the schema

$$(\forall x(Cx \to x \text{ is an ordinal}) \wedge \exists x Cx) \to \exists x(Cx \wedge \forall y(Cy \to y \geq x)),$$

with the order relation appropriately defined, and where C involves only expressions of the unaugmented language of ZFC. The class talk here earns its keep by being brisk and perspicuous. But as Kunen reiterates (p. 26), while

> it is rarely necessary in mathematical arguments about classes to translate away the classes ... it is, however, important to know that this can be done in principle.

And since we can translate away the class talk from the theory ZFC_v – i.e. ZFC with added virtual classes – this theory can prove no more about sets than was already provable in good old ZFC.

VNB: almost virtual classes There are, however, set theories which introduce talk of classes seemingly to handle pluralities of objects which are too many to form a set and where such talk is *not* straightforwardly eliminable. But it is quite a subtle question – and certainly not one we can explore in depth here – how far such theories are really committed to classes as genuine entities over and above sets (for a classic but still very useful discussion, see Fraenkel et al. 1973, Ch. 2 §7, written in fact by Azriel Levy).

Consider, for example, the set theory VNB (due to von Neumann and Bernays, a version of which has been popularized by the very widely used textbook Mendelson 1964 which has gone through numerous editions). This theory is most perspicuously presented as a two-sorted theory with variables for both sets and classes, and which allows quantification over both. VNB's comprehension principle for classes allows us to form class abstracts $\{z \mid C(z)\}$ for the class of sets satisfying C, where C may now contain free class variables as parameters (that wasn't the case in ZFC$_v$, the theory with Quinean merely virtual classes). However, and this is crucial, we still don't allow such conditions C to contain bound class variables.

Now, as Levy points out in Fraenkel et al. (1973, p. 124))

> the conditions $C(z)$ that we allow in [the comprehension principle for classes] differ essentially from pure conditions [i.e. conditions of the original language of ZFC] only in that they may also contain class parameters; once these parameters are given values which are classes determined by pure conditions [the only way in end to make use of the parameters!], the condition $C(z)$ itself becomes equivalent to a pure condition.

Yet Kunen's theory with virtual classes already allows classes determined by pure conditions: so we might reasonably expect that VNB comes to little more than ZFC$_v$. And indeed, as with ZFC$_v$, VNB too is conservative over ZFC: i.e. VNB can prove no more about *sets* than was already provable in ZFC.

There are differences however between VNB and ZFC with or without virtual classes. First, unlike either version of ZFC, VNB is finitely axiomatizable. And second, perhaps surprisingly, VNB exhibits 'speed-up': that is to say, there are infinitely many proofs of ZFC theorems which are exponentially shorter in VNB than they are in ZFC or ZFC$_v$. But although those are not just gains in convenience, arguably these features still don't make VNB's use of classes more than a helpful technical dodge, and so don't really seem to count against Levy's verdict (Fraenkel et al., 1973, p. 131)):

> [O]ne should regard ZF and VNB as essentially the same theory, and the differences between those theories as mere technical matters.

Or here is Potter (2004, p. 307, with omissions) taking a similar line:

> [It] was Hilbert who conceived the general project of instrumentally justifying our use of an extension of an already accepted theory by proving finitistically that it is conservative. . . .
>
> According to Quine's dictum that to be is to be the value of a variable, by adopting the quantified theory of classes we express a commitment to an ontology of classes. But Hilbert's programme suggests an alternative view, according to which the objects of the new theory may be regarded as no more than convenient fictions

> whose use is justified by the conservativeness proof: it is useful to argue as if such entities exist because doing so sometimes permits shorter proofs, but in every such case we know that if required we could in principle purge the proofs of all mention of the ideal elements [in our example, classes] so as to be left with only unobjectionable reference to the real objects [sets] to which our prior theory already committed us.

MK: real classes? Consider next Morse-Kelly set theory – which is what you get if you take VNB but now allow class abstracts for *any* condition C which can be expressed in the language, including conditions which quantify over classes. This theory is no longer conservative over ZFC: the postulation of classes is therefore now more than just a device to make a prior set theory technically smoother in certain respects. Hence it seems that here we really are committed to a two layer universe of collections: old-style sets of sets, then added classes of sets, classes which are too big to themselves be identified with sets. (That fount of all knowledge, Wikipedia, pronounces that the ontology of NBG and of MK is the same: but this seems wrong – as Fraenkel et al. 1973 put it, the former theory doesn't take classes seriously, the latter theory does.)

Now, as with ZFC_v and VNB, in MK too classes have members but are not themselves members (of either sets or of classes). MK therefore avoids an analogue of Russell's paradox appearing at the level of classes: there is no class of all the self-membered classes because there is no class containing any classes. So far, so good.

The trouble is however that if we are allowed to gather sets into collections, some of which are too big to be a set, then – once we are treating classes as genuine entities in their own right – exactly *why* are we not allowed to gather classes into collections too, even if some are too big to be classes themselves? Is it a stably justifiable position both to countenance classes as real entities, and yet also to ban ourselves from considering any collections of them at all? What is the peculiar ontological status of classes which makes them more than fictions, more than the shadows cast by a technical trick, yet also makes such that we can't form even finite collections such as ordered pairs of them? This is puzzling, to say the least.

Classes as big sets Recall that, roughly speaking, an *inaccessible cardinal* is an uncountable cardinal number κ which cannot be 'accessed' from below by applying the basic operations of cardinal arithmetic to smaller cardinals. ZFC cannot prove the existence of inaccessible cardinals. For if κ is indeed an inaccessible cardinal, then the sets introduced below the level V_κ in a cumulative model of set theory will already be a model of ZFC, so if ZFC could prove κ exists, it would prove there is a level V_κ and hence prove itself consistent, which it can't (assuming it is consistent).

So consider now the theory ZFC + *there is an inaccessible cardinal*. This is a proper extension of ZFC, proves ZFC is consistent, and a model of the theory will have 'small' sets which live below level V_κ (for the first inaccessible κ), and then 'big' sets living higher up. (Perhaps we should call the big sets '*potentially* big', as e.g. the singleton of κ is small in terms of cardinality, though it first appears at a rank beyond the inaccessible κ). The resulting universe of sets will be vast: but there is nothing intrinsically mysterious about the status of the 'big' sets – in particular we can collect even big sets living up to a given level $\kappa + \alpha$ into sets living one level yet higher up the hierarchy.

Suppose we call the small sets of ZFC + *there is an inaccessible cardinal* – i.e. the sets which give a model of the intended sort for ZFC – simply *sets* and call the big sets *classes* (some of which will be proper classes, i.e. too big to be put in bijective correspondence with a set). Now at last, it seems, we do have a coherent story which takes classes seriously: our world includes sets and classes of sets which are too big to be sets and classes of classes too – though of course no class of all classes. (Observation: the sets up to $V_{\kappa+1}$ provide a model of MK: but once you countenance $V_{\kappa+1}$ it is difficult to conceive of any reason not to like $V_{\kappa+2}$ etc. Which makes again the point that it is difficult to fix on a sensible conception of a universe of collections which underpins exactly MK and no more.)

8.5 Categories and classes: back to basics

(a) Back to categories. And having now raised issues of 'size' and of collections-too-big-to-be-sets, it is worth briefly backtracking right to the very outset. So recall our definition of the very idea of a category. It began like this:

Definition 1 The data for a category $\mathscr{C}$ comes in two sorts:

(1) *Objects* (which we will typically notate by 'A', 'B', 'C', ...).

(2) *Arrows* (which we typically notate by 'f', 'g', 'h', ...). ...

That is pretty much in accord with e.g. Awodey (2006, p. 4) and Lawvere and Schanuel (2009, p. 21).

But this is not the most common way of putting things. Thus Simmons (2011, p. 2), for example, starts

Definition 1* A category $\mathscr{C}$ consists of

(1) A collection *Obj* of entities called *objects*.

(2) A collection *Arw* of entities called *arrows*. ...

Leinster (2014, p. 10) also talks of categories consisting in *collections*, as does Goedecke (2013). Others, for example Borceux (1994, p. 4) and Adámek et al. (2009, p. 18), talk of a category consisting of a *class* of objects, etc.

We probably shouldn't read very much into the choice of wording, 'collections' vs 'classes'. The real question is: what, if anything, is the difference between talking of a category as having as data some objects (plural) and some arrows (plural), and saying that a category consists in a collection/class (singular) of objects and a collection/class (singular) of arrows?

It obviously depends what we mean here by 'classes'. We know, because many paradigm categories have too many objects to form a set, that some notion of class must be intended here that contrasts with the notion of a set (or with the notion of a set in the usual cumulative hierarchy, at any rate). But that still leaves options. When Defn. 1* is given, are we just employing a fiction of virtual classes and saying no more than is said by Defn. 1 which doesn't talk of collections or classes at all? Or are we buying into some overall theory of sets-and-classes which allows *large* collections, too big to be 'ordinary' sets: and if so, how seriously are we to take these? Two observations:

(1) It is clear that most of those who propose a definition along the lines of Defn. 1* do in fact intend the talk of collections or classes to be construed substantively, as they typically soon go on at least to raise the question of the kind of theory sets-and-classes which is being presupposed (even if they then may do little more than arm-wave or shelve the question).

(2) On the other hand, note that the first six chapters of this introduction to category theory happily worked with Defn. 1, without making any assumption that the objects or the arrows of a category do fall into collections or classes which are distinct entities over and above the objects and arrows. We could have used virtual-class talk to make a few statements along the way marginally snappier: but we didn't, and we lost nothing by eschewing class-talk.

So, it seems that we don't *need* to buy in to a substantive theory of classes *at least at the outset* of our theorizing about categories. The issue then arises: just when, if at all, does the category theorist need to take on the further commitment? And how weighty is this further commitment?

I frankly don't know the detailed answer to this. We have, for example, in this present chapter started talking about categories of categories. Now, small categories can be thought of as living in (or at least as being modelled in) the world of sets, and the collections that constitute a category **Cat** of small categories can perhaps comfortably be thought of as virtual classes. But a locally small category can have too many objects to form a set. So, it seems, the category **CAT** of locally small categories has a class of objects which may themselves comprise classes of objects; this apparently takes us beyond the merely virtual, beyond the reach of VBN or even of MK. But it remains to be seen when **CAT** becomes a load-bearing part of much category-theoretic theorizing which we need to accommodate with a substantive theory of classes.

At this point, then, I can only propose that we keep our wits about us and try to note if and when we do get seriously entangled with a commitment to real-collections-too-big-to-be-sets. Perhaps much the best thing to do is just to continue to develop category theory, talking in a relaxed way about sets, classes (and perhaps even bigger collections) on an as-needed basis. We can then plan to return later to see in retrospect what kind of background theory about sets and classes a category theorist really needs at various stages of the development of her theory if the load-bearing parts of her actual mathematical practice are indeed to be in good order.

If you *do*, however, insist on more up-front discussion of foundations-in-the-theory-of-classes before going any further, then two options are to be found in e.g. Borceux (1994, §1.1) and Adámek et al. (2009, §2). (A comment re Borceux's discussion. His universe $\mathcal{U}$ is a so-called Grothendieck universe; and it can be shown that the level V_κ of the usual cumulative hierarchy for inaccessible κ constitutes a Grothendieck universe. So we are swimming here in the same conceptual waters as in the fourth line on classes which we identified.) For a much more sophisticated considerations of the options looking back from the vantage point of advanced knowledge of category theory, you will perhaps one day want to tackle Shulman (2008).

8.6 Another definition of categories

We've just been remarking that instead of Defn. 1 which talked of a category comprising objects and arrows (plural), it is common to offer Defn. 1* which characterizes a category in terms of a class of objects and a class of arrows. We should note that there is another variant definition:

Definition 1** The data for a category $\mathscr{C}$ comprises:

(1) A class $ob(\mathscr{C})$, whose elements we will call *objects*.
(2) For every $A, B \in ob(\mathscr{C})$, a class $\mathscr{C}(A, B)$, whose elements f we will call *arrows* from A to B. We signify that the arrow f belongs to $\mathscr{C}(A, B)$ by writing $f\colon A \to B$ or $A \xrightarrow{f} B$.
(3) For every $A \in ob(\mathscr{C})$, an arrow $1_A \in \mathscr{C}(A, A)$ called the *identity* on A.
(4) For any $A, B, C \in ob(\mathscr{C})$, a two-place *composition* operation, which takes arrows f, g, where $f\colon A \to B$ and $g\colon B \to C$, to an arrow $g \circ f\colon A \to C$, the composite of f and g.

The axioms are as before.

So the difference is that Defn. 1* has one all-in class of arrows, Defn. 1** has lots of different classes of arrows, one (perhaps empty) for every pair of objects in the category. Obviously if we start from Defn. 1* we can define the class $\mathscr{C}(A, B)$ of arrows from A to B as containing the arrows f from $arw(\mathscr{C})$ such that $src(f) = A \wedge tar(f) = B$.

Now note that on Defn. 1*, the arrows $f\colon A \to B$ and $f'\colon A' \to B'$ cannot be identical items if $A \neq A'$ or $B \neq B'$. For if $src(f) \neq src(f')$, $f \neq f'$; likewise, of course, if $tar(f) \neq tar(f')$, $f \neq f'$. Hence if $A \neq A'$ or $B \neq B'$, $\mathscr{C}(A,B) \neq \mathscr{C}(A',B')$. On the other hand, there's nothing in Defn. 1** that ensures that $\mathscr{C}(A,B)$ and $\mathscr{C}(A',B')$ are disjoint when $A \neq A'$ or $B \neq B'$. That is to say, on the new definition, there's nothing (yet) to stop the arrows $f\colon A \to B$ and $f'\colon A' \to B'$ being the same item even if $A \neq A'$ or $B \neq B'$. That means that our two definitions don't quite line up. What to do?

(i) Some authors simply add to Defn. 1** the requirement that the classes $\mathscr{C}(A,B)$ for different pairs of objects are disjoint.

(ii) Alternatively, we can associate with every category in the sense of the new definition an associated category in the sense of the previous one by putting $arw(\mathscr{C}) = \coprod_{A,B \in ob(\mathscr{C})} \mathscr{C}(A,B)$ – i.e. take the disjoint union of classes of arrows (assume we can make that construction!). Though notice that the round trip – starting with a category in the new sense, constructing (as suggested) the associated category in the original sense and then going back to a category in the new sense using our definition of $\mathscr{C}(A,B)$ from $arw(\mathscr{C})$ – doesn't quite return us to our starting point. We get a category which, in a good sense, 'looks just like' the one we started off with; but it's an isomorphic copy (in a sense to be made clear), not the original. Still, the difference really shouldn't matter.

Among familiar books, Borceux (1994) gives Defn. 1**; so does Adámek et al. (2009), adding 'for technical convenience' the requirement that hom-sets are disjoint, and Leinster (2014) does the same. While Awodey (2006), Goldblatt (2006), Lawvere and Schanuel (2009), Mac Lane (1997), and Simmons (2011) use either Defn. 1 or Defn. 1*. So both styles of definition are current, and hence it is worth explicitly noting the difference. For later convenience, however, we stick to our original definition that requires that classes $\mathscr{C}(A,B)$ for distinct objects are disjoint.

9 The Yoneda embedding

Where do we go next? Here are three large families of topics to explore:

Limits Back in the preamble to Ch. 4, we noted that what's important e.g. about Cartesian products is not how they are implemented, not what a product-object 'consists in', but rather how a product $A \times B$ is related to A and B by pair-forming and pair-unforming morphisms – an observation that is crying out for category-theoretic treatment.

So how *should* we deal with products (and indeed some similar constructions like disjoint sums) now we have some category theory to hand? We will find that these sorts of constructions can all be thought of as involving *limits* (or colimits) in a certain sense we need to explore.

Adjoints Recall the remark from Tom Leinster which we quoted at the outset: 'Category theory takes a bird's eye view of mathematics. From high in the sky ... we can spot patterns that were impossible to detect from ground level.' One of the patterns that is, if not impossible, at least difficult to detect from ground level if you are not imbued with categorial ideas is the ubiquity of so-called *adjunctions* (think of these as a generalization of Galois Connections, if you know what they are). We will want to explore these too.

Representables/Yoneda Lemma Cayley's Theorem in group theory tells us that every group is isomorphic to a permutation group shuffling around the members of a set. In other words, there's a good sense in which any group can given a (relatively!) concrete set-representation. There are other similar classical representation theorems. We can now reflect such results in category theory by investigating functors of a special form from various categories into the category of sets, functors which can be thought of as giving representations of those categories in the world of sets. Key results in this area concern the so-called Yoneda embedding and the associated Yoneda Lemma.

Now, category theory being an enquiry into how mathematical things (in a broad sense) hang together (in a broad sense), it's not surprising to learn that these topics too hang together, being interrelated in complex ways. Which means that there is no obviously best order in which to tackle them. Thus Leinster (2014) has chapters dealing with adjoints, representables, and limits, in that order, before knitting everything together in a final chapter 'Adjoints, representable and limits'. Awodey (2006) reverses this order, and tackles limits first, followed by scattered material on representables with the Yoneda Lemma not appearing until Ch. 8, and then only in his penultimate Ch. 9 do we meet adjoints. In his more elementary text, Simmons (2011) also tackles limits before adjunction, and comparatively downplays representables (and doesn't mention the Yoneda Lemma as such). By contrast Adámek et al. (2009) prove the Yoneda Lemma *very* early in their book, before discussing limits or adjunctions: Borceux (1994) does the same. Finally, to take just one more example, Goldblatt (2006) starts his classic book talking about limits. And it is only four hundred pages later – after we've done some quite detailed exploration of those categories which are toposes (or topoi, if you prefer) – that we get a little on adjoints and on representables.

So we have to make a presentational choice here about the order of exposition. The Cambridge Way has been (briefly) to discuss representables and the Yoneda embedding/Lemma, before tackling limits and adjoints (see this transcription of Peter Johnstone's lectures or Julia Goedecke's notes, and this was also the ordering followed by RL-W's lectures). We'll take the same path here.[1]

9.1 Hom-sets

(a) Let's start by re-introducing a bit of symbolism which appeared at the end of the last chapter (see §8.6, Defn. 1**). We there used $\mathscr{C}(A, B)$ for the class of $\mathscr{C}$-arrows from A to B. Now, if $\mathscr{C}$ is locally small, then by definition there is only a set's worth of arrows from A to B for any choice of $\mathscr{C}$-objects A, B. So in this case, we can think of $\mathscr{C}(A, B)$ as forming a set-sized collection. Which motivates

Definition 39. If $\mathscr{C}$ is locally small, then the *hom-set* $\mathscr{C}(A, B)$ is the set of $\mathscr{C}$-arrows from A to B.

An alternative notation is '$\mathrm{Hom}_{\mathscr{C}}(A, B)$' or simply '$\mathrm{Hom}(A, B)$': the terminology comes from that paradigm case where the set of arrows in question is in fact a set of homomorphisms between two algebraic structures.

But is $\mathscr{C}(A, B)$ a set that lives in **Set**? We haven't fixed which category **Set** is exactly (see the remark back in §2.2). But if you think of this as comprising the sets you know and love from a basic set theory course in the delights of ZFC, then **Set** is a category of pure sets, i.e. of sets whose members, if any, are sets whose members, if any, are sets ... all the way down. So if we think of $\mathscr{C}(A, B)$

[1]Looking back from the vantage point of having got to the end of these Notes, I'd on balance now prefer to do things in a different order. But it would involve a ground-up rewrite. So don't hold your breath waiting!

as living in a category **Set** of pure sets, then the arrows which are members of $\mathscr{C}(A, B)$ will have to be pure sets too. But do we want to be committed to thinking of the arrows of a category $\mathscr{C}$ as always being pure sets?

However, it *is* absolutely standard in category theory to treat $\mathscr{C}(A, B)$ as an object of **Set**, whenever the category $\mathscr{C}$ is small enough. So it seems that we have three options. (i) Continue to regard **Set** as a theory of pure sets, and take $\mathscr{C}(A, B)$ to be a pure set, so arrows of any category (or at least, of any category we are interested in) to be pure sets – which is to build in from the start a thorough-going set-theoretic reductionism about categories. (ii) Take **Set** to be a category of impure sets, where the non-set elements can be arrows in any category. Or (iii) we can take **Set** to be a universe of pure sets, and $\mathscr{C}(A, B)$ to be, in general, an impure set whose members are arrows (which needn't be themselves sets), but then say that it isn't strictly speaking $\mathscr{C}(A, B)$ as originally defined which lives in **Set** but rather a set which represents or models it, whose elements are in a one-to-one correspondence with with the members of $\mathscr{C}(A, B)$ – call this representing set $\mathscr{C}(A, B)$ too, with context deciding when we are talking about the original hom-set and when we are talking about its pure-set representation.

We will have to shelve for now the question of which is the best way to go. Sometimes foundational questions are best tackled late in the game, when we can look back and see what commitments are load-bearing parts of the theory we have developed. I hope to return to the issue in a sequel to these Notes, when I can write with enough benefit of hindsight!

(Note that in his canonical 1997, Saunders Mac Lane initially gives a definition like our Defn. 1 as a definition of what *he* calls metacategories, and then for him a category proper "will mean any interpretation of the category axioms within set theory" – so for Mac Lane and those who follow him, the hom-sets of a category proper, like all the other gadgets of category theory, will unproblematically live in the universe of set theory. Such theorists can cheerfully take option (i). But do we really want or need to suppose that categories are always and everywhere sets? Not perhaps if we have ambitions for category theory as a more democratic way of organizing the mathematical universe, which provides an alternative to set-theoretic imperialism.)

9.2 Hom-functors

(a) Take $\mathscr{C}(A, B)$, the hom-set of $\mathscr{C}$-arrows from A to B. Keep A fixed. Then as we vary X through the objects in $\mathscr{C}$, we get varying $\mathscr{C}(A, X)$.

So: consider the resulting function which sends an object X in $\mathscr{C}$ to the set $\mathscr{C}(A, X)$, a set which we follow standard practice as taking as living in **Set**.

Can we now treat this function on $\mathscr{C}$-objects as the first component of a functor, call it $\mathscr{C}(A, -)$, from $\mathscr{C}$ to **Set**? Well, how could we find a component of the functor to deal with the $\mathscr{C}$-arrows? Such a component is going to need to send an arrow $f\colon X \to Y$ in $\mathscr{C}$ to a **Set**-function from $\mathscr{C}(A,X)$ to $\mathscr{C}(A,Y)$. The obvious candidate for the latter function is the one we can notate as $f \circ -$ that maps any $g\colon A \to X$ in $\mathscr{C}(A,X)$ to $f \circ g\colon A \to Y$ in $\mathscr{C}(A,Y)$. (Note, $f \circ g\colon A \to Y$ has to be in $\mathscr{C}(A,Y)$ because $\mathscr{C}$ is a category which by hypothesis contains $g\colon A \to X$ and $f\colon X \to Y$ and hence must contain their composition.)

It is easy to check that these components add up to a genuine covariant functor – in fact the functoriality in this case just reduces to the associativity of composition for arrows in a category and the basic laws for identity arrows. So far, so good!

Now, start again from the hom-set $\mathscr{C}(A,B)$ but now keep B fixed: then as we vary X through the objects in $\mathscr{C}$, we again get varying hom-sets $\mathscr{C}(X,B)$. Which generates a function which sends an object X in $\mathscr{C}$ to an object $\mathscr{C}(X,B)$ in **Set**. To turn *this* into a functor $\mathscr{C}(-,B)$, we need again to add a component to deal with $\mathscr{C}$-arrows. That will need to send $f\colon X \to Y$ in $\mathscr{C}$ to some function between $\mathscr{C}(X,B)$ to $\mathscr{C}(Y,B)$. But this time, to get functions to compose properly, things will have to go the other way about, i.e. the associated functor will have to send a function $g\colon Y \to B$ in $\mathscr{C}(Y,B)$ to $g \circ f\colon X \to B$ in $\mathscr{C}(X,B)$. So this means that the resulting functor $\mathscr{C}(-,B)$ is a *contravariant* hom-functor.

So, to summarize:

Definition 40. Given a locally small category $\mathscr{C}$, then the associated *covariant hom-functor* $\mathscr{C}(A,-)\colon \mathscr{C} \to$ **Set** is the functor with the following data:

(1) A mapping $\mathscr{C}(A,-)_{ob}$ whose value at the object X in $\mathscr{C}$ is the hom-set $\mathscr{C}(A,X)$.

(2) A mapping $\mathscr{C}(A,-)_{arw}$, whose value at the $\mathscr{C}$-arrow $f\colon X \to Y$ is the set function $f \circ -$ from $\mathscr{C}(A,X)$ to $\mathscr{C}(A,Y)$ which sends an element $g\colon A \to X$ to $f \circ g\colon A \to Y$.

And the associated *contravariant hom-functor* $\mathscr{C}(-,B)\colon \mathscr{C} \to$ **Set** is the functor with the following data:

(3) A mapping $\mathscr{C}(-,B)_{ob}$ whose value at the object X in $\mathscr{C}$ is the hom-set $\mathscr{C}(X,B)$.

(4) A mapping $\mathscr{C}(-,B)_{arw}$, whose value at the $\mathscr{C}$-arrow $f\colon Y \to X$ is the set function $- \circ f$ from $\mathscr{C}(X,B)$ to $\mathscr{C}(Y,B)$ which sends an element $g\colon X \to B$ to the map $g \circ f\colon Y \to B$.

The use of a blank in the notion '$\mathscr{C}(A,-)$' invites an obvious shorthand: instead of writing '$\mathscr{C}(A,-)_{arw}(f)$' to indicate the result of the component of the functor which acts arrows applied to the function f, we will write simply '$\mathscr{C}(A,f)$'. Similarly for the dual.

For the record, we can also talk about a 'bi-functor' $\mathscr{C}(-,-)\colon \mathscr{C}^{op} \times \mathscr{C} \to$ **Set**, contravariant in the first place and covariant in the second. The headline news is that we can think of this as acting on a product category mapping the pair object $\langle A, B\rangle$ in $\mathscr{C}$ to the hom-set $\mathscr{C}(A, B)$, and the pair of morphisms $\langle f : X' \to X, g : Y \to Y'\rangle$ to the morphism between $\mathscr{C}(X, Y)$ and $\mathscr{C}(X', Y')$ that sends $h : X \to Y$ to $h \circ g \circ f : X' \to Y'$. We will return to this if/when we need to say more.

(b) Take a (locally small) category $\mathscr{C}$, and fix on a $\mathscr{C}$-arrow $f : B \to A$. We'll describe how to construct from f a corresponding natural transformation α between the hom-functors $\mathscr{C}(A, –)$ and $\mathscr{C}(B, –)$.

If α is to be a natural transformation, its components must be such that the following diagram commutes, given any arrow $j : X \to Y$:

$$\begin{tikzcd}
\mathscr{C}(A,X) \arrow[r, "\mathscr{C}(A,j)"] \arrow[d, "\alpha_X"] & \mathscr{C}(A,Y) \arrow[d, "\alpha_Y"] \\
\mathscr{C}(B,X) \arrow[r, "\mathscr{C}(B,j)"] & \mathscr{C}(B,Y)
\end{tikzcd}$$

where $\mathscr{C}(C, j)$ is (as just defined) the map which sends an arrow $g : C \to X$ to the map $j \circ g : C \to Y$.

Suppose we set a component $\alpha_Z : \mathscr{C}(A, Z) \to \mathscr{C}(B, Z)$ to be the function that sends an arrow $g : A \to Z$ to the composite $g \circ f : B \to Z$ (so we could also write α_Z as $- \circ f$).

Then our diagram will indeed commute. For going round the top-route takes us from $g : A \to X$ to $j \circ g : A \to Y$ to $(j \circ g) \circ f : B \to Y$; and going round the bottom route takes us from $g : A \to X$ to $g \circ f : A \to Y$ to $j \circ (g \circ f) : B \to Y$.

So in sum, if there is a morphism $f : B \to A$, then there is a corresponding natural transformation $\alpha : \mathscr{C}(A, –) \Rightarrow \mathscr{C}(B, –)$ with components α_Z as defined.

And note: if f is an isomorphism, then each component α_Z (i.e. $- \circ f$) has an inverse (i.e. $- \circ f^{-1}$), so is an isomorphism. Therefore α is then a natural isomorphism.

To sum up this result and introduce some slightly more perspicuous notation:

Theorem 35. *Suppose $\mathscr{C}$ is a locally small category, and $\mathscr{C}(A, –)$, $\mathscr{C}(B, –)$ are hom-functors (for objects A, B in $\mathscr{C}$). Then, given an arrow $f : B \to A$, there exists a corresponding natural transformation $\mathscr{C}(f, –) : \mathscr{C}(A, –) \Rightarrow \mathscr{C}(B, –)$, where for each Z, the component $\mathscr{C}(f, –)_Z : \mathscr{C}(A, Z) \to \mathscr{C}(B, Z)$ sends an arrow $j : A \to Z$ to $j \circ f : B \to Z$. Furthermore, if f is an isomorphism, then $\mathscr{C}(f, –)$ is a natural isomorphism.*

(c) As a quick reality-check, let's for the record show

Theorem 36. *Given a locally small category $\mathscr{C}$ including objects A, B, C, and arrows $f : B \to A$ and $g : C \to B$, then*

(1) $\mathscr{C}(f \circ g, –) = \mathscr{C}(g, –) \circ \mathscr{C}(f, –)$.

(2) $\mathscr{C}(f, -)_A 1_A = f$.

(3) $\mathscr{C}(1_A, -) = 1_{\mathscr{C}(A,-)}$.

Proof. (1) $\mathscr{C}(f \circ g, -)_Z$ sends any arrow $e\colon A \to Z$ to $e \circ (f \circ g)\colon C \to Z$. But $(\mathscr{C}(f,-)_Z(e) = e \circ f$, so $\mathscr{C}(g,-)_Z(\mathscr{C}(f,-)_Z(e)) = (e \circ f) \circ g$. Hence $\mathscr{C}(f \circ g, -)$ and $\mathscr{C}(g,-) \circ \mathscr{C}(f,-)$ agree on all components, so are identical natural transformations.

(2) $\mathscr{C}(f,-)_A$ sends any arrow $j\colon A \to A$ to $j \circ f\colon B \to A$. So in particular it sends 1_A to f.

(3) $\mathscr{C}(1_A,-)_Z$ sends any arrow $j\colon A \to Z$ to itself. While $1_{\mathscr{C}(A,-)}$ is the identity arrow on the object $\mathscr{C}(A,-)$ in the functor category $[\mathscr{C}, \mathbf{Set}]$. Arrows in that functor category are natural transformation: so the component $(1_{\mathscr{C}(A,-)})_Z$ is the identity on $\mathscr{C}(A, Z)$, i.e. sends any arrow $j\colon A \to Z$ to itself. Which shows that $\mathscr{C}(1_A,-)$ and $1_{\mathscr{C}(A,-)}$ agree on all components and therefore are identical. $\square$

(d) The theorems so far have been about covariant hom-functors. We have corresponding duals to Theorems 35 and 36 for contravariant hom-functors. We'll state the first theorem and also the part of the second theorem that we'll need, leaving proofs as routine exercises in dualizing.

Theorem 37. *Suppose $\mathscr{C}$ is a locally small category, and $\mathscr{C}(-, A)$, $\mathscr{C}(-, B)$ are contravariant hom-functors (for objects A, B in $\mathscr{C}$). Then if there exists an arrow $f\colon A \to B$, there is a corresponding natural transformation $\mathscr{C}(-, f)\colon \mathscr{C}(-, A) \Rightarrow \mathscr{C}(-, B)$, where for each Z, the component $\mathscr{C}(-, f)_Z\colon \mathscr{C}(Z, A) \to \mathscr{C}(Z, B)$ sends an arrow $j\colon Z \to A$ to $f \circ j\colon Z \to B$.*

Theorem 38. *Given a locally small category $\mathscr{C}$ including objects A, B, C, and arrows $f\colon A \to B$ and $g\colon B \to C$, then $\mathscr{C}(-, g \circ f) = \mathscr{C}(-, g) \circ \mathscr{C}(-, f)$.*

9.3 Generating natural transformations

The obvious next question to ask is: are *all* possible natural transformations between the hom-functors $\mathscr{C}(A,-)$ and $\mathscr{C}(B,-)$ generated from arrows $f\colon B \to A$ in the way described in Theorem 35.

The answer is give by:

Theorem 39. *Suppose $\mathscr{C}$ is a locally small category, and consider the hom-functors $\mathscr{C}(A, -)$ and $\mathscr{C}(B, -)$, for objects A, B in $\mathscr{C}$. Then if there is a natural transformation $\alpha\colon \mathscr{C}(A, -) \Rightarrow \mathscr{C}(B, -)$, there is a unique arrow $f\colon B \to A$ such that $\alpha = \mathscr{C}(f, -)$, namely $f = \alpha_A(1_A)$.*

Proof. Since α is a natural transformation, the following diagram must commute, for any Z and any $g\colon A \to Z$,

$$\begin{tikzcd}
\mathscr{C}(A,A) \arrow[r, "\mathscr{C}(A,g)"] \arrow[d, "\alpha_A"] & \mathscr{C}(A,Z) \arrow[d, "\alpha_Z"] \\
\mathscr{C}(B,A) \arrow[r, "\mathscr{C}(B,g)"] & \mathscr{C}(B,Z)
\end{tikzcd}$$

where, recalling the definition, $\mathscr{C}(A,g)$ is a map that (among other things) sends an arrow $h\colon A \to A$ to the arrow $g \circ h\colon A \to Z$, and $\mathscr{C}(B,g)$ is a map which sends an arrow $j\colon B \to A$ to the arrow $g \circ j\colon B \to Z$.

Why start with $\mathscr{C}(A,A)$ at the top left? Because we know that such hom-sets must be populated, if only by the identity arrow $1_A\colon A \to A$.

Chase that identity arrow round the diagram from the top left to bottom right nodes. The top route gives us $\alpha_Z(g)$, and the bottom route gives us $\mathscr{C}(B,g)(\alpha_A(1_A))$, and since the diagram commutes these must be equal.

Put $f\colon B \to A =_{\text{def}} \alpha_A(1_A)$. Then

$$\alpha_Z(g) = \mathscr{C}(B,g)(f) = g \circ f = \mathscr{C}(f,–)_Z(g).$$

Therefore since Z and g were arbitrary, we get $\alpha = \mathscr{C}(f,–)$ as required.

Now suppose both f and f' are such that for each Z, $\mathscr{C}(f,–)_Z = \mathscr{C}(f',–)_Z$. Then by Theorem 36 (2)

$$f = \mathscr{C}(f,–)_A(1_A) = \mathscr{C}(f',–)_A(1_A) = f'$$

which shows f's uniqueness. □

Needless to say, there's a dual version of the theorem about contravariant hom-functors. It is another routine exercise in dualizing to prove it:

Theorem 40. *Suppose $\mathscr{C}$ is a locally small category, and consider the hom-functors $\mathscr{C}(–,A)$ and $\mathscr{C}(–,B)$, for objects A,B in $\mathscr{C}$. Then if there is a natural transformation $\alpha\colon \mathscr{C}(–,A) \Rightarrow \mathscr{C}(–,B)$, there is a unique arrow $f\colon A \to B$ such that $\alpha = \mathscr{C}(–,f)$, namely $f = \alpha(1_A)$.*

9.4 The Restricted Yoneda Lemma

For brevity's sake, let's introduce some shorthand:

Definition 41. We use the standard notation '$\mathit{Nat}(F,G)$' to denote the set of natural transformations from F to G in $\mathscr{C}$. And just in this section and the next, we will further abbreviate '$\mathit{Nat}(\mathscr{C}(A,–),\mathscr{C}(B,–))$' by simply '$\mathit{Nat}_{AB}$'.

Since the ambient category $\mathscr{C}$ is being assumed to be locally small, $\mathit{Nat}(F,G)$ is indeed a set.

Now note that a $\mathscr{C}$-arrow $f\colon B \to A$ is of course a member of the hom-set $\mathscr{C}(B,A)$. So, in the proofs of our Theorems 35 and 39 we have in effect defined two families of functions $\mathcal{X}_{AB}$ and $\mathcal{E}_{AB}$, i.e. families of arrows in **Set** indexed by $A,B \in \mathscr{C}$, where

i) $\mathcal{X}_{AB}\colon \mathscr{C}(B,A) \to Nat_{AB}$ sends a function $f\colon B \to A$ to the natural transformation $\mathscr{C}(f,–)$.

ii) $\mathcal{E}_{AB}\colon Nat_{AB} \to \mathscr{C}(B,A)$ sends a natural transformation $\alpha\colon \mathscr{C}(A,–) \Rightarrow \mathscr{C}(B,–)$ to $\alpha_A(1_A)$.

And again, the next question to ask is obvious: are $\mathcal{X}_{AB}$ and $\mathcal{E}_{AB}$ inverses of each other in **Set**? We might expect so.

Let's fix on some particular A and B, and – to improve readability – temporarily drop subscripts on $\mathcal{X}$ and $\mathcal{E}$. Then we note:

(1) Given an arbitrary $f\colon B \to A$,

$$(\mathcal{E} \circ \mathcal{X})f = \mathcal{E}(\mathscr{C}(f,–)) = \mathscr{C}(f,–)_A(1_A) = f$$

whence $\mathcal{E} \circ \mathcal{X} = 1_{\mathscr{C}(B,A)}$.

(2) Given an arbitrary $\alpha\colon \mathscr{C}(A,–) \Rightarrow \mathscr{C}(B–)$,

$$(\mathcal{X} \circ \mathcal{E})\alpha = \mathcal{X}(\alpha_A(1_A)) = \mathscr{C}(\alpha_A(1_A),–)$$

But $\mathscr{C}(\alpha_A(1_A),–)_Z$ sends an arrow $g\colon A \to Z$ to $g \circ \alpha_A(1_A)\colon B \to Z$, and looking again at the last commutative diagram we drew, $g \circ \alpha_A(1_A) = \alpha_Z(g \circ 1_A) = \alpha_Z(g)$. So for each Z, $\mathscr{C}(\alpha_A(1_A),–)_Z = \alpha_Z$. Hence, having identical components, the natural transformations are identical and $(\mathcal{X} \circ \mathcal{E})\alpha = \alpha$. Since that holds for any α, $\mathcal{X} \circ \mathcal{E} = 1_{Nat_{AB}}$.

So $\mathcal{X}_{AB}$ and $\mathcal{E}_{AB}$ are mutual inverses, and hence isomorphisms to and fro between $\mathscr{C}(B,A)$ and Nat_{AB}, i.e. $Nat(\mathscr{C}(A,–),\mathscr{C}(B,–))$. Hence:

Theorem 41 (Restricted Yoneda Lemma)**.** *Suppose $\mathscr{C}$ is a locally small category, and A, B are objects of $\mathscr{C}$. Then $Nat(\mathscr{C}(A,–),\mathscr{C}(B,–)) \cong \mathscr{C}(B,A)$.*

There are, needless to say, dual versions of all this. In particular, for each A, B in $\mathscr{C}$, there is an isomorphism $\mathcal{Y}_{AB}\colon \mathscr{C}(A,B) \to Nat(\mathscr{C}(–,A),\mathscr{C}(–,B))$ which sends a function $f\colon A \to B$ to the natural transformation $\mathscr{C}(–,f)$; and $\mathcal{Y}_{AB}$ has an inverse. Consequently,

Theorem 42 (Restricted Yoneda Lemma)**.** *Suppose $\mathscr{C}$ is a locally small category, and A, B are objects of $\mathscr{C}$. Then $Nat(\mathscr{C}(–,A),\mathscr{C}(–,B) \cong \mathscr{C}(A,B)$.*

The shared label we've given this dual pair of theorems is not standard,[2] but the reason for it will become clear when we meet the full Yoneda Lemma in Ch. 11. The later full version has a reputation for being the first result in category theory whose proof takes some effort to understand. Be that as it may, our current cut-down version at least should seem unproblematic: once we have

[2]Terminology hereabouts varies: I have even seen 'Yoneda Lemma' used to label the earlier key result Theorem 39.

Theorem 35 (which just restated an early result), it's simply a matter of asking the obvious questions and finding the easy answers. And the Restricted Yoneda Lemma is all we need for the main result in this chapter, and for a number of other results which are often said to obtain 'by Yoneda'.

9.5 A new pair of fully faithful functors

Continuing to assume that we are working in a category $\mathscr{C}$ which is locally small, then – falling into the convenient shorthand of writing '$A \in \mathscr{C}$' for 'A is an object of $\mathscr{C}$' –

i) We've associated with each object $A \in \mathscr{C}$ a hom-functor $\mathscr{C}(A, -)$.

ii) For each $A, B \in \mathscr{C}$, we've a bijection $\mathcal{X}_{AB}$ which sends any $f\colon B \to A$ to the natural transformation $\mathscr{C}(f, -)\colon \mathscr{C}(A, -) \Rightarrow \mathscr{C}(B, -)$.

Now, the hom-functors $\mathscr{C}(A, -)$ and $\mathscr{C}(B, -)$ are objects of the functor category $[\mathscr{C}, \mathbf{Set}]$, and each natural transformation $\mathscr{C}(f, -)\colon \mathscr{C}(A, -) \Rightarrow \mathscr{C}(B, -)$ is an arrow in that category. So, given the way the A/B order gets reversed in (ii), the functional associations we've just noted look as if they should add up to a *contravariant* functor from $\mathscr{C}$ to $[\mathscr{C}, \mathbf{Set}]$. Equivalently, it looks as if

Theorem 43. *For any locally small category $\mathscr{C}$, there is a functor we'll label simply $\mathcal{X}\colon \mathscr{C}^{op} \to [\mathscr{C}, \mathbf{Set}]$ with components $\mathcal{X}_{ob}$ and $\mathcal{X}_{are}$ such that*

(1) for any $A \in ob(\mathscr{C}^{op})$, $\mathcal{X}_{ob}(A) = \mathscr{C}(A, -)$,

(2) for any arrow $f \in \mathscr{C}^{op}(A, B)$, i.e. arrow $f\colon B \to A$ in $\mathscr{C}$, $\mathcal{X}_{arw}(f) = \mathscr{C}(f, -)$.

Proof. The values of $\mathcal{X}_{ob}$ and $\mathcal{X}_{arw}$ are indeed respectively objects and arrows in $[\mathscr{C}, \mathbf{Set}]$. So we just need to check the two functorial axioms are indeed satisfied. First, identities are preserved:

$$\mathcal{X}(1_A) = \mathscr{C}(1_A, -) = 1_{\mathscr{C}(A,-)} = 1_{\mathcal{X}(A)}$$

(where the central equation holds by Theorem 36 (3). And second, composition is respected. In other words, for any composable f, g in $\mathscr{C}^{op}$,

$$\mathcal{X}(g \circ^{op} f) = \mathscr{C}(f \circ g, -) = \mathscr{C}(g, -) \circ \mathscr{C}(f, -) = \mathcal{X}(g) \circ \mathcal{X}(f)$$

(where the central equation holds by Theorem 36 (1). □

Once more there is a dual result. Just recall that the contravariant functor $\mathscr{C}(-, A)$ lives as an object in the functor category $[\mathscr{C}^{op}, \mathbf{Set}]$. So the dual result is this:[3]

[3]You need to be alert for variations in notation between different discussions of this material. For example, Barr and Wells (1995, p. 108) uses 'Y'-for-Yoneda to label our functor $\mathcal{X}$ rather than $\mathcal{Y}$.

Theorem 44. *For any locally small category $\mathscr{C}$, there is a functor $\mathcal{Y}\colon \mathscr{C} \to [\mathscr{C}^{op}, \mathbf{Set}]$ with components $\mathcal{Y}_{ob}$ and $\mathcal{Y}_{arw}$ such that*

(1) for any $A \in ob(\mathscr{C})$, $\mathcal{Y}_{ob}(A) = \mathscr{C}(-, A)$.

(2) For any arrow f, $\mathcal{Y}_{arw}(f) = \mathscr{C}(-, f)$.

Theorem 45. *$\mathcal{X}\colon \mathscr{C}^{op} \to [\mathscr{C}, \mathbf{Set}]$ and $\mathcal{Y}\colon \mathscr{C} \to [\mathscr{C}^{op}, \mathbf{Set}]$ are fully faithful functors which are injective on objects.*

Proof. By definition, $\mathcal{X}\colon \mathscr{C}^{op} \to [\mathscr{C}, \mathbf{Set}]$ is fully faithful since each component $\mathcal{X}_{AB}$ is a bijection.

To say $\mathcal{X}$ is injective on objects is to say, of course, that if $A \neq B$, then $\mathcal{X}(A) \neq \mathcal{X}(B)$. Suppose $\mathcal{X}(A) = \mathcal{X}(B)$, i.e. $\mathscr{C}(A,-) = \mathscr{C}(B,-)$. Then $\mathscr{C}(A,-)(C) = \mathscr{C}(B,-)(C)$, i.e. $\mathscr{C}(A,C) = \mathscr{C}(B,C)$. But that can't be so if $A \neq B$, given our assumptions that hom-sets on different pairs of objects are disjoint (see the last sentence of §8.6).

The proof for $\mathcal{Y}\colon \mathscr{C} \to [\mathscr{C}^{op}, \mathbf{Set}]$ is dual. □

Here are two quick corollaries. First, we note that we can improve an earlier crucial theorem, by adding

Theorem 39 (continued) *And if α is an isomorphism, so is f.*

Proof. $\mathcal{X}$ sends f to α; but fully faithful functors reflect isomorphisms. So if if α is an isomorphism, so is f. □

Theorem 46. *For any objects A, B in the locally small category $\mathscr{C}$, $A \cong B$ iff $\mathcal{X}A \cong \mathcal{X}B$, and likewise $A \cong B$ iff $\mathcal{Y}A \cong \mathcal{Y}B$.*

Proof. Suppose $A \cong B$. Then for some isomorphism f, $f\colon B \xrightarrow{\sim} A$. So there is a natural transformation $\mathscr{C}(f,-)\colon \mathscr{C}(A,-) \Rightarrow \mathscr{C}(B,-)$, which by Theorem 35 is an isomorphism. In other words, $\mathscr{C}(f,-)\colon \mathcal{X}A \Rightarrow \mathcal{X}B$ is an isomorphism. Hence $\mathcal{X}A \cong \mathcal{X}B$.

Now suppose $\mathcal{X}A \cong \mathcal{X}B$. So there exists a natural isomorphism $\alpha\colon \mathscr{C}(A,-) \Rightarrow \mathscr{C}(B,-)$. By Theorem 39, α is $\mathscr{C}(f,-)$ for some $f\colon B \to A$, i.e. is $\mathcal{X}f$. But $\mathcal{X}$ is fully faithful, by Theorem 25, hence since $\mathcal{X}f$ is an isomorphism, so is f. Hence $A \cong B$.

That shows $A \cong B$ iff $\mathcal{X}A \cong \mathcal{X}B$. The argument for the functor $\mathcal{Y}$ is parallel. □

9.6 The Yoneda embedding defined

So the situation is this. The functor $\mathcal{Y}$, for example, injects a copy of the $\mathscr{C}$-objects one-to-one into the objects of the functor category $[\mathscr{C}^{op}, \mathbf{Set}]$; and then it faithfully matches up the arrows between $\mathscr{C}$-objects with arrows between the corresponding objects in $[\mathscr{C}^{op}, \mathbf{Set}]$. In other words, $\mathcal{Y}$ *yields an isomorphic copy of* $\mathscr{C}$ *sitting inside the functor category as a full sub-category.*

Why full? Because we know that for any pair of objects $\mathcal{Y}A$ and $\mathcal{Y}B$ in the surrounding category $[\mathscr{C}^{op}, \mathbf{Set}]$, i.e. for any $\mathscr{C}(-, A)$ and $\mathscr{C}(-, B)$, every arrow (natural transformation between them) is of the form $\mathscr{C}(-, f)$, for some $f : A \to B$ in $\mathscr{C}$. But the functor $\mathcal{Y}$ acting on $\mathscr{C}$ provides all these natural transformations.

So, in a phrase, $\mathcal{Y}$ *embeds* a copy of $\mathscr{C}$ in $[\mathscr{C}^{op}, \mathbf{Set}]$. Hence the terminology (in honour of its discoverer):

Definition 42. The full and faithful functor $\mathcal{Y} : \mathscr{C} \to [\mathscr{C}^{op}, \mathbf{Set}]$ is the *Yoneda embedding* of $\mathscr{C}$.

There was a reason, then, behind our use of '$\mathcal{Y}$' for this functor! And indeed the '$\mathcal{Y}$' notation – in upper or lower case, in one font or another – is pretty standard for the Yoneda embedding. However, '$\mathcal{X}$' is just our label for the dual embedding, which doesn't seem to have a standard name or notation, though we'll can usefully call it a Yoneda embedding too. (Alternative notations include e.g. Leinster's 'H_*' for the Yoneda embedding $\mathcal{Y}$, with 'H^*' available for the dual.)

10 An aside on Cayley's Theorem

Let's pause for breath and ask: what makes the Restricted Yoneda Lemma and the related Yoneda embedding interesting?

Here's one rather general thought. Take any locally small category you like. Then the Yoneda embedding tells us how to find a category built from functors-into-**Set**-and-arrows-between-them which looks just like the category we started off with.

Now, this is surely reminiscent of some classical representation theorems which tell us how, given a mathematical structure of a certain type, we can find another structure which lives in the universe of sets and is isomorphic to it. At the simple end of the spectrum there is an observation that we can attribute to Dedekind: any given partially ordered objects are isomorphic to certain corresponding sets ordered by set-inclusion. A significantly more sophisticated result of the same flavour is the Stone Representation Theorem: any Boolean algebra is isomorphic to a field of sets (where a field of sets is a sub-algebra of a canonical power-set alegbra $(\mathscr{P}(X), \overline{}, \cap, \cup, \emptyset, X)$, where X is some set and of course $\overline{A}$ is $X - A$).

In this short chapter, however, we'll concentrate on just one such classical representation theorem, namely ...

10.1 Cayley's Theorem

Theorem 47. *Any group $(G, \cdot)$ is isomorphic to a subgroup of the group $Sym(G)$, i.e. the group of permutations on the set G.*

Proof. (The usual one, just rehearsed as a reminder!) Given any object $g \in G$, we define the set-function $\underline{g}\colon G \to G$ by setting $\underline{g}(x) = g \cdot x$ (i.e. $\underline{g} = \{\langle x, y\rangle \mid x, y \in G \wedge y = g \cdot x\}$).

Evidently any such $\underline{g}$ is surjective: for any $x \in G$, there's an object which $\underline{g}$ sends to x, namely $g^{-1} \cdot x$. And if $\underline{g}(x) = \underline{g}(y)$, then $g \cdot x = g \cdot y$ whence $g^{-1} \cdot g \cdot x = g^{-1} \cdot g \cdot y$, therefore $x = y$. Hence $\underline{g}$ is also an injection and is therefore a bijection on G, i.e. is a permutation of the group objects.

Put $K = \{\underline{g} \mid g \in G\}$. It is now routine to confirm $(K, \circ)$ is a group, and hence a subgroup of $Sym(G)$, where the group operation is composition of functions:

i. Any two functions $\underline{f}$, $\underline{g}$ have a product $\underline{f} \circ \underline{g}$, where $(\underline{f} \circ \underline{g})(x) = f \cdot g \cdot x$.

ii. The function $\underline{e}$ is a group identity, where e is the identity in $(G, \cdot)$.

iii. $\underline{f} \circ (\underline{g} \circ \underline{h}) = (\underline{f} \circ \underline{g}) \circ \underline{h}$ because $f \cdot (g \cdot h) = (f \cdot g) \cdot h$.

iv. We note that $(\underline{g^{-1}} \circ \underline{g})(x) = \underline{g^{-1}}(g \cdot x) = g^{-1} \cdot g \cdot x = x = \underline{e}(x)$. So $\underline{g^{-1}} \circ \underline{g} = \underline{e}$, and similarly $\underline{g} \circ \underline{g^{-1}} = \underline{e}$. So each $\underline{g}$ has an inverse.

It remains to check that the map F defined by $g \mapsto \overline{g}$ is a group isomorphism from $(G, \cdot) \to (K, \circ)$. F is injective. For if $\underline{f} = \underline{g}$, then $\underline{f}(e) = \underline{g}(e)$, so $f \cdot e = g \cdot e$, so $f = g$. Since F is also a surjection just by the definition of K, F (as a map on the carries sets) is an isomorphism.

Also, for any x, $F(f \cdot g)(x) = \underline{(f \cdot g)}(x) = f \cdot g \cdot x = \underline{f}(g \cdot x) = \underline{f}(\underline{g}(x)) = (\underline{f} \circ \underline{g})(x) = (Ff \circ Fg)(x)$, so F indeed respects group structure. $\square$

Now, the modern way is – at least at least officially – to think of a group $(G, \cdot)$ as a set-theoretic structure from the outset; so Cayley's theorem might seem just to tell us that, given one set theoretic structure, we can find another isomorphic one. Big deal! However, that rather disguises what's actually going on.

For various reasons – some good, some rather disreputable – it has become absolutely standard in mathematics to trade in a lot of plural talk (referring to many objects at once) for singular talk (referring to a set of those many objects). For example, we've learnt to slide easily e.g. from talk of the natural numbers (plural) to talk of the set $\mathbb{N}$ (singular). And so instead of stating the Least Number Principle as e.g. 'Given any natural numbers, one of them will be the least' we say 'Any set S, where $S \subseteq \mathbb{N}$, has a least member'. But note that the singular talk about a set here is not yet doing any real work. Indeed, quite a lot of informal set talk is in fact similarly low-level, non-committal stuff – i.e. is talk about merely virtual classes – so is a familiar way of speaking which can however be readily translated away, most naturally into a plural idiom. That applies here, to part of the statement of Cayley's Theorem. Instead of starting 'Any group $(G, \cdot)$, ...' and reading this as referring to a set-theoretic object (e.g. an ordered pair of a set and a set-function), we can capture the core of the theorem like this:

> Suppose we are given some objects and a group operation on them. Then there will be always also be some *sets* (in particular, some set-functions) with a group structure on *them* which form a group isomorphic to the one we started with.

Put this way, stripped of a layer of unnecessary set-idiom, we have (in an intuitive sense) a 'cross-category' result which says that objects with a group structure on them (whatever objects they are) can always be represented by an isomorphic structure living in the world of sets.

10.2 Yoneda meets Cayley

Recall from §2.2 ex.(8) that a monoid can be considered as a category in its own right. This will apply, of course, to those monoids which are groups.

Let's spell this out. Take a group $(G, \cdot)$, where we assume that G, the collection of group elements, is indeed set-sized. And then consider the corresponding category $\mathscr{G}$ with the following data:

i. the sole object of $\mathscr{G}$: choose whatever object you like, and dub it '$\star$'.

ii. the arrows of $\mathscr{G}$ are the *elements* of the group $(G, \cdot)$.

iii. the identity arrow $1_\star$ of $\mathscr{G}$ is the identity element e of the group G.

iv. the composite $g \circ f \colon \star \to \star$ of the two arrows $g, f \colon \star \to \star$ is just $g \cdot f$.

Now, suppose that the *element* a^{-1} is the inverse of a in the group $(G, \cdot)$. Then the *arrow* a^{-1} is the inverse of the arrow a in the category $\mathscr{C}$. Therefore, since every group element in the $(G, \cdot)$ has an inverse, every $\mathscr{C}$-arrow has an inverse. Which proves

Theorem 48. *A group considered as a category is a one-object category all of whose arrows are isomorphisms.*

$\mathscr{G}$ is locally small since its sole potential hom-set $\mathscr{G}(\star, \star)$ is none other than G, which we assumed is indeed set-sized. We can therefore apply the Restricted Yoneda Lemma in one version or the other. And there's only one possible application of each version. Consider the version which tells us that

$$Nat(\mathscr{G}(-, \star), \mathscr{G}(-, \star)) \cong \mathscr{G}(\star, \star).$$

So: what are the natural transformations $\alpha \colon \mathscr{G}(-, \star) \Rightarrow \mathscr{G}(-, \star)$? We can apply Theorem 40: every such α is $\mathscr{G}(-, a)$ for some arrow a in $\mathscr{G}$.

Now, by definition: $\mathscr{G}(-, a)$ sends an arrow $j \colon \star \to \star$ to $a \circ j \colon \star \to \star$. But $\mathscr{G}(\star, \star)$ is just G, and arrows are G-elements, so $\mathscr{G}(-, a)$ acts on G by sending an element j to the element $a \circ j$, which is the function we earlier called $\underline{g}(a)$. But as before, that's a bijective map on G, i.e. a permutation on G. *Therefore our natural transformations are permutations on* G. Hence, so far, we know that G (as a set) is isomorphic to some set of permutations on G.

Moreover, our proof of the Lemma gives us the isomorphism $\mathcal{Y}$, which sends the arrow $a \colon \star \to \star$ to $\mathscr{G}(-, a)$. So by Theorem 38,

$$\mathcal{Y}(a \cdot b) = \mathcal{Y}(a \circ b) = \mathscr{G}(-, a \circ b) = \mathscr{G}(-, a) \circ \mathscr{G}(-, b) = \mathcal{Y}(a) \circ \mathcal{Y}(b).$$

So if we put a group structure on the natural transformations $\mathscr{G}(-, a)$ by using composition as the group operation (check it *is* a group operation in this case!), *our isomorphism $\mathcal{Y}$ preserves group structure.*

So in short, we can more or less immediately read off from the proof of the Restricted Yoneda Lemma that a group $(G, \cdot)$ is isomorphic as a group to a group of permutations on G with composition as the group operation.

Which is why it is often said that the Yoneda Lemma is a generalization of Cayley's Theorem.

11 The Yoneda Lemma

In Chapter 9 we proved that the Yoneda embedding and its dual are indeed embeddings (in a very natural sense) by appeal to a couple of fairly easy results, Theorems 41 and 42. We called the combination of these theorems, for want of a standard label, the Restricted Yoneda Lemma.

Which invites the question: what's the *unrestricted* Yoneda Lemma? This chapter explains.

11.1 Towards the full Yoneda Lemma

Let F be the functor $\mathscr{C}(B,-)$. Then one half of the Restricted version of the Yoneda Lemma tells us that there is an isomorphism between $Nat(\mathscr{C}(A,-),F)$ and $F(A)$. The other half of the Restricted Lemma is the dual of course: for the moment, we'll let it look after itself!

Now, to get from where we are to the Yoneda Lemma proper we need two (again quite easy) observations:

(1) Look again at the ingredients of the proof of the restricted version and ask 'Where did we depend on the fact that the second functor, now notated simply 'F', actually had the form $\mathscr{C}(B,-)$ for some B?' Inspection shows that we didn't! So we in fact have the more general result that for any locally small category $\mathscr{C}$, *any* functor $F\colon \mathscr{C} \to \mathbf{Set}$, and any object $A \in \mathscr{C}$, there is an isomorphism $\mathcal{E}$ between $Nat(\mathscr{C}(A,-),F)$ and $F(A)$.

(2) Note too that our proof provided a *general recipe* for constructing the required isomorphism. Take a locally small category $\mathscr{C}$ and any $\mathscr{C}$-object A, then, without having to invoke any arbitrary choices, our proof fixes inverse isomorphisms $\mathcal{X}_{AF}$ and $\mathcal{E}_{AF}$ between $Nat(\mathscr{C}(A,-),F)$ and $F(A)$ (at least for the case where F is a functor of the form $\mathscr{C}(B,-)$). In an intuitive sense, we've constructed a *natural* isomorphism. And so we should be able to show that there is a *natural isomorphism* in the official, categorial, sense between some relevant functors.

In sum, we will get from the Restricted Yoneda Lemma to the full-dress Yoneda Lemma in two steps, one that involves a generalizing move, and one that involves recasting in category-theoretic terms an intuitive judgement of the naturality of our construction. Neither step involves anything conceptually difficult: we just need to nail down the details. (Some of these proof details are tiresome. By all means just skim over them on a first reading)

11.2 The generalizing move

We continue working in a locally small category $\mathscr{C}$. Let's restate what we know so far, temporarily using 'F' to abbreviate '$\mathscr{C}(B,–)$'.

Theorems 35 and 39 tell us that there's a bijection between $\mathscr{C}$-arrows $f\colon B \to A$ and natural transformation $\mathscr{C}(f,–)\colon \mathscr{C}(A,–) \Rightarrow F$, where for each Z, the component $\mathscr{C}(f,–)_Z$ maps an arrow $g\colon A \to Z$ to $g \circ f\colon B \to Z$.

Now by definition, the functor F maps an arrow $g\colon A \to Z$ to a function which sends an arrow $f\colon B \to A$ to the arrow $g \circ f\colon B \to Z$. Hence $F(g)(f) = g \circ f$.

Putting those two remarks together, we get this: there's a bijection which sends an arrow $f\colon B \to A$ to the natural transformation whose component $\mathscr{C}(f,–)_Z$ maps $g\colon A \to Z$ to $F(g)(f)$.

OK, with that preliminary, here's the announced generalization of the Restricted Lemma where we free up the interpretation of F to encompass any functor from $\mathscr{C}$ to **Set**:

Theorem 49. *For any locally small category $\mathscr{C}$, object $A \in \mathscr{C}$ and functor $F\colon \mathscr{C} \to \mathbf{Set}$, $Nat(\mathscr{C}(A,–), F) \cong F(A)$.*

Proof. Following the construction in the proof leading up to Theorem 41, define θ by requiring $\theta(\alpha) = \alpha_A(1_A)$, for any $\alpha\colon \mathscr{C}(A,–) \Rightarrow F$.

And define ψ by requiring that, for an element $f \in F(A)$, the value of $\psi(f)$ is the natural transformation whose component $\psi(f)_Z$ sends a map $g\colon A \to Z$ to $F(g)(f)$. (In the light of our preliminary observation above, $\psi(f)$ is just a generalized version of $\mathscr{C}(f,–)$ and is readily checked to still be a natural transformation.)

We show that ψ is a two-sided inverse of θ. First, given an arbitrary element $f \in F(A)$,

$$(\theta \circ \psi)f = \psi(f)_A(1_A) = F(1_A)(f) = 1_{FA}(f) = f,$$

and therefore $\theta \circ \psi = 1_{F(A)}$.

Secondly, for $\alpha\colon \mathscr{C}(A,–) \Rightarrow F$, we have $\psi \circ \theta(\alpha)_Z = \psi(\alpha_A(1_A))_Z$, which sends a map $g\colon A \to Z$ to $F(g(\alpha_A(1_A))$. But since α is a natural transformation, this next diagram must commute:

$$\begin{tikzcd}
\mathscr{C}(A,A) \arrow[r, "\mathscr{C}(A,g)"] \arrow[d, "\alpha_A"] & \mathscr{C}(A,Z) \arrow[d, "\alpha_Z"] \\
F(A) \arrow[r, "F(g)"] & FZ
\end{tikzcd}$$

So $F(g(\alpha_A(1_A A)) = \alpha_Z(\mathscr{C}(A,g)(1_A)) = \alpha_Z(g)$.

Which shows that $\psi \circ \theta(\alpha)_Z$ is the map which sends any g to $\alpha_Z(g)$, i.e. is α_Z. So $\psi \circ \theta = 1_{Nat}$.

We have shown, then, that $\theta\colon Nat(\mathscr{C}(A,-),F) \to F(A)$ has two-sided inverse, i.e. is an isomorphism, and so we are done. $\square$

11.3 Making it all natural

A second step takes us to the full Yoneda Lemma. Not only is there an isomorphism we'll now label θ_{AF} from $Nat(\mathscr{C}(A,-),F)$ to $F(A)$, but θ_{AF} is intuitively 'natural' (constructed in a uniform way given A and F, without arbitrary choices). We now want to capture this intuitive remark using our official categorial account of a natural isomorphism.

Here's a reminder:

Definition 27 Given functors $F, G\colon \mathscr{C} \to \mathscr{D}$, we say that $FA \cong GA$ *naturally in* A (or *naturally in* A *in* $\mathscr{C}$) just if F and G are *naturally* isomorphic.

What we want to prove first, keeping F fixed, is that $Nat(\mathscr{C}(A,-),F) \cong F(A)$ naturally in $A \in \mathscr{C}$. Which, by our definition, means we have to establish something along these lines:

Theorem 50. *Let $\mathscr{C}$ be a locally small category, and F a functor $F\colon \mathscr{C} \to \mathbf{Set}$. Then the functors $Nat(\mathscr{C}(\cdot,-),F)$ and F are naturally isomorphic.*

But hold on! We haven't yet met a functor $Nat(\mathscr{C}(\cdot,-),F)\colon \mathscr{C} \to \mathbf{Set}$! What is it? It's the functor N (for short) with the following components:

(i) N sends any $\mathscr{C}$-object A to the set $Nat(\mathscr{C}(A,-),F)$.

(ii) N sends any $\mathscr{C}$-arrow $f\colon A \to B$ to an arrow between $Nat(\mathscr{C}(A,-),F)$ and $Nat(\mathscr{C}(B,-),F)$. Which arrow? The obvious candidate is the one that sends $\alpha\colon \mathscr{C}(A,-) \Rightarrow F$ to $\alpha \circ \mathscr{C}(f,-)\colon \mathscr{C}(B,-) \Rightarrow F$.

Let's check that this definition works! Given $f\colon A \to B$ (NB the order of A and B here!), we have $\mathscr{C}(f,-)\colon \mathscr{C}(B,-) \Rightarrow \mathscr{C}(A,-)$. So yes, $\alpha \circ \mathscr{C}(f,-)$ is a natural transformation (by vertical composition). So now to check N is functorial.

(1) For any $A \in ob(\mathscr{C})$, $N(1_A)$ is the arrow that sends any $\alpha\colon \mathscr{C}(A,-) \Rightarrow F$ to $\alpha \circ \mathscr{C}(1_A,-)\colon \mathscr{C}(A,-) \Rightarrow F$. But $\mathscr{C}(1_A,-) = 1_{\mathscr{C}(A,-)}$ (by Theorem 36). So $N(1_A)$ is the identity arrow on the set $Nat(\mathscr{C}(A,-),F)$, i.e. is $1_{N(A)}$.

(2) For any $\mathscr{C}$-arrows $f\colon A \to B$ and $g\colon B \to C$, $(Ng \circ Nf)\alpha = Ng(\alpha \circ \mathscr{C}(f,-)) = \alpha \circ \mathscr{C}(f,-) \circ \mathscr{C}(g,-) = \alpha \circ \mathscr{C}(g \circ f,-) = N(g \circ f)\alpha$ (by Theorem 36 at the penultimate step). Hence $N(g \circ f) = Ng \circ Nf$.

Hence there really is functor $N = Nat(\mathscr{C}(\cdot,-),F)$ and at least the *statement* of our theorem is in good order. So now we need a ...

Proof. Given any $f\colon A \to B$, consider the following diagram,

$$\begin{tikzcd} Nat(\mathscr{C}(A,-),F) \arrow[r, "Nf"] \arrow[d, "\theta_{AF}"] & Nat(\mathscr{C}(B,-),F) \arrow[d, "\theta_{BF}"] \\ F(A) \arrow[r, "F(f)"] & F(B) \end{tikzcd}$$

Now take any $\alpha\colon \mathscr{C}(A,-) \Rightarrow F$. Then we have:

(1) $\theta_{BF} \circ Nf(\alpha) = \theta_{BF}(\alpha \circ \mathscr{C}(f,-)) = (\alpha \circ \mathscr{C}(f,-))_B(1_B) = \alpha_B(f)$ (for the last equation, compare the end of the proof of Theorem 39).

(2) But also $F(f) \circ \theta_{AF}(\alpha) = F(f)(\alpha_A(1_A)) = \alpha_B \circ \mathscr{C}(A,f)(1_A) = \alpha_B(f)$ (for the middle equation look at the commutative diagram at the end of the proof of Theorem 49, with trivial relabelling).

So our diagram will always commute, and hence there is a natural isomorphism $\theta_F\colon N \Rightarrow F$ with components $(\theta_F)_A = \theta_{AF}$ for each $A \in \mathscr{C}$, and our theorem is proved. □

That captures in categorial terms the intuition that the construction of θ_{AF} depends in a natural way on A; now for the companion intuition that it depends in a natural way on F too. Keeping A fixed, we want to prove $Nat(\mathscr{C}(A,-),F) \cong F(A)$ naturally in F (where F belongs to the functor category $[\mathscr{C}, \mathbf{Set}]$). Which, by definition, means we want to prove something of this shape:

Theorem 51. *Let $\mathscr{C}$ be a locally small category. Then $Nat(\mathscr{C}(A,-),\cdot)$ and $\cdot(A)$ are naturally isomorphic, where these are appropriate functors from $[\mathscr{C}, \mathbf{Set}]$ to* **Set***.*

So of course, our first task, again, is to define the appropriate functors to work with.

We need $Nat(\mathscr{C}(A,-),\cdot)$, or K for short, to act on an object in the functor category $[\mathscr{C}, \mathbf{Set}]$, i.e. on a functor $F\colon \mathscr{C} \to \mathbf{Set}$ and send it to $Nat(\mathscr{C}(A,-),F)$. And how should K act on arrows in the functor category, i.e. on natural transformations $\gamma\colon F \Rightarrow G$ where $\mathscr{C} \overset{G}{\underset{H}{\rightrightarrows}} \mathbf{Set}$? Well, the obvious option is to set the value of $K(\gamma)$ to be the map that sends $\alpha\colon \mathscr{C}(A,-) \Rightarrow F$ to $\gamma \circ \alpha\colon \mathscr{C}(A,-) \Rightarrow G$. It is easily checked that K is then functorial.

Likewise, we need the functor $\cdot(A)$ to act on an object in the functor category, i.e. on a functor $F\colon \mathscr{C} \to \mathbf{Set}$, by sending it to $F(A)$. So we can think of $\cdot(A)$ as a functor which *evaluates* a functor F at the object A, so let's write it more helpfully as ev_A. Then $ev_A(F) = F(A)$, and we can put $ev_A(\gamma) = \gamma_A$ to get a functor, as again is easily checked.

So, having explained our notation in stating our theorem, we now need another ...

Proof. Given any $\gamma\colon F \Rightarrow G$, consider the following diagram,

$$\begin{tikzcd} Nat(\mathscr{C}(A,–),F) \arrow[r, "K(\gamma)"] \arrow[d, "\theta_{AF}"] & Nat(\mathscr{C}(A,–),G) \arrow[d, "\theta_{AG}"] \\ F(A) \arrow[r, "ev_A(\gamma)"] & G(A) \end{tikzcd}$$

Then again this diagram commutes. For take any $\alpha\colon \mathscr{C}(A,–) \Rightarrow F$. Then we have:

(1) $\theta_{AG} \circ K(\gamma)(\alpha) = \theta_{AG}(\gamma \circ \alpha) = (\gamma \circ \alpha)_A(1_A) = \gamma_A(\alpha_A(1_A))$.
(2) But also $ev_A(\gamma) \circ \theta_{AF}(\alpha) = \gamma_A(\alpha_A(1_A))$.

Since the diagram always commutes, there is a natural isomorphism $\theta_A\colon K \Rightarrow ev_A$ with components $(\theta_A)_F = \theta_{AF}$ for each $F \in [\mathscr{C}, \mathbf{Set}]$. So we are done. □

11.4 Putting everything together

So now combine all the ingredients from the last three theorems ...

Cue drum-roll!

... and we at last have the full-dress result:

Theorem 52 (Yoneda Lemma)**.** *For any locally small category $\mathscr{C}$, object $A \in \mathscr{C}$, and functor $F\colon \mathscr{C} \to \mathbf{Set}$, $Nat(\mathscr{C}(A,–),F) \cong F(A)$, both naturally in $A \in \mathscr{C}$ and naturally in $F \in [\mathscr{C}, \mathbf{Set}]$.*

There will evidently be a dual version too (involving contravariant functors in $\mathscr{C}$, i.e. functors in $\mathscr{C}^{op}$):

Theorem 53 (Yoneda Lemma)**.** *For any locally small category $\mathscr{C}$, object $A \in \mathscr{C}$, and functor $F\colon \mathscr{C}^{op} \to \mathbf{Set}$, $Nat(\mathscr{C}(–,A),F) \cong F(A)$, both naturally in $A \in \mathscr{C}$ and naturally in $F \in [\mathscr{C}^{op}, \mathbf{Set}]$.*

Some authors call only the second version the Yoneda Lemma: we'll use the label for both, talking of the covariant and contravariant versions if we need to mark the distinction.

And having done all this work, we see that a further generalization is in principle possible. We've so far been working with locally small categories, i.e. categories whose classes of arrows between pairs of objects are indeed sets which live in **Set**. Suppose we turn our attention to larger categories whose hom-classes (as we could naturally call them) are some bigger collections which live in a suitably well-behaved category SET which allows bigger collections. Then we can re-run our arguments to show that for a category $\mathscr{C}$ with hom-classes in SET, $A \in \mathscr{C}$, and a functor $F\colon \mathscr{C} \to$ SET, then the SET of natural transformations from $\mathscr{C}(A,–)$ to F is in natural isomorphism with $F(A)$.

But we won't delay over this further generalization – indeed, will we have occasion to use it?

12 Representables

12.1 Presheaves, diagrams, representables

Definition 43. Given a category $\mathscr{C}$ then we say

i. A contravariant functor from $\mathscr{C}$ to **Set**, i.e. a functor $F\colon \mathscr{C}^{op} \to \mathbf{Set}$, is a *presheaf* on $\mathscr{C}$.
ii. The presheaves on $\mathscr{C}$ (as objects) and the natural transformations between them (as arrows) form *the presheaf category on* $\mathscr{C}$, denoted $\widehat{\mathscr{C}}$.

Four quick comments on these definitions:

(1) The terminology 'presheaf' comes from an example in topology. But here we can just take it as an arbitrary (though now standard) label.[1]
(2) $\widehat{\mathscr{C}}$ is just a relabelling of the functor category we met in §9.5 and called $[\mathscr{C}^{op}, \mathbf{Set}]$. And so the Yoneda embedding $\mathcal{Y}$ we met there is a functor $\mathcal{Y}\colon \mathscr{C} \to \widehat{\mathscr{C}}$; and in our new notation, assuming the Axiom of Choice, we can say that $\mathscr{C}$ is isomorphic to a full subcategory of $\widehat{\mathscr{C}}$.
(3) Recall $\mathcal{Y}A = \mathscr{C}(-, A)$. Hence $\widehat{\mathscr{C}}(\mathcal{Y}A, F)$ is the hom-class of the presheaf category $\widehat{\mathscr{C}}$ which comprises the arrows of that functor category from $\mathscr{C}(-, A)$ to F, i.e. it is $Nat(\mathscr{C}(-, A), F)$. That's why (one version) of the Yoneda Lemma can be presented like this: on the usual assumptions, $\widehat{\mathscr{C}}(\mathcal{Y}A, F) \cong FA$, naturally in both $A \in \mathscr{C}$ and F in $\widehat{\mathscr{C}}$.

As you would expect, alongside the alternative terminology of 'presheaves' for contravariant functors, there's terminology for the dual covariant functors. Back in §6.2 we said, informally, a functor from one category $\mathscr{J}$ to another category $\mathscr{C}$ projects a 'picture' or 'diagram' of $\mathscr{J}$ into $\mathscr{C}$. That informal way of speaking prefigures some more standard jargon:

[1] Some also talk more generally about a $\mathscr{V}$-valued presheaf on $\mathscr{C}$, i.e. a functor $F\colon \mathscr{C}^{op} \to \mathscr{V}$. An interesting case might be when $\mathscr{V} = \text{SET}$, i.e. is a category with larger collections than sets (see the end of §11.4). Then by talking of SET-valued presheaves we could then extend results about locally small categories and presheaves in our initial sense. But we certainly won't need to pursue this.

Definition 44. Given a category $\mathscr{C}$, we say that a functor into it, i.e. a functor $D\colon \mathscr{J} \to \mathscr{C}$, is a *diagram* – or more explicitly, is a *diagram of shape $\mathscr{J}$ in $\mathscr{C}$.*

(Our definitions reflect that we are going to be particularly interested in contravariant functors *from* a given category $\mathscr{C}$ and covariant functors *into* $\mathscr{C}$.)

How does this last notion, of diagrams-as-functors, relate to the idea(s) of a diagram as originally introduced in §2.3? Think of the case where $\mathscr{J}$ is a very small category, a few objects and arrows. Then the functor $D\colon \mathscr{J} \to \mathscr{C}$ will send $\mathscr{J}$ to a corresponding handful of objects and arrows sitting inside $\mathscr{C}$, i.e. the sort of fragment that we called a diagram in $\mathscr{C}$. Diagrams-as-functors deliver diagrams-in-categories.

Definition 45. If $\mathscr{C}$ is a locally small category, then

i. A (set-valued) presheaf $F\colon \mathscr{C}^{op} \to \mathbf{Set}$ which is naturally isomorphic to the contravariant hom-functor $\mathscr{C}(-, A)$ for some $A \in \mathscr{C}$ is said to be *representable*, and A is said to *represent* it.
ii. Dually, a set-valued diagram $F\colon \mathscr{C} \to \mathbf{Set}$ which is naturally isomorphic to the covariant hom-functor $\mathscr{C}(A, -)$ is also said to be representable, and again A represents it.
iii. Representable functors are often just called *representables.*

We immediately note an important result about representating objects (we state one version, leaving the dual as then obvious):

Theorem 54. *Suppose $F\colon \mathscr{C} \to \mathbf{Set}$ is represented by both $A, B \in \mathscr{C}$. Then $A \cong B$.*

Proof. By our supposition, we have both $F \cong \mathscr{C}(A, -)$ and $F \cong \mathscr{C}(B, -)$, hence $\mathscr{C}(A, -) \cong \mathscr{C}(B, -)$, so $\mathcal{X}A \cong \mathcal{X}B$. Apply Theorem 46. $\square$

12.2 Is the forgetful functor $\mathbf{Mon} \to \mathbf{Set}$ representable?

Trivially, hom-functors themselves are representables. But now let's find some other examples.

Start with a simple example, indeed the very first functor we met back in §4.2. The forgetful functor $F\colon \mathbf{Mon} \to \mathbf{Set}$ sends any monoid $(M, \cdot)$ to its underlying set M, and sends a monoid homomorphism $f\colon (M, \cdot) \to (N, \times)$ to the same function $f\colon M \to N$ in $\mathbf{Set}$.

Question: is there a representing monoid R such that the hom-functor $\mathbf{Mon}(R, -)$ is naturally isomorphic to the forgetful F?

Applying the usual definition, $\mathbf{Mon}(R, -)$ sends a monoid M in $\mathbf{Mon}$ to $\mathbf{Mon}(R, M)$, i.e. to the set of monoid homomorphisms from R to M. And it sends a monoid homomorphism $f\colon M \to N$ to the set-function $f \circ -$ which sends a function $g\colon R \to M$ to the function $f \circ g\colon R \to N$.

And if this functor $\mathbf{Mon}(R, -)$ is to be naturally isomorphic with the forgetful functor F, there will have to be an isomorphism ψ with components such that, for any $f\colon M \to N$ in $\mathbf{Mon}$, the following diagram commutes in $\mathbf{Set}$:

$$\begin{tikzcd} M \arrow[r, "f"] \arrow[d, "\psi_M"] & N \arrow[d, "\psi_N"] \\ \mathbf{Mon}(R, M) \arrow[r, "f \circ -"] & \mathbf{Mon}(R, N) \end{tikzcd}$$

For this to work, we certainly need to choose an R such that (for any monoid M) there is a bijection between M and $\mathbf{Mon}(R, M)$. And presumably, for the needed generality, R will have to be a monoid without too much distinctive structure.

Now, the simplest such 'boring' monoid is the one-element monoid we can suggestively notate $\mathit{0}$ consisting of just an identity element (thinking of the monoid operation as addition, it just has a zero element). But that plainly isn't going to work as a candidate for R (there is only ever a unique member of $\mathbf{Mon}(\mathit{0}, M)$, i.e. the homomorphism that sends the identity element of $\mathit{0}$ to the identity element of M, so in general there is no bijection between $\mathbf{Mon}(\mathit{0}, M)$ and M).

The next simplest monoid is the free monoid with a single generator. This is the monoid we can suggestively label $\mathbb{N}$, which has an identity element '0', a single generator we'll call '1', and all the freely generated 'sums' of these elements. Now consider a homomorphism from $\mathbb{N}$ to M. $0 \in \mathbb{N}$ has to map to the identity element in M; and once we fix that $1 \in \mathbb{N}$ goes to some $m \in M$ that fixes where every element of $\mathbb{N}$ (since every non-zero $\mathbb{N}$ element $1+1+1+\ldots+1$ goes to a corresponding M-element $m \cdot m \cdot m \cdot \ldots \cdot m$).

So consider the map $\psi_M : M \to \mathbf{Mon}(\mathbb{N}, \mathbf{Set})$ which maps m to the homomorphism $\overline{m}\colon \mathbb{N} \to M$ which sends $1 \in \mathbb{N}$ to m. That's well-defined. ψ_M is evidently injective. But ψ_M is also surjective on homomophisms from $\mathbb{N} \to M$ since every such homomorphism must send $1 \in \mathbb{N}$ to some m or other, thereby determining the whole homomorphism.

Hooray! ψ_M is an isomorphism in $\mathbf{Set}$, for any M. And now it is easily seen that our diagram commutes. Chase an element $m \in M$ round the diagram. The route via the north-east node takes gives us $m \longmapsto f(m) \longmapsto \overline{f(m)}$, the other route gives us $m \longmapsto \overline{m} \longmapsto f \circ \overline{m} = \overline{f(m)}$.

Which, in summary, delivers

Theorem 55. *The forgetful functor $F\colon \mathbf{Mon} \to \mathbf{Set}$ is representable, and is represented by $\mathbb{N}$.*

12.3 More examples of representables

Unsurprisingly, there are analogous representation theorems for other forgetful functors. For instance:

Theorem 56. *(1) The forgetful functor $F\colon \mathbf{Grp} \to \mathbf{Set}$ is representable, and is represented by $\mathbb{Z}$.*

(2) The forgetful functor $F\colon \mathbf{Ab} \to \mathbf{Set}$ is representable, and is also represented by $\mathbb{Z}$.

(3) The forgetful functor $F\colon \mathbf{Vect} \to \mathbf{Set}$ (where $\mathbf{Vect}$ is the category of vector spaces over the reals) is representable, and is represented by $\mathbb{R}$.

(4) The forgetful functor $F\colon \mathbf{Top} \to \mathbf{Set}$ is representable, and is represented by the one-point topological space, call it S_0.

To comment on the only last of these, we simply note that a trivial continuous functions with domain S_0 into a space S in effect picks out a single point of S, so the set of arrows $\mathbf{Top}(S_0, S)$ is indeed in bijective correspondence with the set of points of S.

Given such examples, you might be tempted to conjecture that *all* such forgetful functors into **Set** are representable. But not so. Consider **FinGrp**, the category of finite groups. Then

Theorem 57. *The forgetful functor $F\colon \mathbf{FinGrp} \to \mathbf{Set}$ is not representable,*

Proof. Suppose a putative representing group R has r members, and take any group G with $g > 1$ members, where g is coprime with r. Then it is a known result that the only group homomorphism from R to G is the trivial one that sends everything to the identity in G. But then the underlying set of G can't be in bijective correspondence with $\mathbf{FinGrp}(R, G)$ as would be needed to get the analogue of the last commutative diagram for a case involving the group G. □

Let's take another couple of examples. First, a pair of definitions:

Definition 46. i. The (covariant) *powerset functor* $P\colon \mathbf{Set} \to \mathbf{Set}$ maps a set X to its powerset $\mathscr{P}(X)$ and maps a set-function $f\colon X \to Y$ to the function which sends $U \in \mathscr{P}(X)$ to its image $f[U] \in \mathscr{P}(Y)$.

ii. The *contravariant powerset functor* $\overline{P}\colon \mathbf{Set}^{op} \to \mathbf{Set}$ again maps a set to its powerset, and maps a set-function $f\colon Y \to X$ to the function which sends $U \in \mathscr{P}(X)$ to its inverse image $f^{-1}[U] \in \mathscr{P}(Y)$.

We won't pause to check the elementary fact that these are indeed functors, but rather will immediately state

Theorem 58. *The contravariant powerset functor $\overline{P}$ is represented by the set $2 = \{0, 1\}$; but the covariant powerset functor P is not representable.*

Proof. As yet, we don't have any general principles about representables and non-representables which we can invoke to prove theorems such us this. So again we need to labour through by applying definitions and seeing what we get.

If the contravariant functor $\overline{P}$ is to be representable, then there must be a representing set R and a natural isomorphism ψ with components such that, for all set functions $f\colon Y \to X$, the following diagram always commutes:

$$\begin{tikzcd}
\overline{P}X \arrow[rr, "\overline{P}f"] \arrow[d, "\psi_X"] & & \overline{P}Y \arrow[d, "\psi_Y"] \\
\mathbf{Set}(X,R) \arrow[rr, "\mathbf{Set}(f,R)"] & & \mathbf{Set}(Y,R)
\end{tikzcd}$$

Now $\mathbf{Set}(X,R)$ is the set of set-functions from X to R, whose cardinality is $|R|^{|X|}$; and the cardinality of $\overline{P}X$, i.e. $\mathscr{P}(X)$, is $2^{|X|}$. So that forces R to be a two-membered set: so we pick the set $2 = \{0,1\}$.

$\mathbf{Set}(X,2)$ is then the set of characteristic functions for subsets of X, i.e. the set of functions $c_U\colon X \to \{0,1\}$ where $c_U(x) = 1$ iff $x \in U \subseteq X$. So the obvious next move is to take $\psi_X\colon \overline{P}X \to \mathbf{Set}(X,R)$ to be the isomorphism that sends a set $U \in \mathscr{P}(X)$ to its characteristic function c_U.

And with this choice, the diagram always commutes. Chase the element $U \in \overline{P}X$ around. The route via the north-east node takes us from $U \subseteq X$ to $f^{-1}[U] \subseteq Y$ to its characteristic function. i.e. the function which maps $y \in Y$ to 1 iff $f(y) \in U$. Meanwhile, the route via the south-west node takes us first from $U \subseteq X$ to c_U. And then we apply $\mathbf{Set}(f,2)$, which maps $c_U\colon X \to 2$ to $c_U \circ f\colon Y \to 2$, which again is the function which maps $y \in Y$ to 1 iff $f(y) \in U$. Which establishes the first half of the theorem.

For the second half of the theorem, we just note that if we try to run a similar argument for the covariant functor P, we'd need to find a representing set R' such that PX and $\mathbf{Set}(R',X)$ are always in bijective correspondence. But $\mathbf{Set}(R',X)$ is the set of set-functions from R' to X, whose cardinality is $|X|^{|R'|}$, while the cardinality of PX is $2^{|X|}$. And there is no choice of R' which will make these equal for varying X. □

12.4 Universal elements

Take a functor $F\colon \mathscr{C} \to \mathbf{Set}$. In §11.2 we defined the isomorphism ψ between FA and $Nat(\mathscr{C}(A,–),F)$.

Recall, ψ takes an object $a \in FA$ to the natural transformation $\psi(a)\colon \mathscr{C}(A,–) \Rightarrow F$ whose component $\psi(a)_Z$ sends an arrow $f\colon A \to Z$ to $F(f)(a)$.

Now, in §12.1, we said that $F\colon \mathscr{C} \to \mathbf{Set}$ is representable by $A \in \mathscr{C}$ iff it is naturally isomorphic to the hom-functor $\mathscr{C}(A,–)$. So, to be representable by A, there must be (i) some natural transformation between $\mathscr{C}(A,–)$ and F, which in fact (ii) is an isomorphism. Hence there must (i′) be some $a \in FA$ asssociated with a natural transformation $\psi(a)$ such that (ii′) $\psi(a)$ is an isomorphism. But when is $\psi(a)$ an isomorphism? Just when each component $\psi(a)_Z\colon \mathscr{C}(A, Z) \Rightarrow FZ$ is a bijection. That is to say, just when for each $z \in FZ$ there is a unique $f\colon A \to Z$ such that $\psi(a)_Z(f) = z$, i.e. such that $F(f)(a) = z$.

This makes the following notion of interest (even if it doesn't yet explain the label):

Definition 47. A *universal element* of the functor $F\colon \mathscr{C} \to \mathbf{Set}$ is a pair $\langle A, a\rangle$, where $A \in \mathscr{C}$ and $a \in FA$, where for each $Z \in \mathscr{C}$ and $z \in FZ$, there is a unique map $f\colon A \to Z$ such that $F(f)(a) = z$.

Dually, a universal element of the contravariant functor $F\colon \mathscr{C}^{op} \to \mathbf{Set}$ is a pair $\langle A, a\rangle$, where $A \in \mathscr{C}$ and $a \in FA$, where for each $Z \in \mathscr{C}$ and $z \in FZ$, there is a unique map $f\colon Z \to A$ such that $F(f)(a) = z$.

(Note, some call a universal element $\langle A, a\rangle$ of F a representation, others simply call A the representation. Some, e.g. Barr and Wells 1995, p. 111, call just a the universal element.)

We then have the following result, together with its obvious dual:

Theorem 59. *A functor $F\colon \mathscr{C} \to \mathbf{Set}$ is representable by A iff it has a universal element $\langle A, a\rangle$.*

Proof. Our preceding remarks establish the 'only if' direction. So it remains to prove the converse.

Suppose $\langle A, a\rangle$ is a universal element for F. Then, $a \in FA$, and by definition $\psi(a)$ is the natural transformation from $\mathscr{C}(A, –)$ to F whose component $\psi(a)_Z\colon \mathscr{C}(A, Z) \to FZ$ sends a map $f\colon A \to Z$ to $F(f)(a)$.

But then, by the condition on universality, there's a function $\delta(a)_Z$ which sends $z \in FZ$ to the unique $f\colon A \to Z$ in $\mathscr{C}(A, Z)$ where $F(f)(a) = z$. And we can immediately see that $\psi(a)_Z$ and $\delta(a)_Z$ are inverses. So each component $\psi(a)_Z$ is an isomorphism, and hence $\psi(a)$ is a natural isomorphism from $\mathscr{C}(A, –)$ to F witnessing that F is representable by A. $\square$

To tie this with what's gone before, we note that the proof of Theorem 49 includes the claim that $\psi(\alpha_A(1_A))_Z$ sends a map $g\colon A \to Z$ to $F(g(\alpha_A(1_A)))$. Which shows that

Theorem 60. *If functor $F\colon \mathscr{C} \to \mathbf{Set}$ is representable by A in virtue of the natural isomorphism $\alpha\colon \mathscr{C}(A, –) \Rightarrow F$, then F has the universal element $\langle A, \alpha_A(1_A)\rangle$.*

12.5 Categories of elements

Next, we note that the universal elements of F are objects in a broader category:

Definition 48. $\mathbf{Elts}_{\mathscr{C}}(F)$, the *category of elements of the functor $F\colon \mathscr{C} \to \mathbf{Set}$*, has the following data:

(1) Objects are the pairs $\langle A, a\rangle$, where $A \in \mathscr{C}$ and $a \in FA$.
(2) An arrow from $\langle A, a\rangle$ to $\langle B, b\rangle$ is a $\mathscr{C}$-arrow $f\colon A \to B$ such that $F(f)(a) = b$.
(3) The identity arrow on $\langle A, a\rangle$ is 1_A.
(4) Composition of arrows is induced by composition of $\mathscr{C}$-arrows.

It is easily checked that this *is* a category. (Alternative symbolism for the category includes variations on '$\int F$'.)

Now, the standard name 'category of elements of F' is initially puzzling – after all, functors don't in a straightforward sense have elements. But we can perhaps throw some light on the name as follows.

(i) Suppose we are given a category $\mathscr{C}$ whose objects *are* sets (perhaps with some additional structure on them) and whose arrows are functions between sets. Then there will be a derived category – or rather, some derived categories – whose objects are (or involve) *elements* of $\mathscr{C}$'s objects, and whose arrows between these elements are induced by the arrows between the containing sets.

Now such a category can be constructed in more than one way. But if we don't want the derived category to forget about which elements belong to which sets, then a natural way to go would be to say that the objects of the derived category – which could be called the category of elements of $\mathscr{C}$ – are all the pairs $\langle A, a\rangle$ for $A \in \mathscr{C}$, $a \in A$. And then given elements $a \in A$, $b \in B$, whenever there is a $\mathscr{C}$-arrow $f\colon A \to B$ such that $f(a) = b$, we'll say that f is also an arrow from $\langle A, a\rangle$ to $\langle B, b\rangle$ in our new category. This derived category of elements in a sense unpacks what's going on inside the original category $\mathscr{C}$.

(ii) However, in the general case, $\mathscr{C}$'s objects need not be sets so need not have elements. But a functor $F\colon \mathscr{C} \to \mathbf{Set}$ gives us a diagram of $\mathscr{C}$ inside $\mathbf{Set}$, and of course the objects in the resulting diagram of $\mathscr{C}$ *do* have elements. So we can consider the category of elements of F's-diagram-of-$\mathscr{C}$, which – following the template in (i) – has as objects all the pairs $\langle FA, a\rangle$ for $A \in \mathscr{C}$, $a \in FA$. And then given elements $a \in FA$, $b \in FB$, whenever there is a $\mathbf{Set}$-arrow $Ff\colon FA \to FB$ such that $Ff(a) = b$, we'll say that Ff is also an arrow from $\langle FA, a\rangle$ to $\langle FB, b\rangle$ in our new category.

Now, we can streamline that. Instead of taking the objects to be pairs $\langle FA, a\rangle$ take them simply to be pairs $\langle A, a\rangle$ (but where, still, $a \in FA$). And instead of talking of the arrow $Ff\colon FA \to FB$ we can instead talk more simply of $f\colon A \to B$ (but where, still, $Ff(a) = b$). And with that streamlining – lo and behold! – we are back with the category $\mathbf{Elts}_{\mathscr{C}}(F)$, which is isomorphic to category of elements of F's-diagram-of-$\mathscr{C}$, and which – as convention has it – we'll call the category of elements of F, for short.

So the construction of $\mathbf{Elts}_{\mathscr{C}}(F)$ is tolerably natural.

Here is another way of thinking of this category. Let $*$ be some singleton in **Set**. Then what is the comma category $(* \downarrow F)$? Applying the definition of such categories given at the very end of §5.5, the objects of this category are pairs $\langle A, a\rangle$ where $A \in \mathscr{C}$ and $a\colon * \to FA$ is an arrow in **Set**. And the arrows of the category from $\langle A, a\rangle$ to $\langle B, b\rangle$ is a $\mathscr{C}$-arrow $f\colon A \to B$ such that $b = Ff \circ a$.

But *that* is just the definition of $\mathbf{Elts}_{\mathscr{C}}(F)$ except that we have traded in the requirement that a is *member* of FA for the requirement that a is an *arrow* $* \to FA$. But as we well know by now, members of a set are in bijective correspondence with such arrows from a fixed singleton, and from a categorial perspective we can treat members as such arrows (hence our using the same label 'a' here for both). Hence

Theorem 61. *For a given functor $F\colon \mathscr{C} \to \mathbf{Set}$, the category $\mathbf{Elts}_{\mathscr{C}}(F)$ is isomorphic to the comma category $(* \downarrow F)$.*

Having defined a category $\mathbf{Elts}_{\mathscr{C}}(F)$ for universal elements of $F\colon \mathscr{C} \to \mathbf{Set}$ to live in, we can finish by asking: how do we distinguish universal elements from other elements categorially? The answer is immediate from Defn. 47:

Theorem 62. *An object $I = \langle A, a\rangle \in \mathbf{Elts}_{\mathscr{C}}(F)$ is a universal element iff, for every object $E \in \mathbf{Elts}_{\mathscr{C}}(F)$ there is exactly one morphism $f\colon I \to E$, so I is initial in $\mathbf{Elts}_{\mathscr{C}}(F)$.*

But initial objects are unique up to unique isomorphism. Which, recalling what isomorphisms in $\mathbf{Elts}_{\mathscr{C}}(F)$ are, implies

Theorem 63. *If $\langle A, a\rangle$ and $\langle A', a'\rangle$ are universal elements for $F\colon \mathscr{C} \to \mathbf{Set}$, then there is a unique $\mathscr{C}$-isomorphism $f\colon A \to A'$ such that $F(f)(a) = a'$.*

(We can leave it as an exercise to dualize all the propositions in the last two sections.)

13 Initial and terminal objects

After the higher-level abstractions of recent chapters where we have been talking a lot about maps *between* categories (functors) or even about maps between these maps (natural transformations), we now get back to ground-level, and discuss various categorial notions that apply *inside* a category. The notions which we introduce in this and the next two chapters will have a family resemblance to each other, and they will in fact all turn out to be instances of *limits* (or dually, *colimits*) in a technical sense that gets defined in Chapter 16.

13.1 Initial and terminal objects, definitions and examples

We start with the simplest pair of cases:

Definition 49. The object I is an *initial* object of the category $\mathscr{C}$ iff, for every $X \in \mathscr{C}$, there is a unique arrow $I \to X$.

Dually, the object T is a *terminal* object of $\mathscr{C}$ iff, for every $X \in \mathscr{C}$, there is a unique arrow $X \to T$.

Some examples:

(1) In the poset $(\mathbb{N}, \leqslant)$ thought of as a category, zero is trivially the unique initial object and there is no terminal object. The poset $(\mathbb{Z}, \leqslant)$ has neither initial nor terminal objects.

(2) More generally, a poset-treated-as-a-category has an initial object iff the poset has a minimum. Dually for terminal objects/maxima.

(3) In **Set**, the empty set is an initial object (cf. the comment on Ex. 1 in §2.2).

And any singleton set $\{\star\}$ is a terminal object. (For if X has members, there's a unique **Set**-arrow which sends all the members to $\star$; while if X is empty, then there's a unique **Set**-arrow to any set, including $\{\star\}$).

(4) In $\mathbf{Set}_\star$ – the category of pointed sets, non-empty sets equipped with a distinguished member – each singleton is both initial *and* terminal. (A singleton's only member has to be the distinguished member in a pointed set. Arrows in $\mathbf{Set}_\star$ are functions which map distinguished elements to

distinguished element. Hence there can be one and only one arrow from a singleton to some some other pointed set.)

(5) In **Pos**, the category of posets, the empty poset is initial, and any singleton equipped with the only possible order relation on it (the identity relation!) is terminal.

(6) In **Rel**, the category of sets and relations, the empty set is both the sole initial and sole terminal object.

(7) In **Top**, the empty set (considered as a trivial topological space) is the initial object. Any one-point singleton space is a terminal object.

(8) In **Grp**, the trivial one-element group is an initial object (a group has to have at least one object, the identity; now recall that a group homomorphism sends identity elements to identity elements; so there is one and only one homomorphism from the trivial group to any given group G). The same one-element group is also terminal.

(9) In the category **Bool**, the trivial one-object algebra is terminal. While the two-object algebra on $\{0, 1\}$ familiar from propositional logic is initial – for a homomorphism of Boolean algebras from $\{0, 1\}$ to B must send 0 to the bottom object of B and 1 to the top object, and there's a unique map that does that.

(10) Recall: in the slice category $\mathscr{C}/X$ an object is a $\mathscr{C}$-arrow like $f : A \to X$, and an arrow from $f : A \to X$ to $g : B \to X$ is a $\mathscr{C}$-arrow $j : A \to B$ such that $g \circ j = f$ in $\mathscr{C}$. Consider the $\mathscr{C}/X$-object $1_X : X \to X$. A $\mathscr{C}/X$ arrow from $f : A \to X$ to 1_X is a $\mathscr{C}$-arrow $j : A \to X$ such that $1_X \circ j = f$, i.e. such that $j = f$ – which exists and is unique! So 1_X is terminal in $\mathscr{C}/X$.

(11) A universal element is initial in a category of elements $\mathbf{Elts}_{\mathscr{C}}(F)$ (by Theorem 62).

Such various cases show that a category may have zero, one or many initial objects, and (independently of that) may have zero, one or many terminal objects. Further, an object can be both initial and terminal.

There is, incidentally, a standard bit of jargon for the last case:

Definition 50. An object $O \in \mathscr{C}$ is a *null object* of the category $\mathscr{C}$ iff it is both initial and terminal.

13.2 Uniqueness up to unique isomorphism

A category $\mathscr{C}$, as we saw, may have no initial objects, or only one.

But suppose I and J are *both* initial objects in $\mathscr{C}$. By definition there must be unique arrows $f : I \to J$, and $g : J \to I$. Then $g \circ f$ is an arrow from I to itself. Another arrow from I to itself is the identity arrow 1_I. But since I is initial, there can only be one arrow from I to itself, so $g \circ f = 1_I$. Likewise $f \circ g = I_J$.

Hence the unique arrow f is an isomorphism. (Note this pattern of argument: we'll be using it a lot!)

Relatedly, suppose I is initial, and that there is an isomorphism $i\colon I \to J$. Then for any X, there is a unique arrow $f\colon I \to X$, and hence there is an arrow $f \circ i^{-1}\colon J \to X$. Now suppose we also have $g\colon J \to X$. Then $g \circ i\colon I \to X$, and so $g \circ i = f$, hence $(g \circ i) \circ i^{-1} = f \circ i^{-1}$, hence $g = f \circ i^{-1}$. In sum, for any X there is a unique arrow from J to X, thus J is also initial.

These arguments, and their duals, deliver

Theorem 64. *Initial objects, when they exist, are 'unique up to unique isomorphism': i.e. if $I, J \in \mathscr{C}$ are both initial, then there is a unique isomorphism $f\colon I \xrightarrow{\sim} J$. Dually for terminal objects.*

Further, if I is initial and $I \cong J$, then J is initial. Dually for terminal objects.

There is some sense in which category theory doesn't care about distinguishing isomorphic objects (though quite what this remark comes to we ought to reflect on rather more than we do in these Notes!). Hence it is common to introduce notation for an arbitrary initial and terminal objects, as follows:

Definition 51. We use '0' to denote an initial object of $\mathscr{C}$ (assuming one exists), and likewise '1' to denote a terminal object.

Three comments on this terminology:

(i) In **Set**, 0 is $\emptyset$, the only initial object – and $\emptyset$ is also the von Neumann ordinal 0. While the von Neumann ordinal 1 is $\{\emptyset\}$, i.e. a singleton, i.e. a terminal object 1. Which perhaps excuses the recycling of the notation.

(ii) We've also already used '1' in other ways – e.g. back in §5.4, it denoted a one-point topological space. But since such one-point spaces are terminal objects in **Top** our notation remains consistent.

(iii) Often, null objects (objects which are both initial and terminal) are alternatively called 'zero' objects: but that perhaps doesn't sit happily with using '0' for an initial object: for 0 (in the sense of an initial object) typically isn't a zero (in the sense of null) object.

13.3 Elements and generalized elements

(a) There's an obvious intuitive sense in which initial and terminal objects are 'limit cases'. Before turning to consider our next introductory example of a 'limit' in the following chapter, however, let's pause over our first examples a moment longer.

Take **Set**. For every $X \in$ **Set** there is a unique arrow from X *to* 1. What about arrows *from* 1 to X? An arrow $\vec{x}\colon 1 \to X$ is a set-function sending the member of the singleton 1 to some member $x \in X$. Trivially there is exactly one such arrow for any $x \in X$. So, in **Set**, we can think of talk of arrows $\vec{x}\colon 1 \to X$ as the categorial version of talking of elements of X. (We've been here before – see §§2.2, 3.2.)

Which motivates the following more general definition:

Definition 52. In a category $\mathscr{C}$ with a terminal object 1, an *element* or *point* of the $\mathscr{C}$-object X is an arrow $\vec{x}\colon 1 \to X$.

(In fact, the standard terminology for such an element is '*global* element', picking up from a paradigm example in topology – but we won't fuss about that.) Note, by the way, that elements $\vec{x}\colon 1 \to X$ are monic. For suppose $\vec{x} \circ f = \vec{x} \circ g$; then both f and g must be morphisms $Y \to 1$, for the same Y, hence must be the identical since 1 is terminal.

Definition 53. Suppose the category $\mathscr{C}$ has a terminal object, and for any $X, Y \in \mathscr{C}$, and arrows $f, g\colon X \to Y$, $f = g$ iff for all $\vec{x}\colon 1 \to X$, $f \circ \vec{x} = g \circ \vec{x}$. Then $\mathscr{C}$ is said to be *well-pointed.*

Think of it this way: in a well-pointed category, there are enough elements (points) to ensure that arrows which act identically on all relevant elements are indeed identical.

Theorem 65. Set *is well-pointed.* **Grp***, for example, is not.*

Proof. The claim about **Set** is immediate.

In **Grp**, for example, the only homomorphism from 1 (the one-element group) to a group X sends the only element of 1 to the identity element e of X: call this homomorphism $\vec{e}$. So, take two group homorphisms $f, g\colon X \to Y$ where $f \neq g$: still, for the only possible $\vec{e}$, both $f \circ \vec{e}$ and $g \circ \vec{e}$ send the sole element of 1 to the identity element of Y, so are equal. □

(b) We have just seen that, even when an arrow in a category is a function, acting the same way on elements need not imply being the same arrow. But suppose we introduce *this* notion:

Definition 54. A *generalized element* of $X \in \mathscr{C}$ (of shape S) is an arrow $e\colon S \to X$.

Trivially, for relevant $f, g\colon X \to S$, if for all $e\colon S \to X$ of any shape $f \circ e = g \circ e$, then in particular $f \circ I_X = g \circ I_X$, so indeed $f = g$. Hence arrows are indeed identical if and only if the act identically on all *generalized* elements.

Two more remarks. First, we just note that a generalized element of $X \in \mathscr{C}$ is the same as an object of the slice category $\mathscr{C}/X$.

Second, in terms of our new jargon the Yoneda $\mathcal{Y}$ embedding from (a locally small) $\mathscr{C}$ into $[\mathscr{C}^{op}, \mathbf{Set}]$ sends an object $X \in \mathscr{C}$ to $\mathscr{C}(-, X)$, and the latter is the functor which sends an object S to the set of generalized elements of X of shape S. So we can think of the Yoneda embedding as taking us from an object X to something that encodes all the information about what (generalized) elements of that object exist.

14 Products

Our next main topic will be a categorial treatment of products (as in Cartesian products). So the paradigm construction in **Set** takes sets X and Y and forms the set of ordered pairs of their elements. But what are ordered pairs? What is involved in pairing schemes more generally? We'll consider that sweeping question as our route in to a categorial treatment of products.

14.1 Pairing schemes

(a) Suppose for a moment that we are working in a theory of arithmetic and want to start talking about ordered pairs of natural numbers – perhaps we want to go on to use such pairs in constructing integers or rationals. Well, we can represent such pairs without taking on any new commitments by using *code-numbers*. For example, if we want a bijective coding between pairs of naturals and all the numbers, we could adopt the scheme of coding the ordered pair $\langle m, n\rangle$ by the single number $\{(m+n)^2+3m+n\}/2$. Or, if we don't insist on every number coding a pair, we could adopt the simpler policy of putting $\langle m, n\rangle =_{\text{def}} 2^m 3^n$. Relative to a given coding scheme, we can call such code-numbers $\langle m, n\rangle$ *pair-numbers* or, by a slight abuse of terminology, simply *pairs*: and we can refer to m as the first element of the pair, and n as the second element.

Why should this way of handling ordered pairs of natural numbers be regarded as somehow inferior to other, albeit more familiar, coding devices such as explicitly set-theoretic ones?

It might be said that (i) a single pair-number is really neither ordered nor a twosome; (ii) while a number m is a member of (or is one of) the pair m, n, a number can't be a genuine member of a pair-number $\langle m, n\rangle$; and in any case (iii) coding schemes are pretty arbitrary (e.g. we could equally well have used $3^m 5^n$ as a code for the pair m, n).

Which is all true. But of course we can lay *exactly* analogous complaints against e.g. the familiar Kuratowski definition of ordered pairs that we all know and love. This treats the ordered pair m, n as the set $\{\{m\}, \{m, n\}\}$. But (i) that set is not intrinsically ordered (after all, it is a *set*!), nor is it always two-membered (consider the case where $m = n$). (ii) Even when it is a twosome, its members are not the members of the pair: in standard set theories, m cannot be a member of $\{\{m\}, \{m, n\}\}$. And (iii) the construction again involves pretty arbitrary choices: thus $\{\{n\}, \{m, n\}\}$ or $\{\{\{m\}\}, \{\{m, n\}\}\}$ etc., etc., would have done just as well. On these counts, at any rate, coding pairs of numbers by using pair-numbers involves no worse a trick than coding them using Kuratowski's standard gadget.

There is indeed a rather neat symmetry between the adoption of pair numbers as representing ordered pairs of numbers and a familiar procedure adopted by the enthusiast for working in standard ZFC. For remember that pure ZFC knows only about pure sets. So to get natural numbers into the story at all – and hence to get Kuratowski pair-sets of natural numbers – the enthusiast for sets has to choose some convenient sequence of sets to implement the numbers (or to 'stand proxy' for numbers, 'simulate' them, 'play the role' of numbers, or even 'define' them – whatever your favourite way of describing the situation is). But someone who, for her purposes, has opted to play the game this way and is dealing with natural numbers by selecting some convenient sets to implement them, is hardly in a position to complain about someone else who, for his purposes, does the opposite and treats numbers as basic, and deals with ordered pairs of them by choosing some convenient code-numbers to implement *them*. Both theorists are in the implementation game.

It might be retorted that the Kuratowski trick at least has the virtue of being an all-purpose device, available not just when you want to talk about pairs of *numbers*, while e.g. the powers-of-primes coding is of more limited use. Again true. Similarly you can use sledgehammers to crack all sorts of things, while you can only use nutcrackers for nuts. But that's not particularly to the point if it happens to be nuts you currently want to crack, efficiently and with minimum resources. If we want to implement pairs of numbers without ontological inflation – e.g. in pursuing the project of 'reverse mathematics' – then pair-numbers are just the kind of thing we need.

(b) Let's now think a bit more formally about what it takes to have a way of pairing-up an object $x \in X$ with an object $y \in Y$.

We need some objects O to serve as ordered pairs, a pairing function that sends a given x and y to a pair $o \in O$, and (of course!) a couple of functions which allow us to recover x and y from o. And the point we've just been making is that maybe we shouldn't care too much about the 'internal' nature of the objects O, so long as we do have suitable associated pairing and unpairing functions which work in the right way (for example, pairing and then unpairing gets us back to where we started). Which motivates:

Definition 55. Suppose X, Y and O are sets of objects (these can be the same or different). Let $pr\colon X, Y \to O$ be a two-place function, while $\pi_1\colon O \to X$, and $\pi_2\colon O \to Y$, are one-place functions. Then $[O, pr, \pi_1, \pi_2]$ *form a pairing scheme for X with Y* iff

(a) $(\forall x \in X)(\forall y \in Y)(\pi_1(pr(x, y)) = x \wedge \pi_2(pr(x, y)) = y)$,

(b) $(\forall o \in O)\, pr(\pi_1 o, \pi_2 o) = o$.

O will be said to comprise the *pair-objects* of the pairing scheme, with pr the associated *pairing function*, while π_1 and π_2 are unpairing or *projection* functions.

Evidently, if O is the set of naturals of the form $2^m 3^n$ and $pr(m, n) = 2^m 3^n$, with $\pi_1 o$ ($\pi_2 o$) returning the exponent of 2 (3) in the factorization of o, then $[O, pr, \pi_1, \pi_2]$ form a pairing scheme for $\mathbb{N}$ with $\mathbb{N}$.

By the way, don't read too much into the square brackets: they need be read as no more than punctuation. (After all, we are in the business of characterizing pairs; so we don't want to presuppose e.g. that we already know about quadruples!)

Two simple facts about pairing schemes:

i. Different pairs of objects are sent by pr to different pair-objects. For suppose $pr(x, y) = pr(x', y')$. Then by (a) $x = \pi_1(pr(x, y)) = \pi_1(pr(x', y')) = x'$, and likewise $y = y'$.

ii. Note that by (b), pr is surjective. The 'unpairing' or 'projection' functions π_1 and π_2 are also surjective. For given $x \in X$, take any $y \in Y$ and put $o = pr(x, y)$. Then by (a), $x = \pi_1 o$. Likewise, given $y \in Y$ there is an $o \in O$ such that $y = \pi_2 o$.

So, in sum, pairing schemes basically work as you would expect.

As we'd also expect, a given pairing function fixes the two corresponding projection functions, and vice versa, in the following sense:

Theorem 66. *(1) If $[O, pr, \pi_1, \pi_2]$ and $[O, pr, \pi_1', \pi_2']$ are both pairing schemes for X with Y, then $\pi_1 = \pi_1'$ and $\pi_2 = \pi_2'$.*

(2) If $[O, pr, \pi_1, \pi_2]$ and $[O, pr', \pi_1, \pi_2]$ are both pairing schemes, then $pr = pr'$.

Proof. For (1), take any $o \in O$. There is some (unique) x, y such that $o = pr(x, y)$. Hence, applying (a) to both schemes, $\pi_1 o = x = \pi_1' o$. Hence $\pi_1 = \pi_1'$, and similarly $\pi_2 = \pi_2'$.

For (2), take any $x \in X$, $y \in Y$, and let $pr(x, y) = o$, so $\pi_1 o = x$ and $\pi_2 o = y$. Then by (b) applied to the second scheme, $pr'(\pi_1 o, \pi_2 o) = o$. Whence $pr'(x, y) = pr(x, y)$. $\square$

Further, there is a sense in which all schemes for pairing X with Y are equivalent. More carefully,

Theorem 67. *If $[O, pr, \pi_1, \pi_2]$ and $[O', pr', \pi_1', \pi_2']$ are both schemes for pairing X with Y, then there is a unique bijection $f \colon O \xrightarrow{\sim} O'$ such that the following* *diagram commutes:*

$$\begin{tikzcd} X,Y \arrow[r, "pr"] \arrow[dr, "pr'"'] & O \arrow[d, "f"] \\ & O' \end{tikzcd}$$

Putting it another way, there is a unique bijection f such that, if we pair x with y using pr (in the first scheme), use f to send the resulting pair-object o to o', and then retrieve elements using π'_1 and π'_2 (from the second scheme), we get back to the original x and y.

Proof. Define $f\colon O \to O'$ by putting $f(o) = pr'(\pi_1 o, \pi_2 o)$. Then it is immediate that $f(pr(x,y)) = pr'(x,y)$ as required to make the diagram commute.

To show that f is injective, suppose $f(o) = f(o')$, for $o, o' \in O$. Then we have $pr'(\pi_1 o, \pi_2 o) = pr'(\pi_1 o', \pi_2 o')$. Apply π'_1 to each side and then use principle (a), and it follows that $\pi_1 o = \pi_1 o'$. And likewise $\pi_2 o = \pi_2 o'$. Therefore $pr(\pi_1 o, \pi_2 o) = pr(\pi_1 o', \pi_2 o')$. Whence by condition (b), $o = o'$.

To show that f is surjective, take any $o' \in O'$. Then put $o = pr(\pi'_1 o', \pi'_2 o')$. By the definition of f, $f(o) = pr'(\pi_1 o, \pi_2 o)$; plugging the definition of o twice into the right hand side and simplifying using rules (a) and (b) confirms that $f(o) = o'$. Hence $f[O] = O'$.

So f is a bijection with the right properties. And since every $o \in O$ is $pr(x,y)$ for some x, y, the requirement that $f(pr(x,y)) = pr'(x,y)$ fixes f uniquely. $\square$

(c) Here's another simple theorem, to motivate the final definition in this section:

Theorem 68. *Suppose X, Y, O are sets of objects, and the functions $\pi_1 : O \to X$, $\pi_2 : O \to Y$ are such that there is a* unique *function $pr\colon X, Y \to O$ such that (a) $(\forall x \in X)(\forall y \in Y)(\pi_1(pr(x,y)) = x \wedge \pi_2(pr(x,y)) = y)$. Then $[O, pr, \pi_1, \pi_2]$ form a pairing scheme.*

Proof. We argue that the uniqueness of pr ensures that the pairing function is surjective, and then that its surjectivity implies that condition (b) from Defn. 55 holds as well as the given condition (a).

Suppose pr is not surjective. Then for some $o \in O$, there is no $x \in X, y \in Y$ such that $pr(x,y) = o$. So $pr(\pi_1 o, \pi_2 o) = o' \neq o$. Consider then function pr' which agrees with pr on all inputs except that $pr'(\pi_1 o, \pi_2 o) = o$. Then for all cases other than $x = \pi_1 o, y = \pi_2 o$ we still have $\pi_1(pr'(x,y)) = x \wedge \pi_2(pr'(x,y)) = y$, and by construction for the remaining case $\pi_1(pr'(\pi_1 o, \pi_2 o)) = \pi_1 o \wedge \pi_2(pr'(\pi_1 o, \pi_2 o)) = \pi_2 o$. So condition (a) holds for pr', where $pr' \neq pr$. Contraposing, if pr uniquely satisfies the condition, it is surjective.

Because pr is surjective, every $o \in O$ is $pr(x, y)$ for some x, y. But then by (a) $\pi_1 o = x \wedge \pi_2 o = y$, and hence $pr(\pi_1 o, \pi_2 o) = pr(x, y) = o$. Generalizing gives us (b). □

Now, informally, pairing up X with Y through a pairing scheme, gives us a product of X with Y. But we don't want to identify the resulting product simply with the set O (for it depends on the rest of the pairing scheme what role O plays). Our last theorem, however, makes the following an appropriate definition:

Definition 56. If X, Y are sets, then $[O, \pi_1, \pi_2]$ form *a product of X with Y*, where O is a set, and $\pi_1 \colon X \to O, \pi_2 \colon Y \to O$ are functions, so long as there is a unique two-place function $pr \colon X, Y \to O$ such that $(\forall x \in X)(\forall y \in Y)(\pi_1(pr(x, y)) = x \wedge \pi_2(pr(x, y)) = y)$.

Again, don't read too much into the brackets, and cf. Theorem 73.

14.2 Binary products, categorially

We have now characterized pairing schemes and the resulting products they create in terms of a set of objects O being the source and target of some appropriate morphisms, covered by the principles in Defn. 55. How a pairing scheme is concretely implemented – what the objects, what the morphisms are – is usually of little concern: what will matter is that we can show that a proposed scheme fits the definition, so works as advertised. And then we can forget the implementation details.

Which all looks highly categorial in spirit (see the preamble to Ch. 4). But our natural story isn't categorial as it stands. For a crucial ingredient, namely the pairing function $pr \colon X, Y \to O$, is a *two*-place function, while the arrows in a category always have just a single domain. What to do?

Suppose for a moment we are working in a well-pointed category like **Set**, where 'elements' in the sense of Defn. 52 do behave sufficiently like how elements intuitively should behave. In this case, instead of talking of elements $x \in X$ and $y \in Y$, we can talk of two arrows $\vec{x} \colon 1 \to X$ and $\vec{y} \colon 1 \to Y$. And suppose that for every such $\vec{x}$ and $\vec{y}$ there is a *unique* arrow $u \colon 1 \to O$ such the following commutes:

$$\begin{tikzcd}
 & 1 \arrow[dl, "\vec{x}"'] \arrow[d, dashed, "u"] \arrow[dr, "\vec{y}"] & \\
X & O \arrow[l, "\pi_1"'] \arrow[r, "\pi_2"] & Y
\end{tikzcd}$$

Our supposition ensures that there is a unique two-place function $\vec{x}, \vec{y} \mapsto u$ such that we always have $\pi_1 \circ u = \vec{x} \wedge \pi_2 \circ u = \vec{y}$. So in the light of our well-motivated Defn. 56, we can naturally say that $[O, \pi_1, \pi_2]$ form a product of X with Y.

So far so good. But this only works as we want in well-pointed categories with 'enough' elements. However, we know how to generalize to other categories – i.e. replace talk about elements (points) with talk of generalized elements. Which motivates the following definition:

Definition 57. In any category $\mathscr{C}$, *a (binary) product* $[O, \pi_1, \pi_2]$ *for the objects* X *with* Y is an object O together with 'projection' arrows $\pi_1 \colon O \to X, \pi_2 \colon O \to Y$, such that for any object S and arrows $f_1 \colon S \to X$ and $f_2 \colon S \to Y$ there is always a unique arrow arrow $u \colon S \to O$ such that the following diagram commutes:

$$\begin{tikzcd} & S \arrow[dl, "f_1"'] \arrow[d, dashed, "u"] \arrow[dr, "f_2"] & \\ X & O \arrow[l, "\pi_1"'] \arrow[r, "\pi_2"] & Y \end{tikzcd}$$

This unique mediating arrow u is often more helpfully labelled '$\langle f_1, f_2 \rangle$' or simply '(f_1, f_2)' (but don't overinterpret those notations!).

Note, by the way, that we are falling into the following common

Convention. In a category diagram, we use a dashed arrow $\dashrightarrow$ to indicate an arrow which is uniquely fixed by the requirement that the diagram commutes.

Here's a very slightly different way of putting things (sometimes just geometrically distorting/rotating/reflecting a diagram can be illuminating!). Let's say

Definition 58. A *wedge to* X *and* Y (in category $\mathscr{C}$) is an object $S \in \mathscr{C}$ and a pair of arrows $f_1 \colon S \to X$, $f_2 \colon S \to Y$.

Then a wedge $\begin{tikzcd}[row sep=small] & X \\ O \arrow[ur, "\pi_1"] \arrow[dr, "\pi_2"'] & \\ & Y \end{tikzcd}$ is a product for X with Y iff for any other wedge

$\begin{tikzcd}[row sep=small] & X \\ S \arrow[ur, "f_1"] \arrow[dr, "f_2"'] & \\ & Y \end{tikzcd}$ to X and Y, there exists a unique morphism u such that the following

diagram commutes:

$$\begin{tikzcd}
 & & X \\
S \arrow[r, dashed, "u"] \arrow[urr, bend left, "f_1"] \arrow[drr, bend right, "f_2"'] & O \arrow[ur, "\pi_1"] \arrow[dr, "\pi_2"'] & \\
 & & Y
\end{tikzcd}$$

We can say f_1 'factors' as $\pi_1 \circ u$ and f_2 as $\pi_2 \circ u$, and so the whole wedge from S into X and Y *(uniquely) factors through* the product via u. And the definition of a product, now using the notion of wedges, can captured as follows. First, we say:

Definition 59. Given a category $\mathscr{C}$ and objects $X, Y \in \mathscr{C}$, then the derived wedge category $\mathscr{C}_{W(XY)}$ has as objects all wedges $[O, f_1, f_2]$ to X, Y. And an arrow from $[O, f_1, f_2]$ to $[O', f_1', f_2']$ is a $\mathscr{C}$-arrow $g\colon O \to O'$ such that the resulting triangles commute: i.e. $f_j = f_j' \circ g$.

It is easily checked that $\mathscr{C}_{W(XY)}$ is indeed a category, so this definition is in good order. And then our previous definition of a product comes to this:

Definition 57. (Alternative version). A product of X with Y in $\mathscr{C}$ is a terminal object of the derived category $\mathscr{C}_{W(XY)}$.

Let's have some examples.

(1) In **Set**, as you would certainly hope, the usual Cartesian product, the set $X \times Y$ of Kuratowksi pairs $\langle x, y\rangle$ of elements from X and Y, together with the obvious projection functions $\langle x, y\rangle \xmapsto{\pi_1} x$ and $\langle x, y\rangle \xmapsto{\pi_2} y$ form a binary product.

Let's just confirm this. Suppose we are given any set S and functions $f_1\colon S \to X$ and $f_2\colon S \to Y$. Then if, for $s \in S$, we put $u(s) = \langle f_1(s), f_2(s)\rangle$, the diagram evidently commutes. Now trivially, for any pair $p \in X \times Y$, $p = \langle \pi_1 p, \pi_2 p\rangle$. Hence if $u'\colon S \to X \times Y$ is another candidate for completing the diagram, $u(s) = \langle f_1(s), f_2(s)\rangle = \langle \pi_1 u'(s), \pi_2 u'(s)\rangle = u'(s)$. So u is unique.

That paradigm case motivates the following

Notation. We will often use the notation $X \times Y$ for the object O in a binary product $[O, \pi_1, \pi_2]$ for X with Y.

Continuing our examples:

(2) In **Grp**, you can construct a product of the groups $(G, \cdot)$ and $(H, \odot)$ by taking the group $(G \times H, \diamond)$ where $G \times H$ is the usual Cartesian product of the underlying sets and the group operation is defined component-wise, so that $\langle g, h\rangle \diamond \langle g', h'\rangle = \langle g \cdot g', h \odot h'\rangle$, and then equipping this group with

the obvious projection functions from $G \times H$ to G (resp. H) which send $\langle g, h\rangle$ to g (resp. h).

(3) Take a poset $(P, \preccurlyeq)$ considered as a category (so there is an arrow $p \to q$ iff $p \preccurlyeq q$). Then a product of p and q would be an object c such that $c \preccurlyeq p, c \preccurlyeq q$ and such that for any object d with arrows from it to p and q, i.e. any d such that $d \preccurlyeq p, d \preccurlyeq q$, there is a unique arrow from d to c, i.e. $d \preccurlyeq c$. That means the product of p and q must be their greatest lower bound (equipped with the obvious two arrows).

Since pairs of objects in posets need not in general have greatest lower bounds, that goes to show that a category in general need not have products.

(4) Here's a new example of a category, call it $\mathbf{Prop}_L$ – its objects are propositions, wffs of a given first-order language $\mathcal{L}$, and there is a unique arrow from X to Y iff $X \vDash Y$, i.e. iff X semantically entails Y. The reflexivity and transitivity of semantic entailment means we get the identity and composition laws which ensure that this is a category.

In this case, one product of X with Y will be the conjunction $X \wedge Y$ (with the obvious projections $X \wedge Y \to X$, $X \wedge Y \to Y$).

(5) In **Cat**, the category of small categories, the product category $\mathscr{C} \times \mathscr{D}$ with its obvious projection functors, as defined in §5.3 is indeed a product of $\mathscr{C}$ with $\mathscr{D}$

14.3 Uniqueness up to unique isomorphism

We say:

Definition 60. A category $\mathscr{C}$ *has binary products* if for all objects $X, Y \in \mathscr{C}$, there exists a product of X with Y.

But of course, products need not exist for arbitrary objects X and Y in a given category; and when they do, they need not be strictly unique. However, when they exist, they *are* 'unique up to unique isomorphism'. That is to say,

Theorem 69. *If both $[O, \pi_1, \pi_2]$ and $[O', \pi'_1, \pi'_2]$ are products for X with Y in the category $\mathscr{C}$, then there is a unique isomorphism $f\colon O \xrightarrow{\sim} O'$ commuting with the projection arrows.*

Note the statement of the theorem carefully. It is *not* being claimed that there is a unique isomorphism between any product objects O and O'. That's false. (For example, in **Set**, take the product object $X \times X$: there are two isomorphisms between it and itself, given by the maps $\langle x, x'\rangle \mapsto \langle x, x'\rangle$, and $\langle x, x'\rangle \mapsto \langle x', x\rangle$.) There claim is, to repeat, that there is a unique isomorphism between any product objects O and O' which respects their associated projection arrows.

Proof. Start with a simple observation. Since $[O, \pi_1, \pi_2]$ is a product, every wedge factors uniquely through it, including itself. In other words, there is a unique u such that this diagram commutes:

$$\begin{tikzcd} & O \arrow[dl, "\pi_1"'] \arrow[d, dashed, "u"] \arrow[dr, "\pi_2"] & \\ X & O \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & Y \end{tikzcd}$$

But evidently putting 1_O for the central arrow trivially makes the diagram commute. So by the uniqueness requirement we know that

(i) Given an arrow $u\colon O \to O$, if $\pi_1 \circ u = \pi_1$ and $\pi_2 \circ u = \pi_2$, then $u = 1_O$.

Now, since $[O', \pi'_1, \pi'_2]$ is a product, $[O, \pi_1, \pi_2]$ has to uniquely factor through it:

$$\begin{tikzcd} & O \arrow[dl, "\pi_1"'] \arrow[d, dashed, "u"] \arrow[dr, "\pi_2"] & \\ X & O' \arrow[l, "\pi'_1"] \arrow[r, "\pi'_2"'] & Y \end{tikzcd}$$

In other words, there is a unique $f\colon O \to O'$ commuting with the projection arrows, i.e. such that

(ii) $\pi'_1 \circ f = \pi_1$ and $\pi'_2 \circ f = \pi_2$.

And since $[O, \pi_1, \pi_2]$ is also a product, the other wedge has to uniquely factor through *it*. That is to say, there is a unique $g\colon O' \to O$ such that

(iii) $\pi_1 \circ g = \pi'_1$ and $\pi_2 \circ g = \pi'_2$.

Whence,

(iv) $\pi_1 \circ g \circ f = \pi'_1 \circ f = \pi_1$ and $\pi_2 \circ g \circ f = \pi_2$.

But $g \circ f\colon O \to O$, from which it follows – given our initial observation (i) – that

(v) $g \circ f = 1_O$

The situation with the wedges is symmetric so we also have

(vi) $f \circ g = 1_{O'}$

Hence f is an isomorphism. $\square$

14.4 'Universal mapping properties'

Let's pause for a moment. We have defined a binary product for X with Y categorially as a special sort of wedge to X and Y.

Now, that doesn't fix products absolutely – but we have now seen that they are 'unique up to unique isomorphism'. And what makes a wedge a product for X with Y is that it has a certain *universal property* – i.e. *any* other wedge to X and Y factors uniquely through a product wedge via a unique *map*.

We can say, then, that products are defined by a *universal mapping property*. We've already met other examples: terminal and initial objects are defined by how any other object has a unique map to or from them. We'll meet lots more cases soon.

We could attempt a formal definition of what it is to be defined by a universal mapping property. But little would be gained at this stage of the game. Take the notion as an informal gesture towards a common pattern of definition which we can readily recognize when we come across it.

14.5 More properties of binary products

(a) We next verify some more respects in which binary products behave just as you would expect. First, if arrows into a product factor through the product the same way, they are equal. In other words,

Theorem 70. *Given a product* $[X \times Y, \pi_1, \pi_2]$ *and arrows* $S \overset{f}{\underset{g}{\rightrightarrows}} X \times Y$, *then, if* $\pi_1 \circ f = \pi_1 \circ g$ *and* $\pi_2 \circ f = \pi_2 \circ g$, *it follows that* $f = g$.

Proof. A diagram says it all!

$$\begin{tikzcd} & S \arrow[dl, "\pi_1 \circ f / \pi_1 \circ g"'] \arrow[d, shift right, "f"'] \arrow[d, shift left, "g"] \arrow[dr, "\pi_2 \circ f / \pi_2 \circ g"] & \\ X & X \times Y \arrow[l, "\pi_1"'] \arrow[r, "\pi_2"] & Y \end{tikzcd}$$

The wedge $X \leftarrow S \rightarrow Y$ factors uniquely through $X \times Y$, and hence $f = g$. □

Theorem 71. *Given parallel arrows* $S \overset{f_1}{\underset{f_2}{\rightrightarrows}} X$, *with* $f_1 \neq f_2$, *there are (at least) four distinct arrows* $S \rightarrow X \times X$.

Proof. By definition of the product, for each pair of indices $i, j \in \{1, 2\}$ there is a unique map $\langle f_i, f_j \rangle$ which makes the product diagram commute,

$$\begin{tikzcd}
 & S \arrow[dl, "f_i"'] \arrow[d, dashed, "\langle f_i, f_j \rangle"] \arrow[dr, "f_j"] & \\
X & X \times X \arrow[l, "\pi_1"'] \arrow[r, "\pi_2"] & X
\end{tikzcd}$$

It is immediate that if $\langle f_i, f_j \rangle = \langle f_k, f_l \rangle$, then $i = k, j = l$. So each of the four different pairs of indices tally different arrows $\langle f_i, f_j \rangle$. $\square$

Theorem 72. *In a category with a terminal object* 1 *and with the relevant products,*

(1) $1 \times X \cong X \cong X \times 1$
(2) $X \times Y \cong Y \times X$
(3) $X \times (Y \times Z) \cong (X \times Y) \times Z$

Proof. (1) Evidently the wedge $X \xleftarrow{1_X} X \xrightarrow{f} 1$ exists for some unique f since 1 is terminal. Hence if $X \times 1$ is a product, there is a unique u such that this commutes,

$$\begin{tikzcd}
 & & X \\
X \arrow[urr, bend left, "1_X"] \arrow[r, dashed, "u"] \arrow[drr, bend right, "f"'] & X \times 1 \arrow[ur, "\pi_1"] \arrow[dr, "\pi_2"'] & \\
 & & 1
\end{tikzcd}$$

Hence, in particular, $\pi_1 \circ u = 1_X$.

Now consider the arrow $u \circ \pi_1 \colon X \times 1 \to X \times 1$. We have

(a) $\pi_1 \circ (u \circ \pi_1) = (\pi_1 \circ u) \circ \pi_1 = \pi_1$.
(b) $\pi_2 \circ (u \circ \pi_1) = \pi_2$ (both sides being arrows from $X \times 1$ to the terminal object 1, hence necessarily equal).

We now appeal to the principle (i) at the beginning of the proof of Theorem 69 to conclude $u \circ \pi_1 = 1_{X \times 1}$.

Hence u has a two-sided inverse, i.e. is an isomorphism, which gives us $X \cong X \times 1$. Similarly for the other half of (1).

(2) If $[X \times Y, \pi_1 \colon X \times Y \to X, \pi_2 \colon X \times Y \to Y]$ is a product of X with Y, then $[X \times Y, \pi_2, \pi_1]$ is obviously a product of Y with X, and hence – by Theorem 69 – there is a unique isomorphism between the object in that product and the object $Y \times X$ of any other product of Y with X.

(3) It is a just-about-useful reality check to prove this by appeal to our initial definition of a product, using brute force. You are invited to try! But we give a slicker proof of this last part in §14.8. $\square$

(b) We remark in passing on a special case:

Definition 61. In a category with products, the wedge $X \xleftarrow{1_X} X \xrightarrow{1_X} X$ must factor uniquely through the product $X \times X$ via an arrow $u\colon X \to X \times X$. That unique arrow u is *the diagonal morphism* on X, and so we will notate it 'δ_X'.

In **Set**, δ_X is the function that sends an element $x \in X$ to $\langle x, x\rangle$ (hence the label 'diagonal').

(c) Suppose we have a pair of parallel composite arrows built up using the same projection arrow like this: $X \times Y \xrightarrow{\pi_1} X \overset{f}{\underset{g}{\rightrightarrows}} X'$. Then – the thought might go – because the projection arrows here just 'throw away' the second component of pairs living in $X \times Y$, and all the real action happens on X, then if $f \circ \pi_1 = g \circ \pi_1$, we should also have $f = g$. In other words, we might be tempted to suppose that projection arrows in products are always right-cancellable, i.e. epic.

It is worth remarking that this is wrong. Here's a brute-force counterexample. Consider the mini category with just four objects together with the following diagrammed arrows (labelled suggestively but noncommittally), plus all identity arrows, and the necessary two composites.

$$X' \overset{f}{\underset{g}{\leftleftarrows}} X \xleftarrow{\pi_1} V \xrightarrow{\pi_2} Y$$

If that is all the data we have to go on, we can consistently stipulate that in this mini-category $f \neq g$ but $f \circ \pi_1 = g \circ \pi_1$. Now, there is only one wedge of the form $X \longleftarrow\ ?\ \longrightarrow Z$, so trivially all wedges of that shape uniquely factor through it. In other words, the wedge $X \xleftarrow{\pi_1} V \xrightarrow{\pi_2} Y$ is trivially a product and π_1 is indeed a projection arrow. But by construction it isn't epic.

(d) Lastly in this section, we prove a theorem which gives us another characterization of products (at least for locally small categories). First we need a definition.

Definition 62. Suppose $\mathscr{C}$ is locally small, and $O, X, Y \in \mathscr{C}$, with arrows $x\colon O \to X$ and $y\colon O \to Y$. Then $\mathscr{C}(-,X) \times \mathscr{C}(-,Y)\colon \mathscr{C}^{op} \to \mathbf{Set}$ is the functor which sends O to the set $\mathscr{C}(O,X) \times \mathscr{C}(O,Y)$ (invoking the usual cartesian product of sets), and sends an arrow $f\colon O' \to O$ to the function $\langle x, y\rangle \mapsto \langle x \circ f, y \circ f\rangle$.

It is readily checked that this does indeed well-define a functor. Then

Theorem 73. *$[O, \pi_1, \pi_2]$ is a product of X with Y in the locally small $\mathscr{C}$ iff $\langle O, \langle \pi_1, \pi_2\rangle\rangle$ is a universal element of the functor $F = \mathscr{C}(-,X) \times \mathscr{C}(-,Y)$.*

Proof. This is just a straightforward application of Defn. 47. $\langle O, \langle \pi_1, \pi_2 \rangle \rangle$ is a universal element of that functor so long as, for each $S \in \mathscr{C}^{op}$ and each object in FS, i.e. each $\langle f_1, f_2 \rangle$ such that $f_1 \in \mathscr{C}(S, X), f_2 \in \mathscr{C}(S, Y)$, there is a unique map $u \colon S \to O$ (remember the functor is from $\mathscr{C}^{op}$) such that the map $\langle x, y \rangle \mapsto \langle x \circ u, y \circ u \rangle$ sends $\langle \pi_1, \pi_2 \rangle$ to $\langle f_1, f_2 \rangle$, i.e. so long as $\pi_1 \circ u = f_1$, $\pi_2 \circ u = f_2$. Which is exactly the condition for $[O, \pi_1, \pi_2]$ being a product of X with Y. $\square$

(If we really wanted to, we could apply this to bigger categories by using the device of §11.4 and considering functors from $\mathscr{C}^{op}$ to a big category of collections SET. But again, we won't follow up this idea.)

14.6 Two-place functions and arrows from products

(a) Let's step back from category theory just for a moment, and think about a familiar line on multi-place functions.

A one-place total function from numbers to numbers, like the successor function or the square-of function, is a function $f \colon \mathbb{N} \to \mathbb{N}$. What about a two-place total function from numbers to numbers, like addition or multiplication? "Easy-peasy: that is a function of the type $f \colon \mathbb{N}^2 \to \mathbb{N}$."

But hold on! $\mathbb{N}^2$, i.e. $\mathbb{N} \times \mathbb{N}$, is the cartesian product of $\mathbb{N}$ with itself, i.e. is the set of ordered pairs of numbers: and an ordered pair is *one* thing not two things. So a function from $\mathbb{N}^2$ to $\mathbb{N}$ – according to the set-theoretic orthodoxy – is in fact a *unary function* that maps *one* argument, an ordered pair object, to a value, and is *not* (as we initially wanted) a binary function mapping two arguments to a value.

"Ah, don't be so pernickety! Given two objects, we can find a pair-object that codes for them – we usually choose a Kuratowski pair – and we can without loss trade in a function from two objects to a value to a related function from the corresponding pair-object to the same value."

Yes, sure, we *can* do that. And standard notational choices can make the trade invisible. For suppose we adopt the modern convention of using '(m, n)' as our notation for the ordered pair of m with n, then '$f(m, n)$' can be parsed either way, as representing a two-place function $f(\cdot, \cdot)$ with arguments m and n or as a corresponding one-place function $f\cdot$ with the single argument (m, n). But the fact that trade between the two-place and the one-place function is notationally glossed over doesn't mean that it isn't being made. And the fact that the trade *can* be made (even staying within arithmetic, using an arithmetic pairing function) is a *result* and not quite a triviality. So if we are doing things from scratch – including *proving* that there is a pairing scheme that matches two things with one thing in such a way that we can then extract the two objects we started with – then we do need to at least start by talking about two-place functions proper. For example, if we want to confine ourselves to pure arithmetic, we show how to construct a pairing function *from the ordinary school-room two-place addition and multiplication functions*, not from some surrogate one-place functions!

Versions of type theory also trade in two-place functions for unary functions but in a different way, treating addition for example as a function of the type $N \to (N \to N)$; this is a unary function which takes one number (of type N) to output another unary *function* (of type $N \to N$). So this time we get from two numbers as input to a numerical output by feeding the first number to a function which delivers another function as output and then feeding the second number to the second function. This so-called 'currying' trick of course also works to eliminate two-place functions: that is to say, it is adequate for certain particular formal purposes. But is the type-shifting supposed to reveal what two-place functions are 'really'?

Here's a revealing quote from *A Gentle Introduction to Haskell* on the haskell.org site (Haskell being one those programming languages where what we might think of naturally as binary functions are curried):

> Consider this definition of a function which adds its two arguments:
>
> add :: Integer → Integer → Integer
> add x y = x + y

So we have the declaration of type – we are told that add sends a number to a function from numbers to numbers – and we are then told how the curried function acts ... but how? By appeal, of course, to our prior understanding of the familiar school-room two-place function! Ordinary two-place functions are essential rungs on the ladder by which we have to climb to an understanding of what's going on in the likes of Haskell (even if we throw away the ladder after we've climbed it).

(b) How does category theory stand on two-place functions (and multi-place functions more generally)? As we noted before, at ground level, all arrows are from an object (one of them) to an object. So we don't get 'native' binary functions. Nor do we get straightforwardly get currying within a category, at least in the sense that we won't have arrows inside a category from objects to arrows of that category. The obvious trick which allows us to model or represent two-place functions categorially is, rather, a version of the set-theoretic one. We can have an arrow of the kind $f\colon X \times Y \to S$ where, equipping the object with suitable projection arrows, $[X \times Y, \pi_1, \pi_2]$ is a product of X with Y, and such an f will do duty for a two-place function from an object in X and an object in Y to a value in S.

14.7 Maps between two products

(a) Let's now pause to remark on the case where we have two arrows $f\colon X \to X'$, $g\colon Y \to Y'$, and we want to characterize an arrow $f \times g\colon X \times Y \to X' \times Y'$ which works component-wise – i.e., putting it informally, it sends the product of elements x and y to the product of $f(x)$ and $g(y)$. Then, in more categorial terms, we are assuming that $f \times g$ is such that the following diagram commutes:

$$\begin{tikzcd}
X \arrow[d, "f"'] & X \times Y \arrow[l, "\pi_1"'] \arrow[r, "\pi_2"] \arrow[d, "f \times g"] & Y \arrow[d, "g"] \\
X' & X' \times Y' \arrow[l, "\pi_1'"'] \arrow[r, "\pi_2'"] & Y'
\end{tikzcd}$$

Note, however, that the vertical arrow is then a mediating arrow from the wedge $X' \xleftarrow{f \circ \pi_1} X \times Y \xrightarrow{g \circ \pi_2} Y'$ through the product $X' \times Y'$. Therefore $f \times g$ is indeed fixed uniquely, and equals $\langle f \circ \pi_1, g \circ \pi_2 \rangle$ (see Defn. 57). That warrants the following definition as in good order:

Definition 63. Given the arrows $f\colon X \to X'$, $g\colon Y \to Y'$, and the products $[X \times Y, \pi_1, \pi_2]$ and $[X' \times Y', \pi_1', \pi_2']$, then $f \times g\colon X \times Y \to X' \times Y'$ is the unique arrow such that $\pi_1' \circ f \times g = f \circ \pi_1$ and $\pi_2' \circ f \times g = g \circ \pi_2$.

(b) Here's a special case: sometimes we have an arrow $f\colon X \to X'$ and we want to define an arrow from $X \times Y$ to $X' \times Y$ which applies f to the first component of a product and leaves the second alone. Then $f \times 1_Y$ will do the trick.

It is tempting to suppose that if we have parallel maps $f, g\colon X \to X'$ and $f \times 1_Y = g \times 1_Y$, then $f = g$. But this actually fails in some categories – for example, in the toy category we met in §14.5 (c), whose only arrows are as diagrammed

$$\begin{tikzcd}
X' & X \arrow[l, shift right, "f"'] \arrow[l, shift left, "g"] & V \arrow[l, "\pi_1"'] \arrow[r, "\pi_2"] & Y
\end{tikzcd}$$

together with the necessary identities and composites, and where by stipulation $f \neq g$ but $f \circ \pi_1 = g \circ \pi_1$ (and hence $f \times 1_Y = g \times 1_Y$).

(c) Later, we will also need the following (rather predictable) general result:

Theorem 74. *Assume that there are arrows*

$$\begin{tikzcd}
X \arrow[r, "f"] & X' \arrow[r, "j"] & X'' \\
Y \arrow[r, "g"] & Y' \arrow[r, "k"] & Y''
\end{tikzcd}$$

Assume also that there are products $[X \times Y, \pi_1, \pi_2]$, $[X' \times Y', \pi_1', \pi_2']$ and $[X'' \times Y'', \pi_1'', \pi_2'']$. Then $(j \times k) \circ (f \times g) = (j \circ f) \times (k \circ g)$.

Proof. By the defining property of arrow products applied to three different products we get,

$$\pi_1'' \circ (j \times k) \circ (f \times g) = j \circ \pi_1' \circ (f \times g) = j \circ f \circ \pi_1 = \pi_1'' \circ (j \circ f) \times (k \circ g).$$

Similarly

$$\pi_2'' \circ (j \times k) \circ (f \times g) = \pi_2'' \circ (j \circ f) \times (k \circ g)$$

The theorem then immediately follows by Theorem 70. □

14.8 Products generalized

(a) So far we have talked of binary products. But we can generalize in obvious ways. For example,

Definition 64. In any category $\mathscr{C}$, *a ternary product* $[O, \pi_1, \pi_2, \pi_3]$ *for the objects* X_1, X_2, X_3 is an object O together with projection arrows $\pi_i \colon O \to X_i$ $(i = 1, 2, 3)$ such that for any object S and arrows $f_i \colon S \to X_i$ there is always a unique arrow arrow $u \colon S \to O$ such that $f_i = \pi_i \circ u$.

And then, exactly as we would expect, using the same proof ideas as in the binary case, we can prove

Theorem 75. *If both the ternary products* $[O, \pi_1, \pi_2, \pi_3]$ *and* $[O', \pi_1', \pi_2', \pi_3']$ *exist for* X_1, X_2, X_3 *in the category* $\mathscr{C}$*, then there is a unique isomorphism* $f \colon O \xrightarrow{\sim} O'$ *commuting with the projection arrows.*

We now note that if $\mathscr{C}$ has binary products for all pairs of objects, then it has ternary products too, for

Theorem 76. $(X_1 \times X_2) \times X_3$ *together with the obvious projection arrows forms a ternary product of* X_1, X_2, X_3.

Proof. Assume $[X_1 \times X_2, \pi_1, \pi_2]$ is a product of X_1 with X_2, and $[(X_1 \times X_2) \times X_3, \rho_1, \rho_2]$ is a product of $X_1 \times X_2$ with X_3.

Take any object S and arrows $f_i \colon S \to X_i$. By our first assumption, (a) there is a unique $u \colon S \to X_1 \times X_2$ such that $f_1 = \pi_1 \circ u$, $f_2 = \pi_2 \circ u$. So by our second assumption (b) there is then a unique $v \colon S \to (X_1 \times X_2) \times X_3$ such that $u = \rho_1 \circ v$, $f_3 = \rho_2 \circ v$.

Therefore $f_1 = \pi_1 \circ \rho_1 \circ v$, $f_2 = \pi_2 \circ \rho_1 \circ v$, $f_3 = \rho_2 \circ v$

So now consider $[(X_1 \times X_2) \times X_3, \pi_1 \circ \rho_1, \pi_2 \circ \rho_1, \rho_2]$. This, we claim, is indeed a ternary product of X_1, X_2, X_3. We've just proved that S and arrows $f_i \colon S \to X_i$ factor through the product via the arrow v. It remains to confirm v's uniqueness in this new role.

Suppose we have $w \colon S \to (X_1 \times X_2) \times X_3$ where $f_1 = \pi_1 \circ \rho_1 \circ w$, $f_2 = \pi_2 \circ \rho_1 \circ w$, $f_3 = \rho_2 \circ w$. Then $\rho_1 \circ w \colon S \to X_1 \times X_2$ is such that $f_1 = \pi_1 \circ (\rho_1 \circ w)$, $f_2 = \pi_2 \circ (\rho_1 \circ w)$. Hence by (a), $u = \rho_1 \circ w$. But now invoking (b), that together with $f_3 = \rho_2 \circ w$ entails $w = v$. □

Evidently, an exactly similar argument will show that $X_1 \times (X_2 \times X_3)$ together with the obvious projection arrows forms a ternary product of X_1, X_2, X_3. And that, combined with Theorem 75, tells us that $X_1 \times (X_2 \times X_3) \cong (X_1 \times X_2) \times X_3$, so giving us a very much neater proof of the last part of Theorem 72.

(b) What goes for ternary products goes for n-ary products defined in a way exactly analogous to Defn. 64. And if $\mathscr{C}$ has binary products for all pairs of objects it will have quaternary products such as $((X_1 \times X_2) \times X_3) \times X_4$, quinary products, and n-ary products more generally, for any finite $n \geq 2$.

To round things out, how do things go for the nullary and unary cases?

Following the same pattern of definition, a *nullary* product in $\mathscr{C}$ would be an object O together with *no* projection arrows, such that for any object S there is a unique arrow $u\colon S \to O$. Which is just to say that a nullary product is a terminal object of the category.

And a unary product of X would be an object O and a single projection arrow $\pi_1\colon O \to X$ such that for any object S and arrow $f\colon S \to X$ there is a unique arrow $u\colon S \to O$ such that $\pi \circ u = f$. Putting $O = X$ and $\pi = 1_X$ evidently fits the bill. So the basic case of a unary product of X is not quite X itself, but rather X equipped with its identity arrow (and like any product, this is unique up to unique isomorphism). Trivially, all unary products exist in all categories.

In sum, if we say

Definition 65. A category $\mathcal{C}$ *has all finite products* iff $\mathcal{C}$ has n-ary products for any n objects for all $n \geq 0$,

then our preceding remarks establish

Theorem 77. *A category $\mathcal{C}$ has all finite products iff $\mathcal{C}$ has a terminal object and a binary product for any pair of objects.*

(c) We can generalize further in the obvious way, beyond finite products to infinite cases.

Definition 66. In any category $\mathscr{C}$, the product of the $\mathscr{C}$-objects X_i (for indices $i \in I$), if it exists, is an object O together with projection arrows $\pi_i\colon O \to X_i$ (for $i \in I$) such that for any object S and arrows $f_i\colon S \to X_i$ (again for $i \in I$), there is always a unique arrow arrow $u\colon S \to O$ such that $f_i = \pi_i \circ u$. (We can use '$\prod\limits_{i \in I} X_i$' to notate such an O.)

As before, these generalized products will be unique up to unique isomorphism. And we will say

Definition 67. A category $\mathcal{C}$ *has all small products* iff for any set's worth of $\mathscr{C}$-objects – i.e. any objects $\mathscr{C}$-objects X_i, for $i \in I$, where I is some set – the objects in the set have a product.

Obviously, a category which has all small products will a fortiori have all finite products; and equally obviously the reverse does not hold (think, for example, of the category **FinSet** of finite sets).

15 Coproducts and (co)equalizers

Let's note a common terminological device:

Convention. For many kinds of categorially defined widget, a *co-widget* of the category $\mathcal{C}$ is a widget of $\mathcal{C}^{op}$: co-widgets are dual to widgets.

For example, we could (and a few do) call initial objects 'co-terminal'. Likewise we could (and a very few do) call sections 'co-retractions'. True, there is a limit to this sort of thing – no one, as far as I know, talks e.g. of 'co-monomorphisms' (instead of 'epimorphisms'). But the convention is used quite widely. In particular, it is absolutely standard to talk of the duals of products as 'co-products', or indeed more commonly, dropping the hyphen, 'coproducts'.

15.1 Coproducts

The definition of a coproduct is immediately obtained, then, by reversing all the arrows in our definition of products. Thus:

Definition 68. In any category $\mathscr{C}$, *a (binary) coproduct* $[O, \iota_1, \iota_2]$ *for the objects* X *with* Y is an object O together with 'injection' arrows $\iota_1 \colon X \to O, \iota_2 \colon Y \to O$, such that for any object S and arrows $f_1 \colon X \to S$ and $f_2 \colon Y \to S$ there is always a unique arrow arrow $v \colon O \to S$ such that the following diagram commutes:

$$\begin{tikzcd}
 & S & \\
X \arrow[ur, "f_1"] \arrow[r, "\iota_1"'] & O \arrow[u, dashed, "v"] & Y \arrow[ul, "f_2"'] \arrow[l, "\iota_2"]
\end{tikzcd}$$

Note, 'injections' in this sense need not be injective or even monic. The object O in a coproduct for X with Y is often notated '$X \oplus Y$' or '$X \amalg Y$'; and we could notate the mediating arrow v by '$[f_1, f_2]$.

Let's say that objects and arrows arranged as $X \xrightarrow[\iota_1]{} O \xleftarrow[\iota_2]{} Y$ form a *corner* (or we could say 'co-wedge') from X and Y with vertex O. Then a coproduct of X with Y is a corner from X and Y which factors through any other corner from X and Y via a unique map between the vertices of the corners.

Before we ever meet any coproducts, we immediately know that we have the following composite theorem, just by duality:

Theorem 78. *If both the coproducts $[O, \iota_1, \iota_2]$ and $[O', \iota'_1, \iota'_2]$ exist for X and Y in the category $\mathscr{C}$, then there is a unique isomorphism $f \colon O \xrightarrow{\sim} O'$ commuting with the injection arrows.*

And in a category with an initial object 0 and where the coproducts exist,

(1) $0 \oplus X \cong X \cong X \oplus 0$
(2) $X \oplus Y \cong Y \oplus X$
(3) $X \oplus (Y \oplus Z) \cong (X \oplus Y) \oplus Z$

But now let's now have some examples of coproducts. Start with easy cases:

(1) For **Set**, the headline news is that disjoint unions are coproducts.

Given sets X and Y, take the set $X \oplus Y$ with members $\langle x, 0\rangle$ for $x \in X$ and $\langle y, 1\rangle$ for $y \in Y$. And let $\iota_1 \colon X \to X \oplus Y$ be the function $x \mapsto \langle x, 0\rangle$, and similarly let $\iota_2 \colon Y \to X \oplus Y$ be the function $y \mapsto \langle y, 1\rangle$. Then $[x \oplus Y, \iota_1, \iota_2]$ is a coproduct for X with Y.

For take any object S and arrows $f_1 \colon X \to S$ and $f_2 \colon Y \to S$, then define the function $v \colon X \oplus Y \to S$ as sending an element $\langle x, 0\rangle$ to $f_1(x)$ and an element $\langle y, 1\rangle$ to $f_2(y)$. By construction, this will make the diagram commute. Moreover, if v' is another candidate for completing the diagram, $v'(\langle x, 0\rangle) = v' \circ \iota_1(x) = f_1(x) = v(\langle x, 0\rangle)$, and likewise $v'(\langle y, 1\rangle) = v(\langle y, 1\rangle)$, whence $v' = v$, which gives us the desired uniqueness.

(2) Take a poset $(P, \preccurlyeq)$ considered as a category (so there is an arrow $p \to q$ iff $p \preccurlyeq q$). Then a coproduct of p and q would be an object c such that $p \preccurlyeq c, q \preccurlyeq c$ and such that for any object d such that $p \preccurlyeq d, q \preccurlyeq d$ there is a unique arrow from c to d, i.e. $c \preccurlyeq d$. Which means that the coproduct of p and q, if it exists, must be their least upper bound (equipped with the obvious two arrows).

(3) In $\mathbf{Prop}_L$ the disjunction $X \vee Y$ (with the obvious injections $X \to X \vee Y$, $Y \to X \vee Y$) is a coproduct of X with Y.

However things soon get rather less obvious (the details here aren't going to matter at least for now, so by all means skip, but we'll spell out a couple of cases):

(4) In the category **Grp**, coproducts are the so-called 'free products' of groups.

Take the groups $G = (G, \cdot)$, $H = (H, \odot)$ [again, we abuse notation in a familiar way]. If necessary, doctor these to equate their identity elements while ensuring the sets G and H are otherwise disjoint. Form all the finite 'reduced words' $G \star H$ you get by concatenating elements from $G \cup H$, and then multiplying out neighbouring G-elements by $\cdot$ and neighbouring H-elements by $\odot$ as far as you can. Equip $G \star H$ with the operation $\diamond$ of concatenation-of-words-followed-by-reduction. Then $G \star H = (G \star H, \diamond)$ is a group – the free product of the two groups we started with – and there are obvious ‘injection’ group homomorphisms $\iota_1 \colon G \to G \star H$, $\iota_2 \colon H \to G \star H$. Claim: $[G \star H, \iota_1, \iota_2]$ is a coproduct for the groups G and H. That is to say, for any group $K = (K, *)$ and morphisms $f_1 \colon G \to K$, $f_2 \colon H \to K$, there is a unique v such that this commutes:

$$\begin{tikzcd}
 & K & \\
G \arrow[ur, "f_1"] \arrow[r, "\iota_1"'] & G \star H \arrow[u, dashed, "v"'] & H \arrow[ul, "f_2"'] \arrow[l, "\iota_2"]
\end{tikzcd}$$

Put $v \colon G \star H \to K$ to be the morphism that sends a word like $g_1 h_1 g_2 h_2 \cdots g_r$ $(g_i \in G, h_i \in H)$ to $j(g_1) * k(h_1) * j(g_2) * k(h_2) * \cdots * j(g_r)$. By construction, $v \circ \iota_1 = j$, $v \circ \iota_2 = k$. So that makes the diagram commute.

Let v' be any other candidate group homomorphism to make the diagram commute. Then, to take a simple example, consider $gh \in G \star H$. Then $v'(gh) = v'(g) * v'(h) = v'(i_1(g)) * v'(i_2(h)) = f_1(g) * f_2(g) = v(i_1(g)) * v(i_1(g)) = v(i_1(g) * i_1(g)) = v(gh)$. And by induction over the length of words we’ll get $v' = v$. So, as required, v is unique.

(5) So what about coproducts in **Ab**, the category of abelian groups? Since the free product of two abelian groups need not be abelian, the same construction won’t work again as it stands.

OK: hit the construction with the extra requirement that words in $G \star H$ be treated as the same if one can be shuffled into the other (in effect, further reduce $G \star H$ by quotienting out with the obvious equivalence relation). But that means that we can take a word other than the identity, bring all the G-elements to the front, followed by all the H elements: but now multiply out the G-elements and the H-elements and we are left with two-element word gh. So we can equivalently treat the members of our further reduced $G \star H$ as pairs $\langle g, h \rangle$ belonging to $G \times H$. Equip this with the group operation $\diamond$ defined component-wise as before (in §14.2): this gives us an abelian group if G and H are. Take the obvious injections, $g \xrightarrow{\iota_1} \langle g, 1 \rangle$ and $h \xrightarrow{\iota_2} \langle 1, h \rangle$. Then we claim $[G \times H, \iota_1, \iota_2]$ is a coproduct for the abelian groups G and H.

Take any abelian group $K = (K, *)$ and morphisms $f_1 \colon G \to K$, $f_2 \colon H \to K$. Put $v \colon G \times H \to K$ to be the morphism that sends $\langle g, h \rangle$ to $f_1(g) * f_2(h)$. This evidently makes the coproduct diagram (with $G \times H$ for $G \star H$) commute. And a similar argument to before shows that it is unique.

So, in the case of abelian groups, the *same* objects can serve as both products and coproducts, when equipped with (respectively) appropriate projections and injections.

Finally in this section, let's note that the notion of a coproduct generalizes beyond the binary case, just as with products. Thus, exactly as you would expect, we have:

Definition 69. In any category $\mathscr{C}$, the coproduct of the $\mathscr{C}$-objects X_i (for indices $i \in I$) is an object O together with injection arrows $\iota_i\colon X_i \to O$ (for $i \in I$) such that for any object S and suite of arrows $f_i\colon X_i \to O$, there is always a unique arrow arrow $v\colon O \to S$ such that $f_i = v \circ \pi_i$. (We can use '$\coprod\limits_{i \in I} X_i$' to notate such an O.)

By dual arguments to those we've met for products, nullary coproducts are initial objects and unary coproducts are objects equipped with its identity arrow. And we can define what it is for a category to have all finite/small coproducts in the obvious ways.

15.2 Equalizers

We've so far met terminal and initial objects, products and coproducts. Now for our third pair of examples of (co)limits, equalizers and co-equalizers. This third pair of cases is perhaps not so intrinsically interesting; but together with products will play an important role in our general theory of limits.

It was useful, when defining products, to introduce the idea of a 'wedge' (Defn. 58) for a certain small configuration of objects and arrows in a category. Here's a similar definition that is going to be useful in defining the equalisers:

Definition 70. A *fork* (from S through X to Y) consists of arrows $k\colon S \to X$ with $f\colon X \to Y$ and $g\colon X \to Y$, such that $f \circ k = g \circ k$.

So diagrammatically, a fork looks like this: $S \xrightarrow{k} X \overset{f}{\underset{g}{\rightrightarrows}} Y$, with the composite arrows from S to Y being equal.

Now, a product wedge from O to X and Y is a 'limit' wedge such that any other wedge from S to X and Y uniquely factors through it. Likewise, an equalizing fork from E through X to Y is a 'limit' fork such that any other fork from an object S through X to Y uniquely factors through it. That is to say

Definition 71. Let $\mathscr{C}$ be a category and $f\colon X \to Y$ and $g\colon X \to Y$ be a pair of 'parallel' arrows in $\mathscr{C}$. Then the object E and arrow $e\colon E \to X$ form an *equalizer* in $\mathscr{C}$ for those arrows iff $f \circ e = g \circ e$, and for any fork $S \xrightarrow{k} X \overset{f}{\underset{g}{\rightrightarrows}} Y$ there is a unique arrow $u\colon S \to E$ such the following diagram commutes:

$$\begin{tikzcd}
S \arrow[dr, "k"] \arrow[dd, dashed, "u"'] & & \\
& X \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & Y \\
E \arrow[ur, "e"'] & &
\end{tikzcd}$$

Note that, just as with products (see Defn. 59), we can give an alternative definition which defines equalizers in terms of a terminal object in a category of forks:

Definition 72. Given a category $\mathscr{C}$ and parallel arrows $f, g\colon X \to Y$, then the derived fork category $\mathscr{C}_{F(XY)}$ has as objects all forks $S \xrightarrow{\ k\ } X \overset{f}{\underset{g}{\rightrightarrows}} Y$. And an arrow from $S \xrightarrow{k} \cdots$ to $S' \xrightarrow{k'} \cdots$ in $\mathscr{C}_{F(XY)}$ is a $\mathscr{C}$-arrow $g\colon S \to S'$ such that the resulting triangle commutes: i.e. such that $k = k' \circ g$. An equalizer of $f, g\colon X \to Y$ is then some $[E, e]$ such the fork $E \xrightarrow{\ e\ } X \overset{f}{\underset{g}{\rightrightarrows}} Y$ is terminal in $\mathscr{C}_{F(XY)}$.

Before giving some examples let's immediately note that, just as products are unique up to unique isomorphism, equalizers are too. That is to say,

Theorem 79. *If both the equalisers $[E, e]$ and $[E', e']$ exist for $X \overset{f}{\underset{g}{\rightrightarrows}} Y$, then there is a unique isomorphism $j\colon E \xrightarrow{\sim} E'$ commuting with the equalizing arrows, i.e. such that $e = e' \circ j$.*

Plodding proof from first principles. We can use an argument that goes along exactly the same lines as the one we used to prove the uniqueness of products and equalisers. This is of course no accident, given the similarity of the definitions. To spell things out ...

Assume $[E, e]$ equalizes f and g, and suppose $e \circ h = e$. Then observe that the following diagram will commute

$$\begin{tikzcd}
E \arrow[dd, "h"'] \arrow[dr, "e"] & & \\
& X \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & Y \\
E \arrow[ur, "e"'] & &
\end{tikzcd}$$

So $h = 1_E$ makes that diagram commute. But by hypothesis there is a unique arrow $E \to E$ which makes the diagram commute. So we can conclude that if $e \circ h = e$, then $h = 1_E$.

Now suppose $[E', e']$ is also an equalizer for f and g. Then $[E, e]$ must factor uniquely through it. That is to say, there is a (unique) mediating $j\colon E \to E'$ such that $e' \circ j = e$. And since $[E, e]$ must factor uniquely though $[E', e']$ there is a unique k such that $e \circ k = e'$. So $e \circ k \circ j = e$, and hence by our initial conclusion, $k \circ j = 1_E$.

A similar proof shows that $j \circ k = 1_{E'}$. Which makes the unique j an isomorphism. □

Proof using the alternative definition of equalizers. $[E, e]$ and $[E', e']$ are both terminal objects in the fork category $\mathscr{C}_{F(XY)}$. So there is a unique $\mathscr{C}_{F(XY)}$-

isomorphism j between them. But, by definition, this has to be a $\mathscr{C}$-arrow $j\colon E \xrightarrow{\sim} E'$ commuting with the equalizing arrows. And j is easily seen to be an isomorphism in $\mathscr{C}$ too. $\square$

Theorem 80. *If $[E, e]$ constitute an equalizer, then e is a monomorphism.*

Proof. Assume $[E, e]$ equalizes $X \overset{f}{\underset{g}{\rightrightarrows}} Y$, and suppose $e \circ g = e \circ h$, where $D \overset{g}{\underset{h}{\rightrightarrows}} E$. Then the following diagram commutes,

$$\begin{tikzcd}
D \arrow[dd, shift right, "g"'] \arrow[dd, shift left, "h"] \arrow[dr, "e\circ g = e\circ h"] & & \\
& X \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & Y \\
E \arrow[ur, "e"'] & &
\end{tikzcd}$$

So $D \xrightarrow{e\circ g} X \overset{f}{\underset{g}{\rightrightarrows}} Y$ is a fork factoring uniquely through the equalizer, and hence there is only one arrow to do the factoring, i.e. $g = h$. So e is left-cancellable in the equation $e \circ g = e \circ h$; i.e. e is monic. $\square$

Further, in an obvious shorthand,

Theorem 81. *In any category, an epic equalizer is an isomorphism*

Proof. Assume again that $[E, e]$ equalizes $X \overset{f}{\underset{g}{\rightrightarrows}} Y$, so that $f \circ e = g \circ e$. So if e is epic, it follows that $f = g$. Then consider the following diagram

$$\begin{tikzcd}
X \arrow[dd, dashed, "u"'] \arrow[dr, "1_X"] & & \\
& X \arrow[r, "f"', "g"] & Y \\
E \arrow[ur, "e"'] & &
\end{tikzcd}$$

Because e equalises, we know there is a unique u such that (i) $e \circ u = 1_X$.

But then also $e \circ (u \circ e) = 1_X \circ e = e = e \circ 1_E$. Hence, since equalizers are mono by the last theorem, (ii) $u \circ e = 1_E$.

Taken together, (i) and (ii) tell us that e has an inverse. Therefore e is an isomorphism. $\square$

Let's now have some actual examples of equalizers.

(1) Suppose in **Set** we have the functions $X \overset{f}{\underset{g}{\rightrightarrows}} Y$. Then consider the set $E \subseteq X$ such that $x \in E$ implies $fx = gx$, and let $e\colon E \hookrightarrow X$ be the obvious inclusion map. Then $[E, e]$ is an equalizer for f and g.

By construction, $f \circ e = g \circ e$. So suppose $S \xrightarrow{k} X \overset{f}{\underset{g}{\rightrightarrows}} Y$ is any fork. Since $f(k(s)) = g(k(s))$ for each $s \in S$, the set $k[S] = E$: so defining $u\colon S \to k[S]$ to agree with $k\colon S \to X$ on all inputs will make the diagram for equalizers commute.

It remains to show that this is the unique candidate for the function u. But note $k = e \circ u$, and e doesn't change the values of the function (only its codomain), so k and u must indeed agree on all inputs.

(2) Equalizers in categories whose objects are sets-with-structure behave similarly. Take the category **Mon**, for example. Given a pair of monoid homomorphisms $(X, \cdot) \overset{f}{\underset{g}{\rightrightarrows}} (Y, *)$, take the subset E of X on which the functions agree. Evidently E must contain the identity element of X (since f and g agree on this element: being homomorphisms, both must send it to the identity element of Y). And suppose $e, e' \in E$: then $f(e \cdot e') = f(e) * f(e') = g(e) * g(e') = g(e \cdot e')$, which means that E is closed under products of members.

So take E together with the monoid operation from $(X, \cdot)$ restricted to members of E. Then $(E, \cdot)$ is a monoid – for the identity element shared with $(X, \cdot)$ still behaves as an identity, and the operation is still associative. And if we take $(E, \cdot)$ and equip it with the injection homomorphism into $(X, \cdot)$, this will evidently give us an equalizer for $(X, \cdot) \overset{f}{\underset{g}{\rightrightarrows}} (Y, *)$.

(3) Similarly, take **Top**. What is the equalizer for a pair of continuous maps $X \overset{f}{\underset{g}{\rightrightarrows}} Y$? Well, take the subset of (the underlying set of) X on which the functions agree, and give it the subspace topology. This topological space equipped with the injection into X is then the desired equalizer. (This works because of the way that the subspace topology is defined – we won't go into details).

(4) A special case. Suppose we are in **Grp** and have a group homomorphism, $f\colon X \to Y$. There is also another trivial homomorphism $o\colon X \to Y$ which sends any element of the group X to the identity element in Y, i.e. is the composite $X \to 1 \to Y$ of the only possible homomorphisms. Now consider what would constitute an equalizer for f and o.

Suppose K is the kernel of f, i.e. the subgroup of X whose objects are the elements which f sends to the identity element of Y, and let $i\colon K \to X$ be the inclusion map. Then $K \xrightarrow{i} X \overset{f}{\underset{o}{\rightrightarrows}} Y$ is a fork since $f \circ i = o \circ i$.

Let $S \xrightarrow{k} X \overset{f}{\underset{o}{\rightrightarrows}} Y$ be another fork. Now, $o \circ k$ sends every element of S to the unit of Y. Since $f \circ k = o \circ k$, k must send any element of S to some element in the kernel K. So let $k'\colon S \to K$ agree with $k\colon S \to X$ on all arguments.

Then the following commutes:

$$\begin{tikzcd} S \arrow[d, "k'"'] \arrow[dr, "k"] & & \\ {} & X \arrow[r, shift left, "f"] \arrow[r, shift right, "o"'] & Y \\ K \arrow[ur, "i"'] & & \end{tikzcd}$$

And evidently k' is the only possible homomorphism to make the diagram commute.

So the equalizer of f and o is f's kernel K equipped with the inclusion map into the domain of X. Or putting it the other way about, we can define kernels of group homorphisms categorially in terms of equalizers.

(5) Finally we remark that the equalizer of a pair of maps $X \underset{g}{\overset{f}{\rightrightarrows}} Y$ where in fact $f = g$ is simply $[X, 1_X]$.

Consider then a poset $(P, \preccurlyeq)$ considered as a category whose objects are the members of P and where there is a unique arrow $X \to Y$ (for $X, Y \in P$) iff $X \preccurlyeq Y$. So the only cases of parallel arrows from X to Y are cases of equal arrows which then, as remarked, have equalizers. So in sum, a poset category has all possible equalizers.

We could now generalize to define equalizers for more-than-two parallel arrows in obvious ways, and indeed define the trivial unary and null cases. But we don't need to, so we won't.

Instead, we'll make a different point. We saw that in **Set** the equalizer of two parallel arrows from an object X is a certain subset of X equipped with the trivial inclusion function. We now note that the reverse holds too:

Theorem 82. *A subset S of X, equipped with the inclusion map $i\colon S \hookrightarrow X$, is an equalizer in* **Set** *of certain parallel arrows from X.*

Proof. Given $S \subseteq X$, there is a corresponding 'characteristic function' $s\colon X \to \{0, 1\}$ which sends $x \in X$ to 1 ('true').

Another function from X to $\{0, 1\}$ is the cheerfully indiscriminate map t which sends everything in X to 1.

We show that $[S, i]$ is an equalizer for $X \underset{t}{\overset{s}{\rightrightarrows}} \{0, 1\}$. First, it is trivial that $s \circ i = t \circ i$. So we now show that any upper fork in this diagram factors through the lower fork via a unique suitable u:

$$\begin{tikzcd} P \arrow[d, "u"'] \arrow[dr, "f"] & & \\ {} & X \arrow[r, shift left, "s"] \arrow[r, shift right, "t"'] & \{0,1\} \\ S \arrow[ur, "i"'] & & \end{tikzcd}$$

Since $s \circ f = t \circ f$, $f[P] \subseteq S$. So if we put $u\colon P \to S$ to agree with $f\colon P \to X$ on all inputs, then the diagram commutes. And this u is evidently the only possible candidate. □

Which gives us a nice categorial way of dealing with subobjects in **Set**. Later we'll find a different way of talking about subobjects in categories much more generally, and we will then have to investigate its relation to the sort of construction for **Set** that we've just met.

15.3 Co-equalizers

We dualize our definition of an equalizer to get the notion of a co-equalizer. So, as a preliminary, we say

Definition 73. A *co-fork* (from X through Y to S) consists of parallel arrows $f\colon X \to Y$, $g\colon X \to Y$ and an arrow $k\colon Y \to S$, such that $k \circ f = k \circ g$.

(Actually, plain 'fork' is used for the dual too: but the barbarous 'co-fork' keeps things clear.) So diagrammatically, a co-fork looks like this: $X \underset{g}{\overset{f}{\rightrightarrows}} Y \xrightarrow{k} S$, with the composite arrows from X to S being equal.

Definition 74. Let $\mathscr{C}$ be a category and $f\colon X \to Y$ and $g\colon X \to Y$ be a pair of parallel arrows in $\mathscr{C}$. Then the object C and arrow $c\colon Y \to S$ form a *co-equalizer* in $\mathscr{C}$ for those arrows iff $c \circ f = c \circ g$, and for any co-fork from X through Y to S there is a unique arrow $u\colon C \to S$ such the following diagram commutes:

$$\begin{tikzcd} & & S \\ X \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & Y \arrow[ur, "k"] \arrow[dr, "c"'] & \\ & & C \arrow[uu, dashed, "u"'] \end{tikzcd}$$

As we would now expect, by a dual argument, co-equalizers are unique up to a unique isomorphism; and since equalizers are monic, co-equalizers are epic. We won't pause to spell out or prove those results, but turn immediately to give just one central example.

Work in **Set**. The parallel arrows $f, g\colon X \to Y$ determine a relation R on the members of Y where y_1Ry_2 holds when there is an $x \in X$ such that $f(x) = y_1 \wedge g(x) = y_2$.

Now if there is a co-fork $X \underset{g}{\overset{f}{\rightrightarrows}} Y \xrightarrow{k} S$, then if y_1Ry_2 then $k(y_1) = k(y_2)$ (and of course, being mapped to the same value by k is an equivalence relation). So, for every co-fork, there is a corresponding equivalence relation such that being R-related implies being in that equivalence relation. What's the limiting case of such an equivalence relation? It will have to be $R^\sim$, the smallest equivalence relation containing R.

That observation gives us the clue about how to proceed.

Theorem 83. *Given functions $f, g\colon X \to Y$ in* **Set***, let $R^\sim$ be the smallest equivalence relation containing R – where $y_1 R y_2$ iff $(\exists x \in X)(f(x) = y_1 \wedge g(x) = y_2)$.*

Let C be $Y/R^\sim$, i.e. the set of $R^\sim$-equivalence classes of Y; and let c map $y \in Y$ to the $R^\sim$-equivalence class containing y. Then $[C, c]$, so defined, is a co-equalizer for f and g.

Proof. First we need to check that $c \circ f = c \circ g$. But the left-hand side sends $x \in X$ to the $R^\sim$-equivalence class containing $f(x)$ and the right-hand side sends x to the $R^\sim$-equivalence class containing $g(x)$. However, $f(x)$ and $g(x)$ are by definition R-related, and hence are $R^\sim$-related: so by construction they belong to the same $R^\sim$equivalence class.

Now suppose again there is a co-fork $X \overset{f}{\underset{g}{\rightrightarrows}} Y \xrightarrow{k} S$, so $k \circ f = f \circ g$. We next outline a proof that if $y_1 R^\sim y_2$ then $k(y_1) = k(y_2)$.

Start with R and let R' be its reflexive closure. Obviously we'll still have that if $y_1 R' y_2$ then $k(y_1) = k(y_2)$. Now consider R'' the symmetric closure of R': again obviously, we'll still have that $y_1 R'' y_2$ then $k(y_1) = k(y_2)$. Now note that if $y_1 R'' y_2$ and $y_2 R'' y_3$, then $k(y_1) = k(y_3)$. So if we take the transitive closure of R'', we'll still have a relation which, when it holds between some y_1 and y_2, implies that $k(y_1) = k(y_2)$. But the transitive closure of R'' is $R^\sim$.

We have shown, then, that k is constant on members of a $R^\sim$-equivalence class, and so we can well-define a function $u\colon C \to S$ which sends an equivalence class to the value of k on a member of that class, which makes the diagram above commute. Moreover, since c is surjective and C only contains $R^\sim$-equivalence classes, u is the only function for which $u \circ c = k$. $\square$

16 Limits and colimits defined

In this chapter, we first define an official class of *limits*, and the corresponding dual class of *co-limits*. Our examples from the previous chapter will indeed all fall into these classes. We then give a new pair of illustrations, so-called pullbacks and pushouts.

16.1 Defining limits via a 'category of cones'

There's obviously a family resemblance between terminal objects, products, equalizers: roughly, each is a kind of limiting case with unique arrows pointing towards it.

As we've already seen, we can sharpen that up:

(1) A terminal object in $\mathscr{C}$ is ... wait for it! ... terminal in the given category $\mathscr{C}$.

(2) The product of X with Y in $\mathscr{C}$ is a terminal object in the derived category $\mathscr{C}_{W(X,Y)}$ of wedges to X and Y.

(3) The equalizer of parallel arrows through X to Y in $\mathscr{C}$ are (parts of) terminal objects in the derived category $\mathscr{C}_{F(XY)}$ of forks through X to Y.

And that gives us a clue about how to generalize.

Way back in Defn. 3, we characterized a diagram D in a category $\mathscr{C}$ as being simply a bunch of objects with some arrows between some of them. We need some way of indexing these objects to refer to them, so henceforth we'll refer to objects in D by terms like 'D_j' where j is in some appropriate index set. And for convenience, we'll allow double counting, permitting the case where $D_j = D_k$ for different indices. We will also allow the limiting cases(!) where there are no arrows, and even the empty case where there are no objects.

Definition 75. Let D be a diagram in category $\mathscr{C}$. Then a *cone over* D comprises an object $C \in \mathscr{C}$, the *vertex* of the cone, together with $\mathscr{C}$-arrows $c_j \colon C \to D_j$, one for each object D_j in D, such that whenever there is an arrow $d \colon D_k \to D_l$ in D, the following diagram commutes:

$$\begin{tikzcd} & C \arrow[dl, "c_k"'] \arrow[dr, "c_l"] & \\ D_k \arrow[rr, "d"'] & & D_l \end{tikzcd}$$

We use '$[C, c_j]$' as our notation for such a cone.

Think of it diagrammatically like this: arrange the objects in the diagram D in a plane with whatever arrows there are between them. Now sit the object C above the plane, with a quiverful of arrows from C zinging down, one to each object in the plane. Those arrows form a skeletal cone. And the key requirement is that any triangles thus formed with C at the apex must commute.

Now, the cones $[C, c_j]$ over a given diagram D in $\mathscr{C}$ form a category in a very natural way:

Definition 76. Given a diagram D in category $\mathscr{C}$, the derived category $\mathscr{C}_{C(D)}$ – *the category of cones over* D – has the following data:

(1) Its objects are the cones $[C, c_j]$ over D.
(2) An arrow from $[C, c_j]$ to $[C', c'_j]$ is any $\mathscr{C}$-arrow $f\colon C \to C'$ such that $c'_j \circ f = c_j$ for all j. That is to say, in a diagram like

$$\begin{tikzcd} & C \arrow[d, "f"] \arrow[ddl, "c_j"'] \arrow[ddr, "c_k"] & \\ & C' \arrow[dl, "c'_j"] \arrow[dr, "c'_k"'] & \\ D_j & & D_k \end{tikzcd}$$

(imagine a triangle for each object D_i in D) all the triangles commute.

We can leave it as a routine exercise to confirm that $\mathscr{C}_{C(D)}$ is indeed a category. Instead, we next move on to introduce an inviting definition:

Definition 77. A *limit cone for* D in $\mathscr{C}$ is a cone which is a terminal object in $\mathscr{C}_{C(D)}$.

Theorem 84. *Limit cones over a given diagram* D *are unique up to unique isomorphism.*

Proof. Limit cones are unique up to unique isomorphism because they are terminal objects. □

And now let's immediately confirm that our three examples of limits so far are indeed limit cones in the sense just defined.

(1) Let's get the null case out of the way. Start with the empty diagram in $\mathscr{C}$ – *zero* objects and so, necessarily, no arrows. Then a cone over the empty diagram is simply an object C as vertex (there is no further condition to fulfil), and an arrow between such cones is just an arrow between objects in $\mathscr{C}$. So the category of cones over the empty diagram is just the category $\mathscr{C}$ we started with, and a limit cone is just a terminal object in $\mathscr{C}$!

(2) Consider now a diagram which is just *two* objects we'll call 'D_1', 'D_2', still with no arrow between them. Then a cone over such a diagram is just a wedge into D_1, D_2; and a limit cone in the category of such cones (wedges) is simply a product.

(3) Next consider a diagram which is again two objects, but now with two parallel arrows between them, which we can represent $D_1 \overset{d}{\underset{d'}{\rightrightarrows}} D_2$. Then a cone over this diagram is a commuting diagram like this:

$$\begin{tikzcd} & C \arrow[dl, "c_1"'] \arrow[dr, "c_2"] & \\ D_1 \arrow[rr, shift left, "d"] \arrow[rr, shift right, "d'"'] & & D_2 \end{tikzcd}$$

If there is such a diagram, then we must have $d \circ c_1 = d' \circ c_1$: and vice versa, if that identity holds, then we can put $c_2 = d \circ c_1 = d' \circ c_1$ to complete the commutative diagram. Hence we have a cone from the vertex C to our diagram iff $C \xrightarrow{c_1} D_1 \overset{d}{\underset{d'}{\rightrightarrows}} D_2$ is a fork.

Since c_1 fixes what c_2 has to be to complete the cone, we can focus on the cut-down cone consisting of just $[C, c_1]$. In this case, the corresponding cut-down terminal cone is the object and arrow $[E, e]$ such that $E \xrightarrow{e} D_1 \overset{d}{\underset{d'}{\rightrightarrows}} D_2$ is the terminal fork in the category of forks through D_1 to D_2. Hence $[E, e]$ is the equalizer of our diagram.

16.2 Limits defined more directly

(a) We initially defined products without talking of categories of wedges, however; and we first defined equalizers without talking of categories of forks. Similarly, we can of course recast our definition of limits – i.e. limit cones – without going via the notion of a category of cones. Entirely as you would expect, we could simply say

Definition 78 (Alternative)**.** A cone $[L, \pi_j]$ over a diagram D in $\mathscr{C}$ is a limit (cone) iff for any cone $[C, c_j]$ over D there is a unique arrow $u\colon C \to L$ such that, for all indexing objects j in D, $\pi_j \circ u = c_j$.

We can then also give a direct proof, along now familiar lines, from this alternative definition to our preceding

Theorem 84. *Limit cones over a given diagram D are unique up to unique isomorphism.*

Proof. First we note that $[L, \pi_j]$ factors through itself via the identity $1_L : L \to L$. But by definition, a cone over D uniquely factors through the limit, so that means that

(i) if $\pi_j \circ u = \pi_j$ for all j, then $u = 1_L$.

Now suppose $[L', \pi'_j]$ is another limit cone over D. Then $[L', \pi'_j]$ uniquely factors through $[L, \pi_j]$, via some f, so

(ii) $\pi_j \circ f = \pi'_j$ for all j.

And likewise $[L, \pi_j]$ uniquely factors through $[L', \pi'_j]$ via some g, so

(iii) $\pi'_j \circ g = \pi_j$ for all j.

Whence

(iv) $\pi_j \circ f \circ g = \pi_j$ for all j.

Therefore

(v) $f \circ g = 1_L$.

And symmetrically

(vi) $g \circ f = 1_{L'}$.

Whence f is not just unique (by hypothesis, the only way of completing the relevant diagrams) but an isomorphism. □

(b) We can now prove two further simple theorems:

Theorem 85. *Suppose $[L, \pi_j]$ is a limit cone over a diagram D in $\mathscr{C}$, and $[L', \pi'_j]$ is another cone over D which factors through $[L, \pi_j]$ via an isomorphism f. Then $[L', \pi'_j]$ is also a limit cone.*

Proof. Take any cone $[C, c_j]$ over D. We need to show that (i) there is an arrow $v : C \to L'$ such that for all indexing objects j in D, $c_j = \pi'_j \circ v$, and (ii) v is unique.

But we know that there is a unique arrow $u : C \to L$ such that for j, $c_j = \pi_j \circ u$. And we know that $f : L' \to L$ and $\pi'_j = \pi_j \circ f$ (so $\pi_j = \pi'_j \circ f^{-1}$).

Therefore put $v = f^{-1} \circ u$, and that satisfies (i).

Now suppose there is another arrow $v' : C \to L'$ such that $c_j = \pi'_j \circ v'$. Then we have $f \circ v' : C \to L$, and also $c_j = \pi_j \circ f \circ v'$. Therefore $[C, c_j]$ factors through $[L, \pi_j]$ via $f \circ v'$, so $f \circ v' = u$. Whence $v' = f^{-1} \circ u = v$. Which proves (ii). □

Theorem 86. *Suppose $[L, \pi_j]$ is a limit cone over a diagram D in $\mathscr{C}$. Then the cones over D with vertex C correspond one-to-one with $\mathscr{C}$-arrows from C to L.*

Proof. Take any arrow $u\colon C \to L$. If there is an arrow $d\colon D_k \to D_l$ in the diagram D, then (since $[L, \pi_j]$ is a limit), $\pi_l = d \circ \pi_k$, whence $(\pi_l \circ u) = d \circ (\pi_k \circ u)$. Since this holds generally, $[C, \pi_j \circ u]$ is a cone over D. But (again since $[L, \pi_j]$ is a limit) every cone over D with vertex C is of the form $[C, \pi_j \circ u]$ for unique u. Hence there is indeed a one-one correspondence between arrows $u\colon C \to L$ and cones over D with vertex C. (Moreover, since the construction is natural, involving no arbitrary choices, the resulting isomorphism is natural.) $\square$

(c) As a fun exercise and reality check, let's remark that the whole category $\mathscr{C}$ can be thought of as the limiting case of a diagram in itself, and then

Theorem 87. *A category $\mathscr{C}$ has an initial object if and only if $\mathscr{C}$, thought of as a diagram in $\mathscr{C}$, has a limit.*

Proof. Suppose $\mathscr{C}$ has an initial object I. Then for every $\mathscr{C}$-object C, there is a unique arrow π_C. $[I, \pi_C]$ is a cone (since for any arrow $f\colon C \to D$, the composite $f \circ \pi_C$ is an arrow from I to D and hence has to be equal to the unique π_D. Further, $[I, \pi_C]$ is a limit cone. For suppose $[A, a_C]$ is any other cone over the whole of $\mathscr{C}$. Then since it is a cone, the triangle

$$\begin{tikzcd}
 & A \arrow[dl, "a_I"'] \arrow[dr, "a_C"] & \\
I \arrow[rr, "\pi_C"'] & & C
\end{tikzcd}$$

has to commute for all C. But that's just the condition for $[A, a_C]$ factoring through $[I, \pi_C]$ via a_I. And moreover, suppose $[A, a_C]$ also factors through by some u. Then in particular,

$$\begin{tikzcd}
 & A \arrow[dl, "u"'] \arrow[dr, "a_I"] & \\
I \arrow[rr, "1_I"'] & & I
\end{tikzcd}$$

commutes, and so $u = a_C$. So the factoring is unique, and $[I, \pi_C]$ is a limit cone.

Now suppose, conversely, that $[I, \pi_C]$ is a limit cone over the whole of $\mathscr{C}$. Then there is an arrow $\pi_C\colon I \to C$ for each C in $\mathscr{C}$. If we can show it is unique, I will indeed be initial.

Suppose then that there is an arrow $k\colon I \to C$ for a given C. Then since $[I, \pi_C]$ is a cone, the diagram

$$\begin{tikzcd}
 & I \arrow[dl, "\pi_I"'] \arrow[dr, "\pi_C"] & \\
I \arrow[rr, "k"'] & & C
\end{tikzcd}$$

has to commute. Considering the case where $k = \pi_C$, we see that $[I, \pi_C]$ factors through itself via π_I; but it also factors via 1_D, so the uniqueness of factorization entails $\pi_I = 1_D$. Hence the diagram shows that for any $k\colon I \to C$ has to be identical to π_C. So I is initial. $\square$

(d) Let's introduce some standard

Notation. We often denote the *limit object* at the vertex of a limit cone for the diagram D with objects D_j by '$\underset{\leftarrow j}{Lim}\, D_j$'. (This is standard notation: but since limit cones are only unique up to isomorphism, different but isomorphic objects can be denoted in different contexts by '$\underset{\leftarrow j}{Lim}\, D_j$'.)

The 'projection' *arrows* from this limit object to the various objects D_j can naturally be denoted '$\pi_i\colon \underset{\leftarrow j}{Lim}\, D_j \to D_i$' (though many use '$p_i$' rather than '$\pi_i$').

The limit *cone* could therefore be represented by '$[\underset{\leftarrow j}{Lim}\, D_j, \pi_j]$'.

The direction of the arrow under '*Lim*' in this notation is perhaps unexpected, but we just have to learn to live with it.

(e) A final introductory remark. Some prefer to say more austerely that a cone is (not an object-with-a-family-of-arrows but) simply a family of arrows. For example, on such a usage, the limit cone is identified as, simply, (the set of) the arrows $\pi_i\colon \underset{\leftarrow j}{Lim}\, D_j \to D_i$. Now, since we can read off the vertex of a cone as the common domain of all its arrows, there's no loss of information here. It is very largely a matter of convenience whether we speak austerely or explicitly mention the vertex. (One trivial difference is in the way we then have to handle the limiting case of a cone over the null diagram: but we needn't fuss about that: and in what follows, when convenient for brevity's sake, we will occasionally think of cones austerely).

16.3 And another characterization of limits

Recall Theorem 73 which told us that a product can be identified with a universal element for a certain functor. We can now prove a much more general result with the same flavour.

Consider the collection of cones with vertex C over a diagram D which we can denote $Cone(C, D)$. And let's now officially treat cones, at least in small categories, as sets – which ensures that $Cone(C, D)$ lives in **Set**.

We can now define the functor $Cone(-, D)\colon \mathscr{C}^{op} \to \mathbf{Set}$ as follows. $Cone(-, D)$ sends an object $C \in \mathscr{C}$ to $Cone(C, D)$, the set of cones over D with vertex C. And $Cone(-, D)$ sends a $\mathscr{C}$-arrow $f\colon C' \to C$ to the arrow $Cone(f)\colon Cone(C, D) \to Cone(C', D)$, which takes a cone $[C, \pi_j]$ and sends it to $[C', \pi_j \circ f]$. It is easily checked that this is indeed a functor.

Apply Defn. 47. Then a universal element of $Cone(-, D) \colon \mathscr{C}^{op} \to \mathbf{Set}$ is a pair $\langle L, [L, \pi_J] \rangle$, where $L \in \mathscr{C}$ and $[L, \pi_J]$ is in $Cone(L, D)$, the set of cones over D with vertex L. And moreover, we require that for each $C \in \mathscr{C}$ and each cone $[C, c_J]$, there is a unique map $f \colon Z \to L$ such that $Cone(f)[L, \pi_J] = [C, c_J]$, which requires $\pi_J \circ f = c_J$ for each J. But that's just to say that $[L, \pi_J]$ is a limit cone. Hence

Theorem 88. *A limit cone over D in a small category is a universal element for $Cone(-, D)$.*

16.4 Colimits defined

The headline, and thoroughly predictable, story is: reverse the relevant arrows and you get a definition of colimits.

So, dualizing §16.1 and wrapping everything together, we get:

Definition 79. Let D be a diagram in category $\mathscr{C}$. Then a *cocone under D* is an object $C \in \mathscr{C}$, together with an arrow $c_j \colon D_j \to C$ for each object D_j in D, such that whenever there is an arrow $d \colon D_k \to D_l$ in D, the following diagram commutes:

$$\begin{tikzcd} D_k \arrow[rr, "d"'] \arrow[dr, "c_k"'] & & D_l \arrow[dl, "c_l"] \\ & C & \end{tikzcd}$$

The cocones under D form a category with objects the cocones $[C, c_j]$ and an arrow from $[C, c_j]$ to $[C', c'_j]$ being any $\mathscr{C}$-arrow $f \colon C \to C'$ such that $c'_j = f \circ f$ for all indexes j. A colimit for D is an initial object in the category of cocones under D. It is standard to denote the object at the vertex of the colimit cocone for the diagram D by '$\underset{\to j}{Lim}\, D_j$'.

And it is now routine to confirm that our earlier examples of initial objects, coproducts and co-equalizers do count as colimits.

(1) The null case where we start with the empty diagram in $\mathscr{C}$ gives rise to a cocone which which is simply an object in C. So the category of cocones over the empty diagram is just the category $\mathscr{C}$ we started with, and a limit cocone is just an initial object in $\mathscr{C}$!

(2) Consider now a diagram which is just *two* objects we'll call 'D_1', 'D_2', still with no arrow between them. Then a cocone over such a diagram is just a corner from D_1, D_2; and a limit cocone in the category of such cocones is simply a coproduct.

(3) And if we start with the diagram $D_1 \underset{d'}{\overset{d}{\rightrightarrows}} D_2$ then a limit cocone over this diagram gives rise to a co-equalizer.

The arguments for the last two cases are exactly dual to the arguments in §16.1, and we need not pause to spell them out.

16.5 More examples: pullbacks and pushouts

(a) Let's illustrate be exploring another kind of limit and its dual.

We've just met the notion of a co-wedge or, as I prefer to say, a *corner* D in category $\mathscr{C}$ in the context of talking about coproducts. It is a diagram which can be represented like this:

$$\begin{tikzcd} & D_3 \arrow[d, "e"] \\ D_1 \arrow[r, "d"] & D_2 \end{tikzcd}$$

Now, a cone over our corner diagram has a rather familiar shape, i.e. it is a commutative square:

$$\begin{tikzcd} C \arrow[r, "c_3"] \arrow[d, "c_1"] \arrow[dr, "c_2"] & D_3 \arrow[d, "e"] \\ D_1 \arrow[r, "d"] & D_2 \end{tikzcd}$$

Though note, we needn't really draw the diagonal here, for if the sides of the square commute thus ensuring $d \circ c_1 = e \circ c_3$, then we know the diagonal c_2 exists making the triangles commute.

And a limit for this type of cone will be a cone with vertex $L = \underset{\leftarrow j}{Lim}\, D_j$ and projections $\pi_j \colon L \to D_j$ such that for any cone $[C, c_j]$ over D, there is a unique $u \colon C \to L$ such that this whole diagram commutes:

$$\begin{tikzcd} C \arrow[dr, "u"] \arrow[drr, bend left, "c_3"] \arrow[ddr, bend right, "c_1"'] & & \\ & L \arrow[r, "\pi_3"] \arrow[d, "\pi_1"] & D_3 \arrow[d, "e"] \\ & D_1 \arrow[r, "d"] & D_2 \end{tikzcd}$$

(And note that if this commutes, there's just one possible π_2 and c_2 we can draw in which makes it still commute.)

Definition 80. A limit for a corner diagram is a *pullback*. The whole square formed by the original corner and its limit is a *pullback square*.

(b) Let's immediately have some examples of pullback squares living in the category **Set**.

(1) Changing the labelling, consider a corner comprising three sets X, Y, Z and a pair of functions which share the same codomain, thus:

$$\begin{tikzcd} & Y \arrow[d, "g"] \\ X \arrow[r, "f"] & Z \end{tikzcd}$$

We know from the previous diagram that the limit object L must be product-like (with any wedge over X, Y factoring through the wedge with vertex L). Hence to get the other part of the diagram to commute, the pullback square must have at its apex L something isomorphic to $\{\langle x, y\rangle \in X \times Y \mid f(x) = g(y)\}$ with the obvious projection maps to X and Y.

So suppose first that in fact both X and Y are subsets of Z, and f and g are the inclusion functions. And we then get a pullback square

$$\begin{tikzcd} L \arrow[r] \arrow[d] & Y \arrow[d, hook] \\ X \arrow[r, hook] & Z \end{tikzcd}$$

Then $L \cong \{\langle x, y\rangle \in X \times Y \mid x = y\} = \{\langle z, z\rangle \mid z \in X \cap Y\} \cong X \cap Y$. Hence, in **Set**, the intersection of a pair of sets is their pullback object (fixed, as usual, up to isomorphism).

(2) Take another case. Suppose we have a corner as before but with $Y = Z$ and $g = 1_Z$. Then

$$L \cong \{\langle x, z\rangle \in X \times Z \mid f(x) = z\} \cong \{x \mid \exists z f(x) = z\} \cong f^{-1}[Z],$$

i.e. a pullback object for this corner is, up to isomorphism, the inverse image of Z, and we have a pullback square

$$\begin{tikzcd} f^{-1}[Z] \arrow[r] \arrow[d] & Z \arrow[d, "1_Z"] \\ X \arrow[r, "f"] & Z \end{tikzcd}$$

Hence in **Set**, the inverse image of a function is also a pullback object.

(c) Why 'pullback'? Look at that last diagram. We can say that we get to $f^{-1}[Z]$ from Z by pulling back along f – or more accurately, we get to the arrow $f^{-1}[Z] \to X$ by pulling back the identity arrow on Z along f. In this sense,

Theorem 89. *Pulling back a monomorphism yields a monomorphism*

In other words, if we start with the same corner $X \xrightarrow{f} Z \xleftarrow{g} Y$ with g monic, and can pullback g along f to give a pullback square

$$\begin{tikzcd} L \arrow[r, "b"] \arrow[d, "a"] & Y \arrow[d, "g"] \\ X \arrow[r, "f"] & Z \end{tikzcd}$$

then the resulting arrow a is monic. (This does not depend on the character of f.)

Proof. Suppose, for some arrows $C \overset{j}{\underset{k}{\rightrightarrows}} L$, $a \circ j = a \circ k$. Then $g \circ b \circ j = f \circ a \circ j = f \circ a \circ k = g \circ b \circ k$. Hence, given that g is monic, $b \circ j = b \circ k$.

It follows that the two cones over the original corner, $X \xleftarrow{a \circ j} C \xrightarrow{b \circ j} Y$ and $X \xleftarrow{a \circ k} C \xrightarrow{b \circ k} Y$ are in fact the *same* cone, and hence must factor through the limit L via the same unique arrow $C \to L$. Which means $j = k$.

In sum, $a \circ j = a \circ k$ implies $j = k$, so a is monic. □

Here's another result about monomorphisms and pullbacks:

Theorem 90. *The arrow $f\colon X \to Y$ is a monomorphism in $\mathscr{C}$ if and only if the following is a pullback square:*

$$\begin{tikzcd} X \arrow[r, "1_X"] \arrow[d, "1_X"] & X \arrow[d, "f"] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

Proof. Suppose this is pullback diagram. Then any cone $X \xleftarrow{a} C \xrightarrow{b} Y$ over the corner $X \xrightarrow{f} Y \xleftarrow{f} X$ must uniquely factor through the limit with vertex X. That is to say, if $f \circ a = f \circ b$, then there is a u such that $a = 1_X \circ u$ and $b = 1_X \circ u$, hence $a = b$ – so f is monic.

Conversely, if f is monic, then given any cone $X \xleftarrow{a} C \xrightarrow{b} Y$ over the original corner, $f \circ a = f \circ b$, whence $a = b$. But that means the cone factors through the cone $X \xleftarrow{1_X} X \xrightarrow{1_X} X$ via the unique a, making that cone a limit and the square a pullback square. □

(d) We've explained, up to a point, the label 'pullback'. It should now be noted in passing that a pullback is sometimes called a fibered product (or fibre product) because of a construction of this kind on fibre bundles in topology. Those who know some topology can chase up the details.

But here's a way of getting products into the story, using an idea that we already know about. Recall the definition of the slice category $\mathscr{C}/Z$. Its objects are arrows$_{\mathscr{C}}$ $f\colon C \to Z$ for $C \in \mathscr{C}$, and an arrow from $f\colon X \to Z$ to $g\colon Y \to Z$ is an arrow$_{\mathscr{C}}$ $h\colon X \to Y$ such that $f = g \circ h$ in $\mathscr{C}$.

Now the pullback of the corner formed by f and g in $\mathscr{C}$ is a pair of arrows $a\colon L \to X$ and $b\colon L \to Y$ such that $f \circ a = g \circ b$ $(= k)$ and which form a wedge such that any other wedge $a'\colon L' \to X$, $b'\colon L' \to Y$ such that $f \circ a' = g \circ b'$ $(= k')$ factors uniquely through it.

Looked at as a construction in $\mathscr{C}/Z$, this means taking two $\mathscr{C}/Z$-objects f and g and getting a pair of $\mathscr{C}/Z$-arrows $a\colon k \to f$, $b\colon k \to g$ (check that a is indeed a $\mathscr{C}/Z$-arrow from $f \circ a$ as $\mathscr{C}$-arrow to f!). And this pair of arrows forms a wedge such that any other wedge $a'\colon k' \to f$, $b'\colon k' \to g$ factors uniquely through it. In other words, the pullback in $\mathscr{C}$ is a product in $\mathscr{C}/Z$.

Product notation is often used for pullbacks, thus:

$$\begin{tikzcd}
X \times_Z Y \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[d] \\
X \arrow[r] & Z
\end{tikzcd}$$

with the little corner conventionally indicating it is indeed a pullback square

(e) As you would expect, all this dualizes. Suppose we take a wedge D, i.e. a diagram that, with appropriate labelling, looks like this: $D_1 \xleftarrow{d} D_2 \xrightarrow{e} D_3$. A cocone under this diagram is another commutative square (omitting again the diagonal arrow which is fixed by the others).

$$\begin{tikzcd}
D_2 \arrow[r, "e"] \arrow[d, "d"] & D_3 \arrow[d, "c_3"] \\
D_1 \arrow[r, "c_1"] & C
\end{tikzcd}$$

And a limit cocone of this type will be a cocone with apex $L = \underset{\to j}{Lim}\, D_j$ and projections $\pi_j\colon L \to D_j$ such that for any cocone $[C, c_j]$ under D, there is a unique $u\colon L \to C$ such that the obvious dual of the whole pullback diagram above commutes.

Definition 81. A limit for a wedge diagram is a *pushout*.

Now, in **Set**, we get the limit object for a corner diagram $X \xrightarrow{f} Z \xleftarrow{g} Y$ by taking a certain *subset* of a *product* $X \times Y$. Likewise we get the colimit object for a wedge diagram $X \xleftarrow{f} Z \xrightarrow{g} Y$ by taking a certain *quotient* of a *coproduct* $X \amalg Y$. (Recall from our discussion of (co)-equalizers that quotients are categorially dual to subsets in **Set**.) In fact, we need to quotient out by the smallest equivalence relation containing the relation R where $\langle x, 0\rangle R\langle y, 1\rangle$ iff there is a z such that $f(z) = x \wedge g(z) = y$. We won't pause over this. Though it does illustrate how colimits can tend to seem messier than limits.

17 The existence of limits

We have seen, then, that a whole range of familiar constructions from various areas of ordinary mathematics can be regarded as instances of taking limits or colimits of (very small) diagrams in appropriate categories. Examples so far include: forming cartesian products or logical conjunctions, taking disjoint unions or free products, quotienting out by an equivalence relation, taking inverse images.

Now, not *every* familiar kind of construction involves taking (co)limits. We'll later see that exponentiation involves a different idea. But plainly we are mining a very rich seam here.

It would get tedious, however, to explore what it takes for a category to have limits for various further kinds of diagram case-by-case, even if we just stick to very small examples. But fortunately we don't need to do such a case-by-case examination. It turns out that if a category has finite products and has equalizers, then it has, in an obvious sense, *all* finite limits – and the same goes if we replace 'finite' by 'small'.

This chapter explains and proves that important claim. For the first couple of sections, however, we further refine our conception of limits.

17.1 Limits and diagrams-as-functors

So far, our story about limit cones over diagrams has been presented in terms of the idea of a diagram-in-a-category-$\mathscr{C}$ thought of as a (possibly very small) fragment of $\mathscr{C}$ – i.e. just some objects and arrows.

But consider again our three concrete examples of (co)limits so far (apart from the trivial one of limit cones over the null diagram): we've considered limit cones over (1) a discrete two-object diagram, (2) two parallel arrows, and (3) a 'corner', and their duals. Now, it would have made no difference to the stories at all if we'd instead taken the diagrams to be miniature sub-categories, i.e. if we has taken ($1'$) a discrete two object category (just add the necessary identity arrows), ($2'$) a category with two parallel arrows (add identity arrows again), and ($3'$) a category we get again by adding identity arrows, which looks like this:

$$\begin{tikzcd} \bullet \arrow[loop left] \arrow[r] & \star \arrow[loop above] & \diamond \arrow[l] \arrow[loop right] \end{tikzcd}$$

It is easy to see that adding identity arrows to our original diagrams won't affect any of the discussion of cones and cocones.

So from now on, *we are going to concentrate on cases where diagrams have been rounded out with the identity arrows and composites required to form a subcategory.* And next recall that we noted before another way of thinking about such a rounded-out diagram in a category $\mathscr{C}$: i.e. as a 'picture'-in-$\mathscr{C}$ of another category $\mathbf{J}$, with the picture projected onto $\mathscr{C}$ by a functor $D\colon \mathbf{J} \to \mathscr{C}$. See the beginning of §6.2, and then §12.1, Defn. 44 – except that now we follow a common convention and use the notation $\mathbf{J}$ rather than $\mathscr{J}$ when a small – typically *very* small – category is likely to be in focus.

So using this notion of diagrams-as-functors, the corresponding definitions for cones and limit cones over such diagrams go as follows (entirely as you would predict):

Definition 82. Suppose we are given a category $\mathscr{C}$, together with $\mathbf{J}$ a (possibly very small) category, and a diagram-as-functor $D\colon \mathbf{J} \to \mathscr{C}$. Then:

(1) *A cone over D* is an object $C \in \mathscr{C}$, together with an arrow $c_J\colon C \to D(J)$ for each $\mathbf{J}$-object J, such that for any $\mathbf{J}$-arrow $d\colon K \to L$, $c_L = D(d) \circ c_K$. We use $[C, c_J]$ (where 'J' is understood to run over objects in $\mathbf{J}$) for such a cone.

(2) *A limit cone over D* is a cone we can notate $[\underset{\longleftarrow \mathbf{J}}{Lim}\, D, \pi_J]$ such that for every cone $[C, c_J]$ over D, there is a unique arrow $u\colon C \to \underset{\longleftarrow \mathbf{J}}{Lim}\, D$ such that, for all $J \in ob(\mathbf{J})$, $\pi_J \circ u = c_J$.

We can leave the dual cases to take care of themselves, and for the moment just re-iterate the point that not all diagrams in the original sense of fragments-of-a-category $\mathscr{C}$ are diagrams in the sense of pictures of a category $\mathbf{J}$ in $\mathscr{C}$. So not all (limit) cones over a diagram in the more general first sense are (limit) cones over diagrams in the second narrower sense. You need to be aware that different presentations in fact use the different definitions: e.g. compare Borceux (1994) and Leinster (2014). Still, the distinction is one that doesn't seem to weigh much in practice: mere fragments of a category are not of much interest. So we don't miss much if we ignore them. And then we might as well treat diagrams in the rather neat second way, i.e. treat diagrams as functors. As we said, that is what we mostly do from now on.

17.2 Cones as natural transformations (and two functors)

(a) Fix on some small category $\mathbf{J}$. Consider the functor category $[\mathbf{J}, \mathscr{C}]$ whose objects are diagrams-as-functors $D\colon \mathbf{J} \to \mathscr{C}$ and whose arrows are natural transformations between such functors.

One particular kind of object in $[\mathbf{J}, \mathscr{C}]$ is a trivial constant functor such as $\Delta_C \colon \mathbf{J} \to \mathscr{C}$, i.e. the functor that sends every object in $\mathbf{J}$ to the object C and every arrow in $\mathbf{J}$ to 1_C.

Now, what would be a natural transformation from Δ_C to another diagram-as-functor D? Applying the definition, it would be a family α of $\mathbf{J}$ arrows $\alpha_J \colon \Delta_C(J) \to D(J)$ indexed by $J \in \mathbf{J}$, i.e. arrows $\alpha_J \colon C \to D(J)$, such that for every $d \colon K \to L$ in $\mathbf{J}$, the square below always commutes in $\mathscr{C}$. Hence, trivially, so does the triangle:

$$\begin{tikzcd} C \arrow[r, "1_C"] \arrow[d, "\alpha_K"'] & C \arrow[d, "\alpha_L"] \\ D(K) \arrow[r, "D(d)"] & D(L) \end{tikzcd} \quad \Rightarrow \quad \begin{tikzcd} & C \arrow[dl, "\alpha_K"'] \arrow[dr, "\alpha_L"] & \\ D(K) \arrow[rr, "D(d)"] & & D(L) \end{tikzcd}$$

But we recognize that! It means that C together with the α_J form a cone over D. And conversely, of course, the arrows α_J in any cone over D with vertex C form a natural transformation $\alpha \colon \Delta_C \Rightarrow D$. So in sum:

Theorem 91. *A cone over $D \colon \mathbf{J} \to \mathscr{C}$ with vertex C comprises C together with a natural transformation from the trivial functor $\Delta_C \colon \mathbf{J} \to \mathscr{C}$ to D.*

If we think of cones the more austere way (i.e. just as a family of arrows – see the end of §16.2), then we can take $Cone(C, D)$, the collection of cones over D with vertex C, to be simply $[\mathbf{J}, \mathscr{C}](\Delta_C, D)$, i.e. the hom-set of arrows in the functor category $[\mathbf{J}, \mathscr{C}]$ from Δ_C to D.

(b) We can think of the functor Δ_C (living as an object in the functor category $[\mathbf{J}, \mathscr{C}]$) as itself the value at the object C of a functor $\Delta \colon \mathscr{C} \to [\mathbf{J}, \mathscr{C}]$.

To be functorial, how must Δ act on an arrow $f \colon C \to D$ in $\mathscr{C}$? It must send f to an arrow, i.e. a natural transformation, from $\Delta_C \colon \mathbf{J} \to \mathscr{C}$ to $\Delta_D \colon \mathbf{J} \to \mathscr{C}$. But just by definition, a natural transformation α from Δ_C to Δ_D is a suite of arrows α_J indexed by objects $J \in \mathbf{J}$ such that for any $j \colon K \to L$ in $\mathbf{J}$, these diagrams commute

$$\begin{tikzcd} \Delta_C(K) \arrow[r, "\Delta_C(j)"] \arrow[d, "\alpha_K"] & \Delta_C(L) \arrow[d, "\alpha_L"] \\ \Delta_D(K) \arrow[r, "\Delta_D(j)"] & \Delta_D(L) \end{tikzcd} \quad \text{i.e.} \quad \begin{tikzcd} C \arrow[r, "1_C"] \arrow[d, "\alpha_K"] & C \arrow[d, "\alpha_L"] \\ D \arrow[r, "1_D"] & D \end{tikzcd}$$

Therefore every component of α must be equal and we'll have to put all of them equal to f (the only arrow from C to D we are given!). The resulting action of Δ on f is easily checked to be functorial.

(c) While we are considering functors between $\mathscr{C}$ and $[\mathbf{J}, \mathscr{C}]$, let's also pause to note one that goes in the opposite direction. This is the functor $\underset{\leftarrow\mathbf{J}}{\mathit{Lim}}\colon [\mathbf{J}, \mathscr{C}] \to \mathscr{C}$ which exists when, but only when, *every* diagram $D\colon \mathbf{J} \to \mathscr{C}$ has a limit.

As you would expect from its name, the functor $\underset{\leftarrow\mathbf{J}}{\mathit{Lim}}$ sends a diagram D in $[\mathbf{J}, \mathscr{C}]$ to the vertex $\underset{\leftarrow\mathbf{J}}{\mathit{Lim}}\, D$ for some chosen limit cone over D in $\mathscr{C}$. But note however that this functor is not 'naturally' or canonically defined: for in the general case, limits over D are only unique up to isomorphism, so we will indeed have to select a particular limit object $\underset{\leftarrow\mathbf{J}}{\mathit{Lim}}\, D$.

To get a genuine functor, we now need suitably to define $\underset{\leftarrow\mathbf{J}}{\mathit{Lim}}$'s action on arrows. This must send an arrow in $[\mathbf{J}, \mathscr{C}]$, i.e. a natural transformation $\alpha\colon D \Rightarrow D'$ (for functors $D, D'\colon \mathbf{J} \to \mathscr{C}$), to an arrow in $\mathscr{C}$ from $\underset{\leftarrow\mathbf{J}}{\mathit{Lim}}\, D$ to $\underset{\leftarrow\mathbf{J}}{\mathit{Lim}}\, D'$. How can it do this in a natural way? Well, by hypothesis there are limit cones over D and D', respectively $[\underset{\leftarrow\mathbf{J}}{\mathit{Lim}}\, D, \pi_J]$ and $[\underset{\leftarrow\mathbf{J}}{\mathit{Lim}}\, D', \pi'_J]$. So now take any arrow $d\colon K \to L$ living in $\mathbf{J}$ and consider the following diagram:

$$\begin{tikzcd}
& \underset{\leftarrow\mathbf{J}}{\mathit{Lim}}\, D \arrow[dl, "\pi_K"'] \arrow[dr, "\pi_L"] \arrow[dd, dashed, "u_\alpha"] & \\
D(K) \arrow[rr, "D(d)"] \arrow[dd, "\alpha_K"'] & & D(L) \arrow[dd, "\alpha_L"] \\
& \underset{\leftarrow\mathbf{J}}{\mathit{Lim}}\, D' \arrow[dl, "\pi'_K"'] \arrow[dr, "\pi'_L"] & \\
D'(K) \arrow[rr, "D'(d)"] & & D'(L)
\end{tikzcd}$$

The top triangle commutes (because $[\underset{\leftarrow\mathbf{J}}{\mathit{Lim}}\, D, \pi_J]$ is a limit). The lower square commutes by the naturality of α. Therefore the outer pentangle commutes and so, generalizing, $[\underset{\leftarrow\mathbf{J}}{\mathit{Lim}}\, D, \alpha_J \circ \pi_J]$ is a cone over D'. But then *this* cone must factor uniquely through D''s limit cone $[\underset{\leftarrow\mathbf{J}}{\mathit{Lim}}\, D', \pi'_J]$ via some unique $u_\alpha\colon \underset{\leftarrow\mathbf{J}}{\mathit{Lim}}\, D \to \underset{\leftarrow\mathbf{J}}{\mathit{Lim}}\, D'$. The map $\alpha \mapsto u_\alpha$ is then a natural candidate for $\underset{\leftarrow\mathbf{J}}{\mathit{Lim}}$'s action on arrows, and indeed this assignment is easily checked to be yield a functor.

Notation. To decrease future clutter, we'll write '$\underset{\leftarrow\mathbf{J}}{\mathit{Lim}}$ as simply '*Lim*' when context supplies the $\mathbf{J}$. And we'll write $\mathit{Lim}_{\mathscr{C}}$ when we need to emphasize the role of $\mathscr{C}$.

The diagram above can be recycled, by the way, to show

Theorem 92. *Assuming the limit functors exist, then if we have a natural isomorphism $D \cong D'$, $\mathit{Lim}\, D \cong \mathit{Lim}\, D'$.*

Proof. Because we now have an natural isomorphism $D \cong D'$, we can show as above both that there is a unique $u\colon \mathit{Lim}\, D \to \mathit{Lim}\, D'$ and symmetrically that there is a unique $u'\colon \mathit{Lim}\, D' \to \mathit{Lim}\, D$. These compose to give us map $u' \circ u\colon \mathit{Lim}\, D \to \mathit{Lim}\, D$ which must be $1_{\mathit{Lim}\, D}$ by the now familiar argument (the limit cone with vertex $\mathit{Lim}\, D$ can factor through itself by both $u' \circ u$ and $1_{\mathit{Lim}\, D}$, but there is by hypothesis only one way for the limit cone to factor through itself). Likewise, $u \circ u' = 1_{\mathit{Lim}\, D'}$. So u is an isomorphism. □

17.3 Pullbacks, products and equalizers related

We now turn to the main business of the chapter, namely showing that if a category has all finite/small products and equalizers it has all finite/small limits. We could go straight to the abstract general argument (it isn't difficult): but it will hopefully be illuminating if we take things in three gentler stages. As a warm-up exercise, in this section we prove a miniaturized version of our later main result. We will show that, if a category has binary products and equalizers for any pair of parallel arrows then it has pullbacks for any corner.

First, we pause to note an almost-converse of that:

Theorem 93. *Suppose $\mathscr{C}$ has a pullback for any corner and also a terminal object: then $\mathscr{C}$ has (i) binary products, and hence (ii) equalizers.*

Proof. For (i), note that the pullback of the corner $X \xrightarrow{f} 1 \xleftarrow{g} Y$ is just the product $X \times Y$ with the usual projection functions.

For (ii), think of parallel arrows $X \overset{f}{\underset{g}{\rightrightarrows}} Y$ as a wedge $Y \xleftarrow{f} X \xrightarrow{g} Y$, which factors uniquely via $\langle f, g\rangle$ through the product $Y \times Y$.

So now consider the pullback for the corner $X \xrightarrow{\langle f,g\rangle} Y \times Y \xleftarrow{\delta_Y} Y$, where δ_Y is the 'diagonal' arrow (see Defn. 61):

$$\begin{tikzcd}
E \arrow[r, "q"] \arrow[d, "e"'] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[d, "\delta_Y"] \\
X \arrow[r, "\langle f,g\rangle"'] & Y \times Y
\end{tikzcd}$$

Then $[E, e]$ form an equalizer for the parallel pair f, g.

By the commutativity of the pullback square, $\delta_Y \circ q = \langle f, g\rangle \circ e$. That means $\langle q, q\rangle = \langle f \circ e, g \circ e\rangle$. So $f \circ e = g \circ e$. Hence $E \xrightarrow{e} X \overset{f}{\underset{g}{\rightrightarrows}} Y$ is a fork.

Take any other fork $C \xrightarrow{c} X \overset{f}{\underset{g}{\rightrightarrows}} Y$. The wedge $X \xleftarrow{c} C \xrightarrow{f \circ c} Y$ must factor through E (because E is the vertex of the pullback) via a unique v. Which is to say that there is a unique arrow $v\colon C \to E$ through which the second fork factors through the first one, making the fork a limiting case, and $[E, e]$ an equalizer. ☐

And now for the announced warm-up theorem:

Theorem 94. *If a category $\mathscr{C}$ has binary products and equalizers, then it has pullbacks.*

Proof. Suppose we start with an arbitrary corner $X \xrightarrow{f} Z \xleftarrow{g} Y$. Because $\mathscr{C}$ has binary products, there will in particular be a product $X \times Y$ with projections $\pi_1 \colon X \times Y \to X$ and $\pi_1 \colon X \times Y \to Y$. This gives us parallel arrows $X \times Y \overset{f\circ\pi_1}{\underset{g\circ\pi_2}{\rightrightarrows}} Z$. Because $\mathscr{C}$ has equalizers, that parallel pair must have an equalizer $[E, e]$. So we get the square in the following commutative diagram:

$$\begin{tikzcd}
C \arrow[dr, dotted, "?"] \arrow[drrr, bend left, "c_2"] \arrow[dddr, bend right, "c_1"'] & & & \\
& E \arrow[rr, "\pi_2\circ e"] \arrow[dd, "\pi_1\circ e"'] \arrow[dr, "e"] & & Y \arrow[dd, "g"] \\
& & X \times Y \arrow[ur, "\pi_2"'] \arrow[dl, "\pi_1"'] & \\
& X \arrow[rr, "f"] & & Z
\end{tikzcd}$$

Claim: the wedge formed by E with the projections $\pi_1 \circ e$, $\pi_2 \circ e$ is a pullback of the original corner.

Consider any other cone over the original corner, i.e. consider any wedge $X \xleftarrow{c_1} C \xrightarrow{c_2} Y$ with $fc_1 = gc_2$. We need to show that this factors uniquely through E.

Now, that wedge certainly uniquely factors through the product $X \times Y$, so there is a unique $u \colon C \to X \times Y$ such that $c_1 = \pi_1 \circ u$, $c_2 = \pi_2 \circ u$. Hence $f \circ \pi_1 \circ u = g \circ \pi_2 \circ u$. Therefore $C \xrightarrow{u} X \times Y \overset{f\circ\pi_1}{\underset{g\circ\pi_2}{\rightrightarrows}} Z$ is a fork, which must factor uniquely through the equalizer E via some v.

That is to say, there is a $v \colon C \to E$ such that $e \circ v = u$. Hence $\pi_1 \circ e \circ v = \pi_1 \circ u = c_1$. In other words, completing the diagram with $v \colon C \to E$ makes the left triangle commute. Symmetrically, it makes the top triangle commute.

Finally, we need to prove the uniqueness of such a diagram-completing arrow. Suppose $v' \colon C \to E$ also makes $\pi_1 \circ e \circ v' = c_1$, $\pi_2 \circ e \circ v' = c_2$. But $e \circ v' \colon C \to X \times Y$, and we know the wedge $X \xleftarrow{c_1} C \xrightarrow{c_2} Y$ factors uniquely through $X \times Y$, so $e \circ v' = u = e \circ v$. But equalizers are monic, so $v' = v$. ☐

17.4 Set has all finite/small limits

(a) We fix some standard terminology:

Definition 83. Let $\mathscr{C}$ be a category, and $\mathbf{J}$ be a small category. Then:

(1) $\mathscr{C}$ *has all limits of shape* $\mathbf{J}$ iff for all functors $D\colon \mathbf{J} \to \mathscr{C}$ there exists a limit cone over D in $\mathscr{C}$.

(2) $\mathscr{C}$ *has all finite limits* if for any finite category $\mathbf{J}$, $\mathscr{C}$ has limits of shape $\mathbf{J}$. A category with all finite limits is also said to be *finitely complete*.

(3) $\mathscr{C}$ *has all small limits* if for any small category $\mathbf{J}$, $\mathscr{C}$ has limits of shape $\mathbf{J}$. A category with all limits is also said to be *complete*.

(I guess that talk about having 'limits of shape $\mathbf{J}$', though entirely conventional, is rather misplaced. After all, the *limits* won't look like $\mathbf{J}$, rather it is the *diagrams that the limits are over* which will look like $\mathbf{J}$!)

It terms of this jargon, and given what we've seen before, a category's having terminal objects is a matter of having limits of shape $\mathbf{0}$ (where that's the empty category), having binary products is a matter of having limits of shape $\mathbf{T}$ (where that is 'the' two-element discrete category), having equalizers is a matter of having limits of shape $\circlearrowright \bullet \rightrightarrows \star \circlearrowleft$, and having pullbacks is having limits of shape $\bullet \longrightarrow \star \longleftarrow \diamond$. (omitting the identity arrows).

(b) Now revisit the proof of the last theorem where we showed that having binary products and equalizers entails having pullbacks.

We first constructed the product of objects from a corner. But that didn't give us the vertex of the limit cone over the corner. We got to that by taking an equalizer, which supplied us with the limit object and a monic arrow into the product we took. Now in **Set**, monics are injective and – up to isomorphism – a set-with-an-injection-into-a-product can be thought of as a subset-of-that-product. That gives us the clue to how to prove the following in a rather brute-force way:

Theorem 95. **Set** *has all finite limits.*

Proof. Take a finite category $\mathbf{J}$, and a diagram-as-functor $D\colon \mathbf{J} \to \mathbf{Set}$.

First construct the product $[P, p_J]$ of all the objects $D(J)$ for $J \in \mathbf{J}$ (which of course we can do in **Set**). But that's not (in general) going to give us the required vertex E of a limit cone over D [we'll call the vertex 'E' rather than 'L' to hook up nicely with a later argument]. We need again to take a subset of P – i.e. take some subset of the tuples $\vec{x} = \langle x_I \rangle_{I \in \mathbf{J}}$, where each tuple-component x_J is of course a member of the corresponding $D(J)$. *But which subset of P do we want?*

Whenever there is an arrow $f\colon J \to K$ in $\mathbf{J}$ there will be a corresponding arrow $D(f)\colon D(J) \to D(K)$ in **Set**. Hence to get a cone with vertex E (let alone a limit cone) we need E to be equipped with projections $\pi_J \colon E \to D(J)$, $\pi_K \colon E \to D(K)$ that commute with $D(f) \colon D(J) \to D(K)$ – i.e. we require $\pi_K = D(f) \circ \pi_J$.

Now, there's a 'natural' projection from E to D_J for each J. Just have π_J agree with p_J in picking the J-th component of a tuple $\vec{x}$. In effect we just cut down each p_j from the domain P to the domain E. Or to put that more carefully, if $e \colon E \to P$ is the obvious injection map here, then $\pi_J = p_J \circ e$.

To get a cone, we need the relevant projection arrows to commute with any arrow in D. But then, to get $\pi_K = D(f) \circ \pi_J$, for any tuple $\vec{x}$ in E we require that $D(f)(x_J) = x_K$.

So all that tells us one way of getting a cone $[E, \pi_J]$ over $D \colon \mathbf{J} \to \mathbf{Set}$:

(1) Put E to be the set of tuples $\vec{x} = \langle x_I \rangle_{I \in \mathbf{J}}$ such that $x_J \in D(J)$ for all $J \in \mathbf{J}$ and where $D(f)(x_J) = x_K$ for all $f \colon J \to K$ in $\mathbf{J}$.

(2) Put $\pi_J(\vec{x}) = x_J$ (i.e. put $\pi_J = p_J \circ e$, where $e \colon E \to P$ is the injection map).

And since we've constructed this cone by cutting down another limit in the most natural and economical way possible, it morally ought to be a limit cone.

Which it is! Suppose $[C, c_J]$ is any cone over D. Define the map $u \colon C \to E$ as sending $c \in C$ to the tuple $\langle c_J(c) \rangle_{I \in \mathbf{J}}$. Then evidently $[C, c_J]$ factors through $[E, \pi_J]$ via u. And u is unique. For if $[C, c_J]$ also factors through $[E, \pi_J]$ via u', then $\pi_J \circ u' = c_J$, therefore for any c the J-th component of $u'(c) = c_J(c)$, hence $u' = u$. □

Since everything in that proof is finite it also establishes that **FinSet** also has all finite limits. But in **Set** we can of course form Cartesian products for set-indexed collection of sets of arbitrary size. Hence, so long as $\mathbf{J}$ remains set-sized exactly the same proof will establish the stronger result that, unlike with **FinSet**,

Theorem 96. **Set** *has all small limits.*

17.5 The existence of limits, more generally

(a) The proof in the last section almost wrote itself once we had the idea of taking a product $[P, p_J]$ of the objects $D(J)$, and then (pretty much by brute force) extracting a subset of $E \subseteq P$ containing just those tuples which satisfy the condition necessary to get a cone with vertex E over D. However, our proof talked about tuples, subsets etc. in an old-school, non-categorial way. Let's see if we can rethink the proof, continuing to work for the moment in Set but now putting things in more purely categorial terms.

So again we take the product $[P, p_J]$ of the objects of D (we can do this since Set has all finite products). And in headline terms, we know what we need to do next – we must get the subset E by defining it as an equalizer. But an equalizer for what?

Take another look at the proof of Theorem 94. There we started with a corner diagram, i.e. with two arrows sharing a target, $f\colon X \to Z$, $g\colon Y \to Z$. We got parallel arrows which share a source as well as a target by taking a product, thereby getting $X \times Y \overset{f\circ\pi_1}{\underset{g\circ\pi_2}{\rightrightarrows}} Z$. And *then* we could look for an equalizer.

Now, in a diagram D there could be lots of arrows of the kind $d\colon D(K) \to D(L)$, and this time the arrows may well have different targets. But we still want to end up with a pair of parallel arrows with the same target if we are going to be able to take an equalizer. We'll evidently have to go from many targets to a single target by taking a product again. So let's define $[Q, q_L]$ as the product of all the objects $D(L)$ which are targets for arrows in D (again, we can do this since Set has all finite products).

The name of the game is now to define a pair of parallel arrows

$$P \overset{v}{\underset{w}{\rightrightarrows}} Q$$

which we are going to equalize by some $[E, e]$, and we aim to do this so that, in Set, E is the set of tuples which we met in the proof in the last section.

But there are in fact only two naturally arising arrows from P to Q. Consider first the cone with vertex P and with an arrow $p_L\colon P \to D(L)$ for each $D(L)$ which contributes to the product Q. This (by definition of the product $[Q, q_L]$) must factor through the product by a unique mediating arrow v, so that $p_L = q_L \circ v$ for each l.

Consider secondly the cone with vertex P and an arrow $d\circ p_K\colon P \to D(L)$ for each arrow $d\colon D(K) \to D(L)$ in D. This too must factor through the product $[Q, q_L]$ by a unique mediating arrow w, so that $d \circ p_K = q_L \circ w$ for each arrow $d\colon D(K) \to D(L)$.

Since all parallel arrows have equalizers in Set, we can take the equalizer of v and w, namely $[E, e]$. Then E will be, as we want, the subset of tuples of objects from the $D(J)$ such that (for any arrow $d\colon D(K) \to D(L)$) $d(x_K) = x_L$.

And now the big claim: as before, $[E, p_J \circ e]$ will be a limit cone over D. Let's state this as a theorem:

Theorem 97. *Let D be a diagram in* Set. *Let $[P, p_J]$ be the product of the objects $D(J)$ in D, and $[Q, q_L]$ be the product of the objects $D(L)$ which are targets of arrows in D. Then there are arrows*

$$P \overset{v}{\underset{w}{\rightrightarrows}} Q$$

such that the following diagrams commute for each $d\colon D(K) \to D(L)$:

$$\begin{tikzcd} P \arrow[r, "v"] \arrow[dr, "p_L"'] & Q \arrow[d, "q_L"] \\ & D(L) \end{tikzcd} \qquad \begin{tikzcd} P \arrow[r, "w"] \arrow[d, "p_K"'] & Q \arrow[d, "q_L"] \\ D(K) \arrow[r, "d"] & D(L) \end{tikzcd}$$

Let the equalizer of v and w be $[E, e]$. Then $[E, p_J \circ e]$ will be a limit cone over D in Set.

Proof. We have already defined v and w so that the given diagrams commute. So next we confirm $[E, p_J \circ e]$ is a cone. Suppose there is an arrow $d\colon D(K) \to D(L)$. Then we require $d \circ p_K \circ e = p_L \circ e$.

But indeed $d \circ p_K \circ e = q_L \circ w \circ e = q_L \circ v \circ e = p_L \circ e$, where the inner equation holds because e is an equalizer of v and w.

Second we show that $[E, p_J \circ e]$ is a limit. So suppose $[C, c_J]$ is any other cone over D. Then there must be a unique $u\colon C \to P$ such that every c_J factors through the product and we have $c_J = p_J \circ u$.

Since $[C, c_J]$ is a cone, for any $d\colon D(K) \to D(L)$ in D we have $d \circ c_K = c_L$. Hence $d \circ p_K \circ u = p_L \circ u$, and hence for each q_L, $q_L \circ w \circ u = q_L \circ v \circ u$. But then we can apply the obvious generalized version of Theorem 70, and conclude that $w \circ u = v \circ u$. Which means that

$$\begin{tikzcd} C \arrow[r, "u"] & P \arrow[r, shift left, "p"] \arrow[r, shift right, "q"'] & Q \end{tikzcd}$$

is a fork, which must therefore uniquely factor through the equalizer $[E, e]$. That is to say, there is a unique $s\colon C \to E$ such that $u = e \circ s$, and hence for all j, $c_J = p_J \circ u = p_J \circ e \circ s$. That is to say, $[C, c_J]$ factors uniquely through $[E, p_J \circ e]$ via s. Therefore $[E, p_J \circ e]$ is indeed a limit cone. $\square$

(b) That gives us a categorial proof, without mentioning subsets, of our result that **Set** has all (finite) limits. *But now note that we didn't depend on anything but the fact that* **Set** *has (finite) products and equalizers.* Hence, generalizing, we have the originally promised sweeping result:

And similarly, if $\mathbf{J}$ is (not assumed finite) but small, then the argument will again go through so long as this time $\mathscr{C}$ has products for all set-sized collections of objects, or as we say, has all small products. Hence

Theorem 98. *If $\mathscr{C}$ has all small products and has equalizers, then it has all small limits, i.e. is complete.*

Given ingredients from our previous discussions, we can easily show that

Theorem 99. *The categories of structured sets* **Mon***,* **Grp***,* **Ab***,* **Rng** *(among others) are all complete.*

Top too is complete. We have also met categories that are, by contrast, finitely complete but not complete like **FinSet**; while e.g. a poset-as-a-category may lack all many products and hence not be even finitely complete.

17.6 Dualizing again

Needless to say by this stage, our results in the last two sections dualize in obvious ways. Thus we need not delay over the proofs of

Theorem 100. *If $\mathscr{C}$ has initial objects, binary coproducts and co-equalizers, then it has all finite colimits, i.e. is finitely cocomplete. If $\mathscr{C}$ has all coproducts and has co-equalizers, then it has all small colimits, i.e. is cocomplete.*

Theorem 101. **Set** *is cocomplete – as are the categories of structured sets* **Mon***,* **Grp***,* **Ab***,* **Rng***.*

But note that a category can of course be (finitely) complete without being (finitely) cocomplete and vice versa. For a generic source of examples, take a poset $(P, \preccurlyeq)$ considered as a category. This automatically has all equalizers (and coequalizers) – see §15.2 Ex. (5). But it will have other limits (colimits) depending on which products (coproducts) exists, i.e. which sets of elements have supreme (inflame). For a simple case, take a poset with a maximum element and such that every pair of elements has a supremum: then considered as a category it has all finite limits (but maybe not infinite ones). But it need not have a minimal element and/or infima for all pairs of objects: hence it can lack some finite colimits despite having all finite limits.

18 Functors and limits

A functor $F\colon \mathscr{C} \to \mathscr{D}$ must, just in virtue of its functoriality, preserve some aspects of the categorial structure of $\mathscr{C}$ as it sends objects and arrows into $\mathscr{D}$. The 'some' is essential, though. Recall, for example, that functors preserve left and right inverses (sections and retractions) but don't necessarily preserve monomorphisms and epimorphisms. So how do things stand with respect to preserving limits and colimits? And how do functors interact with limits more generally?

18.1 Preserving limits

Start with a natural definition: a functor preserves limits if it sends limits to limits and preserves colimits if it sends colimits to colimits. More carefully,

Definition 84. A functor $F\colon \mathscr{C} \to \mathscr{D}$ *preserves the limit* $[L, \pi_J]$ over $D\colon \mathbf{J} \to \mathscr{C}$ if $[FL, F\pi_J]$ is a limit over $F \circ D\colon \mathbf{J} \to \mathscr{D}$.

A functor $F\colon \mathscr{C} \to \mathscr{D}$ *preserves limits of shape* $\mathbf{J}$ iff, for any diagram $D\colon \mathbf{J} \to \mathscr{C}$, if $[L, \pi_J]$ is a limit cone D, then F preserves it.

A functor which preserves limits of shape $\mathbf{J}$ for all finite small) categories $\mathbf{J}$ is said to *preserve all finite (small) limits.*

Dually for preserving colimits.

Let's immediately look at some examples. Our first two cases might seem a little artificial, but they will serve to illustrate an important point in an elementary way.

(1) Consider the functor $P\colon \mathbf{Set} \to \mathbf{Set}$ which sends the empty set $\emptyset$ to itself and sends every other set to the singleton 1, and acts on arrows in the only possible way if it is to be a functor (i.e. for $A \neq \emptyset$, it sends any arrow $\emptyset \to A$ to the unique arrow $\emptyset \to 1$, it sends the arrow $\emptyset \to \emptyset$ to itself and sends all other arrows to the identity arrow 1_1). Claim: *P preserves binary products but not equalizers.*

For the first half of this claim, we simply consider cases. If neither A nor B is the empty set, then $A \times B$ is not empty either. P then sends the limit wedge $A \leftarrow A \times B \to B$ to $1 \leftarrow 1 \to 1$, and it is obvious that any other wedge $1 \leftarrow L \rightarrow 1$ factors uniquely through the latter. So P sends non-empty products to products.

If A is the empty set and B isn't, $A \times B$ is the empty set too. Then P sends the limit wedge $A \leftarrow A \times B \rightarrow B$ to $\emptyset \leftarrow \emptyset \rightarrow 1$. Since the only arrows in **Set** with the empty set as target have the empty set as source, the only wedges $\emptyset \leftarrow L \rightarrow 1$ have $L = \emptyset$, so trivially factor uniquely through $\emptyset \leftarrow \emptyset \rightarrow 1$. So P sends products of the empty set with non-empty sets to products.

Likewise, of course, for products of non-empty sets with the empty set, and the product of the empty set with itself. So, taking all the cases together, P sends products to products.

Now consider the equalizer in **Set** of the two different maps $1 \overset{f}{\underset{g}{\rightrightarrows}} 2$, where 2 is a two-membered set. Since f and g never agree, their equalizer is the empty set (with the empty inclusion map). But since P sends both the maps f and g to the identity map on 1, the equalizer of Pf and Pg is therefore the set 1 (with the identity map). Which means that the equalizer of $P(f)$ and $P(g)$ is *not* the result of applying P to the equalizer of f and g.

(2) Consider next the functor $Q\colon \mathbf{Set} \rightarrow \mathbf{Set}$ which sends any set X to the set $X \times 2$, and sends any arrow $f\colon X \rightarrow Y$ to $f \times 1_2\colon X \times 2 \rightarrow Y \times 2$ (the latter is of course the function which acts on a pair $\langle x, n\rangle \in X \times 2$ by sending it to $\langle fx, n\rangle$). Claim: *Q preserves equalizers but not binary products.*

Concerning products, if a functor F preserves binary products in **Set**, then by definition $F(X \times Y) \cong FX \times FY$. However, for X, Y finite, we have $Q(X \times Y) = (X \times Y) \times 2 \ncong (X \times 2) \times (Y \times 2) = QX \times QY$.

Now note that the equalizer of parallel arrows $X \overset{f}{\underset{g}{\rightrightarrows}} Y$ is essentially E, the subset of X on which f and g take the same value. And the equalizer of the parallel arrows $QX \overset{Qf}{\underset{Qg}{\rightrightarrows}} QY$ is the subset of $X \times 2$ on which $f \times 1_2$ and $g \times 1_2$ take the same value, which will be $E \times 2$, i.e. QE. So indeed Q preserves equalizers.

What these first two toy examples illustrate is the important point that a functor may preserve some limits but not others – preservation isn't in general an all or nothing business.

However, consider now another example:

(3) The forgetful functor $F\colon \mathbf{Mon} \rightarrow \mathbf{Set}$ sends a terminal object in **Mon**, a one-object monoid, to its underlying singleton set, which is terminal in **Set**. So F preserves limits of the empty shape!

The same functor sends a product $(M, \cdot) \times (N, *)$ in **Mon** to its underlying set of pairs of objects from M and N, which is a product in **Set**. So the forgetful F also preserves limits of the shape of the discrete two object category 2 which looks like $\circlearrowleft \bullet \quad \circlearrowleft \star$.

Likewise for equalizers. As we saw in §15.2, Ex. (2), the equalizer of two parallel monoid homomorphisms $(M, \cdot) \underset{g}{\overset{f}{\rightrightarrows}} (N, *)$ is $(E, \cdot)$ equipped with the inclusion map $E \hookrightarrow M$, where E is the set on which f and g agree. Which means that the forgetful functor takes the equalizer of f and g as monoid homomorphisms to their equalizer as set functions. So F preserves equalizers, i.e. preserves limits of shape $\circlearrowleft \bullet \rightrightarrows \star \circlearrowright$.

So the forgetful $F\colon \mathbf{Mon} \to \mathbf{Set}$ preserves terminal objects, binary products and equalizers. It follows by Theorem 104 in the next section that the functor in fact preserves all limits.

(4) On the other hand, the same forgetful F does not preserve colimits with the 'shape' of the empty category, i.e. initial objects. For a one-object monoid is initial in **Mon** but its underlying singleton set is not initial in **Set**.

F does not preserve coproducts either – essentially because coproducts in **Mon** can be larger than coproducts in **Set**. Recall our discussion in §15.1 of coproducts in **Grp** : similarly, $F(M \oplus N)$, the underlying set of a coproduct of monoids M and N, is (isomorphic to) the set of finite sequences of alternating non-identity elements from M and N. Contrast $FM \oplus FN$, which is just the disjoint union of the underlying sets.

The example generalizes. A forgetful functor from a category of structured sets to **Set** typically preserves limits but not all colimits.

18.2 General results about preservation

We know that limits over a given diagram are, in general, not unique. However, a functor will treat them the same, in this respect:

Theorem 102. *If F preserves some limit over the diagram D, it preserves all limits over that diagram.*

Proof. Suppose $[L, \pi_J]$ is a limit cone over $D\colon \mathbf{J} \to \mathscr{C}$. Then, by Theorem 84, if $[L', \pi'_J]$ is another such cone, there is an isomorphism $f : L' \to L$ in $\mathscr{C}$ such that $\pi'_J = \pi_j \circ f$.

Suppose F preserves $[L, \pi_J]$ so $[FL, F\pi_J]$ is a limit cone over $F \circ D$. Then F sends $[L', \pi'_J]$ to $[FL', F\pi'_J] = [FL', F\pi_J \circ Ff]$. But this factors through $[FL, F\pi_J]$ via the isomorphism $Ff\colon FL' \to FL$. Hence, by Theorem 85, $[FL', F\pi'_J]$ is also a limit over $F \circ D$. In other words, F preserves $[L', \pi'_J]$ too. ☐

Next, we consider what happens if we apply isomorphic functors to the same limit:

Theorem 103. *Suppose the parallel functors $F, G\colon \mathscr{C} \to \mathscr{D}$ are naturally isomorphic. Then if F preserves a given limit so does G.*

This is one of those straightforward apply-the-definitions proofs that looks even simpler if we can wave our hands at diagrams grown in real time on a blackboard, using different coloured chalks! ...

Proof. Let $[L, \pi_J]$ be a limit cone for $D\colon \mathbf{J} \to \mathscr{C}$. Then for any $f\colon J \to K$ in $\mathbf{J}$, the upper diagram commutes in $\mathscr{C}$. The actions of F and G send this triangle to the two commuting triangles in the lower diagram, and the assumumed natural isomorphism $\alpha\colon F \Rightarrow G$ gives us *three* naturality squares, resulting in a commuting prism in $\mathscr{D}$:

$$\begin{tikzcd}
& L \arrow[dl, "\pi_J"'] \arrow[dr, "\pi_K"] & \\
DJ \arrow[rr, "Df"'] & & DK
\end{tikzcd}$$

$$\begin{tikzcd}
& FL \arrow[ddl, "F\pi_J"'] \arrow[ddr, "F\pi_K"] \arrow[drr, leftrightarrow, "\alpha_L"] & & & \\
& & & GL \arrow[ddl, "G\pi_J"'] \arrow[ddr, "G\pi_K"] & \\
FDJ \arrow[rr, "FDf"'] \arrow[drr, leftrightarrow, "\alpha_{DJ}"'] & & FDK \arrow[drr, leftrightarrow, "\alpha_{DK}"] & & \\
& & GDJ \arrow[rr, "GDf"'] & & GDK
\end{tikzcd}$$

So now consider any cone $[C, c_J]$ over GD with vertex C. Each tall triangle such as the one below commutes:

$$\begin{tikzcd}
& & & C \arrow[dll, dashed, "u"'] \arrow[d, dashed, "v"] \arrow[dddl, "c_J"'] \arrow[dddr, "c_K"] & \\
& FL \arrow[ddl, dotted] \arrow[ddr, dotted] \arrow[drr, dotted, leftrightarrow] & & GL \arrow[ddl, dotted] \arrow[ddr, dotted] & \\
& & & & \\
FDJ \arrow[rr, dotted] \arrow[drr, dotted, leftrightarrow] & & FDK \arrow[drr, dotted, leftrightarrow] & & \\
& & GDJ \arrow[rr, "GDf"'] & & GDK
\end{tikzcd}$$

Using the commuting base square of the prism, we can evidently extend each leg c_J of the cone by composition with α_{DJ}^{-1} to get a cone $[C, \alpha_{DJ}^{-1} \circ c_J]$ over FD.

Now *suppose that F preserves the limit* $[L, \pi_J]$, so that $[FL, F\pi_J]$ is a limit cone. Then our cone over FD must factor through this limit cone via a unique $u\colon C \to FL$. But, chasing arrows round the diagram, using the sloping sides of the prism, that implies in turn that $[C, c_J]$ over GD factors through $[GL, G\pi_J]$ via $v = \alpha_L \circ u$. And $[C, c_J]$ can't factor through a distinct v': or else there would be a distinct $u' = \alpha_L^{-1} \circ v'$ which makes everything commute, which is impossible by the uniqueness of u.

Hence, in sum, any $[C, c_J]$ factors through $[GL, G\pi_J]$ via a unique v, and therefore $[GL, G\pi_J]$ is a limit cone. *So G also preserves the limit* $[L, \pi_J]$. □

Theorem 104. *If $\mathscr{C}$ is (finitely) complete, and a functor $F\colon \mathscr{C} \to \mathscr{D}$ preserves terminal objects, binary products and equalizers, then F preserves all (finite) limits.*

Proof. Suppose $\mathscr{C}$ is finitely complete. Then any limit cone $[C, c_J]$ over a diagram $D\colon \mathbf{J} \to \mathscr{C}$ is uniquely isomorphic to some limit cone $[C', c'_J]$ constructed from equalizers and finite products built in turn from terminal objects and binary products, by Theorem **??**. Since F preserves terminal objects, binary products and equalizers, it sends the construction for $[C', c'_J]$ to a construction for a limit cone $[FC', Fc'_J]$ over $F \circ D\colon \mathbf{J} \to \mathscr{D}$. But F preserves isomorphisms, so $[FC, Fc_J]$ will be isomorphic to $[FC', Fc'_J]$ and hence is also a limit cone over $F \circ D\colon \mathbf{J} \to \mathscr{D}$.

The proof for the case where $\mathscr{C}$ is complete is exactly similar. □

Theorem 105. *Suppose that $\mathscr{C}$ has all limits of shape $\mathbf{J}$. Then for any $D\colon \mathbf{J} \to \mathscr{C}$ which the functor $F\colon \mathscr{C} \to \mathscr{D}$ preserves,*

$$(*) \qquad F(\underset{\leftarrow \mathbf{J}}{Lim}\, D) \cong \underset{\leftarrow \mathbf{J}}{Lim}(F \circ D).$$

In brief: F commutes with $\underset{\leftarrow \mathbf{J}}{Lim}$.

Proof. Since $\mathscr{C}$ has all limits of shape $\mathbf{J}$, the functor $\underset{\leftarrow \mathbf{J}}{Lim}$ as introduced in §17.2 (c) is well-defined. And if F preserves a limit cone over $D\colon \mathbf{J} \to \mathscr{C}$ with vertex $\underset{\leftarrow \mathbf{J}}{Lim}\, D$, then F sends that limit cone to some limit cone over $F \circ D$. But that latter limit must be isomorphic via a unique isomorphism to the cone which exists with vertex $\underset{\leftarrow \mathbf{J}}{Lim}(F \circ D)$. □

The reverse doesn't hold, however: commuting with limits of shape $\mathbf{J}$ in the sense captured by $(*)$ *doesn't* always imply preserving such limits.

Here's an example. Take the category $\mathbf{C}$ of countably infinite sets and functions between them. And let 2 be the discrete two-object category with objects $0, 1$. A limit cone over any $D\colon 2 \to \mathbf{C}$ is therefore a product of $D0$ and $D1$, and all such limits exist.

But now consider the 'add one' functor $A\colon \mathbf{C} \to \mathbf{C}$ which sends a $\mathbf{C}$-object X to $X^+ = X \cup \{X\}$ and sends the $\mathbf{C}$-arrow $f\colon X \to Y$ to the function $f^+\colon X^+ \to Y^+$ which agrees with f on elements of X and sends X itself to Y.

Trivially,

$$A(\underset{\leftarrow 2}{Lim}\, D) \cong \underset{\leftarrow 2}{Lim}(A \circ D)$$

because all objects in $\mathbf{C}$ are isomorphic. But A doesn't preserve products. For A sends the standard product cone $[D0 \times D1, \pi_i]$ over D to the cone $[(D0 \times D1)^+, \pi_i^+]$ over $A \circ D$. But it is easy to see that *that* cone doesn't factor via an isomorphism through the standard product cone $[D0^+ \times D1^+, \pi_i]$ so it isn't a product over $A \circ D$.

Caccamo and Winskel (2004) consider what we need to add to $(*)$ to enable us to conclude that F preserves limits of shape $\mathbf{J}$. They offer a surprisingly lengthy proof that if the isomorphism in $(*)$ is *natural* in D in $[\mathbf{J}, \mathscr{C}]$, then F preserves the limits. But we won't actually rely on this announced result here.

We will just recall a slogan from elementary analysis: 'continuous functions commute with limits'. Which explains a bit of standard terminology you might come across:

Definition 85. A functor which commutes with limits of shape $\mathbf{J}$ for all small categories $\mathbf{J}$ is said to be *continuous*. And dually, a functor which commutes with colimits is said to be *cocontinuous*.

18.3 Two more useful results about preservation

For future use, we remark in passing on two results about preservation of limits. The first is easy:

Theorem 106. *The functor $F\colon \mathscr{C} \to \mathscr{D}$ preserves pullbacks it preserves monomorphisms (i.e. sends monos to monos). Dually, if F preserves pushouts it preserves epimorphisms.*

Proof. We need only prove the first part. By Theorem 90, if $f\colon X \to Y$ in $\mathscr{C}$ is monic then it is part of the pullback square on the left:

$$\begin{array}{ccc} X & \xrightarrow{1_X} & X \\ \downarrow{\scriptstyle 1_X} & & \downarrow{\scriptstyle f} \\ X & \xrightarrow{f} & Y \end{array} \quad \Rightarrow \quad \begin{array}{ccc} FX & \xrightarrow{1_{FX}} & FX \\ \downarrow{\scriptstyle 1_{FX}} & & \downarrow{\scriptstyle Ff} \\ FX & \xrightarrow{Ff} & FY \end{array}$$

By assumption F sends a pullback squares to pullback squares, so the square on the right is also a pullback square. So by Theorem 90 again, Ff is monic too. $\square$

For the second theorem we need, recall from §5.5 the definition of a comma category $(A \downarrow G)$ from given a functor $G\colon \mathscr{B} \to \mathscr{A}$ and an object $A \in \mathscr{A}$. This has as objects pairs of the form $\langle B, f\rangle$ (with $B \in \mathscr{B}$ and $f\colon A \to GB$ in $\mathscr{A}$); and an arrow from $\langle B, f\rangle$ to $\langle B', f'\rangle$ is a $\mathscr{B}$-arrow $j\colon B \to B'$ such that $f' = Gj \circ f$. Then we have:

Theorem 107. *Suppose we have a functor $G\colon \mathscr{B} \to \mathscr{A}$ and an object $A \in \mathscr{A}$. Then if $\mathscr{B}$ has limits of shape $\mathbf{J}$ and G preserves them, then $(A \downarrow G)$ also has limits of shape $\mathbf{J}$.*

Proof. For convenience, we introduce the forgetful functor $U\colon (A \downarrow G) \to \mathscr{B}$ which acts in the obvious way, i.e. it sends an $(A \downarrow G)$-object $\langle B, f\rangle$ to B, and sends an $(A \downarrow G)$-arrow $j\colon B \to B'$ to itself.

Take any diagram $D\colon \mathbf{J} \to (A \downarrow G)$ where for $J \in \mathbf{J}$, $D(J) = \langle D_J, f_J\rangle$, and for $d\colon J \to K$ in $\mathbf{J}$, $D(d)\colon D_J \to D_K$ is an arrow such that $f_K = GD(d) \circ f_J$. The target is to show that D has a limit.

Start with the functor $U \circ D\colon \mathbf{J} \to \mathscr{B}$. We know that *this* has a limit (by our hypothesis), call it $[L, \pi_J]$, where for any $d\colon J \to K$, $\pi_K = UD(d) \circ \pi_J$, i.e. $\pi_K = D(d) \circ \pi_J$. And since G preserves limits, we know that $[GL, G\pi_J]$ is a limit cone for $GUD\colon \mathbf{J} \to \mathscr{A}$.

Now take $[A, f_J]$. This is also a cone over GUD. Why? Each f_J is an arrow from A to GD_j i.e to $GUD(J)$. And we know that for each $d\colon J \to K$, $f_K = GUD(d) \circ f_J$.

Hence $[A, f_J]$ factors uniquely through the limit cone $[L, \pi_J]$: i.e. there is a unique $u\colon A \to GL$ such that for all J, $f_J = G\pi_J \circ u$. Which, by definition of arrows in the comma category, means that for each J, π_J is an arrow from $\langle L, u\rangle$ to $\langle D_J, f_J\rangle$ in $(A \downarrow G)$.

Now we observe that these arrows π_J give us a cone with vertex $\langle L, u\rangle$ over D, since as we saw, for any $d\colon J \to K$, $\pi_K = D(d) \circ \pi_J$. If we can show that this cone is a limit cone, then we are done.

Suppose there is another cone over D with vertex $\langle B, v\rangle$ and arrows $b_J : \langle B, v\rangle \to \langle D_J, f_J\rangle$ in $(A \downarrow G)$. Here b_J is an arrow $b_J : B \to D_J$ in $\mathscr{B}$, and given $d\colon J \to K$ in $\mathbf{J}$, $b_K = D(d) \circ b_J$. But that also makes $[B, b_J]$ a cone over $U \circ D$. So $[B, b_J]$ must factor though the limit $[L, \pi_J]$ via a unique $k\colon B \to L$: there is a unique k such that, for each J, $b_J = \pi_J \circ k$.

We are almost there! We need to prove that the cone we mentioned with vertex $\langle B, v\rangle$ factors uniquely through the cone with vertex $\langle L, u\rangle$, i.e. the arrows b_J factor uniquely through the arrows π_J. But that's what we've just shown: so we are done. □

We remark that nothing in this last proof in fact depends on $\mathbf{J}$ being small: the result therefore applies to limits of any size or shape.

18.4 Reflecting limits

Definition 86. A functor $F\colon \mathscr{C} \to \mathscr{D}$ *reflects limits of shape* $\mathbf{J}$ iff, given a cone $[C, c_J]$ over a diagram $D\colon \mathbf{J} \to \mathscr{C}$, then if $[FC, Fc_J]$ is a limit cone over $F \circ D\colon \mathbf{J} \to \mathscr{D}$, $[C, c_J]$ is already a limit cone over D.

Reflecting colimits is defined dually.

Let's again immediately turn to some examples. We can cook up more toy cases to show, e.g., that reflecting products needn't go along with reflecting equalizers and vice versa (do it!). But we have:

(1) The forgetful functor $F\colon \mathbf{Mon} \to \mathbf{Set}$ reflects all limits. Similarly for some other forgetful functors from familiar categories of structured sets to **Set**.

However, be careful! For we also have . . .

(2) The forgetful functor $F\colon \mathbf{Top} \to \mathbf{Set}$ which sends topological space to its underlying set *preserves* all limits but does not *reflect* all limits. Here's a case involving binary products. Suppose X and Y are a couple of spaces with a coarse topology, and let Z be the space $FX \times FY$ equipped with a finer topology. Then, with the obvious arrows, $X \leftarrow Z \to Y$ is a wedge to X, Y but not the limit wedge in **Top**: but $FX \leftarrow FX \times FY \to FY$ is a limit wedge in **Set**.

(3) Neither of the forgetful functors $F\colon \mathbf{Mon} \to \mathbf{Set}$ or $F\colon \mathbf{Top} \to \mathbf{Set}$ preserves all finite colimits.

Here, though, a couple of theorems that tell more us about functors that do reflect limits:

Theorem 108. *Suppose $F\colon \mathscr{C} \to \mathscr{D}$ preserves limits. Then if $\mathscr{C}$ is complete and F reflects isomorphisms, then F reflects limits.*

Proof. Since $\mathscr{C}$ is complete there exists a limit cone $[B, b_J]$ over a diagram $D\colon \mathbf{J} \to \mathscr{C}$, and so – since F preserves limits – $[FB, Fb]$ is a limit cone over $F \circ D\colon \mathbf{J} \to \mathscr{D}$.

Now suppose that there is a cone $[C, c_J]$ over a diagram D such that $[FC, Fc_J]$ is another limit cone over $F \circ D$. Now $[C, c_J]$ must uniquely factor through $[B, b_J]$ via a map $f\colon C \to B$. Which means that $[FC, Fc_J]$ factors through $[FB, Fb]$ via Ff. However, since these are by hypothesis both limit cones over $F \circ D$, Ff must be a isomorphism. Hence, since F reflects isomorphisms, f must be an isomorphism. So $[C, c_J]$ must be a limit cone by Theorem 85. □

Theorem 109. *Suppose $F\colon \mathscr{C} \to \mathscr{D}$ is full and faithful. Then F reflects limits.*

Proof. Suppose $[C, c_J]$ is a cone over a diagram $D\colon \mathbf{J} \to \mathscr{C}$, and $[FC, Fc_J]$ is a limit cone over $F \circ D\colon \mathbf{J} \to \mathscr{D}$.

Now take any other cone $[B, b_J]$ over D. Then F sends this to a cone $[FB, Fb_J]$ which must uniquely factor through the limit cone $[FC, Fc_J]$ via some $u\colon FB \to FC$ which makes $Fb_J = Fc_J \circ u$ for each $J \in \mathbf{J}$. Since F is full and faithful, $u = Fv$ for some unique $v\colon B \to C$ such that $b_J = Fc_J \circ u$ for each J. So $[FB, Fb_J]$ factors uniquely through $[C, c_J]$. Which confirms that $[C, c_J]$ is a limit cone. □

18.5 Creating limits

Alongside the rather natural notions of preserving and reflecting limits, we meet a third notion :

Definition 87. A functor $F\colon \mathscr{C} \to \mathscr{D}$ *creates limits of shape* $\mathbf{J}$ iff, for any diagram $D\colon \mathbf{J} \to \mathscr{C}$, if $[M, m_J]$ is a limit cone over $F \circ D$, there is a unique cone $[C, c_J]$ over D such that $[FC, Fc_J] = [M, m_J]$, and moreover $[C, c_J]$ is a limit cone.

Creating colimits is defined dually.

[Note: some define creation of limits by only requiring that $[FC, Fc_J]$ is isomorphic to $[M, m_J]$ in the obvious sense.]

Roughly speaking: reflection is a condition on those limit cones over $F \circ D$ which take the form $[FC, Fc_J]$, creation is a similar condition on *any* limit cone over $F \circ D$. So as you would predict,

Theorem 110. *If the functor $F\colon \mathscr{C} \to \mathscr{D}$ creates limits of shape $\mathbf{J}$, it reflects them.*

Proof. Suppose $[FC, Fc_J]$ is a limit cone over $F \circ D\colon \mathbf{J} \to \mathscr{C}$. Then for some unique cone $[C', c'_J]$, $[FC, Fc_J] = [FC', Fc'_J]$, and also $[C', c'_J]$ is a limit cone. By the uniqueness requirement, $[C', c'_J] = [C, c_J]$ (since the latter certainly fits the condition!). Hence $[C, c_J]$ is a limit cone. □

Theorem 111. *Suppose $F\colon \mathscr{C} \to \mathscr{D}$ is a functor, that $\mathscr{D}$ has limits of shape $\mathbf{J}$ and F creates such limits. Then $\mathscr{C}$ has limits of shape $\mathbf{J}$ and F preserves them.*

Proof. Take any diagram $D\colon \mathbf{J} \to \mathscr{C}$. Then there is a limit $[M, m_J]$ over $F \circ D$ (since $\mathscr{D}$ has all limits of shape $\mathbf{J}$). Hence (since F creates limits), there is a limit cone $[C, c_J]$ over D where this is such that $[FC, Fc_J]$ is $[M, m_J]$ and hence is a limit cone too. □

18.6 Hom-functors preserve limits

For the rest of this chapter $\mathscr{C}$ is a locally small category, and as usual $\mathbf{J}$ is small. The next main theorem locates one rich source of limit-preserving functors:

Theorem 112. *The covariant hom-functor $\mathscr{C}(A, –)\colon \mathscr{C} \to \mathbf{Set}$, for any A in the category $\mathscr{C}$, preserves all limits that exist in $\mathscr{C}$.*

And let's go first for a straight just-apply-the-definitions-and-see-what-happens demonstration (it's a useful reality check to run through the details):

Proof. We'll first check that $\mathscr{C}(A, –)\colon \mathscr{C} \to \mathbf{Set}$ sends a cone over the diagram $D\colon \mathbf{J} \to \mathscr{C}$ to a cone over $\mathscr{C}(A, –) \circ D\colon \mathbf{J} \to \mathbf{Set}$.

A cone has a vertex C, and arrows $C_J\colon C \to DJ$ for each $J \in \mathbf{J}$, where for any $f\colon J \to K$ in $\mathbf{J}$, so for any $Df\colon DJ \to DK$, $c_K = Df \circ c_J$.

Now, acting on objects, $\mathscr{C}(A,–)$ sends C to $\mathscr{C}(A,C)$ and sends DJ to $\mathscr{C}(A,DJ)$. And acting on arrows, $\mathscr{C}(A,–)$ sends $c_J\colon C \to DJ$ to the set function $c_J \circ –$ which takes $g\colon A \to C$ and outputs $c_J \circ g\colon A \to DJ$; and it sends $Df\colon DJ \to DK$ to the set-function $Df \circ –$ which takes $h\colon A \to DJ$ and outputs $Df \circ h\colon A \to DK$.

Diagrammatically, then, the functor sends the triangle on the left to the one on the right:

$$\begin{tikzcd} & C \arrow[dl, "c_J"'] \arrow[dr, "c_K"] & & \Rightarrow \\ DJ \arrow[rr, "Df"'] & & DK \end{tikzcd}$$

$$\begin{tikzcd} & \mathscr{C}(A,C) \arrow[dl, "c_J \circ -"'] \arrow[dr, "c_K \circ -"] & \\ \mathscr{C}(A,DJ) \arrow[rr, "Df \circ -"'] & & \mathscr{C}(A,DK) \end{tikzcd}$$

And assuming $c_K = Df \circ c_J$, we have $c_K \circ – = (Df \circ c_J) \circ – = Df \circ (c_J \circ –)$: hence, if the triangle on the left commutes, so does the triangle on the right. Likewise for other such triangles. Which means that if $[C, c_J]$ is a cone over D, then $[\mathscr{C}(A,C), c_J \circ –]$ is indeed a cone over $\mathscr{C}(A,–) \circ D$.

So far, so good! It remains, then, to show that in particular $\mathscr{C}(A,–)$ sends limit cones to limit cones.

Suppose then that $[L, \pi_J]$ is a limit cone in $\mathscr{C}$ over D. The functor $\mathscr{C}(A,–)$: $\mathscr{C} \to \mathbf{Set}$ sends the commuting triangle on the left in the next diagram to the commuting triangle at the bottom of the right triangle in the next diagram. And we now suppose that $[M, m_j]$ is any other cone over the image of D:

$$\begin{tikzcd} & L \arrow[dl, "\pi_J"'] \arrow[dr, "\pi_K"] & \\ DJ \arrow[rr, "Df"'] & & DK \end{tikzcd} \qquad \begin{tikzcd} & M \arrow[ddl, "m_J"'] \arrow[ddr, "m_K"] & \\ & \mathscr{C}(A,L) \arrow[dl, "\pi_J \circ -"] \arrow[dr, "\pi_K \circ -"'] & \\ \mathscr{C}(A,DJ) \arrow[rr, "Df \circ -"'] & & \mathscr{C}(A,DK) \end{tikzcd}$$

Hence $m_K = (Df \circ –) \circ m_J$. Now remember that M lives in $\mathbf{Set}$: so take a member x. Then $m_J(x)$ is a particular arrow in $\mathscr{C}(A,DJ)$, so we have $m_J(x)\colon A \to DJ$. Likewise we have $m_K(x)\colon A \to DK$. But $m_K(x) = Df \circ m_J(x)$. Which means that for all f the outer triangles on the left below commute and so $[A, m_J(x)]$ is a cone over D. And this must factor uniquely through an arrow $u(x)$ as follows:

$$\begin{tikzcd}
& A \arrow[d, "u(x)"] \arrow[ddl, "m_J(x)"'] \arrow[ddr, "m_K(x)"] & \\
& L \arrow[dl, "\pi_J"] \arrow[dr, "\pi_K"'] & \\
DJ \arrow[rr, "Df"'] & & DK
\end{tikzcd}$$

$$\begin{tikzcd}
& M \arrow[d, "u"] \arrow[ddl, "m_J"'] \arrow[ddr, "m_K"] & \\
& \mathscr{C}(A, L) \arrow[dl, "\pi_J \circ -"] \arrow[dr, "\pi_K \circ -"'] & \\
\mathscr{C}(A, DJ) \arrow[rr, "Df \circ -"'] & & \mathscr{C}(A, DK)
\end{tikzcd}$$

So $u(x)$ is an arrow from A to L, i.e. an element of $\mathscr{C}(A, L)$. So consider the map $u\colon M \to \mathscr{C}(A, L)$ which sends x to $u(x)$. Since $m_J(x) = \pi_J \circ u(x)$ for each x, $m_J = (\pi_J \circ -) \circ u$. Since this applies for each J, So $[M, m_j]$ factors through the image of the cone $[L, \pi_J]$ via u.

Suppose there is another map $v\colon M \to \mathscr{C}(A, L)$ such that we also have each $m_J = (\pi_J \circ -) \circ v$. Then again take an element $x \in M$: then $m_J(x) = \pi_J \circ v(x)$. So again, $[A, m_J(x)]$ factorizes through $[L, \pi_J]$ via $v(x)$ – which, by the uniqueness of factorization through limits, means that $v(x) = u(x)$. Since that obtains for all $x \in M$, $v = u$. Hence $[M, m_j]$ factors uniquely through the image of $[L, \pi_J]$. Since $[M, m_j]$ was an arbitrary cone, we have therefore proved that the image of the limit cone $[L, \pi_J]$ is also a limit cone. □

As a simple corollary of this theorem, we can infer

Theorem 113. *A representable functor $F\colon \mathscr{C} \to \mathbf{Set}$ preserves all limits that exist in $\mathscr{C}$.*

Proof. By Defn. 45, $F\colon \mathscr{C} \to \mathbf{Set}$ is said to be representable if for some A, F is naturally isomorphic to the hom-functor $\mathscr{C}(A, -)\colon \mathscr{C} \to \mathbf{Set}$. We can then use Theorem 103 to conclude that since $\mathscr{C}(A, -)$ preserves whatever limits exist, so does F. □

18.7 More on hom-functors and limits

Our proof of Theorem 112, as we announced, involved working straight from definitions. We could also get to the same result by remarking that if there is a limit cone over $D\colon \mathbf{J} \to \mathscr{C}$, then it can be constructed from suitable products and equalizers (as indicated by the proof of Theorem 98). So to show hom-functors preserve those limits which exists it is enough to show that they preserve products and equalizers which exist. See e.g. Awodey (2006, p. 106) for this line of argument.

But we won't pause over this, but rather consider another way of using the proof of the existence of all small limits in **Set** to show that hom-functors commute with such limits (we can then use the result we announced at the end of §18.2 to recover the claim that hom-functors preserve limits). Note, then, two preliminary results which flow easily from previous theorems:

Theorem 114. *Suppose $D\colon \mathbf{J} \to \mathscr{C}$ is a diagram and A an object in $\mathscr{C}$. Then*

$$Cone(A, D) \cong \underset{\leftarrow \mathbf{J}}{Lim}(\mathscr{C}(A, -) \circ D).$$

Proof. We have already shown that **Set** has all small limits, and we are assuming that $\mathbf{J}$ is small, so the functor $\underset{\leftarrow \mathbf{J}}{Lim}$ here is well-defined.

We know from the proof of Theorem 95 what limits for a functor $F\colon \mathbf{J} \to \mathbf{Set}$ look like. The vertex of the limit cone is the set of tuples $\langle x_I \rangle_{I \in \mathbf{J}}$ such that $x_J \in FJ$ for all J in $\mathbf{J}$ and where $(Ff)x_J = x_K$ for all $f\colon J \to K$ in $\mathbf{J}$.

So the vertex of the limit cone for $\mathscr{C}(A,-) \circ D$ is the set of tuples $\langle x_I \rangle_{I \in \mathbf{J}}$ such that $x_J \in \mathscr{C}(A, DJ)$ and $Df \circ x_J = x_K$. But those last conditions hold just when the x_J in a tuple comprise the arrows forming a cone with vertex A over D. Hence each tuple either is a cone (in the austere sense of cone) or is in a trivial natural isomorphism with a cone with vertex A over D.

Hence $\underset{\leftarrow \mathbf{J}}{Lim}(\mathscr{C}(A,-) \circ D)$, the set of such tuples, is isomorphic (indeed naturally isomorphic) to the set of cones $Cone(A, D)$. □

Theorem 115. *Suppose $D\colon \mathbf{J} \to \mathscr{C}$ is a diagram and A an object in the category $\mathscr{C}$ which has all limits of shape $\mathbf{J}$. Then*

$$Cone(A, D) \cong \mathscr{C}(A, \underset{\leftarrow \mathbf{J}}{Lim}\, D).$$

Proof. Let $[\underset{\leftarrow \mathbf{J}}{Lim}\, D, \pi_J]$ be a limit cone over D. Then there is a one-one correspondence between arrows $f \in \mathscr{C}(A, \underset{\leftarrow \mathbf{J}}{Lim}\, D)$, i.e. arrows $f\colon A \to \underset{\leftarrow \mathbf{J}}{Lim}\, D$ and cones $[A, \pi_J \circ f] \in Cone(A, D)$ by Theorem 86. □

Putting those two theorems together we immediately get:

Theorem 116. *Suppose $D\colon \mathbf{J} \to \mathscr{C}$ is a diagram and A an object in the category $\mathscr{C}$ which has all limits of shape $\mathbf{J}$. Then*

$$\mathscr{C}(A, \underset{\leftarrow \mathbf{J}}{Lim}\, D) \cong \underset{\leftarrow \mathbf{J}}{Lim}(\mathscr{C}(A, -) \circ D).$$

Indeed, the isomorphism here is natural. Put $F = \mathscr{C}(A, -)$, and we have $F(\underset{\leftarrow \mathbf{J}}{Lim}\, D) \cong \underset{\leftarrow \mathbf{J}}{Lim}(F \circ D)$, so the hom-functor F commutes with limits. And because the isomorphism is a natural one we can recover the claim the covariant hom-functor F preserves the limits over D.

19 Exponentials

There is more to be said about how limits can be carried *between* categories by functors, and we will return to the theme later (after we have met the pitoval idea of adjoint functors in Chapters 22 and 23). Next, however, we introduce another categorial notion that – like limit-notions – applies *within* a category and that is also defined in terms of a 'universal mapping property'.

19.1 Exponentials defined

(a) It is standard to use the notation 'C^B' in set theory to denote the set of functions $f\colon B \to C$. But why is the exponential notation apt?

Here is one reason. 'C^n' is of course natural notation for the n-times Cartesian product of C with itself, i.e. the set of n-tuples of elements from C. But recall that an n-tuple of C-elements can be regarded as equivalent to a function from an indexing set n, i.e. from the set $\{0, 1, 2, \ldots, n-1\}$, to C. Therefore C^n, the set of n-tuples, can indeed be thought of as C^n, re-defined as the set of functions $f\colon n \to C$. And is then natural to extend this notation to the case where the indexing set B is no longer a number.

Four initial observations, still in informal set-theory:

(1) For all sets B, C there is a set C^B.

(2) There is a two-place evaluation function $ev(\cdot\,,\cdot)$ which takes an element $f \in C^B$ and an element $b \in B$, evaluates f for the selected argument, and returns the value $f(b) \in C$.

(3) Take any set A, and any two-place function $g(\cdot\,,\cdot)$ that maps an element of A and an element of B to some value in C. Then, fixing an element $a \in A$ determines a one-place function $g(a,\cdot)\colon B \to C$.

(4) So, for any such $g(\cdot\,,\cdot)$ there is a unique associated function, its *exponential transpose* $\overline{g}\colon A \to C^B$, which sends $a \in A$ to $g(a,\cdot)\colon B \to C$, so that $ev(\overline{g}(a), b) = g(a, b)$.

These observations indicate how to characterize the 'exponential' object C^B in **Set**. We simply need to remember that categorially we regiment two-place functions as arrows from products (see §14.6).

Hence, we can say this. In **Set**, for all B, C, there is an object C^B and an arrow $ev\colon C^B \times B \to C$ such that for any arrow $g\colon A \times B \to C$, there is a *unique* $\overline{g}\colon A \to C^B$ (g's exponential transpose) which makes the following diagram commute:

$$\begin{tikzcd} A \times B \arrow[d, dashed, "\overline{g}\times 1_B"'] \arrow[dr, "g"] & \\ C^B \times B \arrow[r, "ev"'] & C \end{tikzcd}$$

The product arrow $\overline{g} \times 1_B$ here, which acts componentwise on pairs in $A \times B$, is defined categorially using the construction in §14.7.

(b) Now generalize in the obvious way:

Definition 88. Assume $\mathscr{C}$ is a category with binary products. Then $[C^B, ev]$, with C^B a $\mathscr{C}$-object and $ev\colon C^B \times B \to C$, forms *the exponential* of C by B in $\mathscr{C}$ iff the following holds, with all the mentioned objects and arrows being in $\mathscr{C}$: for every object A and arrow $g\colon A \times B \to C$, there is a unique arrow $\overline{g}\colon A \to C^B$ such that $ev \circ \overline{g} \times 1_B = g$ (i.e. such that the diagram above commutes).

Note that, as with products, the square-bracket notation here is just punctuation for readability's sake. More importantly, note that as we take different objects B, C the evaluation arrows ev associated with the varying C^B will be different, having different sources and targets.

Definition 89. A category $\mathscr{C}$ *has all exponentials* iff for all objects $B, C \in \mathscr{C}$, there is a corresponding exponential $[C^B, ev]$.

(c) Exponentials aren't limits. But just as a limit cone over D is defined in terms of every cone over D 'factoring through' the limit via a unique arrow, so an exponential of C with B is defined in terms of every arrow from a product-with-B to C 'factoring through' the exponential via a unique arrow. In short, limits and exponentials alike are defined in terms of every relevant item factoring through via a unique map. That's why the properties of being a limit and being an exponential are standardly *both* said to be examples of *universal mapping properties*.

19.2 Exponentials are unique

Defn. 88 talks of 'the' exponential of C with B. This is legitimate because exponentials – as we might expect – are unique, at least up to unique isomorphism:

Theorem 117. *In a category $\mathscr{C}$ with exponentiation, if given objects B, C have exponentials $[E, ev]$ and $[E', ev']$, then there is a unique isomorphism between E and E' compatible with the evaluating arrows.*

Proof from first principles. Two commmuting diagrams encapuslate the core of the argument, which parallels the proof of Theorem 69:

$$
\begin{tikzcd}
E \times B \arrow[d, dashed, "1_E \times 1_B"'] \arrow[dr, "ev"] & \\
E \times B \arrow[r, "ev"'] & C
\end{tikzcd}
\qquad
\begin{tikzcd}
E \times B \arrow[d, dashed, "\overline{ev} \times 1_B"'] \arrow[dr, "ev"] & \\
E' \times B \arrow[r, "ev'"] \arrow[d, dashed, "\overline{ev'} \times 1_B"'] & C \\
E \times B \arrow[ur, "ev"'] &
\end{tikzcd}
$$

The diagram on the left reminds us that, since $[E, ev]$ is an exponential of C by B there is a unique arrow $e\colon E \to E$ such that $ev \circ e \times 1_B = ev$. Evidently, $e = 1_E$.

The diagram on the right then reminds us that $[E, ev]$ and $[E', ev']$ factor uniquely through each other, and hence

$$ev \circ \overline{ev'} \times 1_B \circ \overline{ev} \times 1_B = ev.$$

Applying Theorem 74, it follows that

$$ev \circ (\overline{ev'} \circ \overline{ev}) \times 1_B = ev.$$

And now applying the uniqueness result from the first diagram

$$\overline{ev'} \circ \overline{ev} = e = 1_E.$$

Similarly, by interchanging E and E' in the second diagram, we get

$$\overline{ev} \circ \overline{ev'} = 1_{E'}.$$

Whence $\overline{ev}\colon E \to E'$ is an isomorphism. $\square$

That was straightforward: but there is perhaps some interest in noting the possibility of a couple of other proofs calling on more general results (even if all three proofs are quite closely related):

Proof via universal elements. Consider the (contravariant) functor $\mathscr{C}(- \times B, C)$ which sends an object $A \in \mathscr{C}$ to the hom-set of arrows from $A \times B \to C$ and sends an arrow $f\colon A' \to A$ to the map $- \circ f \times 1_B$ (i.e. the map which takes an arrow $j\colon A \times B \to C$ and yields the arrow $j \circ f \times 1_B\colon A' \times B \to C$).

Now apply Defn. 47, the definition of universal element (changing bracketing notation). Then a universal element of $\mathscr{C}(- \times B, C)$ is a pair $[E, ev]$, with $E \in \mathscr{C}$ and $ev \in \mathscr{C}(E \times B, C)$, such that for every A and every $g \in \mathscr{C}(A \times B, C)$, there is a unique $\overline{g}\colon A \to E$ such that $\mathscr{C}(- \times B, C)(\overline{g})(ev) = g$, i.e. $ev \circ \overline{g} \times 1_B = g$.

So, just by definition, the exponential $[E, ev]$ is a universal element of $\mathscr{C}(- \times B, C)$. But by Theorem 63, universal elements are unique up to unique isomorphism. $\square$

Looking ahead: proof via adjoints. Exponentials are right adjoints (§22.2, Ex.(8)), and adjoints are unique up to unique isomorphism (Theorem 143). $\square$

19.3 Examples of exponentials

Defns. 88 and 89 were purpose-built to ensure that **Set** counts as having all exponentials. Let's quickly consider a few other examples of categories with all exponentials.

(1) In **Set**, the categorial exponential of C by B is C^B (in the standard set-theoretic sense) equipped with the set function ev as described above. But this construction applies equally in **FinSet**, the category of finite sets, since the set C^B is finite if both B and C are finite, and hence C^B is also in **FinSet**. Therefore **FinSet** has all exponentials.

(2) In §14.2 (3) we met the category $\mathbf{Prop}_L$ whose objects are wffs of a given first-order language $\mathcal{L}$, and where there is a unique arrow from A to B iff $A \vDash B$. Assuming $\mathcal{L}$ has the usual rules for conjunction and implication, then for any B, C, the exponential object C^B is the conditional $B \to C$, and the evaluation arrow $ev : C^B \times B \to C$ just reflects the modus ponens entailment $B \to C, B \vDash C$.

Why does this work? Recall that products in $\mathbf{Prop}_L$ are conjunctions. And note that, given $A \wedge B \vDash C$, then by the standard rules $A \vDash B \to C$ and hence – given $B \vDash B$ – $A \wedge B \vDash (B \to C) \wedge B$. We therefore get the required commuting diagram of this shape,

$$\begin{tikzcd} A \wedge B \arrow[d, dashed] \arrow[dr] & \\ (B \to C) \wedge B \arrow[r] & C \end{tikzcd}$$

where the down arrow is the product of the implication arrow from A to $B \to C$ and the trivial entailment from B to B.

(3) Relatedly, take a Boolean algebra $(B, \neg, \wedge, \vee, 0, 1)$, and put $a \leqslant b =_{\mathrm{def}} (a \wedge b) = a$ for all $a, b \in B$. Then, treated as a partially ordered set with that order, the Boolean algebra corresponds to a poset category, with a unique arrow between a and b when $a \leqslant b$. In this category, $a \wedge b$, with the only possible projection arrows, is the categorial product of a, b

Such a poset category based on a Boolean algebra has an exponential object for each pair of objects, namely (to use a suggestive notation) $b \Rightarrow$

$c =_{\text{def}} \neg b \vee c$, with the evaluation arrow ev the unique arrow corresponding to $(b \Rightarrow c) \wedge b \leqslant c$.

To check this claim, we need first to show that we have indeed well-defined the evaluation arrow ev for every b, c, i.e. show that we always have $(b \Rightarrow c) \wedge b \leqslant c$. However, as we want,

$$(\neg b \vee c) \wedge b = (\neg b \wedge b) \vee (c \wedge b) = 0 \vee (c \wedge b) = (c \wedge b) \leqslant c$$

by Boolean rules and the definition of $\leqslant$.

Second, we need to verify that the analogous diagram to the last one commutes, which crucially involves showing that if $a \wedge b \leqslant c$ then $a \wedge b \leqslant (b \Rightarrow c) \wedge b$. That's more Boolean algebra, which can be left as a brain-teaser.

So Boolean-algebras-treated-as-poset-categories have all exponentials. Working through the details, we find that the required proofs *don't* call on the Boolean principle $\neg\neg a = a$, so the claim about Boolean algebras can be strengthened to the claim that Heyting-algebras-treated-as-poset-categories have all exponentials (where a Heyting algebra is, in effect, what you get when you drop the 'double negation' rule from the Boolean case).

At some later point, we'll want to return to these important examples from logic. For the moment, however, we will mention just one more category where there are exponentials for every pair of objects:

(4) Recall **Cat**, the category whose objects are small categories and whose arrows are functors between them. Then the exponential object for the **Cat**-objects $\mathscr{C}, \mathscr{D}$ is the functor category $[\mathscr{C}, \mathscr{D}]$ which is also small so is an object in **Cat**.

The underlying idea is that, just as (in **Set**) a function from $A \times B$ to C induces a function from A to a function-from-B-to-C, so (in **Cat**) a functor from the product category $\mathscr{A} \times \mathscr{B}$ to C will induce a functor that sends A to a functor-from-B-to-C. And just as (in **Set**) there is an evaluation function which, as it were, enables us to reverse the effect, so too there is such an evaluation functor living in **Cat**. The devil, of course, is in the details – which we needn't pause over here.

19.4 Exponentiation as a functor

Suppose we are working in a category $\mathscr{C}$ which has all exponentials (and all binary products). Suppose we have a map $f : C \to C'$ between a couple of $\mathscr{C}$-object. Pick another object B in the category. There is then a commuting diagram which looks like this:

$$\begin{tikzcd}
C^B \times B \arrow[r, "ev"] \arrow[d, dashed, "\overline{(f \circ ev)} \times 1_B"'] & C \arrow[d, "f"] \\
C'^B \times B \arrow[r, "ev'"'] & C'
\end{tikzcd}$$

Why so? There is a composite arrow $f \circ ev\colon C^B \times B \to C'$. And since $[C'^B, ev']$ is an exponential, there is therefore a unique transpose $\overline{f \circ ev}\colon C^B \to C'^B$ which makes the diagram commute.

In this way, there is a natural association between the arrows $f\colon C \to C'$ and, for fixed B, $\overline{f \circ ev}\colon C^B \to C'^B$. Indeed, this association is functorial:

Theorem 118. *For any $B \in \mathscr{C}$, there is a corresponding exponentiation functor $(-)^B\colon \mathscr{C} \to \mathscr{C}$ which sends an object C to C^B, and sends an arrow $f\colon C \to C'$ to $\overline{f \circ ev}\colon C^B \to C'^B$.*

Proof. We need to show that this putative functor preserves identities and respects composition.

The first is easy. $(1_C)^B$ is by definition $\overline{1_C \circ ev}\colon C^B \to C^B$, so we have

$$\begin{tikzcd}
C^B \times B \arrow[r, "ev"] \arrow[d, dashed, "(1_C)^B \times 1_B"'] & C \arrow[d, "1_C"] \\
C^B \times B \arrow[r, "ev"'] & C
\end{tikzcd}$$

But evidently, the arrow $1_{C^B} \times 1_B$ on the left would also make the diagram commute. So by the requirement that there is unique filling for $- \times 1_B$ which makes the square commute, $(1_C)^B = 1_{C^B}$, as required for functoriality.

Second, we need to show that given arrows $f\colon C \to C'$ and $g\colon C' \to C''$, $(g \circ f)^B = g^B \circ f^B$. Consider the following diagram where the top square, bottom square, and (outer, bent) rectangle commute:

$$\begin{tikzcd}
C^B \times B \arrow[r, "ev"] \arrow[d, dashed, "f^B \times 1_B"] \arrow[dd, dashed, bend right=60, "(g \circ f)^B \times 1_B"'] & C \arrow[d, "f"] \\
C'^B \times B \arrow[r, "ev'"] \arrow[d, dashed, "g^B \times 1_B"] & C' \arrow[d, "g"] \\
C''^B \times B \arrow[r, "ev''"'] & C''
\end{tikzcd}$$

By Theorem 74 $(g^B \times 1_B) \circ (f^B \times 1_B) = (g^B \circ f^B) \times 1_B$. Hence $(g^B \circ f^B) \times 1_B$ is another arrow that makes a commuting rectangle. So again by the requirement that there is unique filling for $- \times 1_B$ which makes the square commute, $(g \circ f)^B = g^B \circ f^B$. $\square$

19.5 Exponentiation and hom-sets

(a) Suppose $\mathscr{C}$ is a category with all exponentials. Then by definition there is always a map sending $g\colon A \times B \to C$ to a unique $\overline{g}\colon A \to C^B$. This map from an arrow to its transpose is injective; for if $\overline{g} = \overline{h}$, then $g = ev \circ (\overline{g} \times 1_B) = ev \circ (\overline{h} \times 1_B) = h$. Further, the map is surjective; for take any $k\colon A \to C^B$, then if we put $g = ev \circ (k \times 1_B)$, $\overline{g}$ is the unique map such that $ev \circ (\overline{g} \times 1_B) = g$, so $\overline{g} = k$. Hence we have a bijection between arrows like g and arrows like $\overline{g}$. Assume that we are working in a locally small category or are otherwise happy to put this in terms of hom-sets (or hom-classes, if your prefer): then we can say

$$\mathscr{C}(A \times B, C) \cong \mathscr{C}(A, C^B).$$

But we can say more. This isomorphism between arrows and their transposes is set up quite generally, in a systematic way: we will therefore expect naturality. We can therefore expect to establish, in particular,

Theorem 119. *In a locally small category $\mathscr{C}$ with all exponentials, for all $A, B, C \in \mathscr{C}$, $\mathscr{C}(A \times B, C) \cong \mathscr{C}(A, C^B)$ naturally in A, and naturally in C.*

The proofs of the two naturality claims here involve defining some relevant functors and then indulging in a fun bit of diagram-chasing. This is, I guess, worth doing as a reality check. But by all means skip the details.

Proof of naturality in A. We need to consider the two (contravariant) functors $F = \mathscr{C}(- \times B, C)$ and $G = \mathscr{C}(-, C^B)$ from $\mathscr{C}$ to **Set**.

G is a straightforward hom-functor, while as we saw near the end of §19.2, F sends an object A to $\mathscr{C}(A \times B, C)$, and sends an arrow $f\colon A' \to A$ to the map we can represent $- \circ (f \times 1_B)\colon (A \times B, C) \to (A' \times B, C)$.

We have to show that there is a natural isomorphism from F to G (and hence $FA \cong GA$ naturally in A). So we need to find a family of isomorphisms α_K for $K \in \mathscr{C}$ such that for every $f\colon A' \to A$ in $\mathscr{C}$, the following diagram commutes in **Set**:

$$\begin{tikzcd}
FA \arrow[r, "Ff"] \arrow[d, "\alpha_A"] & FA' \arrow[d, "\alpha_{A'}"] \\
GA \arrow[r, "Gf"] & GA'
\end{tikzcd}$$

In other words, unpacking the F/G notation, this must always commute:

$$\begin{tikzcd}
\mathscr{C}(A \times B, C) \arrow[r, "- \circ (f \times 1_B)"] \arrow[d, "\alpha_A"] & \mathscr{C}(A' \times B, C) \arrow[d, "\alpha_{A'}"] \\
\mathscr{C}(A, C^B) \arrow[r, "- \circ f"] & \mathscr{C}(A', C^B)
\end{tikzcd}$$

The obvious candidate for the component α_A is, of course, the isomorphism which sends an arrow $g \in \mathscr{C}(A \times B, C)$ to its transpose $\overline{g} \in \mathscr{C}(A, C^B)$. But will that make the diagram commute?

Chase an arrow $g \in \mathscr{C}(A \times B, C)$ round the diagram both ways, taking the α components, as suggested, to send arrows to transposes. Then the diagram will commute if $\overline{g} \circ f = \overline{g \circ (f \times 1_B)}$.

But now consider this further diagram:

$$\begin{tikzcd}
A' \times B \arrow[r, "f \times 1_B"] \arrow[dr, dashed, "\overline{g \circ (f \times 1_B)} \times 1_B"'] & A \times B \arrow[d, dashed, "\overline{g} \times 1_B"'] \arrow[dr, "g"] & \\
& C^B \times B \arrow[r, "ev"'] & C
\end{tikzcd}$$

By definition, $\overline{g \circ (f \times 1_B)} \colon A' \to C^B$ is the unique arrow that when plugged into $- \times 1_B$ makes the rhombus commute.

But the right-hand triangle commutes, so it follows that $(\overline{g} \times 1_B) \circ (f \times 1_B)$ is another arrow from $A' \times B$ to $C^B \times B$ which makes the rhombus commute. However, by Theorem 74, $(\overline{g} \times 1_B) \circ (f \times 1_B) = (\overline{g} \times f) \times 1_B$,. Hence $\overline{g} \times f$ when plugged into $- \times 1_B$ makes the rhombus commute. Which establishes that $\overline{g} \circ f = \overline{g \circ (f \times 1_B)}$. □

Proof of naturality in C. We need this time to consider the two (covariant) functors $F = \mathscr{C}(A \times B, -)$ and $G = \mathscr{C}(A, -^B)$, again from $\mathscr{C}$ to **Set**.

F is a straightforward hom-functor. G is defined in the obvious way, given our discussion of the functor $(-)^B$ in previous section.

To spell that out, G sends an object C to $\mathscr{C}(A, C^B)$. And how does G act on an arrow $f \colon C \to C'$? It must yield a map from $\mathscr{C}(A, C^B)$ to $\mathscr{C}(A, C'^B)$, i.e. a map which sends an arrow $g \colon A \to C^B$ to an arrow $g' \colon A \to C'^B$. A map can do that by composing g with the arrow $f^B \colon C^B \to C'^B$, where as before $f^B = \overline{f \circ ev}$. So in sum $G(f) = \overline{f \circ ev} \circ -$. It is then easily checked that G will indeed be functorial.

We have to show that there is a natural isomorphism from F to G (and hence $FC \cong GC$ naturally in C). So we need to find a family of isomorphisms α_K for $K \in \mathscr{C}$ such that for every $g \colon C \to C'$ in $\mathscr{C}$, the following diagram commutes in **Set**:

$$\begin{tikzcd}
FC \arrow[r, "Ff"] \arrow[d, "\alpha_C"] & FC' \arrow[d, "\alpha_{C'}"] \\
GC \arrow[r, "Gf"] & GC'
\end{tikzcd}$$

In other words, unpacking the new F/G notation, this must always commute:

$$\begin{tikzcd}
\mathscr{C}(A \times B, C) \arrow[r, "f \circ -"] \arrow[d, "\alpha_C"] & \mathscr{C}(A \times B, C') \arrow[d, "\alpha_{C'}"] \\
\mathscr{C}(A, C^B) \arrow[r, "\overline{(f \circ ev)} \circ -"] & \mathscr{C}(A, C'^B)
\end{tikzcd}$$

Again, suppose we take the component α_C to be the isomorphism which sends an arrow $g \in \mathscr{C}(A \times B, C)$ to its transpose $\overline{g} \in \mathscr{C}(A, C^B)$. Will that make the diagram commute?

Chase an arrow $g \in \mathscr{C}(A \times B, C)$ round the diagram both ways. Then the diagram will commute if $\overline{f \circ ev} \circ \overline{g} = \overline{f \circ g}$.

Now consider the following diagram:

$$\begin{tikzcd}
A \times B \arrow[d, dashed, "\overline{g} \times 1_B"] \arrow[dr, "g"] \arrow[dd, dashed, bend right=60, "\overline{f \circ g} \times 1_B"'] & \\
C^B \times B \arrow[r, "ev"] \arrow[d, dashed, "\overline{f \circ ev} \times 1_B"] & C \arrow[d, "f"] \\
C'^B \times B \arrow[r, "ev'"] & C'
\end{tikzcd}$$

Note the composite $f \circ g \colon A \times B \to C'$. By the definition of $[C'^B, ev']$ as an exponential, there is a unique arrow $\overline{f \circ g}$ such that

$$ev' \circ \overline{f \circ g} \times 1_B = f \circ g.$$

But since the top triangle and the bottom square also commute, we have

$$f \circ g = ev' \circ (\overline{f \circ ev} \times 1_B) \circ (\overline{g} \times 1_B) = ev' \circ (\overline{f \circ ev} \times \overline{g}) \times 1_B.$$

Hence, by the uniqueness requirement, we get $\overline{f \circ ev} \circ \overline{g} = \overline{f \circ g}$, and we are done. □

(b) Theorem 119 is the main result of this section; but now we draw out an important implication.

Continue to assume that $\mathscr{C}$ has all exponentials, but now suppose in addition that $\mathscr{C}$ has a terminal object 1. Then our theorem tells us that

$$\mathscr{C}(B, C) \cong \mathscr{C}(1 \times B, C) \cong \mathscr{C}(1, C^B).$$

So, as in **Set**, there will again be a one-one correspondence between the arrows $B \to C$ of $\mathscr{C}$ and arrows $1 \to C^B$, i.e. elements of C^B. So we can then think of C^B which lives *inside* $\mathscr{C}$ as an internal counterpart to the hom-set $\mathscr{C}(B, C)$ which lives in **Set**, i.e. normally lives *outside* $\mathscr{C}$. If you like, via C^B, $\mathscr{C}$ 'knows something' about its own hom-sets.

Note too that if a category has all exponentials, this iterates. **Set** has a rich hierarchy of hom-sets: as we would informally put it, there are sets of functions from objects to objects, but there are also sets of functions from objects to functions, sets of functions from functions to functions, sets of functions from objects to functions-from-functions-to-functions, and so on upwards. Similarly in a category with all exponentials there are exponentials corresponding internally to basic hom-sets, and exponenentials of exponentials, corresponding to hom-sets of hom-sets, and so it goes. In this way, having all exponentials again gives a certain richness to a category.

19.6 Cartesian closed categories

These rich categories which have all exponentials (and that presupposes binary prodcuts) and also a terminal object (and hence *all* finite products) are important enough to deserve a standard label:

Definition 90. A category $\mathscr{C}$ is a *Cartesian closed category* iff it has all finite products and all exponentials.

Our examples above of categories with plentiful exponentials are evidently all Cartesian closed.[1]

And in these categories, exponentials indeed behave as exponentials 'ought'. In particular,

Theorem 120. *If $\mathscr{C}$ is a Cartesian closed category, then for all $A, B, C \in \mathscr{C}$*

(1) $A^1 \cong A$,

(2) $1^A \cong 1$,

(3) $(A \times B)^C \cong A^C \times B^C$.

And if $\mathscr{C}$ also has an initial object 0*, then*

(4) $A^0 \cong 1$,

(5) if there is an arrow $A \to 0$, then $A \cong 0$,

(6) there exists an arrow $1 \to 0$ iff $\mathscr{C}$ is equivalent to the trivial category one-object, one-arrow category $\mathbf{1}$.

[1]Terminological aside: some call a category with all finite products a *Cartesian* category – but this term is also used in other ways so is probably best avoided. By contrast, the notion of a Cartesian closed category has a settled usage.

Proof. (1) Recall that $D \cong D \times 1$. So we have $\mathscr{C}(D, A) \cong \mathscr{C}(D \times 1, A) \cong \mathscr{C}(D, A^1)$. Apply Theorem 46 and infer $A^1 \cong A$.

(2) $\mathscr{C}(D \times A, 1) \cong \mathscr{C}(D, 1^A)$. Since 1 is terminal, the left hom-set has a unique member for every $D \in \mathscr{C}$, hence so does the right hom-set, and hence 1^A is terminal too.

(3) We can prove this from first principles (try it!); but we will return to give a slicker proof later.

(4) Recall that $0 \cong D \times 0$. So we have $\mathscr{C}(0, A) \cong \mathscr{C}(D \times 0, A) \cong \mathscr{C}(D, A^0)$. Since 0 is initial, the left hom-set always has a unique member, hence so does the right and hence A^0 is terminal.

(5) By assumption, we have a wedge $A \xleftarrow{1_A} A \xrightarrow{f} 0$, and this will factor uniquely through the product $A \times 0$, as in

$$\begin{tikzcd}
& A \arrow[dl, "1_A"'] \arrow[d, dashed, "\langle 1_A, f\rangle"] \arrow[dr, "f"] & \\
A & A \times 0 \arrow[l, "\pi_1"'] \arrow[r, "\pi_2"] & 0
\end{tikzcd}$$

So $\pi_1 \circ \langle 1_A, f\rangle = 1_A$. But $A \times 0 \cong 0$, so $A \times 0$ is an initial, so there is a unique arrow $A \times 0 \to A \times 0$, namely $1_{A\times 0}$. Hence (travelling round the left triangle) $\langle 1_A, f\rangle \circ \pi_1 = 1_{A\times 0}$. Therefore $\langle 1_A, f\rangle \colon A \to A \times 0$ has a two-sided inverse. Whence $A \cong A \times 0 \cong 0$.

(6) One direction is trivial. For the other, suppose there is an arrow $: 1 \to 0$. Then, for any A there must be a composite arrow $A \longrightarrow 1 \xrightarrow{f} 0,$, hence by (5), $A \cong 0$. So every object in the category is isomorphic – the skeleton of the category is **1**. $\square$

Here's an application of that last result: the category **Grp** is *not* Cartesian closed. For we noted the one-element group is both initial and terminal in the category **Grp**, so $1 \cong 0$. Hence there is an arrow $1 \to 0$ in **Grp**, but this is far from being equivalent to a trivial category!

20 Galois connections

We will return to say more about functors and limits and exponentials too after we have introduced the next Big Idea from category theory that we need to get our heads around – namely, the notion of pairs of *adjoint functors* and the *adjunctions* they form.

Rather than dive straight in to the general story, however, we are going to look first at a (very) restricted class of cases. These are the so-called Galois connections, which are in effect adjunctions between two categories which are posets. So, in this chapter, we will discuss Galois connections in a pre-category-theoretic way, as a way of introducing us to some key themes in an uncluttered context. In the next chapter – just because it has some stand-alone interest – we will pause to see the idea of Galois connection at work in a famous special case of interest to logicians. The general story about adjoints and adjunctions then starts in Chapter 22.

20.1 (Probably quite unnecessary) reminders about posets

Recall: The set P equipped with the binary relation $\preccurlyeq$, which we denote $(P, \preccurlyeq)$, is a poset just in case $\preccurlyeq$ is a partial order – i.e., for all $x, y, z \in S$, (i) $x \preccurlyeq x$, (ii) if $x \preccurlyeq y$ and $y \preccurlyeq z$ then $x \preccurlyeq z$, (iii) if $x \preccurlyeq y$ and $y \preccurlyeq x$ then $x = y$. (We will, as appropriate, recruit '$\sqsubseteq$', '$\leqslant$', '$\subseteq$' as other symbols for partial orders.)

Reversing a partial order gives us another partial order. Hence reversing the order in a poset $\mathscr{P} = (P, \preccurlyeq)$ gives us a dual poset $\mathscr{P}^{op} = (P, \succcurlyeq)$ defined in the obvious way.

There is a related notion of a strict poset defined in terms of the notion of a strict partial order $\prec$, where $x \prec y$ iff $x \preccurlyeq y \wedge x \neq y$ (and so $x \preccurlyeq y$ iff $x \prec y \vee x = y$). It is just a matter of convenience whether we concentrate on the one flavour of poset or the other. We can take it that you are already familiar with a variety of examples of 'naturally occurring' posets of both flavours.

The following notions will also be familiar, in one terminology or another:

Definition 91. Suppose that $(P, \preccurlyeq)$ and $(Q, \sqsubseteq)$ are two posets. Let $f : P \to Q$ be a map between their carrier sets. Then

(1) f is monotone just in case, for all $x, y \in P$, if $x \preccurlyeq y$ then $f(x) \sqsubseteq f(y)$;

(2) f is an order-embedding just in case, for all $x, y \in P$, $x \preccurlyeq y$ iff $f(x) \sqsubseteq f(y)$;

(3) f is an order-isomorphism iff f is a surjective order-embedding.

Some obvious remarks about these notions:

i. Monotone maps compose to give monotone maps and composition is associative. Likewise for order-embeddings and order-isomorphisms.

ii. As you'd expect from the label, order-embeddings are injective. Keeping the same notation, suppose $f(x) = f(y)$ and hence both $f(x) \sqsubseteq f(y)$ and $f(y) \sqsubseteq f(x)$. Then, if f is an embedding, $x \preccurlyeq y$ and $y \preccurlyeq x$, and hence $x = y$.

iii. If $f[P]$ is P's image under f, an order-embedding $f\colon (P, \preccurlyeq) \to (Q, \sqsubseteq)$ is an order-isomorphism from $(P, \preccurlyeq)$ to $(f[P], \sqsubseteq)$.

iv. An order-isomorphism is bijective, and therefore is an isomorphism as a set-function. Order-isomorphisms have unique inverses which are also order-isomorphisms.

v. Posets are deemed isomorphic if there is an order-isomorphism between them.

If $(P, \preccurlyeq)$ is a poset and $X \subseteq P$, then a maximum of X (with respect to the inherited order $\preccurlyeq$) is defined in the obvious way: m is a maximum of X iff $m \in X \wedge (\forall x \in X) x \preccurlyeq m$. Maxima are unique – for if $m, m' \in X$ are both maxima, $m' \preccurlyeq m$ and similarly $m \preccurlyeq m'$ and hence $m = m'$.

If $X \subseteq P$ we say that $(X, \preccurlyeq)$ is a sub-poset of $(P, \preccurlyeq)$ – and note here that we will not routinely fuss to distinguish a relation defined over P from the restriction of that relation to X.

Definition 92. Suppose S is a set, $\mathcal{P}(S)$ is its powerset, and $\Pi \subseteq \mathcal{P}(S)$ is some set of subsets of S. Then Π ordered by inclusion, i.e. $(\Pi, \subseteq)$, is an *inclusion poset.*

Theorem 121. *Every poset is isomorphic to an inclusion poset.*

Proof. Take the poset $(P, \preccurlyeq)$. For each $p \in P$, now form the set containing it and its $\preccurlyeq$-predecessors $\pi_p = \{x \in P \mid x \preccurlyeq p\}$. Let Π the set of all π_p for $p \in P$. Then $(\Pi, \subseteq)$ is an inclusion poset.

Define $f\colon P \to \Pi$ by putting $f(p) = \pi_p$. Then f is very easily seen to be a bijection, and also $p \preccurlyeq q$ iff $\pi_p \preccurlyeq \pi_q$. So f is an order-isomorphism. $\square$

20.2 A motivating example

This section is going to be informal, sketching one situation in which a Galois connection can naturally arise. We'll give some much more carefully presented examples in the next section, and then a detailed treatment of a version of this first informal example in the next chapter. But this rather arm-waving first introduction may help.

Suppose we have a poset $\mathscr{P} = (P, \preccurlyeq)$ where we can think of the members of P as 'concrete' items, and $\preccurlyeq$ as a 'part-of' relation: for example, perhaps the concrete items are sets of natural numbers, and then the ordering is simply set-inclusion. And now suppose that there is another poset $\mathscr{Q} = (Q, \sqsubseteq)$ such that we can think of the members of Q as various 'abstract' representations applying to those concrete items, and $\sqsubseteq$ is the relation 'more specific than'. To pursue the same example, perhaps the representations are the likes of *primes*, *evens*, *multiples of three*, *even multiples of three*, *powers of two*, ..., and we take e.g. *primes* to apply to a set of numbers if each number in the set is prime: a less inclusive abstraction precedes a more inclusive one.

Now, in 'nice' cases, $\mathscr{P}$ and $\mathscr{Q}$ can have enough lattice structure for the following to hold. (i) There is an abstraction function $f_* \colon P \to Q$ which maps each $p \in P$ to a corresponding 'best', i.e. most specific, abstract characterization $f_*(p) \in Q$ (for example, it maps a set of numbers to the most specific abstraction they all satisfy). And (ii) there is also a concretization function $f^* \colon Q \to P$ which maps an abstract characterization $q \in Q$ to its 'best', i.e. most inclusive, concrete exemplar $f_*(q) \in P$ (for example it maps *powers of two* to the largest set of numbers in P which are powers of two).

In general f_* and f^* will not be isomorphisms (indeed, we typically won't want them to be – abstract representations may be useful precisely because they *don't* track all the fine differences to be found among their concrete exemplars). But the mapping functions will have the following properties:

(1) f_* and f^* are monotone. If $p \preccurlyeq p'$, so the concrete p is just some of q, then $f_*(p) \sqsubseteq f_*(q)$ – i.e. the best representation of p will be at least as specific as the best representation of p'. Likewise, if $q \sqsubseteq q'$, so the abstract representation q' is less restrictive than q, then $f^*(q) \preccurlyeq f^*(q)$ – i.e. the most inclusive exemplar of q' must include the most inclusive exemplar of q.

(2) For all $p \in P$, $q \in Q$, $p \preccurlyeq f^*(f_*(p))$ and $f_*(f^*(q)) \sqsubseteq q$. If we take the best available abstract representation in Q of p, this representation may also apply to a $\preccurlyeq$-larger p' in P, so the concretization of $f_*(p)$ will then take us back (at least) to the larger p'. Likewise, if we take the most generous concrete exemplar in P of the representation q, and then look for the most restrictive representation in Q of *that*, we might end up with something more specific than q.

(3) For all $p \in P$, $q \in Q$, $f_*(p) \sqsubseteq q$ iff $p \preccurlyeq f^*(q)$. Suppose that the most specific representation of p is more specific than the representation q, then the concrete item p has to be no more than part of what is best represented by q; and vice versa.

Now we've described the case decidedly informally: though in the context of theoretical computer science we meet formalized versions of the ideas of concrete items and their abstract representations, where just the described pattern of mappings provably holds. And in the next chapter we will meet a related example from logic. But we needn't go into more details for the moment. For as we will see over the coming couple of sections, we in fact repeatedly encounter elsewhere the same situation of having a pair of posets $\mathscr{P} = (P, \preccurlyeq)$, $\mathscr{Q} = (Q, \sqsubseteq)$ and a pair of functions $f_* : P \to Q$ and $f^* : Q \to P$ for which conditions (1) to (3) hold.

So now let's generalize. However, as we'll see, conditions (1) to (3) are not independent. The first two together imply the third, and vice versa. Simply because it is prettier, we'll first plump for a general definition in terms of condition (3), relabeled.

20.3 Galois connections defined

Definition 93. Suppose that $\mathscr{P} = (P, \preccurlyeq)$ and $\mathscr{Q} = (Q, \sqsubseteq)$ are two posets, and let $f_* : P \to Q$ and $f^* : Q \to P$ be a pair of functions such that for all $p \in P$, $q \in Q$,

$$\text{(G)} \qquad f_*(p) \sqsubseteq q \text{ iff } p \preccurlyeq f^*(q).$$

Then (f_*, f^*) form a *Galois connection* between $\mathscr{P}$ and $\mathscr{Q}$. And if (f_*, f^*) form such a connection, f_* is said to be the *left adjoint* of f^*, and f^* the right adjoint of f_*.

To remember which way round the stars go in our convention, think that the left adjoint f_* with the lower star appears on the left or 'lower' side of the ordering sign in (G), and f^* appears on the upper side. (Though just to be annoying, at least as many authors use f_* and f^* the other way about.) Talk of adjoints here seems to have been originally borrowed from the old theory of Hermitian operators, where in e.g. a Hilbert space with inner product $\langle \cdot, \cdot \rangle$ the operators A and A^* are said to be adjoint when we have, generally, $\langle Ax, y \rangle = \langle x, A^*y \rangle$. The formal analogy is evident.

The first discussion of a version of such a connection (f_*, f^*) – and hence the name – is to be found in Evariste Galois's work in what has come to be known as Galois theory, a topic beyond our purview here. And there are plenty of serious mathematical examples (e.g. from number theory, abstract algebra and topology) of two posets with a Galois connection between them. But we really don't want to get bogged down in unnecessary mathematics at this early stage; so for the moment let's just give some more very simple cases, to add to our informally described motivating one:

(1) Suppose f is an order isomorphism between $(P, \preccurlyeq)$ and $(Q, \sqsubseteq)$: then f^{-1} is an order-isomorphism in the reverse direction. Take $p \in P, q \in Q$: then trivially $f(p) \sqsubseteq q$ iff $f^{-1}(f(p)) \sqsubseteq f^{-1}(q)$ iff $p \sqsubseteq f^{-1}(q)$, so (f, f^{-1}) form a Galois connection.

(2) Take $\mathscr{N} = (\mathbb{N}, \leqslant)$ and $\mathscr{Q} = (\mathbb{Q}^+, \leqslant)$, i.e. the naturals and the non-negative rationals in their standard orders. Let $i \colon \mathbb{N} \to \mathbb{Q}^+$ be the injection function which maps a natural number to the corresponding rational integer, and let $f \colon \mathbb{Q}^+ \to \mathbb{N}$ be the 'floor' function which maps a rational to the natural corresponding to its integral part. Then (i, f) is a Galois connection between the posets $\mathscr{N}$ and $\mathscr{Q}$. And if $c \colon \mathbb{Q}^+ \to \mathbb{N}$ is the 'ceiling' function which maps a rational to the smallest integer which is at least as big, then (c, i) is a Galois connectoin between $\mathscr{Q}$ and $\mathscr{N}$.

(3) Let $f \colon X \to Y$ be some function between two sets X and Y. It induces a function $F \colon \mathcal{P}(X) \to \mathcal{P}(Y)$ between their powersets which sends $A \subseteq X$ to $f[A]$, and another function $F^{-1} \colon \mathcal{P}(Y) \to \mathcal{P}(X)$ which sends $B \subseteq Y$ to its pre-image under f, $F^{-1}[B] = \{x \in X \mid f(x) \in B\}$. Then (F, F^{-1}) is a Galois connection between the inclusion posets $(\mathcal{P}(X), \subseteq)$ and $(\mathcal{P}(Y), \subseteq)$.

(4) Take any poset $(P, \preccurlyeq)$, and let $(1, =)$ be a one-object poset with the only possible partial order relation. Let $t \colon P \to 1$ be the only possible function, the trivial one which sends every element of P to the single element 0 of 1. Suppose $t^* \colon 1 \to P$ is its right adjoint. Then for any p, $t(p) = 0$ iff $p \preccurlyeq t^*(0)$. So t has a right adjoint just in case P has a maximum, and then t^* sends 1's only element to it. Dually, t has a left adjoint just in case P has a minimum, and then the left adjoint t_* sends 1's only element to *that*.

(5) Our next example is from elementary logic. Let $\mathcal{L}$ be a formal language equipped with a proof-system (it could be classical or intuitionistic, or indeed any other logic with normally-behaved conjunction and conditional connectives and a sensible deducibility relation): $\alpha \vdash \beta$ notates that there is an $\mathcal{L}$-proof from premiss α to conclusion β. Then let $|\alpha|$ be the equivalence class of $\mathcal{L}$-wffs interderivable with α. Take E to be set of all such equivalence classes, and put $|\alpha| \trianglelefteq |\beta|$ in E iff $\alpha \vdash_L \beta$. Then it is easily checked that $(E, \trianglelefteq)$ is a poset.

Now consider the following two functions between $(E, \trianglelefteq)$ and itself. Fix γ to be some $\mathcal{L}$-wff. Then let f_γ send the equivalence class $|\alpha|$ to the class $|(\gamma \wedge \alpha)|$, and let f^γ send $|\alpha|$ to the class $|(\gamma \to \alpha)|$.

Given our normality assumption, $(\gamma \wedge \alpha) \vdash_L \beta$ if and only if $\alpha \vdash_L (\gamma \to \beta)$. Hence $|(\gamma \wedge \alpha)| \trianglelefteq |\beta|$ iff $|\alpha| \trianglelefteq |(\gamma \to \beta)|$. That is to say $f_\gamma(|\alpha|) \trianglelefteq |\beta|$ iff $|\alpha| \trianglelefteq f^\gamma(|\beta|)$. Hence (f_γ, f^γ) form a Galois connection between $(E, \trianglelefteq)$ and itself. So, in a slogan, 'Conjunction is left adjoint to conditionalization'.

(6) Our last example for the moment is again from elementary logic. Let $\mathcal{L}$ now be a first-order language, and consider the set of $\mathcal{L}$-wffs with at most the variables $\vec{x}$ free.

We will write $\varphi(\vec{x})$ for a formula in in this class, $|\varphi(\vec{x})|$ for the class of formulae interderivable with $\varphi(\vec{x})$, and $E_{\vec{x}}$ for the set of such equivalence classes of formulae with at most $\vec{x}$ free. Using $\trianglelefteq$ as in the last example, $(E_{\vec{x}}, \trianglelefteq)$ is a poset for any choice of variables $\vec{x}$.

We now consider two maps between the posets $(E_{\vec{x}}, \trianglelefteq)$ and $(E_{\vec{x},y}, \trianglelefteq)$. In other words, we are going to be moving between (equivalence classes of) formulae with at most $\vec{x}$ free, and (equivalence classes of) formulae with at most $\vec{x}, y$ free – where y is a new variable not among the $\vec{x}$.

First, since every wff with at most the variables $\vec{x}$ free also has at most the variables $\vec{x}, y$ free, there is a trivial map $f_*\colon E_{\vec{x}} \to E_{\vec{x},y}$ that sends the class of formulas $|\varphi(\vec{x})|$ in $E_{\vec{x}}$ to the same class of formulas which is also in $E_{\vec{x},y}$.

Second, we define the companion map $f^*\colon E_{\vec{x},y} \to E_{\vec{x}}$ that sends $|\varphi(\vec{x}, y)|$ in $E_{\vec{x},y}$ to $|\forall y \varphi(\vec{x}, y)|$ in $E_{\vec{x}}$.

Then (f_*, f^*) form a Galois connection. For that is just to say

$$f_*(|\varphi(\vec{x})|) \trianglelefteq |\psi(\vec{x}, y)| \quad \text{iff} \quad |\varphi(\vec{x})| \trianglelefteq f^*(|\psi(\vec{x}, y)|).$$

Which just reflects the familiar logical rule that

$$\varphi(\vec{x}) \vdash \psi(\vec{x}, y) \text{ iff } \varphi(\vec{x}) \vdash \forall y \psi(\vec{x}, y),$$

so long as y is not free in $\varphi(\vec{x})$. Hence, in another slogan, universal quantification is right-adjoint to a certain trivial operation of adding a dummy variable.

And we can exactly similarly show that existential quantification is left-adjoint to the same operation.

Our first example, then, shows that Galois connections exist and are at least as plentiful as order-isomorphisms: such an isomorphism has a right and left adjoint which are the same (i.e. are the inverse). The second and fourth cases show that posets that aren't order-isomorphic can in fact still be Galois connected. The third case shows that posets can have many Galois connections between them (as any $f\colon X \to Y$ generates a connection between the inclusion posets on the powersets of X and Y). The fourth example gives a case where a function has both a left and a right adjoint which are different. The fourth and sixth cases give a couple of illustrations of how a significant construction (taking maxima, forming a universal quantification respectively) can be regarded as adjoint to some quite trivial operation. The fifth example, like the third, shows that even when the Galois-connected posets are isomorphic (in the fifth case trivially so, because they are identical!), there can be a pair of functions which aren't isomorphisms but which also go to make up a connection between the posets. And the fifth and sixth examples already show that Galois connections are of interest to logicians.

20.4 Galois connections re-defined

Here is the promised alternative definition for a Galois connection:

Definition 94 (Alternative)**.** Suppose that $\mathscr{P} = (P, \preccurlyeq)$ and $\mathscr{Q} = (Q, \sqsubseteq)$ are posets, and let $f_*\colon P \to Q$ and $f^*\colon Q \to P$ be a pair of functions such that for all $p \in P$, $q \in Q$,

(1) f_* and f^* are both monotone,

(2) for all $p \in P$, $q \in Q$, $p \preccurlyeq f^*(f_*(p))$ and $f_*(f^*(q)) \sqsubseteq q$.

Then (f_*, f^*) is a Galois connection between $\mathscr{P}$ and $\mathscr{Q}$.

This alternative definition coincides with our original one, for we have the following simple theorem:

Theorem 122. *(f_*, f^*) is a Galois connection by our original definition if and only if it is a Galois connection by the alternative definition.*

Proof. (*If*) Assume conditions (1) and (2) of the alternative definition both hold. And suppose $f_*(p) \sqsubseteq q$. Since by (1) f^* is monotone, $f^*(f_*(p)) \preccurlyeq f^*(q)$. But by (2) $p \preccurlyeq f^*(f_*(p))$. So, since $\preccurlyeq$ is transitive, $p \preccurlyeq f^*(q)$.

That establishes one half of the biconditional (G) $f_*(p) \sqsubseteq q$ iff $p \preccurlyeq f^*(q)$. The proof of the other half is dual.

(*Only if*) Suppose (G) is true. Then in particular, $f_*(p) \sqsubseteq f_*(p)$ iff $p \preccurlyeq f^*(f_*(p))$. Since $\sqsubseteq$ is reflexive, $p \preccurlyeq f^*(f_*(p))$. Similarly for the other half of (2).

Now, suppose also that $p \preccurlyeq p'$. Then since we've just shown $p' \preccurlyeq f^*(f_*(p'))$, we have $p \preccurlyeq f^*(f_*(p'))$. But by (G) we have $f_*(p) \sqsubseteq f_*(p')$ iff $p \preccurlyeq f^*(f_*(p'))$. Whence, $f_*(p) \sqsubseteq f_*(p')$ and f_* is monotone. Similarly for the other half of (1). $\square$

In the light of condition (1) of our equivalent alternative, Galois connections as we have now twice defined them are sometimes called *monotone* connections. Now note that if (f_*, f^*) is a (monotone) connection between $\mathscr{P} = (P, \preccurlyeq)$ and $\mathscr{Q} = (Q, \sqsubseteq)$ so that $f_*(p) \sqsubseteq q$ iff $p \preccurlyeq f^*(q)$ then there is an *antitone* connection between $\mathscr{P}$ and the dual poset $\mathscr{Q}^{op} = (Q, \trianglelefteq)$, meaning a pair of functions (f_*, f^*) such that $q \trianglelefteq f_*(p)$ iff $p \preccurlyeq f^*(q)$ (since, trivially, $x \trianglelefteq y \equiv y \sqsubseteq x$).

The only point of mentioning antitone connections here is to note that (a) you might sometimes encounter the idea of a connection defined the antitone way (which indeed is how it appears in Galois's work), but (b) since a antitone connection between two posets is just a monotone connection between the first poset and the second's dual, we just don't need separate stories about the two types of connection, and without loss of generality we can concentrate entirely on connections presented the monotone way.

20.5 Some basic results about Galois connections

In this section, we prove that Galois connections behave as we would hope. We start by proving that such connections compose in an obvious way to produce new connections:

Theorem 123. *Let (f_*, f^*) be a Galois connection between the posets $\mathscr{P} = (P, \preccurlyeq)$ and $\mathscr{Q} = (Q, \sqsubseteq)$, and let (g_*, g^*) be a Galois connection between the posets $\mathscr{Q}$ and $\mathscr{R} = (R, \subseteq)$. Then $(g_* \circ f_*, f^* \circ g^*)$ is a Galois connections between $\mathscr{P}$ and $\mathscr{R}$.*

Proof. By assumption, (G) for any $p \in P, q \in Q$ we have $f_*(p) \sqsubseteq q$ iff $p \preccurlyeq f^*(q)$, and (G′) for any $q \in Q, r \in R$ we have $g_*(q) \sqsubseteq r$ iff $q \preccurlyeq g^*(r)$.

So take any $p \in P, r \in R$. (G) implies that $f_*(p) \sqsubseteq g^*(r)$ iff $p \preccurlyeq f^*(g^*(r))$ while (G′) implies that $g_*(f_*(p)) \sqsubseteq r$ iff $f_*(p) \preccurlyeq g^*(r)$ – whence $g_*(f_*(p)) \sqsubseteq r$ iff $p \preccurlyeq f^*(g^*(r))$, which is what we needed to show. □

Next we show that the components of a Galois connection are indeed connected in the sense that if a Galois connection $(f, \cdot)$ between two posets exists, then f uniquely fixes what its right adjoint is; and if a connection $(\cdot, f)$ exists, f uniquely fixes its left adjoint.

Theorem 124. *If (f_*, f^1) and (f_*, f^2) are both Galois connections between the posets $(P, \preccurlyeq)$ and $(Q, \sqsubseteq)$, then $f^1 = f^2$. Likewise, if (f_1, f^*) and (f_2, f^*) are both Galois connections between the same posets, then $f_1 = f_2$.*

Proof. Take the basic principle (G) applied to the connection (f_*, f^1), put $p = f^2(q)$, and we get $f_*(f^2(q)) \sqsubseteq q$ iff $f^2(q) \preccurlyeq f^1(q)$.

But by Theorem 122, applied to the connection (f_*, f^2), we have $f_*(f^2(q)) \sqsubseteq q$. So we can infer that, indeed, $f^2(q) \preccurlyeq f^1(q)$.

Interchanging the two connections we can similarly show that $f^1(q) \preccurlyeq f^2(q)$. And hence $f^1(q) = f^2(q)$. But q was arbitrary, so indeed $f^1 = f^2$ as was to be shown.

The other part of the theorem is proved similarly. □

Careful, though! This theorem does not say that, for any f_* which maps between $(P, \preccurlyeq)$ and $(Q, \sqsubseteq)$, there must actually exist a unique corresponding f^* in the reverse direction such that (f_*, f^*) form a Galois connection (we'll note counterexamples in a moment). Nor does it say that when there is a Galois connection between the posets, it is unique (our toy examples have already shown that that is false too). The claim is only that, if you are given a possible left adjoint – or a possible right adjoint – there can be at most one candidate for its companion to complete a connection.

Given that adjoint functions determine each other, we naturally seek an explicit definition of one in terms of the other. Here it is:

Theorem 125. *If (f_*, f^*) is a Galois connection between the posets $(P, \preccurlyeq)$ and $(Q, \sqsubseteq)$, then*

(1) $f^*(q) =$ the maximum of $\{p \in P \mid f_*(p) \sqsubseteq q\}$,

(2) $f_*(p) =$ the minimum of $\{q \in Q \mid p \preccurlyeq f^*(q)\}$.

Proof. Note that Theorem 122 yields (a) $p \preccurlyeq f^*(f_*(p))$, (b) $f_*(f^*(q)) \sqsubseteq q$, and (c) f^* is monotone.

Suppose $u \in \{p \in P \mid f_*(p) \sqsubseteq q\}$. Then $f_*(u) \sqsubseteq q$. By (c), $f^*(f_*(u)) \preccurlyeq f^*(q)$. Whence from (a), $u \preccurlyeq f^*(q)$.

That shows $f^*(q)$ is an upper bound for the set $\{p \in P \mid f_*(p) \sqsubseteq q\}$. But by (b) we have $f^*(q) \in \{p \in P \mid f_*(p) \sqsubseteq q\}$. So (1) $f^*(q)$ is indeed the maximum of the set.

The proof of (2) is simply dual. □

It follows that if $f_* \colon P \to Q$ is a function such that the set $\{p \in P \mid f_*(p) \sqsubseteq q\}$ doesn't have a maximum for each $q \in Q$, then there can't be a corresponding right adjoint $f^* \colon Q \to P$. To revert to a previous example, if we take the posets $(P, \preccurlyeq)$ and $(1, =)$ and let $f_* \colon P \to 1$ be only possible such function, then $\{p \in P \mid f_*(p) = 1\} = P$, so if P has no maximum, f_* has no right adjoint.

Now recall again the posets $\mathscr{N} = (\mathbb{N}, \leqslant)$ and $\mathscr{Q} = (\mathbb{Q}^+, \leqslant)$ with the injection map $i \colon \mathbb{N} \to \mathbb{Q}^+$ and floor function $f \colon \mathbb{Q}^+ \to \mathbb{N}$ which maps a rational to the natural corresponding to its integral part. Then we remarked before that (i, f) is a Galois connection between $\mathscr{N}$ and $\mathscr{Q}$. Now we note that (f, i) is *not* a Galois connection between $\mathscr{Q}$ and $\mathscr{N}$. Indeed, there can be no such connection of the form $(f, \cdot)$. For $f(p) = 1$ iff $1 \leqslant p < 2$, and hence there is no maximum member of $\{p \in \mathbb{Q}^+ \mid f(p) \leqslant 1\}$. Hence there is no right adjoint to f.

Generalizing, we have the following:

Theorem 126. *Galois connections are not necessarily symmetric. That is to say, given (f_*, f^*) is a Galois connection between the posets $\mathscr{P}$ and $\mathscr{Q}$, it does not follow that (f^*, f_*) is a connection between $\mathscr{Q}$ and $\mathscr{P}$.*

Finally in this section, we note the following key result:

Theorem 127. *If (f_*, f^*) is a Galois connection between the posets $(P, \preccurlyeq)$ and $(Q, \sqsubseteq)$, then $f_* \circ f^* \circ f_* = f_*$ and $f^* \circ f_* \circ f^* = f^*$.*

Proof. We know that, for any $p \in P$, $p \preccurlyeq f^*(f_*(p))$, and also f_* is monotone. So $f_*(p) \sqsubseteq f_*(f^*(f_*(p)))$.

But an instance of the fundamental relation (G) yields $f_*(f^*(f_*(p))) \sqsubseteq (f_*(p)$ iff $f^*(f_*(p)) \preccurlyeq f^*((f_*(p))$. The r.h.s. is trivially true, so indeed $f_*(f^*(f_*(p))) \sqsubseteq (f_*(p)$.

The antisymmetry of $\sqsubseteq$ then entails $f_*(f^*(f_*(p))) = f_*(p)$. Since p was arbitrary, that gives us one half of our result, and the other half is dual. □

20.6 Fixed points and closures

Theorem 128. *If (f_*, f^*) is, again, a Galois connection between the posets $(P, \preccurlyeq)$ and $(Q, \sqsubseteq)$, then*

(1) $p \in f^*[Q]$ if and only if p is a fixed point of $f^* \circ f_*$, and $q \in f_*[P]$ if and only if q is a fixed point of $f_* \circ f^*$,

(2) $f^*[Q] = (f^* \circ f_*)[P]$ and $f_*[P] = (f_* \circ f^*)[Q]$.

Proof. (1) Suppose $p \in f^*[Q]$. Then for some $q \in Q$, $p = f^*(q)$ and hence $f^*(f_*(p)) = f^*(f_*(f^*(q))) = f^*(q) = p$, so p is a fixed point of $f^* \circ f_*$.

Conversely suppose p is a fixed point of $f^* \circ f_*$, so $f^*(f_*(p)) = p$. Then p is the value of $f^*(q)$ for $q = f_*(p)$, and hence $p \in f^*[Q]$.

So $p \in f^*[Q]$ if and only if p is a fixed point of $f^* \circ f_*$, and that's half of (1). The other half is dual.

(2) We have just seen that if $p \in f^*[Q]$ then $p = f^*(f_*(p))$ so $p \in (f^* \circ f_*)[P]$. Therefore $f^*[Q] \subseteq (f^* \circ f_*)[P]$.

Conversely, suppose $p \in (f^* \circ f_*)[P]$, then as we saw before $p \in f^*[Q]$. Therefore $(f^* \circ f_*)[P] \subseteq f^*[Q]$.

Hence $f^*[Q] = (f^* \circ f_*)[P]$, and that's half of (2). The other half is dual. $\square$

Now let's introduce a new bit of abbreviatory notation:

Definition 95. Suppose $\mathfrak{f} = (f_*, f^*)$ is a Galois connection between the posets $\mathscr{P} = (P, \preccurlyeq)$ and $\mathscr{Q} = (Q, \sqsubseteq)$. Put $P^{\mathfrak{f}} = (f^* \circ f_*)[P]$ and $Q_{\mathfrak{f}} = (f_* \circ f^*)[Q]$. Then we define $\mathscr{P}^{\mathfrak{f}} = (P^{\mathfrak{f}}, \preccurlyeq)$ and $\mathscr{Q}_{\mathfrak{f}} = (Q_{\mathfrak{f}}, \sqsubseteq)$ – so these are sub-posets of $\mathscr{P}$ and $\mathscr{Q}$ respectively.

Theorem 129. *If $\mathfrak{f} = (f_*, f^*)$ is a Galois connection between the posets $\mathscr{P} = (P, \preccurlyeq)$ and $\mathscr{Q} = (Q, \sqsubseteq)$, then $\mathscr{P}^{\mathfrak{f}}$ and $\mathscr{Q}_{\mathfrak{f}}$ are order-isomorphic.*

Proof. By the previous theorem, $\mathscr{P}^{\mathfrak{f}} = (f^*[Q], \preccurlyeq)$ and $\mathscr{Q}_{\mathfrak{f}} = (f_*[P], \sqsubseteq)$. We show that f_* restricted to $f^*[Q]$ provides the desired order isomorphism.

Note first that if $p \in f^*[Q]$ then, for some q, $p = f^*(q)$ and hence $f_*(p) = f_*(f^*(q)) \in Q_{\mathfrak{f}} = f_*[P]$. So f_* as required sends elements of $f^*[Q]$ to elements of $f_*[P]$. Moreover every element of $f_*[P]$ is $f_*(u)$ for some $u \in f^*[Q]$. For if $q \in f_*[P]$, then for some p, $q = f_*(p) = f_*(f^*(f_*(p))) = f_*(u)$ where $u = f^*(f_*(p)) \in f^*[Q]$.

So f_* restricted to $f^*[Q]$ is onto $f_*[P]$. It remains to show that it is an order-embedding. We know that f_* will be monotone, so what we need to prove is that, if $p, p' \in f^*[Q]$ and $f_*(p) \sqsubset f_*(p')$, then $p \preccurlyeq p'$.

But if $f_*(p) \sqsubset f_*(p')$, then by the monotonicity of f^*, $f^*(f_*(p)) \preccurlyeq f^*(f_*(p'))$. Recall, though, that $p, p' \in f^*[Q]$ are fixed points of $f^* \circ f_*$. Hence $p \preccurlyeq p'$ as we want. $\square$

We know that a pair of posets which have a Galois connection between them needn't be isomorphic overall, and can – so to speak – be lop-sidedly related. But this nice theorem says that they must, for all that, contain a pair of isomorphic sub-posets (and typically, more than just the one-object posets which are trivially isomorphic).

Definition 96. Suppose $\mathscr{P} = \langle P, \preccurlyeq \rangle$ is a poset; then a *closure function* on $\mathscr{P}$ is a function $c\colon P \to P$ such that, for all $p, p' \in P$,

(1) $p \preccurlyeq c(p)$;

(2) if $p \preccurlyeq p'$, then $c(p) \preccurlyeq c(p')$, i.e. c is monotone;

(3) $c(c(p)) = c(p)$ i.e. c is idempotent.

Roughly speaking, then, a closure function c maps a poset 'upwards' to a subposet which then stays fixed under further applications of c.

Theorem 130. *If $\langle f_*, f^* \rangle$ is a Galois connection between $\mathscr{P} = \langle P, \leqslant \rangle$ and some poset, then $f^* \circ f_*$ is a closure function for $\mathscr{P}$.*

Proof. We quickly check that the three conditions for closure apply. (i) is given by Theorem 122. (ii) is immediate as $f^* \circ f_*$ is a composition of monotone functions. And for (iii), we know that $f_* \circ f^* \circ f_* = f_*$, and hence $(f^* \circ f_*) \circ (f^* \circ f_*) = f^* \circ f_*$. □

20.7 One way a Galois connection can arise

The last three sections have been about Galois connections in general, and reveal that they have a perhaps surprisingly rich structure. In this final section, we note one particular way in which connections can arise.

Theorem 131. *Let R be a binary relation between members of X and members of Y. Define $f_R\colon \mathcal{P}(X) \to \mathcal{P}(Y)$ by putting $f_R(A) = \{b \mid (\forall a \in A)aRb\}$, and similarly define $f^R\colon \mathcal{P}(Y) \to \mathcal{P}(X)$ by putting $f_R(B) = \{a \mid (\forall b \in B)aRb\}$.*

Then (f_R, f^R) is a Galois connection between the inclusion posets $(\mathcal{P}(X), \subseteq)$ and $(\mathcal{P}(Y), \supseteq)$ – note the order reversal.

Proof. We just have to prove that principle (G) holds, i.e. for any $A \subseteq X$, $B \subseteq Y$, $f_R(A) \supseteq B$ iff $A \subseteq f^R(B)$.

But simply by applying definitions we see $f_R(A) \supseteq B$ iff $(\forall b \in B)(\forall a \in A)aRb$ iff $(\forall a \in A)(\forall b \in B)aRb$ iff $A \subseteq f^R(B)$. □

Let's say that Galois connection produced in this way is *relation-generated*. Galois's original classic example was of this kind. And the modern classic example which is the topic of the next chapter is relation-generated too.

21 An aside on syntax and semantics

In his famous Dialectica paper 'Adjointness in foundations' (1969), F. William Lawvere writes of 'the familiar Galois connection between sets of axioms and classes of models, for a fixed [signature]'. We are now in a position to explain this connection and draw out some consequences. (It is related too to the motivating example in §20.2.)

21.1 Structures and models

First, in this section, some quick reminders. Let $\mathcal{L}$ be a formal language. Then a set of $\mathcal{L}$-*axioms* in the wide sense that Lawvere is using is just any old set of $\mathcal{L}$-sentences. And by talk of 'models', Lawvere in fact just means *structures* apt for interpreting a language with $\mathcal{L}$'s signature. So the claim is that we can make a Galois connection between sets of sentences and classes of structures.

Take a familiar type of example. Suppose $\mathcal{L}$ is a classical first-order language. $\mathcal{L}$ will have its distinctive non-logical vocabulary. The particular choice of symbolism is unimportant, of course – what really matters is $\mathcal{L}$'s signature, which specifies $\mathcal{L}$ as having a certain fixed number of constants, a certain fixed number of predicates (each with a given arity), and a certain fixed number of function-symbols (again each with a given arity). A structure apt for interpreting $\mathcal{L}$ will have a domain, with distinguished elements assigned to each constant, sets of n-tuples from the domain assigned to the n-ary predicates, and appropriate sets of $n{+}1$-tuples from the domain assigned to the n-ary functions. We can then define what it is for an $\mathcal{L}$-sentence φ to be *true* with respect to the $\mathcal{L}$-structure σ in a now entirely familiar way.

But nothing that follows will in fact depend on $\mathcal{L}$'s being specifically classical or being first-order. We just need there to be some conception of a class of $\mathcal{L}$-structures apt for interpreting $\mathcal{L}$-sentences, and the idea of a particular $\mathcal{L}$-sentence being true with respect to a structure (and later we make some minimal assumptions about consistency and negation).

For the record, we recall some standard notation:

Definition 97. Let $\mathcal{L}$ be a given formal language, φ be an $\mathcal{L}$-sentence, Γ be a set of $\mathcal{L}$-sentences, and let σ be $\mathcal{L}$-structure, i.e. a structure apt for interpreting

$\mathcal{L}$. Then

(1) If φ is true w.r.t. σ, we say σ is a model for φ and write '$\sigma \vDash \varphi$'.

(2) σ is a model for Γ – notated '$\sigma \vDash \Gamma$' – iff, for every $\varphi \in \Gamma$, $\sigma \vDash \varphi$.

(3) If every model for Γ is a model for φ, we say Γ semantically entails φ and write '$\Gamma \vDash \varphi$'.

(4) If, in particular, if every model of φ makes ψ true too, we write '$\varphi \vDash \psi$' (rather than $\{\varphi\} \vDash \psi$).

Note the second definition – the modern conventional one – contrasts with Lawvere's wider usage of 'model' for structures more generally. The overloading of the symbol '$\vDash$' is of course entirely standard.

21.2 The Galois connection between syntax and semantics

Now we make the Galois connection between syntax and semantics which Lawvere refers to. We continue to assume that $\mathcal{L}$ is a formal language, and $\mathcal{L}$-structures are those which are apt for interpreting it. Then we offer the following further definitions:

Definition 98. Let L be the set of sentences of the language $\mathcal{L}$ and let S be the set of $\mathcal{L}$-structures. Then

(1) $\mathscr{L}$ is the poset $(\mathcal{P}(L), \subseteq)$,

(2) $\mathscr{S}$ is the poset $(\mathcal{P}(S), \supseteq)$,

(3) $f_* : \mathcal{P}(L) \to \mathcal{P}(S)$ is such that, for $\Gamma \subseteq L$, $f_*(\Gamma) = \{\sigma \mid (\forall \varphi \in \Gamma)\, \sigma \vDash \varphi\}$,

(4) $f^* : \mathcal{P}(S) \to \mathcal{P}(L)$ is such that, for $\Sigma \subseteq S$, $f^*(\Sigma) = \{\varphi \mid (\forall \sigma \in \Sigma)\, \sigma \vDash \varphi\}$.

(Ok, there's a bug lurking here which will probably immediately strike you: but bear with me, and I'll return to the point.)

The definitions here should all strike you as entirely natural ones. $\mathscr{L}$ is the inclusion poset on sets-of-$\mathcal{L}$-sentences – an obvious syntactic construction to think about. $\mathscr{S}$ is the dual inclusion poset on sets-of-$\mathcal{L}$-structures – apart from the reversing the order, an equally obvious semantic construction to think about. Then f_* is the obvious 'find the models' function. It takes a bunch of $\mathcal{L}$-sentences Γ and returns all the models of Γ. In the other direction, f^* is the equally obvious natural 'find all the true sentences' function. It takes a bunch of $\mathcal{L}$-structures and returns the set of $\mathcal{L}$-sentences that are true in all of those structures.

We then have

Theorem 132. *(f_*, f^*) is a Galois connection between $\mathscr{L}$ and $\mathscr{S}$.*

Proof. Use Theorem 131 – putting the generating relation R between a sentence and a structure to be the converse of $\vDash$. □

21.3 Drawing some consequences

Terrific! Now we can just turn the handle, and apply all those general theorems about Galois connections from the last chapter to our special case of the connection between 'syntax' $\mathscr{L}$ and 'semantics' $\mathscr{S}$. So let's collect together some of the implications:

Theorem 133. *Let $\mathscr{L} = (\mathcal{P}(L), \subseteq)$, $\mathscr{S} = (\mathcal{P}(S), \supseteq)$ and the Galois connection (f_*, f^*) between them be as defined in Defn. 98. Then*

(1) f_* is monotone, i.e. if $\Gamma \subseteq \Delta$ then $f_*(\Gamma) \supseteq f_*(\Delta)$,

(2) for any $\Gamma \subseteq L$, $\Gamma \subseteq (f^* \circ f_*)[\Gamma]$,

(3) $f_* \circ f^* \circ f_* = f_*$,

(4) $\Gamma \in f^*[\mathcal{P}(S)]$ iff Γ is a fixed point of $f^* \circ f_*$.

And dually,

(5) f^* is monotone, i.e. if $\Sigma \supseteq \Xi$ then $f^*(\Sigma) \subseteq f^*(\Xi)$,

(6) for any $\Sigma \subseteq S$, $(f_* \circ f^*)[\Sigma] \supseteq \Sigma$,

(7) $f^* \circ f_* \circ f^* = f^*$,

(8) $\Sigma \in f_*[\mathcal{P}(L)]$ iff Σ is a fixed point of $f_* \circ f^*$.

The proofs of all these claims are already to hand from the last chapter.

But what do they really mean? Well, result (1) just reminds us that if the set of sentences Γ is contained in the set of sentences Δ then the set of models of Γ contains the models of Δ. Which is simply the point that, as we expand a set of sentences, we can't increase the number of ways of making them all true together.

What about results (2) to (4)? To get a handle on these, let's consider the significance of the composite map $f^* \circ f_*$. By definition, $f_*[\Gamma]$ is the set of all structures which are models for Γ. So $(f^* \circ f_*)[\Gamma]$ returns the set of all sentences which are true in all those structures which are models for Γ. That is to say, $\varphi \in (f^* \circ f_*)[\Gamma]$ iff every interpretation which makes all of Γ true makes φ true too. Hence $(f^* \circ f_*)[\Gamma]$ is exactly the set of semantic logical consequences of Γ.

This means we are in very familiar territory. Recalling some standard definitions, we have

Definition 99. (1) A set of $\mathcal{L}$-sentences Γ is *closed under (semantic) logical consequence* just in case, if $\Gamma \vDash \varphi$, then $\varphi \in \Gamma$.

(2) An *$\mathcal{L}$-theory* Θ is a set of $\mathcal{L}$-sentences closed under logical consequence.

(3) The theory Θ *with axioms* Γ is the smallest theory containing Γ.

'Theories' are, of course, equally often defined as sets of sentences closed under some syntactic deducibility relation rather than as sets closed under semantic consequence (though in a first-order context the definitions come to the same because of the completeness theorem). So let's emphasize: it is the semantic relation that is in play in our definition here.

And in these terms, then, our previous remarks show that if Θ is a theory, then $(f^* \circ f_*)[\Theta]$ is Θ (closing under logical consequence a set of sentences already closed under logical consequences yields the same set of sentences). And hence

Theorem 134. $(f^* \circ f_*)[\Gamma]$ *is the theory with axioms* Γ*.*

Proof. We've seen that $(f^* \circ f_*)[\Gamma]$ is closed under logical consequence and is hence a theory. So we just need to confirm that $(f^* \circ f_*)[\Gamma]$ is the smallest theory containing Γ. But suppose Θ is a theory and $\Gamma \subseteq \Theta$. Then by the monotonicity of the composite function, $(f^* \circ f_*)[\Gamma] \subseteq (f^* \circ f_*)[\Theta] = \Theta$. □

Looked at through the lens of this result, parts (2) to (4) of our previous theorem now become familiar near-trivia. Thus (2) just says again that, given a set of sentences as axioms, the theory with those axioms includes the axioms! (3) says that if you first round out a bunch of sentences Γ by taking their consequences to get the theory $f^*(f_*(\Gamma))$ and then look for the theory's models, you'll get the same upshot as if you'd just looked for the models of Γ straight off. And (4) tells us that if Γ is a set of just those sentences made true across a bunch of structures, then it is closed under logical consequence.

And what about the other parts of Theorem 133? (5) just confirms our expectations again: it says that if the set of structures Σ contains Ξ, then the set of truths verified by all the structures in Σ is contained in the set of truths verified by all the structures in Ξ. But as for the other results, what's their significance? What does the map $f_* \circ f^*$ do for us?

Well, $f^*[\Sigma]$ is the set of sentences from L made true by every structure in Σ. So $f^*[\Sigma]$ is a theory (that's because the logical consequences of any sentences in $f^*[\Sigma]$ will also be made true by every structure in Σ, so $f^*[\Sigma]$ is closed under logical consequence). And hence $(f_* \circ f^*)[\Sigma]$ is in turn the set of all structures which are models for that theory. This links up with another familiar pair of ideas:

Definition 100. (1) A set of $\mathcal{L}$-structures Σ is *axiomatized* by the $\mathcal{L}$-theory Θ just in case $\sigma \in \Sigma$ if and only if σ is a model of Θ.

(2) A set of $\mathcal{L}$-structures Σ is *axiomatizable* iff there is an $\mathcal{L}$-theory which axiomatizes it.

And by a proof dual to that of the previous theorem

Theorem 135. $(f_* \circ f^*)[\Sigma]$ *is the smallest axiomatizable set of structures containing* Σ*.*

And looked at through the lens of *this* result, the last three parts of Theorem 133 again turn into near-trivia. Thus, (6) just says that the smallest axiomatizable class containing Σ contains Σ; (7) says, in effect, that if you first round out a bunch of models Σ to get the minimal axiomatizable class containing it and then look an axiomatization for those models, you get to the same place as if you'd just looked for the theory of Σ straight off; and (8) tells us that if Σ is a set of just those structures that make true some bunch of sentences, then it is axiomatizable set.

21.4 Is this all trivial, then?

'Hold on! Is that it? Those implications of Theorem 133 really *are*, as frankly acknowledged, familiar near-trivialities! Have we laboured so hard to bring forth such a mouse?'

An understandable first reaction, perhaps, but one that entirely misses the point. We are not in the business here of aiming to prove exciting new results about theories and the structures they are about: rather we are trying to fit very familiar old thoughts (at least, old thoughts well known to logicians) into a very much more general order-theoretic framework (less familiar, perhaps, to logicians). Look at it this way. Start from the fundamental *true-of* relation which can obtain between an $\mathcal{L}$-sentence and an $\mathcal{L}$- structure. This, as we've seen, immediately generates a certain Galois connection (f_*, f^*) between two naturally ordered posets whose elements are sets of sentences and sets of structures. And this by itself *already* dictates that e.g. the composite map $f^* \circ f_*$ has to have a special significance as a closure function. So in this way, the notion of the theory $f^* \circ f_*[\Gamma]$ with axioms Γ, with all the properties we now expect of that notion, emerges as generated by a basic construction that appears all over the place in mathematics.

This setting of familiar ideas into a wider abstract framework – so we get to see local results in one domain as in fact exemplifying a very general pattern that can be found across many domains – is, as we are seeing, characteristic of modern mathematics in general and of category theory in particular. And showing in this way how the local fits into a general pattern is one kind of explanatory exercise, revealing how the particular case exemplifies a 'natural kind' of phenomenon in perhaps surprising ways.

21.5 Another implication

There is perhaps not a great deal more juice we can usefully squeeze out of the idea of a Galois connection applied to the simple observation that the 'true of' relation generates such a connection between sets of $\mathcal{L}$–sentences and sets of $\mathcal{L}$–structures. But let's make one more observation.

Applying Theorem 129 and using the same notation, we have

Theorem 136. *If $\mathfrak{f} = (f_*, f^*)$ is the defined Galois connection between the posets $\mathscr{L}$ and $\mathscr{S}$, then $\mathscr{L}^{\mathfrak{f}}$ and $\mathscr{S}_{\mathfrak{f}}$ are order-isomorphic, and (the restriction* *of) f_* provides an order-isomorphism between them.*

$\mathscr{L}^{\mathfrak{f}}$ is the poset $((f^* \circ f_*)[\mathcal{P}(L)], \subseteq)$, which is the poset of theories built in the language $\mathcal{L}$, partially ordered by inclusion. This poset evidently has a maximum, namely the inconsistent theory containing all $\mathcal{L}$-sentences. Then, at the top of the poset in the ordering, just under the maximum, will be those theories Θ such that adding even one more new axiom to Θ which isn't already in the set takes us back to the maximal, inconsistent, theory. Such a Θ is consistent at least in Post's sense of not containing all $\mathcal{L}$-sentences. And, assuming the language has a negation operator obeying minimal normal constraints, Θ is also negation-complete, i.e. for every $\mathcal{L}$-sentence φ, either $\Theta \vDash \varphi$ or $\Theta \vDash \neg\varphi$. (For suppose otherwise. Then since $\Theta \nvDash \varphi$, $\varphi \notin \Theta$. And since $\Theta \nvDash \neg\varphi$, $\Theta \cup \{\varphi\}$ is consistent, i.e. we can add a new axiom to Θ without getting inconsistency, contrary to assumption.) And as we go down the poset $\mathscr{L}^{\mathfrak{f}}$ further in the ordering, then we move from the inconsistent theory, through the negation-complete theories, on to more and more partial theories, till we get down to the theory consisting of just logical truths of $\mathcal{L}$ (which belong to any $\mathcal{L}$-theory) as the minimum.

Now, our theorem tells us that f_* is an order-isomorphism between $\mathscr{L}^{\mathfrak{f}}$, our poset of theories, and a corresponding poset of axiomatizable-sets-of-structures, $\mathscr{S}_{\mathfrak{f}}$. And how is that second poset built up? Well, start with the maximum of $\mathscr{L}^{\mathfrak{f}}$, the inconsistent theory: then f_* will map that across to the maximum of $\mathscr{S}_{\mathfrak{f}}$, namely the empty set of structures (remember which way up the ordering is on the structures side of the Galois connection). And then, just below the maximum of $\mathscr{L}^{\mathfrak{f}}$, f_* maps each consistent negation-complete theory to the corresponding set of its models. But the models of a negation-complete theory Θ agree on the truth-value of every $\mathscr{L}$-sentence (for they make φ true if $\Theta \vDash \varphi$ and φ false if $\Theta \vDash \neg\varphi$ and one of those must hold). So f_* maps the negation-complete theories to sets of elementarily equivalent structures, in the model-theorist's sense (and the map is onto and injective).

Then, as we go further down the poset $\mathscr{L}^{\mathfrak{f}}$ we get more and more partial theories which settle the truth-values of more and more limited classes of sentences. And f_* maps these partial theories to more inclusive bunches of structures, i.e. elements of $\mathscr{S}_{\mathfrak{f}}$, which need agree on less and less. Again, a familiar story really: but its contours are rather neatly brought out by thinking in terms of the Galois connection between syntax and semantics.

21.6 A bug?

Suppose that $\mathscr{L}$ is, for example, a standard first-order language. Reflect that *any* non-empty set can in principle be made into a structure for interpreting $\mathscr{L}$. Just take the set as the domain, select out enough elements (repetitions are allowed!) to assign to constants as denotations, and similarly define such n-ary relations and functions over the set as are needed to interpret a language with the given signature. And now think again about our definition of the poset $\mathscr{S}$.

Its carrier 'set' is supposed to be the powerset of the set S of $\mathscr{L}$-structures. But S is in this case as big as the universe of all sets, period. So the carrier 'set' for $\mathscr{S}$, on standard views, is then too big to be a kosher set. It's a collection whose members are proper classes of structures – which is no doubt why Lawvere did here talk of 'classes of models'. So $\mathscr{S}$ can't, it seems, really be a po*set*. Here then is the potential bug I alluded to earlier. What to do?

The short answer, for our purposes, is not worry! We looked before at some options for dealing with sets that are seemingly 'too big' (see §8.4). Let's mention just two possible lines again, at different ends of the spectrum.

1. At the non-committal end, maybe we can think of the posets we seem to have been talking about as no more than virtual collections. For it seems that, for most of our discussion, we might have used a plural idiom instead and simply talked about having some objects and a partial ordering defined over them (and then about selections from these objects, also thought of plurally, and partial orderings over *them*). The suggestion is, then, that to get the elements of the theory of order under way, and to talk about Galois connections in particular, we don't really need to think of the various objects we are putting into an order as themselves, taken all together, constituting a new object, the set of them.

2. Going to the other extreme, we could buy into a big set ontology, but take it that the structures talked about by logicians normally involve sets-smaller-than-some-inaccessible (surely that's enough for most purposes!), and then the posets we've been dealing with are just yet bigger sets.

Let's be cheerfully agnostic about whether either of these, or some other line, indicates the best direction to go. However things come out in the wash, hopefully the neat insight that there is a Galois connection between syntax and semantics won't get swept away on a technicality!

22 Adjoints introduced

Recall that quotation from Tom Leinster which we gave at the very outset:

> Category theory takes a bird's eye view of mathematics. From high in the sky, details become invisible, but we can spot patterns that were impossible to detect from ground level. (Leinster, 2014, p. 1)

Perhaps the most dramatic patterns that category theory newly reveals are those which involve *adjunctions*. As Mac Lane famously puts it (1997, p. vii) the slogan is "Adjoint functors arise everywhere." In the last two chapters, we have seen a restricted version of the phenomenon (well known before category theory). But category theory enables us to generalize radically.

22.1 Adjoint functors: a first definition

(a) Let $\mathscr{P}$ now be (not the poset itself but) the category corresponding to the poset $(P, \preccurlyeq)$. So the objects of $\mathscr{P}$ are the members of P, and there is a $\mathscr{P}$-arrow $p \to p'$ (for $p, p' \in P$), which we can identify with the pair $\langle p, p' \rangle$, if and only if $p \preccurlyeq p'$. Similarly let $\mathscr{Q}$ be the category corresponding to the poset $(Q, \sqsubseteq)$.

Now, changing symbolism just a little, a Galois connection between the posets $(P, \preccurlyeq)$ and $(Q, \sqsubseteq)$ is a pair of functions $f\colon P \to Q$ and $g\colon Q \to P$ such that

(i) f and g are monotone, and

(ii) $f(p) \sqsubseteq q$ iff $p \preccurlyeq g(q)$ for all $p \in P, q \in Q$.

(Well, we know condition (ii) implies condition (i), but it is helpful now to make it explicit.) However, monotone functions f, g between posets give rise to functors F, G between the corresponding categories – see §4.2, Ex. (F4). Thus the monotone function $f\colon P \to Q$ gives rise to the functor $F\colon \mathscr{P} \to \mathscr{Q}$ which sends the object p in $\mathscr{P}$ to $f(p)$ in $\mathscr{Q}$, and sends an arrow $p \to p'$ in $\mathscr{P}$, i.e. the pair $\langle p, p' \rangle$, to the pair $\langle f(p), f(p') \rangle$ which is an arrow in $\mathscr{Q}$. Similarly, $g\colon Q \to P$ gives rise to a functor $G\colon \mathscr{Q} \to \mathscr{P}$.

So (ii) means that our adjoint *functions*, i.e. the Galois connection (f, g) between the posets $(P, \preccurlyeq)$ and $(Q, \sqsubseteq)$, gives rise to a pair of *functors* (F, G) between the poset categories $\mathscr{P}$ and $\mathscr{Q}$, one in each direction, such that there is a (unique) arrow $Fp \to q$ in $\mathscr{Q}$ iff there is a corresponding (unique) arrow $p \to Gq$ in $\mathscr{P}$. This sets up an isomorphism between the hom-sets $\mathscr{Q}(Fp, q)$ and $\mathscr{P}(p, Fq)$, for each $p \in \mathscr{P}, q \in \mathscr{Q}$.

Of course, for a particular choice of p, q, this will be a rather trivial isomorphism, as the homsets in this case are either both empty or both single-membered. But what isn't trivial is that the existence of the isomorphism arises *systematically* from the Galois connection, in a uniform and natural way. And we now know how to put that informal claim into more formal category-theoretic terms: we have a *natural isomorphism* here, i.e. $\mathscr{Q}(Fp, q) \cong \mathscr{P}(p, Fq)$ *naturally* in $p \in \mathscr{P}, q \in \mathscr{Q}$.

(b) Now we generalize this last idea in the obvious way, and also introduce some absolutely standard notation:

Definition 101. Suppose $\mathscr{A}$ and $\mathscr{B}$ are categories and $F\colon \mathscr{A} \to \mathscr{B}$ and $G\colon \mathscr{B} \to \mathscr{A}$ are functors. Then F *is left adjoint to* G *and* G *is right adjoint to* F, notated $F \dashv G$, iff

$$\mathscr{B}(F(A), B) \cong \mathscr{A}(A, G(B))$$

naturally in $A \in \mathscr{A}$, $B \in \mathscr{B}$. We also write $\mathscr{A} \underset{G}{\overset{F}{\underset{\longleftarrow}{\overset{\longrightarrow}{\perp}}}} \mathscr{B}$ when this situation obtains, or $F \dashv G\colon \mathscr{A} \to \mathscr{B}$, and we say that F and G (together with the associated isomorphism between the relevant hom-sets) form an *adjunction*.

Here, and onwards through our discussions of adjunctions, we'll take it that there is no problem in talking about the relevant hom-*sets* (either because the categories are small enough, or because we are taking a relaxedly inclusive line on what counts as 'sets').

There is an additional fairly standard bit of notation to indicate the action of the natural isomorphism between the hom-sets in an adjunction:

Definition 102. Given the situation just described, and an arrow $f\colon F(A) \to B$, then one direction of the natural bijection between the hom-sets sends that arrow to its *transpose* $\overline{f}\colon A \to G(B)$; likewise the inverse bijection associates an arrow $g\colon A \to G(B)$ to its transpose $\overline{g}\colon F(A) \to B$.

(Another common notation distinguishes $f^\flat$ for our $\overline{f}$ and $g^\sharp$ for our $\overline{g}$, and this notation might be preferable in principle since transposing by 'sharpening' and 'flattening' are indeed different operations. But the double use of the overlining notation is standard, and is slick.)

Evidently, transposing twice takes us back to where we started: $\overline{\overline{f}} = f$ and $\overline{\overline{g}} = g$.

22.2 Examples

As we'd expect from our discussion of Galois connections, given the existence of an adjoint connection $F \dashv G$ we can deduce a range of additional properties of the adjoint functors and of the operation of transposition. But before exploring this any further in the abstract, let's have some more examples of adjunctions (to add to those generated by Galois connections).

For a warm-up exercise, we start with a particularly easy case:

(1) Consider any (non-empty!) category $\mathscr{A}$ and the one object category $\mathbf{1}$ (comprising just the object $\bullet$ and its identity arrow). There is a unique functor $F\colon \mathscr{A} \to \mathbf{1}$. Questions: when does F have a right adjoint $G\colon \mathbf{1} \to \mathscr{A}$? what about a left adjoint?

If G is to be a right adjoint, remembering that $FA = \bullet$ for any $A \in \mathscr{A}$, we require

$$\mathbf{1}(\bullet, \bullet) \cong \mathscr{A}(A, G\bullet),$$

for any A. The hom-set on the left contains just the identity arrow. So that can only be in bijection to the hom-set on the right, for each A, if there is always a *unique* arrow $A \to G\bullet$, i.e. if $G\bullet$ is terminal in $\mathscr{A}$.

In sum, F has a right adjoint $G\colon \mathbf{1} \to \mathscr{A}$ just in case G sends $\mathbf{1}$'s unique object to $\mathscr{A}$'s terminal object: no terminal object, no right adjoint.

Dually, F has a left adjoint if and only if $\mathscr{A}$ has an initial object.

This toy example reminds of what we have already seen in the special case of Galois connections, namely that a functor may or may not have a right adjoint, and independently may or may not have a left adjoint, and if both adjoints exist they may be different. But let's also note that we have here a first indication that adjunctions and limits can interact in interesting way: in this case, indeed, we could *define* terminal and initial objects for a category $\mathscr{A}$ in terms of the existence of right and left adjoints to the functor $F\colon \mathscr{A} \to \mathbf{1}$. We will return to this theme.

Now for a couple of more substantive examples. And to speed things along, we will procede informally: we won't in this section actually prove that the relevant hom-sets in our various examples are naturally isomorphic in the official formal sense, but rather we will take it as enough to find a bijection which can be evidently set up in a systematic and intuitively natural way, without arbitrary choices.

(2) Let's next consider the forgetful functor $U\colon \mathbf{Top} \to \mathbf{Set}$ which sends each topological space to its underlying set of points, and sends any continuous function between topological spaces to the same function thought of as a set-function. Questions: does this have a left adjoint? a right adjoint?

If U is to have a left adjoint $F\colon \mathbf{Set} \to \mathbf{Top}$, then for any set S and for any topological space (T, O) – with T a set of points and O a topology (a suitable collection of open sets) – we require

$$\mathbf{Top}(F(S),(T,O)) \cong \mathbf{Set}(S, U(T,O)) = \mathbf{Set}(S,T),$$

where the bijection here needs to be a natural one.

Now, on the right we have the set of *all* functions $f\colon S \to T$. So that needs to be in bijection with the set of all *continuous* functions from FS to (T,O). How can we ensure this holds in a systematic way, for any S and (T,O)? Well, suppose that for any S, F sends S to the topological space (S,D) which has the discrete topology (i.e. all subsets of S count as open). It is a simple exercise to show that *every* function $f\colon S \to T$ then counts as a continuous function $f\colon (S,D) \to (T,O)$. So the functor F which assigns a set the discrete topology will indeed be left adjoint to the forgetful functor.

Similarly, the functor $G\colon \mathbf{Set} \to \mathbf{Top}$ which assigns a set the indiscrete topology (the only open sets are the empty set and S itself) is right adjoint to the forgetful functor U.

(3) Let's now take another case of a forgetful functor, this time the functor $U\colon \mathbf{Mon} \to \mathbf{Set}$ which forgets about monoidal structure. Does U have a left adjoint $F\colon \mathbf{Set} \to \mathbf{Mon}$. If $(M,\cdot)$ is a monoid and S some set, we need

$$\mathbf{Mon}(FS,(M,\cdot)) \cong \mathbf{Set}(S, U(M,\cdot)) = \mathbf{Set}(S,M).$$

The hom-set on the right contains all possible functions $f\colon S \to M$. How can these be in one-one correspondence with the monoid homomorphisms from FS to $(M,\cdot)$?

Arm-waving for a moment, suppose FS is some monoid with a lot of structure (over and above the minimum required to be a monoid). Then there may be few if any monoid homomorphisms from FS to $(M,\cdot)$. Therefore, if there are potentially to be *lots* of such monoid homorphisms, one for each $f\colon S \to M$, then FS will surely need to have minimal structure. Which suggests going for broke and considering the limiting case, i.e. the functor F which sends a set S to $(S^*,*)$, the *free* monoid on S which we met back in §4.2, Ex. (F11). Recall, the objects of $(S^*,*)$ are sequences of S-elements (including the null sequence) and its monoid operation is concatenation.

There is an obvious map α which takes an arrow $f\colon S \to M$ and sends it to $\overline{f}\colon (S^*,*) \to (M,\cdot)$, where $\overline{f}$ sends the empty sequence of S-elements to the unit of M, and sends the finite sequence $x_1 * x_2 * x_3 * \ldots * x_n$ to the M-element $fx_1 \cdot fx_2 \cdot fx_3 \cdot \ldots \cdot fx_n$. So defined, $\overline{f}$ respects the unit and the monoid operation and so is a monoid homomorphism.

There is an equally obvious map β which takes an arrow $g\colon (S^*,*) \to (M,\cdot)$ to the function $\overline{g}\colon S \to M$ which sends an element $x \in S$ to $g\langle x\rangle$ (i.e. to g applied to the one-element list containing x).

Evidently α and β are inverses, so form a bijection, and their construction is quite general (i.e. can be applied to any set S and monoid $(M, \cdot)$). Which establishes that, as required $\mathbf{Mon}(FS, (M, \cdot)) \cong \mathbf{Set}(S, M)$.

So in sum, the free functor F which takes a set to the free monoid on that set is left adjoint to the forgetful function U which sends a monoid to its underlying set.

Now recall Theorem 124: if a function f has a left or right adjoint to make up a Galois connection, then that adjoint is unique. An analogous uniqueness result applies to adjoints more generally: if a functor has a left adjoint, then it is unique up to isomorphism, and likewise right adjoints (when they exist) are unique up to isomorphism. So we can say that the functor which assigns a set the indiscrete topology is in fact *the* right adjoint to the functor which forgets topological structure, and we can say that the functor sending a set to the free monoid on that set is *the* left adjoint of the forgetful functor on monoids. However, the uniqueness theorem for adjoints takes a bit of work; so we'll delay the proof until the next chapter, §**??**. For the moment, then, we'll officially continue simply to talk of one functor being left (right) adjoint to another without making explicit uniqueness claims.

Our example involving monoids is actually typical of a whole cluster of cases. A left adjoint of the trivial forgetful functor from some class of algebraic structures to their underlying sets is characteristically provided by the non-trivial functor that takes us from a set to a free structure of that algebraic kind. Thus we have, for example,

(4) The forgetful functor $U\colon \mathbf{Grp} \to \mathbf{Set}$ has as a left adjoint the functor $F\colon \mathbf{Set} \to \mathbf{Grp}$ which sends a set to the free group on that set (i.e. the group obtained from a set S by adding just enough elements for it to become a group while imposing no constraints other than those required to ensure we indeed have a group).

What about *right* adjoints to our last two forgetful functors?

(5) We will later show that the forgetful functor $U\colon \mathbf{Mon} \to \mathbf{Set}$. has no right adjoint by a neat proof in §24.3. But here's a more arm-waving argument. U would have a right adjoint $G\colon \mathbf{Set} \to \mathbf{Mon}$ just in case $\mathbf{Set}(M, S) = \mathbf{Set}(U(M, \cdot), S) \cong \mathbf{Mon}((M, \cdot), GS)$, for all monoids $(M, \cdot)$ and sets S. But this requires the monoid homomorphisms from $(M, \cdot)$ to GS always to be in bijection with the set-functions from M to S. But that's not possible (consider keeping the sets M and S fixed, but changing the possible monoid operations with which M is equipped).

Similarly the forgetful functor $U\colon \mathbf{Grp} \to \mathbf{Set}$ has no right adjoint.

(6) There are however examples of 'less forgetful' algebraic functors which have both left and right adjoints. Take the functor $U\colon \mathbf{Grp} \to \mathbf{Mon}$ which forgets about group inverses but keeps the monoidal structure. This has

a left adjoint $F\colon \mathbf{Mon} \to \mathbf{Grp}$ which converts a monoid to a group by adding inverses for elements (and otherwise making no more assumptions that are needed to get a group). U also has a right adjoint $G\colon \mathbf{Mon} \to \mathbf{Grp}$ which rather than adding elements subtracts them by mapping a monoid to the submonoid of its invertible elements (which can be interpreted as a group).

Let's quickly check just the second of those claims. We have $U \dashv G$ so long as

$$\mathbf{Mon}(U(K, \times), (M, \cdot)) \cong \mathbf{Grp}((K, \times), G(M, \cdot)),$$

for any monoid $(M, \cdot)$ and group $(K, \times)$. Now we just remark that every element of $(K, \times)$-as-a-monoid is invertible and a monoid homomorphism sends invertible elements to invertible elements. Hence a monoid homomorphism from $(K, \times)$-as-a-monoid to $(M, \cdot)$ will in fact also be a group homomorphism from $(K, \times)$ to the submonoid-as-a-group $G(M, \cdot)$.

(7) Recall the functor $F\colon \mathbf{Set} \to \mathbf{Rel}$ which 'forgets' that arrows are functional (see §4.2, Ex. (F7)). And now we introduce a powerset functor $P\colon \mathbf{Rel} \to \mathbf{Set}$ defined as follows:

a) P sends a set A to its powerset $\mathcal{P}(A)$, and

b) P sends a relation R in $A \times B$ to the function $f_R\colon \mathcal{P}(A) \to \mathcal{P}(B)$ which sends $X \subseteq A$ to $Y = \{b \mid (\exists x \in X)Rxb\} \subseteq B$.

Claim: $F \dashv P$.

We observe that there is a (natural!) isomorphism which correlates a relation R in $A \times B$ with a function $f\colon A \to \mathcal{P}(B)$ where $f(x) = \{y \mid Rxy\}$ and so Rxy iff $y \in f(x)$. This gives us a isomorphism $\mathbf{Rel}(FA, B) \cong \mathbf{Set}(A, PB)$ which can be checked to be natural in $A \in \mathbf{Set}$ and $B \in \mathbf{Rel}$.

And now for some cases not involving forgetful functors:

(8) Suppose $\mathscr{C}$ is a category with exponentiation (and hence with products). Then, in a slogan, exponentiation by B is right adjoint to taking the product with B.

To see this, we define a pair of functors from $\mathscr{C}$ to itself. First, there is the functor $- \times B\colon \mathscr{C} \to \mathscr{C}$ which sends an object A to the product $A \times B$, and sends an arrow $f\colon A \to A'$ to $f \times 1_B\colon A \times B \to A' \times B$.

Second there is the functor $(-)^B\colon \mathscr{C} \to \mathscr{C}$ which sends an object C to C^B, and sends an arrow $f\colon C \to C'$ to $\overline{f \circ ev}\colon C^B \to C'^B$ as defined in the proof of Theorem 119. It is easily checked that $(-)^B$ satisfies the conditions for functoriality.

By the theorem just mentioned, $\mathscr{C}(A \times B, C) \cong \mathscr{C}(A, C^B)$ naturally in A and C. Hence $(- \times B) \dashv (-)^B$.

(9) Recall Defn. 19 which defined the product of two categories. Given a category $\mathscr{C}$ there is a trivial diagonal functor $\Delta\colon \mathscr{C} \to \mathscr{C} \times \mathscr{C}$ which sends a $\mathscr{C}$-object A to the pair $\langle A, A\rangle$, and sends a $\mathscr{C}$-arrow f to the pair of

arrows $\langle f, f\rangle$. What would it take for this functor to have a right adjoint $G\colon \mathscr{C}\times\mathscr{C}\to\mathscr{C}$? We'd need

$$(\mathscr{C}\times\mathscr{C})(\langle A, A\rangle, \langle B, C\rangle) \cong \mathscr{C}(A, G\langle B, C\rangle)$$

naturally in $A\in\mathscr{C}$ and in $\langle B, C\rangle\in\mathscr{C}\times\mathscr{C}$. But by definition the left hand hom-set is $\mathscr{C}(A,B)\times\mathscr{C}(A,C)$. But then if we can take G to be the product functor that sends $\langle B, C\rangle$ to the product object $B\times C$ in $\mathscr{C}$ we'll get an obvious natural isomorphism

$$\mathscr{C}(A,B)\times\mathscr{C}(A,C)\cong\mathscr{C}(A,B\times C).$$

So in sum, $\Delta\colon\mathscr{C}\to\mathscr{C}\times\mathscr{C}$ has a right adjoint if $\mathscr{C}$ has binary products.

(10) For topologists, let's simply mention another example of a case where the adjoint of a trivial functor is something much more substantial. The inclusion functor from **KHaus**, the category of compact Hausdorff spaces, into **Top** has a left adjoint, namely the Stone-Čech compactification functor.

22.3 Naturality

We said: $F\dashv G\colon\mathscr{A}\to\mathscr{B}$ just in case

$$\mathscr{B}(F(A), B)\cong\mathscr{A}(A, G(B))$$

holds naturally in $A\in\mathscr{A}$, $B\in\mathscr{B}$. Let's now be more explicit about what the official naturality requirement comes to.

By Defn. 27, the required bijection holds naturally in B (to take that case first) just if the two hom-functors $\mathscr{B}(F(A),-)$ and $\mathscr{A}(A,G(-))$ are naturally isomorphic. By Defn. 23, that means there have to be isomorphisms $\varphi_B : \mathscr{B}(F(A),B)\to\mathscr{A}(A,G(B))$, one for each B, such that for every $h\colon B\to B'$, the usual naturality square always commutes:

$$\begin{tikzcd}
\mathscr{B}(F(A),B) \arrow[r, "\mathscr{B}(F(A),h)"] \arrow[d, "\varphi_B"] & \mathscr{B}(F(A),B') \arrow[d, "\varphi_{B'}"] \\
\mathscr{A}(A,G(B)) \arrow[r, "\mathscr{A}(A,G(h))"] & \mathscr{A}(A,G(B'))
\end{tikzcd}$$

But how does the covariant hom-functor $\mathscr{B}(F(A),-)$ operate on $h\colon B\to B'$? As we saw in §9.2, it sends h to $h\circ -$, i.e. to that function which composes h with an arrow from $\mathscr{B}(F(A),B)$ to give an arrow in $\mathscr{B}(F(A),B')$. Similarly, $\mathscr{A}(A,G(-))$ will send h to $Gh\circ -$.

So consider an arrow $f\colon F(A)\to B$ living in $\mathscr{B}(F(A),B)$. The naturality square now tells us that for any $h\colon B\to B'$, $\varphi_{B'}(h\circ f) = Gh\circ\varphi_B(f)$.

But (by the definition of transposition!), the components of φ send an arrow to its transpose. So we have shown the first part of the following theorem. And the second part of this theorem follows by a dual argument, in which some arrows get reversed because the relevant hom-functors in this case are contravariant.

Theorem 137. *Given $F \dashv G\colon \mathscr{A} \to \mathscr{B}$, then*

(1) for any $f\colon F(A) \to B$ and $h\colon B \to B'$, $\overline{h \circ f} = Gh \circ \overline{f}$,

(2) for any $g\colon A \to G(B)$ and $k\colon A' \to A$, $\overline{g \circ k} = \overline{g} \circ Fk$, i.e. $\overline{\overline{g} \circ Fk} = g \circ k$.

Inspecting the proof, we see that there is an obvious converse to this theorem. Given functors $F\colon \mathscr{A} \to \mathscr{B}$ and $G\colon \mathscr{B} \to \mathscr{A}$ such that there is always a bijection between $\mathscr{B}(F(A), B)$ and $\mathscr{A}(A, G(B))$ then, if conditions (1) and (2) hold, the bijections (for different As and Bs) will assemble into natural transformations, so that $\mathscr{B}(F(A), B) \cong \mathscr{A}(A, G(B))$ holds naturally in $A \in \mathscr{A}$, $B \in \mathscr{B}$, and hence $F \dashv G$.

22.4 An alternative definition

We now know what it takes for a pair of functors to be adjoint to each other, and we have given various examples of adjoint pairs (to add to the special cases from the previous two chapters where the adjunctions are Galois connections).

Now, our first definition of adjunctions was inspired by our original definition of Galois connections in §20.3. But we gave an alternative definition of such connections in §20.4. This too can be generalized to give a second definition of adjunctions. In this section we show how, and prove that the new definition is equivalent to our first one. (This alternative definition will turn out to look somewhat more complicated, but it is useful in practice – though for the moment our prime aim is to bring out something of the structural richness of adjunctions.)

A Galois connection between the posets $(P, \preccurlyeq)$, $(Q, \sqsubseteq)$, according to the alternative definition, comprises a pair of functions $f\colon P \to Q$ and $g\colon Q \to P$ such that

(i) f and g are monotone,

(ii) $p \preccurlyeq g(f(p))$ for all $p \in P$, and

(iii) $f(g(q)) \sqsubseteq q$ for all $q \in Q$.

Since the composition of monotone functions is monotone, (ii) and (iii) are in fact easily seen to be equivalent to

(ii′) if $p \preccurlyeq p'$, then $p \preccurlyeq p' \preccurlyeq g(f(p'))$ and $p \preccurlyeq g(f(p)) \preccurlyeq g(f(p'))$,

(iii′) if $q \sqsubseteq q'$, then $f(g(q)) \sqsubseteq q \sqsubseteq q'$ and $f(g(q)) \sqsubseteq f(g(q')) \sqsubseteq q'$.

As before, let $\mathscr{P}$ be the category corresponding to the poset $(P, \preccurlyeq)$, and recall that there is an arrow $p \to p'$ in $\mathscr{P}$ just when $p \preccurlyeq p'$ in the poset $(P, \preccurlyeq)$. Likewise for $\mathscr{Q}$ corresponding to $(Q, \sqsubseteq)$. And again as before, note that the monotone functions f, g between the posets give rise to functors F, G between the corresponding categories. Hence, in particular, the composite monotone function $g \circ f$ gives rise to a functor $G \circ F \colon \mathscr{P} \to \mathscr{P}$, and likewise $f \circ g$ gives rise to a functor $F \circ G \colon \mathscr{Q} \to \mathscr{Q}$.

Now, (ii′) corresponds in $\mathscr{P}$ to the claim that the following diagram always commutes:

$$\begin{tikzcd} p \arrow[r] \arrow[d] & p' \arrow[d] \\ (G \circ F)p \arrow[r] & (G \circ F)p' \end{tikzcd}$$

(We needn't label the arrows as in the poset category $\mathscr{P}$ arrows between objects are unique when they exist.)

Dropping the explicit sign for composition of functors for brevity's sake, let's define $\eta_p \colon 1_{\mathscr{P}} \Rightarrow GF$ to be the arrows $p \to GFp$, one for each $p \in \mathscr{P}$. Then our commutative diagram version of (ii′) can be revealingly redrawn as follows:

$$\begin{tikzcd} 1_{\mathscr{P}}\, p \arrow[r] \arrow[d, "\eta_p"] & 1_{\mathscr{P}}\, p' \arrow[d, "\eta_{p'}"] \\ GFp \arrow[r] & GFp' \end{tikzcd}$$

This commutes for all p, p'. So applying Defn. 24, this is just to say that the η_p assemble into a natural transformation $\eta \colon 1_{\mathscr{P}} \Rightarrow GF$ in $\mathscr{P}$.

Likewise, (iii) and hence (iii′) correspond to the claim that there is a natural transformation $\varepsilon \colon FG \Rightarrow 1_{\mathscr{Q}}$ in $\mathscr{Q}$.

(a) So far so good. We have here the initial ingredients for an alternative definition for an adjunction between functors $F \colon \mathscr{A} \to \mathscr{B}$ and $G \colon \mathscr{B} \to \mathscr{A}$: we will require there to be a pair of natural transformations $\eta \colon 1_{\mathscr{A}} \Rightarrow GF$ and $\varepsilon \colon FG \Rightarrow 1_{\mathscr{B}}$.

However, as we'll see, this isn't yet quite enough. But the additional ingredients we want are again suggested by our earlier treatment of Galois connections. Recall from Theorem 127 that if (f, g) is a Galois connection between $(P, \preccurlyeq)$ and $(Q, \sqsubseteq)$, then we immediately have the key identities

(iv) $f \circ g \circ f = f$, and

(v) $g \circ f \circ g = g$.

By (iv), $fp \preccurlyeq (f \circ g \circ f)p \preccurlyeq f(p)$ in $(P, \preccurlyeq)$. Hence in $\mathscr{P}$ the following diagram commutes for each p:

$$\begin{tikzcd} Fp \arrow[r] \arrow[dr] & FGFp \arrow[d] \\ & Fp \end{tikzcd}$$

Here, the diagonal arrow is the identity 1_{Fp}. The downward arrow is ε_{Fp} (the component of ε at Fp). And the horizontal arrow is $F\eta_p$. So we have $\varepsilon_{Fp} \circ F\eta_p = 1_{Fp}$ for each p.

Or what comes to the same, in the functor category $[\mathscr{P}, \mathscr{Q}]$ this diagram commutes[1]

$$\begin{tikzcd} F \arrow[r, "F\eta"] \arrow[dr, "1_F"'] & FGF \arrow[d, "\varepsilon F"] \\ & F \end{tikzcd}$$

For remember whiskering(!), discussed in §6.5: the components $F(\eta_p)$ assemble into the natural transformation we symbolized '$F\eta$', and the components ε_{Fp} assemble into the natural transformation we symbolized 'ε_F'. And then recall from §6.4 that 'vertical' composition of natural transformations between e.g. the functors $F\colon \mathscr{A} \to \mathscr{B}$ and $FGF\colon \mathscr{A} \to \mathscr{B}$ is defined component-wise. So, for each p,

$$(\varepsilon F \circ F\eta)_p = \varepsilon_{Fp} \circ F\eta_p = 1_{Fp} = (1_F)_p,$$

where 1_F is the natural transformation whose component at p is 1_{Fp}. Since all components are equal, the left-most and right-most natural transformations in that equation are equal and our diagram commutes.

Exactly similarly, from (v) we infer that $G\varepsilon_q \circ \eta_{Gq} = 1_{Gq}$. In other words, the next diagram commutes in $[\mathscr{Q}, \mathscr{P}]$:

$$\begin{tikzcd} G \arrow[r, "\eta G"] \arrow[dr, "1_G"'] & GFG \arrow[d, "G\varepsilon"] \\ & G \end{tikzcd}$$

(b) And *now* we can put everything together to give us our second definition for adjoint functors:

[1]Notational fine print: our convention has been to use single arrows to represent arrows inside particular categories, and double arrows to represent natural transformations between functors across categories. We are now dealing with natural-transformations-thought-of-as-arrows-within-a-particular-functor-category. Some use double arrows for diagrams in a functor category, to remind us these are natural transformations (between functors relating some other categories); some use single arrows because these are being treated as arrows (in the functor category). I'm jumping the second way, following the majority and also getting slightly cleaner diagrams.

Definition 103 (Alternative)**.** Suppose $\mathscr{A}$ and $\mathscr{B}$ are categories and $F\colon \mathscr{A} \to \mathscr{B}$ and $G\colon \mathscr{B} \to \mathscr{A}$ are functors. Then F *is left adjoint to* G *and* G *is right adjoint to* F, notated $F \dashv G$, iff

(i) there are natural transformations $\eta\colon 1_{\mathscr{A}} \Rightarrow GF$ and $\varepsilon\colon FG \Rightarrow 1_{\mathscr{B}}$ such that

(ii) $\varepsilon_{FA} \circ F\eta_A = 1_{FA}$ for all $A \in \mathscr{A}$, and $G\varepsilon_B \circ \eta_{GB} = 1_{GB}$ for all $B \in \mathscr{B}$; or equivalently

(ii′) the following *triangle identities* hold in the functor categories $[\mathscr{A}, \mathscr{B}]$ and $[\mathscr{B}, \mathscr{A}]$ respectively:

$$\begin{tikzcd} F \arrow[r, "F\eta"] \arrow[dr, "1_F"'] & FGF \arrow[d, "\varepsilon F"] \\ & F \end{tikzcd} \qquad \begin{tikzcd} G \arrow[r, "\eta G"] \arrow[dr, "1_G"'] & GFG \arrow[d, "G\varepsilon"] \\ & G \end{tikzcd}$$

Note, η and ε are standardly called the *unit* and *counit* of the adjunction.

It remains to show that Defn. 101 and Defn. 103 are equivalent:

Theorem 138. *For given functors $F\colon \mathscr{A} \to \mathscr{B}$ and $G\colon \mathscr{B} \to \mathscr{A}$, $F \dashv G$ holds by our original definition iff it holds by the alternative definition.*

Proof (If). Suppose there are natural transformations $\eta\colon 1_{\mathscr{A}} \Rightarrow GF$ and $\varepsilon\colon FG \Rightarrow 1_{\mathscr{B}}$ for which the triangle identities hold.

Take any f in $\mathscr{B}(F(A), B)$. Then $\eta_A\colon A \to GF(A)$ and $G(f)\colon GF(A) \to GB$ compose. And so we can define $\varphi_{AB}\colon \mathscr{B}(F(A), B) \to \mathscr{A}(A, G(B))$ by putting $\varphi_{AB}(f) = G(f) \circ \eta_A$.

Likewise, we can define $\psi_{AB}\colon \mathscr{A}(A, G(B)) \to \mathscr{B}(F(A), B)$ by putting $\psi_{AB}(g) = \varepsilon_B \circ F(g)$ for any $g\colon A \to G(B)$.

Keep A fixed: then, as we vary B, the various components φ_{AB} assemble into a natural transformation $\varphi_A\colon \mathscr{B}(F(A), -) \Rightarrow \mathscr{A}(A, G(-))$. That's because the naturality square

$$\begin{tikzcd} \mathscr{B}(F(A), B) \arrow[r, "h \circ -"] \arrow[d, "\varphi_{AB}"] & \mathscr{B}(F(A), B') \arrow[d, "\varphi_{AB'}"] \\ \mathscr{A}(A, G(B)) \arrow[r, "Gh \circ -"] & \mathscr{A}(A, G(B')) \end{tikzcd}$$

commutes for every $h\colon B \to B'$, i.e. for every f in $\mathscr{B}(F(A), B)$ we have

$$\varphi_{AB'}(h \circ f) = G(h \circ f) \circ \eta_A = Gh \circ (Gf \circ \eta_A) = Gh \circ \varphi_{AB}(f)$$

which holds by the functoriality of G.

Now keep B fixed: then, as we vary A, the various components φ_{AB} assemble into a natural transformation $\varphi_B\colon \mathscr{B}(F(-), B) \Rightarrow \mathscr{A}(-, G(B))$ between the two *contravariant* functors. That's because the required naturality square

$$\begin{tikzcd}
\mathscr{B}(F(A), B) \arrow[r, "- \circ Fg"] \arrow[d, "\varphi_{AB}"] & \mathscr{B}(F(A'), B) \arrow[d, "\varphi_{A'B}"] \\
\mathscr{A}(A, G(B)) \arrow[r, "- \circ g"] & \mathscr{A}(A', G(B))
\end{tikzcd}$$

commutes for every $g\colon A' \to A$. Why so? Because for every f in $\mathscr{B}(F(A), B)$, we have

$$\varphi_{A'B}(f \circ Fg) = G(f \circ Fg) \circ \eta_{A'} = Gf \circ (GFg \circ \eta_{A'}) = Gf \circ (\eta_A \circ g) = \varphi_{AB}(f) \circ g$$

which holds by the functoriality of G and the naturality of η.

Similarly if we keep A fixed, the various ψ_{AB} assemble into a natural transformation $\psi_A\colon \mathscr{A}(A, G(-)) \Rightarrow \mathscr{B}(F(A), -)$; and if we keep B fixed, the various ψ_{AB} assemble into $\psi_B\colon \mathscr{A}(-, G(B)) \Rightarrow \mathscr{B}(F(-), B)$.

We now need to show that these natural transformations are isomorphisms, from which the desired result will follow: i.e. $\mathscr{B}(F(A), B) \cong \mathscr{A}(A, G(B))$ naturally in $A \in \mathscr{A}$ and in $B \in \mathscr{B}$.

We show each φ_{AB} and ψ_{AB} are mutually inverse. Take any $f\colon FA \to B$. Then

$$\begin{aligned}
\psi_{AB}(\varphi_{AB}(f)) &= \psi_{AB}(G(f) \circ \eta_A) && \text{by definition of } \phi \\
&= \varepsilon_B \circ F(G(f) \circ \eta_A)) && \text{by definition of } \psi \\
&= \varepsilon_B \circ FGf \circ F\eta_A && \text{by functoriality of } F \\
&= f \circ \varepsilon_{FA} \circ F\eta_A && \text{by naturality square for } \varepsilon \\
&= f \circ 1_{FA} && \text{by triangle equality} \\
&= f
\end{aligned}$$

Hence $\psi_{AB} \circ \varphi_{AB} = 1$ (note how we did need to appeal to the added triangle equality, not just functoriality and the naturality of ε). Likewise $\psi_{AB} \circ \varphi_{AB} = 1$. □

Proof (Only if). Suppose $\mathscr{B}(F(A), B) \cong \mathscr{A}(A, G(B))$ naturally in $A \in \mathscr{A}$ and in $B \in \mathscr{B}$. We need to define a unit and counit for the adjunction, and show they satisfy the triangle equalities.

Take the identity map 1_{FA} in $\mathscr{B}(FA, FA)$. The natural isomorphism defining the adjunction sends 1_{FA} to a map we will hopefully call $\eta_A\colon A \to GF(A)$.

We first show that the components η_A do indeed assemble into a natural transformation from $1_{\mathscr{A}}$ to GF. So consider the following two diagrams:

$$\begin{tikzcd}
FA \arrow[r, "Ff"] \arrow[d, "1_{FA}"] & FA' \arrow[d, "1_{FA'}"] \\
FA \arrow[r, "Ff"] & FA'
\end{tikzcd}
\qquad
\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, "\eta_A"] & A' \arrow[d, "\eta_{A'}"] \\
GFA \arrow[r, "GFf"] & GFA'
\end{tikzcd}$$

Trivially, the diagram on the left commutes for all $f\colon A \to A'$. That is to say, $Ff \circ 1_{FA} = 1_{FA'} \circ Ff$. Transposition must evidently preserve identities. So $\overline{Ff \circ 1_{FA}} = \overline{1_{FA'} \circ Ff}$. But by the first of the naturality requirements in §22.3, $\overline{Ff \circ 1_{FA}} = GFf \circ \overline{1_{FA}} = GFf \circ \eta_A$. And by the other naturality requirement, $\overline{1_{FA'} \circ Ff} = \overline{\overline{\eta_{A'}} \circ Ff} = \eta_{A'} \circ f$. So we have $GFf \circ \eta_A = \eta_{A'} \circ f$ and the diagram on the right commutes for all f. Hence the components η_A do indeed assemble into a natural transformation.

Similarly the same natural isomorphism in the opposite direction sends 1_{GB} to its transpose $\varepsilon_B\colon FG(B) \to B$, and the components ε_B assemble into a natural transformation from FG to $1_{\mathscr{B}}$.

Now consider these three diagrams:

$$\begin{tikzcd} A \arrow[r, "\eta_A"] \arrow[d, "\eta_A"'] & GFA \arrow[d, "1_{GFA}"] \\ GFA \arrow[r, "1_{GFA}"] & GFA \end{tikzcd} \qquad \begin{tikzcd} FA \arrow[r, "F\eta_A"] \arrow[d, "1_{FA}"'] & FGFA \arrow[d, "\varepsilon_{FA}"] \\ FA \arrow[r, "1_{FA}"] & FA \end{tikzcd}$$

The diagram on the left trivially commutes. Transpose it via the natural isomorphism that defines the adjunction and use the naturality requirements again; we find that the diagram on the right must also commute. So $\varepsilon_{FA} \circ F\eta_A = 1_{FA}$ for all $A \in \mathscr{A}$ – which gives us one of the triangle identities. The other identity we get dually. □

We are done. But although the strategies for proving the equivalence of our definitions are entirely straightforward, checking the details was a bit tedious and required keeping our wits about us. So let's pause before resuming in the next chapter the exploration of adjunctions.

22.5 Adjoints and equivalent categories

Our second definition of an adjunction should remind you strongly of our earlier characterization of what it takes to for categories to be equivalent. We should pause to say something about this.

We can slightly recast our definitions to highlight the parallelism:

Definition 32* An equivalence between categories $\mathscr{A}$ and $\mathscr{B}$ is a pair of functors $F\colon \mathscr{A} \to \mathscr{B}$ and $G\colon \mathscr{B} \to \mathscr{A}$ and a pair of natural *isomorphisms* $\eta\colon 1_{\mathscr{A}} \Rightarrow GF$ and $\varepsilon\colon FG \Rightarrow 1_{\mathscr{B}}$.

Definition 103* An adjunction between categories $\mathscr{A}$ and $\mathscr{B}$ is a pair of functors $F\colon \mathscr{A} \to \mathscr{B}$ and $G\colon \mathscr{B} \to \mathscr{A}$ and a pair of natural *transformations* $\eta\colon 1_{\mathscr{A}} \Rightarrow GF$ and $\varepsilon\colon FG \Rightarrow 1_{\mathscr{B}}$ such that $\varepsilon_{FA} \circ F\eta_A = 1_{FA}$ for all $A \in \mathscr{A}$, and $G\varepsilon_B \circ \eta_{GB} = 1_{GB}$ for all $B \in \mathscr{B}$.

Since transformations need not be isomorphisms, an adjunction needn't be an equivalence (and indeed we have met lots of examples of adjunctions between non-equivalent categories). In the other direction, an isomorphism needn't satisfy the triangle identities, so an equivalence needn't be an adjunction either. However, we *do* have the following result:

Theorem 139. *If there is an equivalence between $\mathscr{A}$ and $\mathscr{B}$ constituted by a pair of functors $F\colon \mathscr{A} \to \mathscr{B}$ and $G\colon \mathscr{B} \to \mathscr{A}$ and a pair of natural isomorphisms $\eta\colon 1_{\mathscr{A}} \Rightarrow GF$ and $\gamma\colon FG \Rightarrow 1_{\mathscr{B}}$, then there is an adjunction $F \dashv G$ with unit η and counit ε (defined in terms of γ and η), and further there is also an adjunction $G \dashv F$.*

In other words, take an equivalence, fix one of the natural transformations, but tinker (if necessary) with the other, and we get an adjunction. Further we can construct an adjunction in the opposite direction.

Proof. Define the natural transformation ε by composition as follows:

$$\varepsilon\colon FG \xRightarrow{FG\gamma^{-1}} FGFG \xRightarrow{(F\eta G)^{-1}} FG \xRightarrow{\ \gamma\ } 1_{\mathscr{B}}$$

Since η and γ are isomorphisms, and by Theorem 20 whiskering natural isomorphisms yields another natural isomorphism, the inverses mentioned here must exist.

So we just need to establish that, with ε so defined, we get the usual triangle identities $\varepsilon_{FA} \circ F\eta_A = 1_{FA}$ for all $A \in \mathscr{A}$, and $G\varepsilon_B \circ \eta_{GB} = 1_{GB}$ for all $B \in \mathscr{B}$.

So, firstly, for any A, we need the composite arrow (*)

$$FA \xrightarrow{F\eta_A} FGFA \xrightarrow{(FG\gamma^{-1})_{FA}} FGFGFA \xrightarrow{(F\eta G)^{-1})_{FA}} FGFA \xrightarrow{\gamma_{FA}} FA$$

to equal the identity arrow on FA (recall, the component of a 'vertical' composite of natural transformations for FA is the composite of the components of the individual transformations).

We begin be noting that, for any $A \in \mathscr{A}$, the following square commutes by the naturality of η:

$$\begin{tikzcd}
A \arrow[r, "\eta_A"] \arrow[d, "\eta_A"'] & GFA \arrow[d, "\eta_{GFA}"] \\
GFA \arrow[r, "GF\eta_A"] & GFGFA
\end{tikzcd}$$

So we have $\eta_{GFA} \circ \eta_A = GF\eta_A \circ \eta_A$. But since η_A is an isomorphism, it is epic (right-cancellable), so we have $\eta_{GFA} = GF\eta_A$ for all A. Similarly, we have $\gamma^{-1}_{FGB} = (FG\gamma^{-1})_B$ for all $B \in \mathscr{B}$.

So now consider the following diagram:

$$\begin{tikzcd}
FA \arrow[r, "F\eta_A"] \arrow[d, "(\gamma^{-1})_{FA}"] & FGFA \arrow[d, "(\gamma^{-1})_{FGFA}=(FG\gamma^{-1})_{FA}"] \\
FGFA \arrow[r, "FGF\eta_A"] \arrow[d, "1_{FGFA}"] & FGFGFA \arrow[dl, "(F\eta G)^{-1}_{FA}"] \\
FGFA \arrow[d, "\gamma_{FA}"] & \\
FA &
\end{tikzcd}$$

The top square commutes, being a standard naturality square. (Fill in the schema of Defn. 24 by putting the natural transformation $\alpha = \gamma^{-1}\colon 1_{\mathscr{B}} \to FG$, and put f to be the function $F\eta_A\colon FA \to FB$.) And the triangle below commutes because $FGF\eta_A = F\eta_{GFA}$ from the equation above and $F\eta_{GFA} = (F\eta G)_{FA}$ (since $\eta_{GFA} = (\eta G)_{FA}$), so the arrows along two sides are simply inverses, and therefore compose to the identity.

The whole diagram therefore commutes. The arrows on longer circuit from top-left to bottom form the composite (*). The arrows on the direct route from top to bottom compose to the identity 1_{FA}. The composites are equal and hence we have established that the first triangle identity holds.

The second triangle identity holds by a similar argument.

Hence $F \dashv G$. And finally we note that if we put $\eta' = \gamma^{-1}$ and $\gamma' = \eta^{-1}$, and put $F' = G$, $G' = F$, the same line of proof shows that $F' \dashv G'$, and so $G \dashv F$. $\square$

23 Adjoints further explored

We have given a pair of definitions of adjoint functors, mirroring the two alternative definitions of Galois connections. We showed the definitions to be equivalent, and met some initial examples of adjunctions.

In this chapter, after a couple of preliminary sections, we continue to generalize some of the most basic results we found for Galois connections to adjunctions more generally.

23.1 Adjunctions reviewed

Let's gather together what we know about adjunctions so far.

Suppose $F\colon \mathscr{A} \to \mathscr{B}$ and $G\colon \mathscr{B} \to \mathscr{A}$ are functors. Then F is left-adjoint to G (equivalently, G is right-adjoint to F), in symbols $F \dashv G$, or more fully $F \dashv G\colon \mathscr{A} \to \mathscr{B}$, iff the following conditions all hold together:

(1) $\mathscr{B}(FA, B) \cong \mathscr{A}(A, GB)$ naturally in $A \in \mathscr{A}$, $B \in \mathscr{B}$ – the isomorphism in each direction is said to send an arrow f in one hom-set to its transpose $\overline{f}$ in the other.

(2) There are natural transformations $\eta\colon 1_{\mathscr{A}} \Rightarrow GF$ and $\varepsilon\colon FG \Rightarrow 1_{\mathscr{B}}$ such that $\varepsilon_{FA} \circ F\eta_A = 1_{FA}$ for all $A \in \mathscr{A}$, and $G\varepsilon_B \circ \eta_{GB} = 1_{GB}$ for all $B \in \mathscr{B}$. η is said to be the unit, ε the counit of the adjunction.

(3) The component $\eta_A\colon A \to GFA$ of the natural transformation η can be identified as the transpose of $1_{FA}\colon FA \to FA$ under the natural isomorphism between $\mathscr{B}(FA, FA)$ and $\mathscr{A}(A, GFA)$. Likewise, the component ε_B is the transpose of 1_{GB} under the natural isomorphism between $\mathscr{A}(GB, GB)$ and $\mathscr{A}(FGB, B)$.

(4) The inverse isomorphisms from $\mathscr{B}(FA, B)$ to $\mathscr{A}(A, GB)$ and back can be identifed as $G(-)\circ\eta_A\colon \mathscr{B}(FA, B) \xrightarrow{\sim} \mathscr{A}(A, GB)$ and $\varepsilon_B\circ F(-)\colon \mathscr{A}(A, GB) \xrightarrow{\sim} \mathscr{B}(FA, B)$.

(5) For any $f\colon FA \to B$ and $h\colon B \to B'$, $\overline{h \circ f} = Gh \circ \overline{f}$; and for any $g\colon A \to GB$ and $k\colon A' \to A$, $\overline{g \circ k} = \overline{g} \circ Fk$, i.e. $\overline{\overline{g} \circ Fk} = g \circ k$.

These conditions are not independent, however: (1) and (2) are equivalent, and both then imply (3) to (5).

23.2 Two more definitions!

Using (2) and (4) in our conditions on adjunctions, if follows that if $F \dashv G$, then there is a natural transformation $\eta\colon 1_{\mathscr{A}} \Rightarrow GF$ which has the following 'universal mapping property': for any $g\colon A \to G(B)$ there is a unique associated $f\colon F(A) \to B$ such that $g = G(f) \circ \eta_A$.

It is worth noting that we can also prove the converse here, so we get a biconditional:

Theorem 140. *Given functors $F\colon \mathscr{A} \to \mathscr{B}$ and $G\colon \mathscr{B} \to \mathscr{A}$, then $F \dashv G$ iff (i) there is a natural transformation $\eta\colon 1_{\mathscr{A}} \Rightarrow GF$, for which (ii) for any $g\colon A \to G(B)$ in $\mathscr{A}$ there is a unique $f\colon F(A) \to B$ in $\mathscr{B}$ such that $g = G(f) \circ \eta_A$.*

Proof for 'if'. First use clause (i) and define $\varphi_{AB}\colon \mathscr{B}(F(A), B) \to \mathscr{A}(A, G(B))$ by putting $\varphi_{AB}(f) = G(f) \circ \eta_A$.

By same proof as for Theorem 138, when we keep A fixed the various components φ_{AB} assemble into a natural transformation $\varphi_A\colon \mathscr{B}(F(A), -) \Rightarrow \mathscr{A}(A, G(-))$. And when we keep B fixed, the various components φ_{AB} assemble into a natural transformation $\varphi_B\colon \mathscr{B}(F(-), B) \Rightarrow \mathscr{A}(-, G(B))$.

Further, by the uniqueness clause (ii) the components φ_{AB} are bijections, so the natural transformations are indeed natural isomorphisms. Therefore $\mathscr{B}(F(A), B) \cong \mathscr{A}(A, G(B))$ naturally in $A \in \mathscr{A}$, $B \in \mathscr{B}$. $\square$

Our theorem has a dual companion of course:

Theorem 141. *Given functors $F\colon \mathscr{A} \to \mathscr{B}$ and $G\colon \mathscr{B} \to \mathscr{A}$, then $F \dashv G$ iff (i) there is a natural transformation $\varepsilon\colon FG \Rightarrow 1_{\mathscr{B}}$, for which (ii) for any $f\colon F(A) \to B$ there is a unique $g\colon A \to G(B)$ such that $f = \varepsilon_B \circ F(g)$.*

Evidently, we could have recruited either of these companion theorems as the basis of two further alternative definitions for $F \dashv G$ – as, for example, in (Awodey, 2006, §9.1).

23.3 Adjunctions compose

Recall Theorem 123: in a different notation, if (f, g) is a Galois connection between the posets $\mathscr{P}$ and $\mathscr{Q}$, and (f', g') is a Galois connection between the posets $\mathscr{Q}$ and $\mathscr{R}$, then $(f' \circ f, g \circ g')$ is a Galois connections between $\mathscr{P}$ and $\mathscr{R}$.

Adjunctions similarly compose:

Theorem 142. *Given* $\begin{tikzcd} \mathscr{A} \arrow[r, shift left, "F"] \arrow[r, phantom, "\perp"] & \mathscr{B} \arrow[l, shift left, "G"] \end{tikzcd}$ *and* $\begin{tikzcd} \mathscr{B} \arrow[r, shift left, "F'"] \arrow[r, phantom, "\perp"] & \mathscr{C} \arrow[l, shift left, "G'"] \end{tikzcd}$*, then* $\begin{tikzcd} \mathscr{A} \arrow[r, shift left, "F'F"] \arrow[r, phantom, "\perp"] & \mathscr{C} \arrow[l, shift left, "GG'"] \end{tikzcd}$.

Proof via homsets. Since $F' \dashv G'$, we have $\mathscr{C}(F'FA, C) \cong \mathscr{B}(FA, G'C)$, naturally in A – by Theorem 22 (3) – and also naturally in C.

Also, since $F \dashv G$, we have $\mathscr{B}(FA, G'C) \cong \mathscr{A}(A, GG'C)$, naturally in A and in C.

So by Theorem 22 (2), $\mathscr{C}(F'FA, C) \cong \mathscr{A}(A, GG'C)$ naturally in A and in C. Hence $F'F \dashv GG'$ □

That was quick and easy. But there is perhaps some additional fun to be had by working through another argument:

Proof by units and counits. Since $F \dashv G$, there are a pair of natural transformations $\eta\colon 1_{\mathscr{A}} \Rightarrow GF$ and $\varepsilon\colon FG \Rightarrow 1_{\mathscr{B}}$, satisfying the usual triangle identities.

Since $F' \dashv G'$, there are natural transformations $\eta'\colon 1_{\mathscr{B}} \Rightarrow G'F'$ and $\varepsilon'\colon F'G' \Rightarrow 1_{\mathscr{C}}$, again satisfying the triangle identities.

We now define two more natural transformations by composition,

$$\eta'' : \quad 1_{\mathscr{A}} \xRightarrow{\ \eta\ } GF \xRightarrow{G\eta' F} GG'F'F$$

$$\varepsilon'' : \quad F'FGG' \xRightarrow{F'\varepsilon G'} F'G' \xRightarrow{\ \varepsilon'\ } 1_{\mathscr{C}}$$

To show $F'F \dashv GG'$ it suffices to check that η'' and ε'' also satisfy the triangle identities.

Consider, then, the following diagram:

$$\begin{tikzcd}
F'F \arrow[r, "F'F\eta"] \arrow[dr, "1_{F'F}"'] & F'FGF \arrow[r, "F'FG\eta'F"] \arrow[d, "F'\varepsilon F"] & F'FGG'F'F \arrow[d, "F'\varepsilon G'F'F"] \\
& F'F \arrow[r, "F'\eta'F"] \arrow[dr, "1_{F'F}"'] & F'G'F'F \arrow[d, "\varepsilon'F'F"] \\
& & F'F
\end{tikzcd}$$

'Whiskering' the triangle identity $\varepsilon F \circ F\eta = 1_F$ by F' shows that the top left triangle commutes. And whiskering the identity $\varepsilon' F' \circ F'\eta' = 1_{F'}$ on the other side by F shows that the bottom triangle commutes.

Further, the square commutes. For by either the naturality of ε or the naturality of η', the following square commutes in the functor category:

$$\begin{tikzcd}
FG \arrow[r, "FG\eta'"] \arrow[d, "\varepsilon"] & FGG'F' \arrow[d, "\varepsilon G'F'"] \\
1 \arrow[r, "\eta"] & G'F'
\end{tikzcd}$$

And whiskering again gives the commuting square in the big diagram. [Exercise: check the claims about whiskering and the naturality square.]

So the whole big diagram commutes, and in particular the outer triangle commutes. But that tells us that $\varepsilon'' F'F \circ F'F\eta'' = 1_{FF'}$ – which is one of the desired triangle identities for η'' and ε''.

The other identity follows similarly. $\square$

23.4 The uniqueness of adjoints

Now recall Theorem 124. This tells us that if (f, g) and (f, g') are both Galois connections between the posets $\mathscr{P}$ and $\mathscr{Q}$, then $g = g'$. Likewise, if (f, g) and (f', g) are both Galois connections between the same posets, then $f = f'$.

The corresponding result for adjunctions more generally is this:

Theorem 143. *Adjoints are unique up to natural isomorphism. If $F \dashv G$ and $F \dashv G'$ then $G \cong G'$. If $F \dashv G$ and $F' \dashv G$ then $F \cong F'$.*

Proof. Assume we have $F \dashv G\colon \mathscr{A} \to \mathscr{B}$ and $F \dashv G'\colon \mathscr{A} \to \mathscr{B}$. Then

$$\mathscr{A}(A, GB) \cong \mathscr{B}(FA, B) \cong \mathscr{A}(A, G'B)$$

naturally in $A \in \mathscr{A}$, $B \in \mathscr{B}$. It follows, using Theorem 22, that

$$(*) \quad \mathscr{A}(A, GB) \cong \mathscr{A}(A, G'B)$$

naturally in A and B.

(*)'s naturality in A means that $\mathscr{A}(-, GB) \cong \mathscr{A}(-, G'B)$, i.e. $\mathcal{Y}GB \cong \mathcal{Y}G'B$, where $\mathcal{Y}$ is the Yoneda embedding. And then, by Theorem 46, $GB \cong G'B$.

Moreover, this holds naturally in B – intuitively, because the isomorphism is generated systematically from the isomorphism in (*) which is also natural in B – so $G \cong G'$.

To confirm this, note that $\mathcal{Y}$ sends the diagram on the left in $\mathscr{A}$ to the diagram on the right in **Set** for any $f\colon B \to B'$:

$$\begin{tikzcd} GB \arrow[r, "Gf"] \arrow[d, "\beta_B"] & GB' \arrow[d, "\beta_{B'}"] \\ G'B \arrow[r, "G'f"] & G'B' \end{tikzcd} \qquad \begin{tikzcd} \mathscr{A}(-, GB) \arrow[r, "\mathscr{A}(-,Gf)"] \arrow[d, "\alpha_B"] & \mathscr{A}(-, GB') \arrow[d, "\alpha_{B'}"] \\ \mathscr{A}(-, G'B) \arrow[r, "\mathscr{A}(-,G'f)"] & \mathscr{A}(-, G'B') \end{tikzcd}$$

where the αs are components of the natural transformation required by the naturality of $(*)$ in B, and $\beta_B = \alpha_B(1_{GB})$ by appeal to Theorem 40. But $\mathcal{Y}$ is an embedding, remember, so each diagram commutes if and only if the other does does. However, the diagram on the right commutes for all $f\colon B \to B'$ by the naturality in B; hence the diagram on the left does too (embeddings must evidently preserve commutativity relations). So the β assemble into a natural transformation between G and G'.

The proof of the second half of the theorem is dual. $\square$

We should note too an obvious companion theorem:

Theorem 144. *If $F \dashv G$ and $G \cong G'$ then $F \dashv G'$. Likewise, if $F \dashv G$ and $F \cong F'$ then $F' \dashv G$.*

Proof. By definition, given $F \dashv G \colon \mathscr{A} \to \mathscr{B}$, we have $\mathscr{B}(FA, B) \cong \mathscr{A}(A, GB)$ naturally in $A \in \mathscr{A}, B \in \mathscr{B}$.

But given $G \cong G'$, then it is almost immediate that $\mathscr{A}(A, GB) \cong \mathscr{A}(A, G'B)$, again naturally in $A \in \mathscr{A}, B \in \mathscr{B}$.

Hence by Theorem 22 again, $\mathscr{B}(FA, B) \cong \mathscr{A}(A, G'B)$, still naturally in $A \in \mathscr{A}, B \in \mathscr{B}$. Which means that $F \dashv G'$.

The other half of the theorem is dual. □

23.5 How left adjoints can be defined in terms of right adjoints

Theorem 124 states that each component of a Galois connection uniquely fixes the other. So we would hope to be able to explicitly define one such component in terms of the other, and Theorem 125 in fact tells us how to do this. For example, assuming there is a connection (f, g) between the posets $(P, \preccurlyeq)$ and $(Q, \sqsubseteq)$, we can define the left adjoint in terms of the right by setting $f(p)$ to be the minimum of $\{q \in Q \mid p \preccurlyeq g(q)\}$ for every $p \in P$.

We have now shown, more generally, that each component of an adjunction uniquely fixes the other, at least up to isomorphism. We would expect that we can, similarly, characterize one functor in an adjunction in terms of its partner. So let's consider, in particular, how a left adjoint might be defined in terms of its right partner. (There will of course also be a dual story to be told about how right adjoints can be defined in terms of left ones. We can cheerfully leave spelling out the dual constructions and arguments as an exercise.)

Functions in Galois connections between posets correspond to adjoint functors between poset categories (see §22.1). And a minimum for the poset $\{q \in Q \mid p \preccurlyeq g(q)\}$ corresponds to an initial object for the poset-as-category (see §13.1). So this suggests that we might be able to characterize a left adjoint as the initial object of some suitable category.

And this is indeed more or less the case. Suppose $F \colon \mathscr{A} \to \mathscr{B}$ and $G \colon \mathscr{B} \to \mathscr{A}$ are functors such that $F \dashv G$. Now consider the comma category $(A \downarrow G)$, for $A \in \mathscr{A}$ – we met this construction at the end of §5.5. To recap,

(a) the objects of $(A \downarrow G)$ are pairs $\langle B, f \rangle$ where B is a $\mathscr{B}$-object and $f : A \to GB$ is an arrow in $\mathscr{A}$,

(b) an arrow in $(A \downarrow G)$ from $\langle B, f \rangle$ to $\langle B', f' \rangle$ is a $\mathscr{B}$-arrow $j \colon B \to B'$ making the following commute:

$$\begin{tikzcd}[row sep=small] & GB \arrow[dd, "Gj"] \\ A \arrow[ur, "f"] \arrow[dr, "f'"'] & \\ & GB' \end{tikzcd}$$

The definitions for the identity arrows and for composition of arrows in $(A \downarrow G)$ are the obvious ones.

Theorem 145. *Given an adjunction* $\mathscr{A} \underset{G}{\overset{F}{\underset{\longleftarrow}{\overset{\longrightarrow}{\perp}}}} \mathscr{B}$, *the pair* $\langle FA, \eta_A \rangle$ *is initial in* $(A \downarrow G)$ *for any* $A \in \mathscr{A}$.

Proof. Let $\langle B, f \rangle$ be any object of $(A \downarrow G)$. We need to show that there is a unique arrow in $(A \downarrow G)$ from $\langle FA, \eta_A \colon A \to GFA \rangle$ to $\langle B, f \rangle$. That is to say, there must be (i) an arrow $j \colon FA \to B$ such that $f = Gj \circ \eta_A$, i.e.

$$\begin{tikzcd}[row sep=small] & GFA \arrow[dd, "Gj"] \\ A \arrow[ur, "\eta_A"] \arrow[dr, "f"'] & \\ & GB \end{tikzcd}$$

commutes, and (ii) this arrow must be unique. But we've already proved *that* – see one half of Theorem 140. ☐

We have a converse result too:

Theorem 146. *Given functors* $\mathscr{A} \underset{G}{\overset{F}{\rightleftarrows}} \mathscr{B}$ *, then if (C)* $\eta \colon 1_{\mathscr{A}} \to GF$ *is a natural transformation and the pair* $\langle FA, \eta_A \rangle$ *is initial in* $(A \downarrow G)$ *for every* $A \in \mathscr{A}$*, then* $F \dashv G$*.*

So that tells us how to characterize a left adjoint for G when it exists, since left adjoints are unique up to isomorphism, i.e. as a functor F satisfying condition (C).

Proof. Suppose η is natural transformation, and that $\langle FA, \eta_A \rangle$ is initial in $(A \downarrow G)$ for every $A \in \mathscr{A}$. Then for every $f \colon A \to GB$ there is a unique $j \colon FA \to B$ such that $f = Gj \circ \eta_A$. Apply the other half of Theorem 140. ☐

But there was no fun in that instant proof. So, as an instructive and amusing exercise in diagram chasing here is

Another proof, by constructing a counit for η from first principles. We need to find a natural transformation $\varepsilon \colon FG \Rightarrow 1_{\mathscr{B}}$ such that η and ε satisfy the triangle equalities, i.e. such that $\varepsilon_{FA} \circ F\eta_A = 1_{FA}$ for all $A \in \mathscr{A}$, and $G\varepsilon_B \circ \eta_{GB} = 1_{GB}$ for all $B \in \mathscr{B}$.

Taken any $B \in \mathscr{B}$. By hypothesis $\langle FGB, \eta_{GB}\rangle$ is initial in $(GB \downarrow G)$, so there is a unique arrow to the object $\langle B, 1_{GB}\rangle$. Call this unique arrow (hopefully!) ε_B. Then just by its definition, for any B we have (*):

$$\begin{tikzcd} & GFGB \arrow[d, "G\varepsilon_B"] \\ GB \arrow[ur, "\eta_{GB}"] \arrow[r, "1_{GB}"'] & GB \end{tikzcd}$$

which gives us one lot of the triangle identities for free. So it remains to show that (i) we also have the other triangle identities, and (ii) the components ε_B do indeed assemble into a natural transformation. Try before reading on!

For (i), we need to show that the following diagram commutes:

$$\begin{tikzcd} & FGFA \arrow[d, "\varepsilon_{FA}"] \\ FA \arrow[ur, "F\eta_A"] \arrow[r, "1_{FA}"'] & FA \end{tikzcd}$$

Since $\eta\colon 1_{\mathscr{A}} \Rightarrow GF$ is natural, for every $f\colon A \to A'$ with $A, A' \in \mathscr{A}$, there is a commuting naturality square. Take in particular the case where $f = \eta_A(!)$. Then paste on the commuting triangle of the type (*), with $B = FA$, to get the commuting rhombus:

$$\begin{tikzcd} A \arrow[r, "\eta_A"] \arrow[d, "\eta_A"] & GFA \arrow[d, "\eta_{GFA}"] \arrow[dr, "1_{GFA}"] & \\ GFA \arrow[r, "GF\eta_A"] & GFGFA \arrow[r, "G\varepsilon_{FA}"] & GFA \end{tikzcd}$$

Composing arrows, using the functoriality of G, and re-arranging we get the commuting triangle on the left:

$$\begin{tikzcd} & GFA \arrow[d, "G(\varepsilon_{FA} \circ F\eta_A)"] \\ A \arrow[ur, "\eta_A"] \arrow[r, "\eta_A"'] & GFA \end{tikzcd} \qquad \begin{tikzcd} & GFA \arrow[d, "Gj"] \\ A \arrow[ur, "\eta_A"] \arrow[r, "\eta_A"'] & GFA \end{tikzcd}$$

Now, $\langle FA, \eta_A\rangle$ is initial in $(A \downarrow G)$ so there must be a *unique* arrow j from the initial object to itself such the triangle on the right commutes. But evidently $j = 1_{FA}$ makes the triangle commute. But so, as we've just seen, does $j = \varepsilon_{FA} \circ F\eta_A$. Hence $\varepsilon_{FA} \circ F\eta_A = 1_{FA}$. Which establishes (i).

To establish (ii) – the naturality of $\varepsilon\colon FG \Rightarrow 1_{\mathscr{B}}$, when assembled from the components ε_B – we need to show that for any $g\colon B \to B'$, the following commutes (**):

$$\begin{tikzcd}
FGB \arrow[r, "FGg"] \arrow[d, "\varepsilon_B"] & FGB' \arrow[d, "\varepsilon_{B'}"] \\
B \arrow[r, "g"] & B'
\end{tikzcd}$$

We again start by taking a naturality square for η, this time for $f = Gg\colon GB \to GB'$, and then paste on a commuting triangle of type $(*)$, to get the commuting rhombus

$$\begin{tikzcd}
GB \arrow[r, "Gg"] \arrow[d, "\eta_{GB}"] & GB' \arrow[d, "\eta_{GB'}"] \arrow[dr, "1_{GB'}"] & \\
GFGB \arrow[r, "GFGg"] & GFGB' \arrow[r, "G\varepsilon_{B'}"] & GB'
\end{tikzcd}$$

Again composing arrows, using the functoriality of G, and re-arranging we get the commuting triangle on the left:

$$\begin{tikzcd}
 & GFGB \arrow[dd, "G(\varepsilon_{B'} \circ FGg)"] \\
GB \arrow[ur, "\eta_{GB}"] \arrow[dr, "Gg"'] & \\
 & GB'
\end{tikzcd}
\qquad
\begin{tikzcd}
 & GFGB \arrow[d, "G\varepsilon_B"] \arrow[dd, bend left, "G(g \circ \varepsilon_B)"] \\
GB \arrow[ur, "\eta_{GB}"] \arrow[r, "1_{GB}"] \arrow[dr, "Gg"'] & GB \arrow[d, "Gg"] \\
 & GB'
\end{tikzcd}$$

On the right, we've pasted together (*) with a trivially commuting triangle, and then composed the downwards arrows to give the big triangle. However, by assumption, $\langle FGB, \eta_{GB}\rangle$ is initial in the comma category $(GB \downarrow G)$, so there is a *unique* arrow j to $\langle B', g\rangle$ such that $g = Gj \circ \eta_{GB}$. Whence $\varepsilon_{B'} \circ FGg = j = g \circ \varepsilon_B$, proving (**) commutes and establishing (ii). □

Here's a nice corollary:

Theorem 147. *Suppose $G\colon \mathscr{B} \to \mathscr{A}$ is a functor. If the derived comma category $(A \downarrow G)$ has an initial object for every $A \in \mathscr{A}$, then G has a left adjoint.*

Proof. Choose an initial object for each $(A \downarrow G)$: it is a pair that we will write (hopefully!) as $\langle FA, \eta_A\rangle$, with $FA \in \mathscr{B}$, and $\eta_A\colon A \to GFA$.

So we now define a functor $F\colon \mathscr{A} \to \mathscr{B}$ which sends an object $A \in \mathscr{A}$ to this $FA \in \mathscr{B}$. How should F act on an arrow $f\colon A \to A'$? It must yield an arrow from FA to FA'. But since $\langle FA, \eta_A\rangle$ is initial, we know that there is exactly one arrow in $(A \downarrow G)$ from $\langle FA, \eta_A\rangle$ to $\langle FA', \eta_{A'} \circ f\rangle$. That is to say, there is a unique $g\colon FA \to FA'$ such that $\eta_{A'} \circ f = Gg \circ \eta_A$. Put $Ff = g$, and it is easy enough to check that F is functorial.

So now consider this diagram:

$$\begin{array}{ccc} A & \xrightarrow{f} & A' \\ \Big\downarrow{\scriptstyle \eta_A} & & \Big\downarrow{\scriptstyle \eta_{A'}} \\ GFA & \xrightarrow{GFf} & GFA' \end{array}$$

We've defined Ff to make this commute. But this is a naturality square showing that the components η_A assemble into a natural transformation $\eta\colon 1_{\mathscr{A}} \to GF$.

So, in sum, $F\colon \mathscr{A} \to \mathscr{B}$ as defined is such that (C), $\eta\colon 1_{\mathscr{A}} \to GF$ is a natural transformation and the pair $\langle FA, \eta_A \rangle$ is initial in $(A \downarrow G)$ for every $A \in \mathscr{A}$. Hence, by the previous theorem, $F \dashv G$. $\Box$

23.6 Another way of getting new adjunctions from old

We've already met one way of getting new adjunctions from old, i.e. simple composition. Finally in this chapter, we now introduce another.

Definition 104. Given a functor $F\colon \mathscr{C} \to \mathscr{D}$ and small category $\mathbf{J}$, then the functor $[\mathbf{J}, F]\colon [\mathbf{J}, \mathscr{C}] \to [\mathbf{J}, \mathscr{D}]$ sends a functor $K\colon \mathbf{J} \to \mathscr{C}$ to $F \circ K\colon \mathbf{J} \to \mathscr{D}$.

Strictly speaking that's an incomplete definition. We need to specify not just how $[\mathbf{J}, F]$ acts on objects in $[\mathbf{J}, \mathscr{C}]$ (i.e. acts on functors), but how it acts on arrows (i.e. on natural transformations). But the needed completion, as often in defining functors, writes itself. For what is the obvious way for $[\mathbf{J}, F]$ to act on a natural transformation from K to K' with components $\alpha_J\colon KJ \to K'J$ (for $J \in \mathbf{J}$ and functors $K, K'\colon \mathbf{J} \to \mathscr{C}$)? By sending it, of course, to the natural transformation from $F \circ K$ to $F \circ K'$ with components $F\alpha_J\colon FKJ \to FK'J$. Full functoriality is then immediate.

We can now state our result about how a given adjunction between functors F and G generates a new adjunction between new-style functors $[\mathbf{J}, F]$ and $[\mathbf{J}, G]$:

Theorem 148. *If $F \dashv G\colon \mathscr{C} \to \mathscr{D}$ then $[\mathbf{J}, F] \dashv [\mathbf{J}, G]\colon [\mathbf{J}, \mathscr{C}] \to [\mathbf{J}, \mathscr{D}]$.*

Proof. Take functors $K\colon \mathbf{J} \to \mathscr{C}$, $L\colon \mathbf{J} \to \mathscr{D}$. Then take any natural transformation $\beta\colon FK \Rightarrow L$ living as an arrow in $[\mathbf{J}, \mathscr{D}]$. This has components $\beta_J\colon FKJ \to LJ$ living in $\mathscr{D}(FKJ, LJ)$. By the adjunction $F \dashv G$ these components are in a natural bijection with arrows $\alpha_J\colon KJ \to GLJ$ living in $\mathscr{C}(KJ, GLJ)$, and these assemble into a natural transformation $\alpha\colon K \Rightarrow GL$ which lives in $[\mathbf{J}, \mathscr{C}]$ (the adjunction is easily checked to associate naturality squares with naturality squares). In this way we set up a natural one-to-one correspondence between natural transformations like α and β.

So we have established that there is, naturally, a bijection

$$[\mathbf{J}, \mathscr{D}]([\mathbf{J}, F]K, L) \cong [\mathbf{J}, \mathscr{C}](K, [\mathbf{J}, G]L),$$

which proves $[\mathbf{J}, F] \dashv [\mathbf{J}, G]$. $\Box$

24 Adjoint functors and limits

We now turn to some key results which tell us how adjoint functors interact with limits. A key bit of news is that right adjoints preserve limits: and dually, exactly as you would now expect, left adjoints preserve co-limits.

24.1 Limit functors as adjoints

(a) Suppose the category $\mathscr{C}$ has all limits of shape $\mathbf{J}$. Three observations:

(1) By Theorem 86, the cones over $D\colon \mathbf{J} \to \mathscr{C}$ with vertex C correspond one-to-one with $\mathscr{C}$-arrows from C to $\underset{\leftarrow \mathbf{J}}{Lim}\, D$.

(2) But by Theorem 91, the set of cones over $D\colon \mathbf{J} \to \mathscr{C}$ with vertex C is the hom-set $[\mathbf{J}, \mathscr{C}](\Delta(C), D)$. Here $\Delta\colon \mathscr{C} \to [\mathbf{J}, \mathscr{C}]$ is the functor introduced just after that theorem, which sends an object $C \in \mathscr{C}$ to the constant functor $\Delta_C\colon \mathbf{J} \to \mathscr{C}$. (For convenience, understand the cones here austerely).

(3) The set of $\mathscr{C}$-arrows from C to the limit vertex $Lim\, D$ is $\mathscr{C}(C, Lim(D))$, where $Lim\colon [\mathbf{J}, \mathscr{C}] \to \mathscr{C}$ is the functor introduced in §17.2 (c), a functor that exists if $\mathscr{C}$ has all limits of shape $\mathbf{J}$ and that sends a diagram D of shape $\mathbf{J}$ in $\mathscr{C}$ to some limit object in $\mathscr{C}$.

So, in summary, still assuming that $\mathscr{C}$ has all limits of shape $\mathbf{J}$, the situation is this. We have a pair of functors $\mathscr{C} \underset{Lim}{\overset{\Delta}{\rightleftarrows}} [\mathbf{J}, \mathscr{C}]$ such that

$$[\mathbf{J}, \mathscr{C}](\Delta(C), D) \cong \mathscr{C}(C, Lim(D)).$$

Moreover, the isomorphism that is given in our proof of Theorem 86 arises in a natural way, without making any arbitrary choices.[1] So, we can take it that the isomorphism is natural in $C \in \mathscr{C}$ and $D \in [\mathbf{J}, \mathscr{C}]$. Hence Δ has a right adjoint, and one such right adjoint is Lim.

We now argue in the opposite direction starting from the assumption that the diagram Δ has a right adjoint, call it L.

[1] Careful: there were arbitrary choices made in determining what Lim does. But once Lim is fixed, the isomorphism arises naturally.

Suppose that D is a diagram $D\colon \mathbf{J} \to \mathscr{C}$. Applying Theorem 141 about a universal mapping property of adjunctions, for any arrow $c\colon \Delta(C) \to D$ in $[\mathbf{J}, \mathscr{C}]$ – in other words for any cone over D with vertex C – there is a unique arrow $u\colon C \to L(D)$ in $\mathscr{C}$, such that $c = \varepsilon_D \circ \Delta(u)$, where ε is the co-unit of the adjunction.

By the definition of Δ, $\Delta(u)$ is the natural transformation from Δ_C to $\Delta_{L(D)}$ with every component equal to u.

And by §23.1 (3), ε_D is the transpose of $1_L(D)$, i.e. is some arrow $\pi\colon \Delta L(D) \to D$ in $[\mathbf{J}, \mathscr{C}]$, i.e. is some particular cone π over D with vertex L_D.

Taken component-wise, the equation $c = \varepsilon_D \circ \Delta(u)$ tells us that for each $J \in \mathbf{J}$, $c_J = \pi_J \circ u$. In other words any cone c factors through our cone π via the unique u. Hence the cone π with vertex $L(D)$ and projection arrows π_J is a limit cone for D. However, D was any diagram $D\colon \mathbf{J} \to \mathscr{C}$. Therefore $\mathscr{C}$ has all limits of shape $\mathbf{J}$.

Summing up, we get the following nice theorem:

Theorem 149. *If category $\mathscr{C}$ has all limits of shape $\mathbf{J}$, then Δ has a right adjoint, and indeed $\Delta \dashv Lim$. Conversely, if Δ has any right adjoint, then $\mathscr{C}$ has all limits of shape $\mathbf{J}$.*

(b) Keeping $\mathbf{J}$ fixed, we can make Δ's dependence on $\mathscr{C}$ explicit by writing $\Delta_{\mathscr{C}}\colon \mathscr{C} \to [\mathbf{J}, \mathscr{C}]$. Similarly we can explicitly write $Lim_{\mathscr{C}}\colon [\mathbf{J}, \mathscr{C}] \to \mathscr{C}$. Then we have the following easy corollary of the last theorem:

Theorem 150. *Suppose the categories $\mathscr{B}$ and $\mathscr{C}$ have all limits of shape $\mathbf{J}$. Then if $G\colon \mathscr{C} \to \mathscr{B}$ is a right adjoint (i.e. has a left adjoint), $G \circ Lim_{\mathscr{C}} \cong Lim_{\mathscr{B}} \circ [\mathbf{J}, G]$.*

Proof. Let $F\colon \mathscr{B} \to \mathscr{C}$ be left adjoint to G, and consider this pair of diagrams:

$$\begin{tikzcd} \mathscr{B} \arrow[r, "F"] \arrow[d, "\Delta_{\mathscr{B}}"'] & \mathscr{C} \arrow[d, "\Delta_{\mathscr{C}}"] \\ {[\mathbf{J}, \mathscr{B}]} \arrow[r, "{[\mathbf{J}, F]}"] & {[\mathbf{J}, \mathscr{C}]} \end{tikzcd} \qquad \begin{tikzcd} \mathscr{B} & \mathscr{C} \arrow[l, "G"] \\ {[\mathbf{J}, \mathscr{B}]} \arrow[u, "Lim_{\mathscr{B}}"] & {[\mathbf{J}, \mathscr{C}]} \arrow[l, "{[\mathbf{J}, G]}"] \arrow[u, "Lim_{\mathscr{C}}"'] \end{tikzcd}$$

Claim: the left-hand diagram commutes. (i) On the south-west path, an object $B \in \mathscr{B}$ is sent by $\Delta_{\mathscr{B}}$ to the functor $\Delta_B\colon \mathbf{J} \to \mathscr{B}$ which sends every object to B and every arrow to 1_B; and this is sent in turn by $[\mathbf{J}, F]$ to the functor which sends every object to FB and every arrow to 1_{FB}, i.e. the functor Δ_{FB}. (ii) On the north-east path, an object $B \in \mathscr{B}$ is sent by F to FB, and this is sent by $\Delta_{\mathscr{C}}$ to the functor Δ_{FB} again.

Now, given the assumption that $\mathscr{B}$ and $\mathscr{C}$ have all limits of shape $\mathbf{J}$, $\Delta_{\mathscr{B}}$ and $\Delta_{\mathscr{C}}$ have right adjoints $Lim_{\mathscr{B}}$ and $Lim_{\mathscr{C}}$. And since $F \dashv G$, $[\mathbf{J}, F] \dashv [\mathbf{J}, G]$, by Theorem 148.

So our right-hand diagram records the adjoints of the functors in the left-hand diagram. We now know that the composite left-adjoint functors $\Delta_{\mathscr{C}} \circ F$ and

$[\mathbf{J}, F] \circ \Delta_{\mathscr{B}}$ are the same. By Theorem 142 about the composition of adjunctions, their right-adjoints are $G \circ Lim_{\mathscr{C}}$ and $Lim_{\mathscr{B}} \circ [\mathbf{J}, G]$. And these composites, being right adjoint to the same functor, must be naturally isomorphic by Theorem 143. □

24.2 Right adjoints preserve limits

We can usefully begin by restating part of a key definition and reminding ourselves of a basic theorem:

Definition 84 A functor $G\colon \mathscr{C} \to \mathscr{B}$ *preserves limits of shape* $\mathbf{J}$ iff, for any diagram $D\colon \mathbf{J} \to \mathscr{C}$, if $[L, \pi_J]$ is a limit cone over D, then $[GL, G\pi_J]$ is a limit cone over $G \circ D\colon \mathbf{J} \to \mathscr{B}$.

Theorem 112 *The covariant hom-functor $\mathscr{C}(A, –)\colon \mathscr{C} \to$ **Set**, for any A in the category $\mathscr{C}$, preserves all limits that exist in $\mathscr{C}$.*

Now, this theorem is easily seen to imply the following:

Theorem 151. *Any set-valued functor $G\colon \mathscr{C} \to$ **Set** which is a right adjoint (i.e. has a left adjoint) preserves all limits that exist in $\mathscr{C}$.*

Proof. Suppose we have a functor F such that $F \dashv G$. Then

$$GA \cong \mathbf{Set}(1, GA) \cong \mathscr{C}(F1, A)$$

with both isomorphisms natural in A (the first relies on the familiar association in **Set** between elements of a set and arrows from a terminal object into that set). Hence G is naturally isomorphic to the hom-functor $\mathscr{C}(F1, –)$. But the latter preserves limits, by Theorem 112. Hence so does G, by Theorem 103. □

We now show that there is in fact nothing special here about set-valued functors. In fact, Theorem 150 already tells us that in the case where the categories $\mathscr{B}$ and $\mathscr{C}$ have all limits of shape $\mathbf{J}$, then if $G\colon \mathscr{C} \to \mathscr{B}$ is a right adjoint, G commutes with limits and hence by the result announced at the end of §18.2, G preserves limits. But we can do better, drop the assumption that the relevant categories are complete, and prove quite generally:

Theorem 152. *If the functor $G\colon \mathscr{C} \to \mathscr{B}$ is a right adjoint (i.e. has a left adjoint), it preserves all limits that exist in $\mathscr{C}$.*

Proof from basic principles about limits and adjoints. Suppose that G has the left adjoint $F\colon \mathscr{B} \to \mathscr{C}$; and suppose also that the diagram $D\colon \mathbf{J} \to \mathscr{C}$ has a limit cone $[L, \pi_J]$ in $\mathscr{C}$.

Then $[GL, G\pi_J]$ is certainly a cone over $G \circ D$ in $\mathscr{B}$. We need to show, however, that it is a *limit* cone. That is to say, we need to show that, if we take any cone $[B, b_J]$ over $G \circ D$, there is a unique $u\colon B \to GL$ such that (i) for all $J \in \mathbf{J}$, $b_J = G\pi_J \circ u$.

Well, take such a cone $[B, b_J]$ over $G \circ D$. Then, going back in the other direction, $[FB, \overline{b_J}]$ is a cone over D in $\mathscr{C}$, where $\overline{b_J}\colon FB \to D_J$ is the transpose of $b_J\colon B \to GD_J$ under the adjunction.

Why is $[FB, \overline{b_J}]$ a cone? Suppose we have an arrow $d\colon D_K \to D_K$. Then by assumption, since $[B, b_J]$ is a cone over $G \circ D$, $b_K = Gd \circ b_J$. Hence $\overline{b_K} = \overline{Gd \circ b_J} = d \circ \overline{b_J}$, with the second equation by Theorem 137 (1). Which indeed makes $[FB, \overline{b_J}]$ a cone too.

And now we add that $[FB, \overline{b_J}]$ must factor through $[L, \pi_J]$ via a unique $v\colon FB \to L$ such that (ii) for all $J \in \mathbf{J}$, $\overline{b_J} = \pi_J \circ v$.

So the state of play is: we have found a unique $v\colon FB \to L$; we want to find a suitable $u\colon B \to GL$. The hopeful thought is that one will turn out to be the transpose of the other under the adjunction.

The adjunction means that $\mathscr{C}(FB, C) \cong \mathscr{B}(B, GC)$ naturally in C. Which in turn means that the following square commutes, for any $\pi_J\colon L \to D_J$:

$$\begin{tikzcd}
\mathscr{C}(FB, L) \arrow[r, "\pi_J \circ -"] \arrow[d] & \mathscr{C}(FB, D_J) \arrow[d] \\
\mathscr{B}(B, GL) \arrow[r, "G\pi_J \circ -"] & \mathscr{B}(B, GD_J)
\end{tikzcd}$$

where the vertical arrows are components of the natural transformation which sends an arrow to its transform. Chase the arrow $v\colon FB \to L$ round the diagram in both directions and we get $G\pi_J \circ \overline{v} = \overline{\pi_J \circ v}$. Therefore, using (ii), if we put $u = \overline{v}$, we indeed get as required that (i) for all $J \in \mathbf{J}$, $b_J = G\pi_J \circ u$.

It just remains to confirm u's uniqueness. Suppose that $[B, b_J]$ factors through $[GL, G\pi_J]$ by some $u' = \overline{w}$. Then for all $J \in \mathbf{J}$, $b_J = G\pi_J \circ \overline{w}$. We show as before that $\overline{b_J} = \pi_J \circ w$, whence $[FB, \overline{b_J}]$ factors through $[L, \pi_J]$ via w. By the uniqueness of factorization, $w = v$ again. □

A more compressed proof. Again, suppose that G has the left adjoint $F\colon \mathscr{B} \to \mathscr{C}$; and suppose also that the diagram $D\colon \mathbf{J} \to \mathscr{C}$ has a limit cone $[L, \pi_J]$ in $\mathscr{C}$. Then, using the notation '$\mathscr{C}(X, D)$' as shorthand for the functor $\mathscr{C}(X, -) \circ D$, we have

$$\begin{aligned}
\mathscr{B}(B, GL) &\cong \mathscr{C}(FB, L) \\
&\cong \mathit{Lim}\,\mathscr{C}(FB, D) \\
&\cong \mathit{Lim}\,\mathscr{B}(B, GD) \\
&\cong \mathit{Cone}(B, GD).
\end{aligned}$$

all naturally in B. So the functor $\mathit{Cone}(-, GD)$, being naturally isomorphic to $\mathscr{B}(-, GL)$ is representable, and is represented by GL, and therefore has a universal element of the form $\langle GL, g\rangle$. But such a universal element is a limit cone with vertex GL. Hence G preserves the limit $[L, \pi_J]$. □

But compression doesn't always make for illumination, and our second proof (see Leinster 2014, p. 158; compare Awodey 2006, pp. 225–6) needs some commentary.

The first line of course comes from the adjunction, and the second from the fact that the hom-functor $\mathscr{C}(FB,-)$ preserves limits, by Theorem 112. The move from the third to the fourth line is by Theorem 114. And the arguments at the end about representability, universal elements and limits appeal to Theorems 59 and 88.

So that leaves the move from the second to the third line, which obviously invokes the adjunction between F and G again. We know that $\mathscr{C}(FB,X) \cong \mathscr{B}(B,GX)$ naturally in X, i.e. $\mathscr{C}(FB,-)$ is naturally isomorphic to $\mathscr{B}(B,G-)$, hence by whiskering, $\mathscr{C}(FB,-)\circ D$ is naturally isomorphic to $\mathscr{B}(B,G-)\circ D$. Now apply Theorem 92 and we can conclude that $\mathit{Lim}\,\mathscr{C}(FB,D) \cong \mathit{Lim}\,\mathscr{B}(B,GD)$.

Which all goes to combine a bunch of earlier results into a neat package: but my own feeling is that the first direct proof from the underlying principles reveals better what is really going on here.

24.3 Some examples

Right adjoints preserve limits. Dually, of course, left adjoints preserve colimits (we surely needn't pause at this stage in the game to state the duals of the theorems in the last couple of sections!). So we now mention just a few elementary examples of (co)limit preservation – and also some examples where we can argue from non-preservation to the non-existence of adjoints.

(1) Back in §18.1, Ex. (3) we noted that the forgetful functor $U\colon \mathbf{Mon} \to \mathbf{Set}$ preserves limits. But we now have another proof: U has a left adjoint (by §22.2, Ex. (3)) i.e. it *is* a right adjoint, so indeed must preserve limits.

There are other examples of this kind, involving a forgetful functor $U\colon \mathbf{Alg} \to \mathbf{Set}$, where $\mathbf{Alg}$ is a category of sets equipped with some algebraic structure for U to ignore. Such a forgetful U standardly has a left adjoint, so must preserve whatever limits exist in the relevant $\mathbf{Alg}$.

Further, a left-adjoint to U must preserve existing colimits in $\mathbf{Set}$. But $\mathbf{Set}$ has *all* colimits; so that this indeed requires the left-adjoints in such cases to be rather lavish constructions (as we saw them to be).

(2) Consider exponentials again.

We noted that if $\mathscr{C}$ is a category with exponentiation, and hence with products, exponentiation is right adjoint to taking products: $(-\times B) \dashv (-)^B$.

Since the functor $(-)^B$ is a right adjoint, it preserves such limits as exist in $\mathscr{C}$. So take in particular the functor $A\colon 2 \to \mathscr{C}$ (where as before 2 is the discrete two object category with objects $0, 1$). Then $A_0 \times A_1$ is the vertex of a limit over A. Hence $(A_0 \times A_1)^B$ is the vertex of a limit over $(-)^B \circ A$. But the canonical limit over that composite functor is $A_0^B \times A_1^B$. Hence $(A_0 \times A_1)^B \cong A_0^B \times A_1^B$

Since the functor $- \times B$ is a left adjoint, it preserves such colimits as exist in $\mathscr{C}$. Assume $\mathscr{C}$ has coproducts. Then, by a similar argument, $(A_0 +$

$A_1) \times B \cong (A_0 \times B) + (A_1 \times B)$.

(3) Take the discussion in §20.3, Ex. (6) where we looked at the Galois connection between two functions between posets of equivalence classes of wffs, with the left adjoint a trivial 'add a dummy variable' map, and the right adjoint provided by applying a universal quantifier. This carries over to an adjunction of functors between certain poset categories. Since quantification is a right adjoint, it preserves limits, and in particular preserves products, which are conjunctions in this category. Which reflects the familiar logical truth that $\forall x(Px \land Qx) \equiv (\forall x Px \land \forall x Qx)$.

(4) Claim: the forgetful functor $F\colon \mathbf{Grp} \to \mathbf{Set}$ has no right adjoint. Proof: the trivial one-object group is initial in **Grp**; but a singleton is not initial in **Set**; so there is a colimit which F doesn't preserve and it therefore cannot be a left adjoint.

(5) Theorem 14 tells us that the forgetful functor $F\colon \mathbf{Mon} \to \mathbf{Set}$ fails to preserve all epimorphisms. By Theorem 106 this implies that F doesn't preserve all pushouts, and hence doesn't preserve all colimits. Hence this forgetful functor too can't be a left adjoint. Compare the arm-waving argument to the same conclusion in §22.2. Ex. (5).

24.4 The Adjoint Functor Theorems

Right adjoints preserve limits. What about the converse? If a functor preserves limits must it be a right adjoint? Well, given some results already to hand, we can easily prove the following:

Theorem 153. *If the category $\mathscr{B}$ has all limits, and the functor $G\colon \mathscr{B} \to \mathscr{A}$ preserves them, then G is a right adjoint.*

Proof. If $\mathscr{B}$ has all limits and G preserves them, then for any $A \in \mathscr{A}$, $(A \downarrow G)$ has all limits (by Theorem 107, and the remark immediately after its proof).

So any $(A \downarrow G)$ in particular has a limit for the big diagram-as-part-of-a-category consisting of the whole of $(A \downarrow G)$ – or in terms of diagrams-as-functors, it has a limit for the identity functor $1_{(A\downarrow G)}$. Hence by Theorem 87, each $(A \downarrow G)$ has an initial object. Hence by Theorem 147, there is a functor $F\colon \mathscr{A} \to \mathscr{B}$ such that $F \dashv G$. $\square$

And now we see the proof, we see that the condition that $\mathscr{B}$ has *all* limits overshoots: the result will go through so long as $\mathscr{B}$ has sufficiently large limits, enough to guarantee that all the functors $1_{(A\downarrow G)}$ have a limit.

This theorem looks neat but is in fact not very useful. Having all sufficiently large limits is a hard condition to fulfil. More precisely, we have

Theorem 154. *If a category $\mathscr{C}$ has limits for diagrams over all categories of size up to the size of the collection of $\mathscr{C}$'s arrows, then $\mathscr{C}$ has at most one arrow between any two objects.*

For example, the condition of having small limits is not satisfied by typical small categories – in the terminology of §2.2, Ex. (10), a complete small category has to be a *pre-order* category.

Proof. Let $\mathbf{J}$ be a discrete category of the same cardinality as the set of arrows of $\mathscr{C}$. Let $D\colon \mathbf{J} \to \mathscr{C}$ be the diagram which sends every object in $\mathbf{J}$ to B. By hypothesis, D has a limit, namely the product $\prod\limits_{J \in \mathbf{J}} D(J)$ (so this is the product of B with itself, $\mathbf{J}$-many times).

Suppose there are objects $A, B \in \mathscr{C}$ with arrows $f_1, f_2\colon A \to B$ where $f \neq g$. Simple cardinality considerations show that this further supposition leads to contradiction. Which proves the theorem.

We start by asking: how many different arrows $A \to \prod\limits_{J \in \mathbf{J}} D(J)$ are there? Theorem 71 showed that if $\mathbf{J}$ is the discrete two object category, then there are four such arrows. Generalizing the proof in the obvious way shows that if $|\mathbf{J}|$ is the cardinality of the objects of $\mathbf{J}$, there are $2^{|\mathbf{J}|}$ different arrows from $A \to \prod\limits_{J \in \mathbf{J}} D(J)$.

Hence our suppositions imply that there is a subset of the arrows in $\mathscr{C}$ whose cardinality is strictly greater than the cardinality of the set of arrows in $\mathscr{C}$. Contradiction. □

So, in sum, Theorem 153 is of very limited application. If we want a more widely useful result of the form 'Given such-and such conditions on the functor $G\colon \mathscr{B} \to \mathscr{A}$ and the categories it relates, then G is a right adjoint', we'll need to consider a new bunch of conditions.

Here are two such theorems of rather wider application (the labels are standard):

Theorem 155 (The General Adjoint Functor Theorem)**.** *If category $\mathscr{B}$ is a locally small category with all small limits, and the functor $G\colon \mathscr{B} \to \mathscr{A}$ is such that for each $A \in \mathscr{A}$, the category $(A \downarrow G)$ has a weakly initial set, then G is a right adjoint iff it preserves all small limits.*

Theorem 155* (The General Adjoint Functor Theorem, alternative version)**.** *If category $\mathscr{B}$ is a locally small category with all small limits, and $G\colon \mathscr{B} \to \mathscr{A}$ is a functor, then G is a right adjoint iff it preserves all small limits and satisfies the solution set condition.*

Theorem 156 (The Special Adjoint Functor Theorem)**.** *If the categories $\mathscr{A}$ and $\mathscr{B}$ are locally small, and $\mathscr{B}$ has all small limits, is well powered, and has* *a coseparating set of objects, then G is a right adjoint iff it preserves all small limits.*

But to investigate these theorems properly would require not just explaining the concepts 'weekly initial set', 'solution set condition', 'well powered' and 'coseparating' and then doing the proofs, but also explaining what might motivate the conditions our new concepts are used to state, and also explaining why the resulting theorems, with just those conditions in play, might be of interest and use. That's a non-trivial expositional task, and one I am going to shirk in this current version of these Notes. If you want to follow up the technical details, which aren't particularly difficult, I can refer you to for example Leinster (2014, pp. 159–164, 171–173) and Awodey (2006, §9.8). But I'm not sure I yet have a sufficiently good grip on the place of these theorems in the scheme of things to give an illuminating account of the motivations here.

Indeed, the Adjoint Functor Theorems arguably sit at the boundary between basic category theory and the beginnings of more serious stuff. So given the intended limited remit of *these* Notes, this is in any case the point at which I should probably stop.

And what comes next?

"So [said the doctor]. Now vee may perhaps to begin. Yes?"
Philip Roth, *Portnoy's Complaint.*

We have only touched on some of the very basics of category theory. But that's fine: the aim of the exercise here has been to take some elementary steps at a gentle pace, and to make things as clear as I currently can do.

We have now got to a pretty natural place to pause and draw breath. But if you want to begin more serious study of category theory, where next? Here are just some initial pointers.

First, to consolidate and deepen the treatment of the topics we have touched on in these notes, the best place to go is Tom Leinster's terrific *Basic Category Theory* (2014) which covers the same topics faster, in a different order, and with more detail. At a similar level of sophistication – rather less well organized, though taking in some related topics of special interest to logicians – there is also Steve Awodey's *Category Theory* (2006).

Moving on to new, more advanced topics, the next steps along one way forward are those traditionally traced out in the Cambridge Part III course: see for example Julia Goedecke's excellent notes 'Category Theory' (2013). Here, after speeding for three chapters over the ground covered in these notes, she has chapters on Monads (Awodey too finishes his book with a brief treatment), and on Abelian Categories. And then, with that further preparation, you could begin to tackle Saunders Mac Lane's tougher classic *Categories for the Working Mathematician* (1997).

Or you could jump from where we are straight to the beginnings of the usual companion Cambridge course, on topos theory. Here one introductory option is Colin McLarty's *Elementary Categories, Elementary Toposes* (1992); Parts I and II are a speedy review of some of the material we have covered here, again useful for consolidation, and then Parts III and IV are about toposes (or topoi, if you like). But I still prefer Robert Goldblatt's wonderfully lucid *Topoi: The Categorial Analysis of Logic* (originally 1979, reprinted 2006).

You will also surely want to read at a fairly early stage *Sets for Mathematicians* by William Lawvere and Robert Rosebrugh (2003), which gives a category-theoretic approach to basic set theory.

If you get through a selection of those, and/or some of the on-line materials linked at this Category Theory page, you'll then discover for yourself where you want to explore next if you have caught the categorial bug ...

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